



महाराष्ट्र गृहनिर्माण
विकास महामंडळ
(MAHAHOUSING)
(गृहनिर्माण विभाग, महाराष्ट्र शासन)

**MAHARASHTRA HOUSING DEVELOPMENT CORPORATION LIMITED MUMBAI
(GOVERNMENT OF MAHARASHTRA UNDERTAKING)**

TENDER FORM

**Construction of Affordable Housing Project of 1824 EWS Tenements Under Pradhan
Mantri Awas Yojana (U) on Kh. No 59/2 & 60, Bhilgaon, Tal. Kamptee, Dist. Nagpur
(Package – 2)**

**Price: Rs. 5900/- (Rupees Five Thousand Nine Hundred only) including 18% GST
(Non -refundable)**

-OFFICE-

**Maharashtra Housing Development Corporation Limited
Corporate Office: Office No. 32, 3rd floor, Sakhar Bhavan,
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INDEX

Name of Work: Construction of Affordable Housing Project of 1824 EWS Tenements Under Pradhan Mantri Awas Yojana (U) including Allied Infrastructure Amenities on plot no KH 59/2 & KH 60 for PMAY Scheme at Nagpur of village Kamptee. (Package - 2)

Chapter No.	Clause No.	Particulars	Page No.
		DISCLAIMER	8
		GLOSSARY	9
1		E-Tender Notice	13
	1.1	E-Tender Notice.....	13
	1.2	Eligibility Criteria:	16
	1.3	Instructions to Tenderers	19
	1.4	Preparation of Bid/ Tendering Procedure	24
	1.5	Manner of Submission of Tender and its Accompaniments:.....	29
	1.6	Opening of Tenders:	29
	1.7	Award of Work:	32
	1.8	Bid Security:	33
2		Additional Instructions to Tenderer	37
	2.1	Definitions:	37
	2.2	Errors, Omissions, Discrepancies and Interpretations:.....	39
	2.3	Tenderer to Study the Site Conditions and Tender Document:	39
	2.4	Brief Scope of Work:	40
	2.5	Assistance in Procuring Priorities Noc's, Permits, etc:	41
	2.6	Co-ordination with Agencies:	41
	2.7	Relation with Public Authorities:	41
	2.8	Treasure-Trove:	41
	2.9	Patented Device:	42
	2.10	Explosives:	42
	2.11	Quarries.....	42
	2.12	Accident:.....	43
	2.13	Police Protection.....	43
	2.14	Protection / Shifting of Utilities:	44
	2.15	Working Methods and Construction Programme.....	44
	2.16	The Client and his Representative:	45
	2.17	Management of Works by the Contractor:.....	46
	2.18	Site Order Book:	48
	2.19	Line Out of the Project:	49
	2.20	Testing and Rejection of Work:	50
	2.21	Completion of Project:.....	50

2.22	Handing Over of work:	51
2.23	Disputes Resolution in Bidding Process:	51
3	Scope of Civil Work in the Project	52
3.1	Scope of Civil Work	52
3.2	Work Specifications:	58
3.3	Materials and Samples	61
3.4	Infrastructure at Site and up to Site:	65
3.5	Completion and Handing Over:	65
3.6	Green Building (Deleted)	66
3.7	Defective Work and Defect Liability Period (DLP):	66
4	Design Criteria for Structural Designs of Buildings	68
4.1	General Requirements:	68
4.2	Restriction on Type of Structures:	68
4.3	Requirements and Data:	69
4.4	Verification of Survey Data:	70
4.5	Obligatory Provisions:	71
4.6	Open Footings / Columns:	73
4.7	Expansion Joints:	74
4.8	Reinforcement Steel:	74
4.9	A Certification regarding Centering and Steel Checking:	74
4.10	Non-Destructive Testing:	74
4.11	Structural stability certificate	74
4.12	Specifications/Instructions for Alternative Building Technologies	74
5	Scope of Electrical Work	76
5.1	Brief Scope of Electrical Work:	76
5.2	General Conditions:	76
5.3	Scope of Electrical Work:	77
5.4	Execution:	78
5.5	Completion and Handing Over:	80
5.6	Work Specifications:	80
5.7	Permanent Electrical Supply Arrangement and Infrastructure:	81
5.8	Materials and Samples:	82
6	Schedule– C	83
I.	Building Work – Civil	83
II.	Infrastructure works – Civil	94
III.	Building Internal Plumbing & Sanitation	123
IV.	Building Plumbing & Sanitation and External works	142
V.	Internal Electrical works	150
VI.	Building Internal Electrical Item Wise Specification	163
VII.	Infra Electrical Item Wise Specification	178

VIII.	Firefighting Item Wise Specification	214
IX.	External Water Supply, Drainage, Pumps and Panels	221
	PART B - Alternative Building Technologies.....	235
7	List of Approved Material for Civil and Interior Works	238
8	List of Approved Material for Electrical Works	245
9	Payment Schedule	252
9.1	Preamble	252
9.2	Initial Measurement of Work:	252
9.3	Payments:.....	252
9.4	Final Bill:	254
9.5	Payment for Additional Work:.....	255
9.6	Payment Schedule.....	256
10	Form of Offer	261
10.1	Memorandum of Understanding:.....	261
10.2	Proforma of Agreement.....	263
11	Price Variation Clause (Price Adjustment)	266
12	Conditions of Contract.....	267
12.1	Definition of Work/Project:	267
12.2	Acceptance of Conditions & Conditional Tenders Liable for Rejection:	267
12.3	Contractor's Invalid Conditions	267
12.4	Security Deposit.....	267
12.5	Termination of Contract	268
12.6	Action for Unsatisfactory Progress of any Particular Portion of the Work	269
12.7	Compensation if action not taken under Clause 12.5 AND 12.6.	270
12.8	Power Regarding Contractor's Plant.....	270
12.9	Compensation Under Workmen's Compensation Act.....	271
12.10	Compensation Under Medical Aid, Safety & Apprentices Act.....	271
12.11	Insurance Cover for the Work - Delated	272
12.12 A.	Compensation for Delay of Work:	274
12.12 B	Incentive Clause for Early Completion of Work:.....	274
12.12 C	Mobilization Advance:	275
12.12 D	Interest & Recovery:.....	275
12.13	Action to Recover Liquidated Damages.	275
12.14	Action & Compensation for Inferior Quality of Materials / Workmanship:.....	276
12.15	Penalties/ Fine	277
12.16	Works to be executed in accordance with Specifications Drawings, Orders, Etc.	277
12.17	Alternation in drawings or specifications, etc. not to invalidate contracts:.....	278
12.18	No Claim or Compensation for Alteration in or Restriction of Work:	279
12.19	Claim for Compensation for Delay in Starting the Work	280
12.20	Action when Contractor(S) becomes Defaulter:	280

12.21	Action where no Specifications	281
12.22	Time Limit for Unforeseen Claims:	281
12.23	Claim for Quantities (Variation) & Variation not to Vitate Contract	281
12.23 A)	Sufficiency of Schedule of Quantities:	281
12.24	Extension of Time on Account of Alterations Etc:	282
12.25	Priority of Documents.....	283
12.26	Method of Payment:.....	283
12.27	Advances:.....	283
12.28	Bills to be on Printed Form.....	283
12.29	Bills to be Submitted Monthly.....	284
12.30	Payment at Reduced Rates	284
12.31	Payment on Interim Certificate to be regarded as Advances.....	284
12.32	Completion Certificate in the form of Final Bill:.....	285
12.33	Final Certificate:.....	285
12.34	Maintenance and Repairs during Guarantee Period:.....	286
12.35	Decision of PMC/Client shall be Final:	286
12.36	Direction and Control of PMC/Client	286
12.37	Control over Works & Removal of Employee:	287
12.38	Work to be open to Inspection.	288
12.39	Notice to be given before work is covered up.....	288
12.40	Quality Assurance Plan.....	288
12.41	Testing of Material or any type of Concrete Element:.....	289
12.42	Minimum Age and the Payments of Fair Wages:	289
12.43	Work on Sundays or Weekly Off:	290
12.44	Employment of Scarcity Labour.	290
12.45	Employment of Local Labour:	290
12.46	Employment of Female Labour	290
12.47	Employment of Famine Labour:	290
12.48	Carpenter/Black-Smith Etc.....	291
12.49	Minimum Wages.....	291
12.50	Government Circular No. Cat-1284/ (120)/Bldg.2 DT.14.8.85.	291
12.51	Cess on Building & Construction Labour Welfare.....	292
12.52	Conditions for Malaria Eradication Anti Malaria and other Health Measures	292
12.53	Responsibilities of Contractor(S) for Damage by Fire, etc.:	293
12.54	Measure for Prevention of Fire.....	293
12.55	Liabilities of Contractor(S) for any Damages done in/out side of the Work Area:	293
12.56	Contractor Liable for Damage done and for Imperfections.	294
12.57 A)	Contractor to Supply Plant, Ladders Scaffoldings, etc.	295
12.57 B)	Contractor to Supply Hoisting Machine	296
12.57 C)	Contractor to Supply Fencing and Light	297

12.58	Sum Payable by Way of Compensation	298
12.59	Entering upon or Commencing any Portion of Work.....	298
12.60	Changes in Constitution of Firm to be Notified	298
12.61	Work not to be Sub-Late:	298
12.62	Guarantee for Durability of the Structure for Design Life:	299
12.63	Taxes Rates and Cess:.....	299
12.64	Quarry Fees, Royalties, etc.:.....	299
12.65	Durability of the Structure:.....	299
12.66	Settlement of Disputes:.....	300
12.67	Jurisdiction.....	301
13	Additional General Specification.....	302
13.1	Preamble	302
13.2	Pre-Construction Activities	302
13.3	General Specifications for Interior Furnishing Work	303
13.4	Non-Destructive Tests:	306
13.5	Street Light:	307
13.6	Additional Technical Specifications for STP	308
13.7	Additional Technical Specifications for WTP	313
13.8	Additional Technical Specifications for OWC	314
14	Labour Regulations	316
14.1	Preamble:	316
14.2	Employees State Insurance Contribution:	317
14.3	Employees Provident Fund and Misc. Provisions Act 1952:.....	317
14.4	Weekly Day of Rest:	317
14.5	Display of Notice Regarding Wages, Weekly Day, Day of Rest Etc.:	318
14.6	Fixation of Wage Periods:.....	318
14.7	Register of Workmen:	318
14.8	Employment Card:	318
14.9	Register of Wages Etc.:	318
14.10	Preservation of Registers:.....	319
14.11	Appeal against Decisions of Inspecting Officer:	319
14.12	Representation of Parties:	319
14.13	Inspection of Registers and other Documents:.....	319
14.14	Amendments:	319
14.15	Model Rules for Labour Welfare:.....	320
14.16	Accommodation for Labour:	320
14.17	Scale of Accommodation in Latrines and Urinals:	320
14.18	Latrines and Urinals:	320
14.19	Construction of Latrines:	321
14.20	Provision of Shelter During Rest:	321

14.21	Creches:	321
14.22	Canteen	322
14.23	Enforcement:	322
14.24	Interpretation:	322
15	Construction Safety Practices.....	327
15.1	Safety Code:.....	327
16	Standard Format	328
16.1	(Format of Declaration)	328
16.2	Declaration of the Contractors:	329
16.3	Indemnity Bond:.....	330
16.4	Form of Bank Guarantee Bond (EMD)	332
16.5	Form of Bank Guarantee Bond:(SD)	334
16.6	Guarantee Bond for Anti-termite / Water Proofing Treatment.....	336
16.7	A Check list for Final Bill Processing:.....	337
16.8	Check list for Handing over of the Project:	364
17	Special Condition of Contract	366
17.1	General.....	366
17.2	Special Condition of Contract for MEP	366
18	Forms.....	376
18.1	Form I A: Details of list of works in hand as on submission of Tender	376
18.2	Form I B: List and Details of Tendered Works.....	377
18.3	Form II: List of Plant and Machinery:.....	378
18.4	Form III: Details of work of similar type and magnitude:	379
18.5	Form IV: Details if Technical Personals available.....	380
18.6	Form No- V: Statement Showing Payment Received	381
19	Drawings	382
19.1	Contract drawings: -.....	382
19.2	Documentation:.....	382

DISCLAIMER

The information contained in this Request for Proposals document ("TENDER") or subsequently provided to Applicants, whether verbally or in documentary or any other form by or on behalf of the Maharashtra Housing Development Corporation Limited, Maharashtra or any of its employees or advisers, is provided to Applicants on the terms and conditions set out in this TENDER and such other terms and conditions subject to which such information is provided.

This TENDER is not an agreement and is neither an offer nor invitation by the MHDC to the prospective Applicants or any other person. The purpose of this TENDER is to provide interested parties with information that may be useful to them in the formulation of their Proposals pursuant to this TENDER. This TENDER includes statements, which reflect various assumptions and assessments arrived at by the MHDC in relation to the Consultancy. Such assumptions, assessments and statements do not purport to contain all the information that each Applicant may require. This TENDER may not be appropriate for all persons, and it is not possible for the MHDC, its employees or advisers to consider the objectives, technical expertise and particular needs of each party who reads or uses this TENDER. The assumptions, assessments, statements and information contained in this TENDER, may not be complete, accurate, adequate or correct. Each Applicant should, therefore, conduct its own investigations and analysis and should check the accuracy, adequacy, correctness, reliability and completeness of the assumptions, assessments and information contained in this TENDER and obtain independent advice from appropriate sources.

Information provided in this TENDER to the Applicants is on a wide range of matters, some of which depends upon interpretation of law. The information given is not an exhaustive account of statutory requirements and should not be regarded as a complete or authoritative statement of law. The MHDC accepts no responsibility for the accuracy or otherwise for any interpretation or opinion on the law expressed herein.

The MHDC, its employees and advisers make no representation or warranty and shall have no liability to any person including any Applicant under any law, statute, rules or regulation or tort, principles of restitution or unjust enrichment or otherwise for any loss, damages, cost or expense which may arise from or be incurred or suffered on account of anything contained in this TENDER or otherwise, including the accuracy, adequacy, correctness, reliability or completeness of the TENDER and any assessment, assumption, statement or information contained therein or deemed to form part of this TENDER or arising in any way in this Selection Process.

The MHDC also accepts no liability of any nature whether resulting from negligence or otherwise, howsoever caused, arising from reliance of any Applicant upon the statements contained in this TENDER.

The MHDC may in its absolute discretion, but without being under any obligation to do so, update, amend or supplement the information, assessment or assumption contained in this TENDER.

The issue of this TENDER does not imply that the MHDC is bound to select an Applicant or to appoint the Selected Applicant, as the case may be, for the Consultancy and the MHDC reserves the right to reject all or any of the Proposals without assigning any reasons whatsoever.

The Applicant shall bear all its costs associated with or relating to the preparation and submission of its Proposal including cost which are not limited to preparation such as copying, postage, delivery fees, expenses associated with any demonstrations or presentations which may be required by the MHDC or any other costs incurred in connection with or relating to its Proposal. All such costs and expenses will remain with the Applicant and the MHDC shall not be liable in any manner whatsoever for the same or for any other costs or other expenses incurred by an Applicant in preparation or submission of the Proposal, regardless of the conduct or outcome of the Selection Process.

Signature of Bidder

Date:

For MHDC

Date:

GLOSSARY

A Glossary of the different acronyms & terms used in the document is given below with the reference of individual acronyms/terms. These are to be used in relation to the interpretation of the acronyms/terms mentioned in the TENDER.

ACRONYM	REFERENCE TO
AC	Alternating Current
ACC	Associate Cement Company
BBM	Bituminous Bound Macadam
BM	Bituminous Macadam
CAR	Contractors All Risk
CC	Cement Concrete
CC	Completion Certificate
CD	Cross Drainage
CFL	Compact Fluorescent Lamp
CIDCO	City Industrial Development Corporation
CM	Cement Mortar
CPVC	Chlorinated Polyvinyl Chloride
CPWD	Central Public Works Department
DB	Department Distribution Board
DI	Ductile Iron
DILR	District Inspector of Land Record
DLP	Defect Liability Period
DP	Development Plan
EMD	Earnest Money Deposit
ESR	Elevated Storage Reservoir
FRC	Fibre Reinforced Concrete
FRP	Fibre Reinforced Polymer
GI	Galvanized Iron
GRP	Glass Reinforced Polymer
GSR	Ground Storage Reservoir
Govt	Government
HDPE	High Density Polyethylene
HSD	High Speed Diesel
HT	High Tension
ISNP	Indian Standard Normal Pressure.
KVA	Kilovolt-ampere
KW	Kilowatt
KV	Kilovolt
LCD	Liquid Crystal Display
LT	Low-tension
MB	Measurement Book
MCB	Miniature Circuit Breaker

MES	Military Engineering Service
MHDC	Maharashtra Housing Development Corporation Limited
MHADA	Maharashtra Housing & Area Development Authority
MJP	Maharashtra Jeevan Pradhikaran
MPCB	Maharashtra Pollution Control Board
MMRDA	Maharashtra Metropolitan Regional Development Authority
MOU	Memorandum of Understanding
MS	Mild Steel
MSEDCL	Maharashtra State Electrical Distribution Company Limited
MSRDC	Maharashtra State Road Development Corporation
MT	Metric Ton
NBC	National Building Code
OC	Occupation Certificate
OPC	Ordinary Portland Cement
PMC	Project Management Consultant
POL	Petrol Oil Lubricant
PPC	Portland Pozzolana Cement
PRB	Permanent Register of building of PWD
PVC	Polyvinyl Chloride
PWD	Public Works Department
RA Bills	Running Account Bills
RCC	Reinforced Cement Concrete
RMT	Running Meter
SD	Security Deposit
SPNDB	Single Pole & Neutral Distribution Board
SWG	Stone Ware Glazed
TBM	Temporary Benchmark
TDR	Tax Deduction Receipt
TDS	Tax Deduction @Sources
TILR	Taluka Inspector of Land Record
TMT	Thermo-Mechanical Treated
TPNDB	Triple Pole & Neutral Distribution Board
TW	Teak Wood
UCR	Uncoursed Rubble Masonry
UPVC	Ultra-Polyvinyl Chloride
VAT	Value Added Tax
WBM	Water Bound Macadam
WC	Water Closet

ई-निविदा सूचना

कामाचे नाव:- खा.क्र. ५९/२ व ६०, मौजे भिलगाव, ता. कामटी, जि. नागपूर येथील भूखंडावर प्रधानमंत्री आवास योजनेअंतर्गत (शहरी) परवडणा-या घरांचा १८२४ अर्थिक दृष्ट्या दुर्बल गटातील सदनाकांचा गृहप्रकल्प. (पॅकेज-२)

महाराष्ट्र गृहनिर्माण विकास महामंडळाची दि. १६.०५.२०२३ रोजीची कार्यालयीन जावक क्र. महाहौसिंग/तांत्रिक/४३६/२०२३ अन्वये निवडसूचीमध्ये सुचिबद्ध (Empanelled) असलेल्या योग्य त्या कंत्राटदारांकडून खालील कामांकरिता ई-निविदा प्रणालीद्वारे ऑनलाईन पद्धतीने मागविण्यात येत आहे. ई-निविदा प्रक्रिया खालीलप्रमाणे:-

ई-निविदा सूचना व दिनांक	०५/२३-२४, दिनांक ०६.०६.२०२३
कामाचे नाव	खा.क्र. ५९/२ व ६०, मौजे भिलगाव, ता. कामटी, जि. नागपूर येथील भूखंडावर प्रधानमंत्री आवास योजनेअंतर्गत (शहरी) परवडणा-या घरांचा १८२४ अर्थिक दृष्ट्या दुर्बल गटातील सदनाकांचा गृहप्रकल्प. (पॅकेज-२)
अंदाजित किंमत (GST वगळून)	रु. १८२,९३,७९,३५९/-
काम पूर्ण करण्याचा कालावधी	१५ महिने (मान्सून सह)
इसारा रक्कम	रु. ९१,४६,८९७/- इसारा रक्कम फक्त ऑनलाईन पद्धतीनेच
सुरक्षा ठेव	ई-निविदाधारकाने देऊ केलेल्या आणि महाहौसिंगने स्विकृत केलेल्या ई-निविदा किंमतीचा १%
संकेतस्थळावरून ई-निविदा उपलब्ध कालावधी	दि.०६.०६.२०२३ वेळ ११.०० पासून ते दि.२७.०६.२०२३ वेळ १६.०० पर्यंत
ऑनलाईन ई-निविदा सादर करण्याचा अंतिम कालावधी	दि.२७.०६.२०२३ वेळ १६.०० पर्यंत
निविदा पुर्व बैठकीतील शंका सादर करण्याचा अंतिम कालावधी	दि. १३.०६.२०२३ वेळ १६.०० पर्यंत (ई-मेल:-mhdc.mahahousing@maharashtra.gov.in)
निविदा पुर्व बैठक	दि.१४.०६.२०२३ वेळ १५.००, महाहौसिंग कार्यालय, मुंबई
ई-निविदेचा तांत्रिक लिफाफा उघडण्याचा दिनांक	दि.२९.०६.२०२३ वेळ १७.०० (शक्य झाल्यास)
ई-निविदेचा आर्थिक लिफाफा उघडण्याचा दिनांक	तांत्रिक लिफाफ्यातील पात्र कंत्राटदारांना कळविण्यात येईल.
ई-निविदा शुल्क	रु.५,९००/- न परतावा रक्कम (GST सहीत) ऑनलाईन पद्धत
ई-निविदा भरण्याचे संकेतस्थळ	https://mahatenders.gov.in

टिप:-

- सर्व पात्र /इच्छुक निविदाकारांनी निविदापत्रक डाऊनलोड करण्यासाठी व निविदा प्रक्रियेत भाग घेण्यासाठी ई-निविदा प्रणालीच्या <https://mahatenders.gov.in/> वर enrolled करणे आवश्यक आहे.
- निविदाकारांना वर नमुद केलेल्या संकेतस्थळावर ऑनलाईन निविदा सादर करणे संदर्भात काही शंका/अडचणी असल्यास त्यांनी खालील दुरध्वनी क्रमांकावर संपर्क साधावा (०१२०-४००१ ००२, ०१२०-४००१ ००५, ०१२०-६२७७ ७८७).
- निविदाकारांनी निविदा संदर्भात सर्व दस्तऐवज ऑनलाईन सादर करणे अनिवार्य राहिल.
- इतर अटी व शर्ती ई-निविदा नमुन्यामध्ये पहावयास मिळतील. सदर कामांची एक अथवा सर्व ई-निविदा कोणतेही कारण न देता रद्द करण्याचे अधिकार सक्षम अधिकारी यांनी राखून ठेवलेले आहेत.
- ई निविदेची किंमत (ना परतावा) व इसारा रक्कम फक्त ऑनलाईन पद्धतीनेच स्विकारली जाईल.

**AJIT
MALHARI
KAWADE**
Digitally signed
by AJIT MALHARI
KAWADE
Date: 2023.06.06
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(अजित कवडे)
व्यवस्थापकीय संचालक तथा मुकाअ
महाराष्ट्र गृहनिर्माण विकास महामंडळ

SHORT E - TENDER NOTICE

Name of Work: Construction of Affordable Housing Project of 1824 EWS Tenements Under Pradhan Mantri Awas Yojana (Urban) including Allied Infrastructure Amenities on Land Bearing Kh. No 59/2 & 60 at Bhilgaon, Tal. Kamtee, Dist. Nagpur. (Package-2)

The Maharashtra Housing Development Corporation Ltd. (MHDC) invites tenders from MHDC empanelled agencies/contractor/company which are empanelled in MHDC vide Office Order No. **MHDC/Tech/436/2023 Dt 16.05.2023** uploaded on MHDC website. Programme of bidding shall be as follows

Tender Notice No. & Date	05/23-24, dated 06.06.2023
Name of the work	Construction of Affordable Housing Project of 1824 EWS Tenements Under Pradhan Mantri Awas Yojana (U) including Allied Infrastructure Amenities on Land Bearing Kh. No 59/2 & 60 at Bhilgaon, Tal. Kamtee, Dist. Nagpur. (Package-2)
Estimated Cost Put to Tender (Excluding GST)	Rs. 182,93,79,359/-
Time of completion	15 Months including Monsoon
Earnest Money Deposit	Rs. 91,46,897/- should be paid through online gateway mode
Security Deposit	1 % (One Percent) of the tendered cost offered by Tenderer and Accepted by Client.
Downloading of blank Tender from Website	Dt. 06.06.2023 from 17.00 Hrs. Till Dt. 27.06.2023 up to 16:00 Hrs.
Last date of Online submission of Tender	Dt. 27.06.2023 up to 16:00 Hrs.
Last date of receipt of Pre-Bid Queries	(Pre-Bid queries shall reach this office on or before Dt 13.06.2023 till 16:00 Hrs. on mhdc.mahahousing@maharashtra.gov.in)
Pre-Bid Meeting	Dt 14.06.2023 at 15.00 Hrs. at MHDC Office, Mumbai
Technical Bid Opening Date	Dt 29.06.2023 at 17:00 Hrs. (if possible)
Financial Bid Opening Date	It will be communicated at the time of opening of Eligibility-cum-Technical Bid
Blank Tender Fees to be paid in DD Form before the last date of submission of tenders	Rs 5,900/- Non-Refundable (Inclusive of GST) should be paid through online gateway mode
Website for Downloading & Uploading	https://mahatenders.gov.in

Note:-

1. Detailed Tender Notice and Blank Tender form can be downloaded from the <https://mahatenders.gov.in>
2. Instructions to the Contractors/Bidders for the e-submission of the bids online through this tender site: <https://mahatenders.gov.in>
3. For any assistance on the use of Electronic Tendering System, the Users may call the following numbers: Helpdesk No.- 0120-4001 002, 0120-4200 462, 0120-4001 005, 0120-6277 787 E-Tender Portal: - <https://mahatenders.gov.in>
4. The MHDC reserves the right to accept or reject any or all the tenders without assigning any reason thereof.
5. The cost of Blank tender form Fees and EMD to be paid via. Online mode only.

AJIT MALHARI KAWADE

Digitally signed by AJIT MALHARI KAWADE

Date: 2023.06.06 10:49:27 +05'30'

(Ajit Kawade)
Managing Director & CEO
Maharashtra Housing Development Corporation

1 E-Tender Notice

1.1 E-Tender Notice

Maharashtra Housing Development Corporation Ltd., Mumbai, (hereinafter referred to as MHDC / Maha Housing / Client / Corporation), invites Online e-tenders from MHDC empanelled agencies / contractors who are empanelled in MHDC and eligible contractor who fulfils the eligibility criteria as specified under clause no. 1.2.

I) Name of Work: Construction of Affordable Housing Project of 1824 EWS Tenements Under Pradhan Mantri Awas Yojana (U) on Kh. No 59/2 & 60, Bhilgaon, Tal. Kamptee, Dist. Nagpur (Package – 2)

II) Land / Plot Details:

KH. No. 59/2 & 60 of Village Mauje, Kamptee Nagpur. (39, 200 Sq. Mt).

III) Details of the Project:

As per layout and plans enclosed with tender.

Total Built-up area/ construction area of the project as per the Table below.

IV) Estimated Cost of the Project:

Estimated Cost of the project is **Rs 1,82,93,79,359/- (Excluding 18% GST)**

BUILT-UP AREA / CONSTRUCTION AREA STATEMENT

EWS - TYPE 1							
Sr.No.	Floor	No. of Floors	Total no. of Units	RERA Carpet Area per Unit Sq. Mts	Total RERA Carpet Area (Sq. Mts.)	Total Construction Area (Sq. Mts.)	Total BUA (sq. m)
1	Ground Floor	1	0	0	0	813.01	130.03
2	Typical Floor (1st-7th, 9th-12th, 14th)	12	240	29.07	6,976.8	9,756.12	9601.8

3	8th Refuge Floor	1	18	29.07	523.26	813.01	735.24
4	13th Refuge Floor	1	18	29.07	523.26	813.01	735.24
	Total	15	276		8,023.32	12,195.15	11,202.31
EWS - TYPE 2							
Sr.No.	Floor	No. of Floors	Total no. of Units	RERA Carpet Area per Unit Sq. Mts	Total RERA Carpet Area (Sq. Mts.)	Total Construction Area (Sq. Mts.)	Total BUA (sq. m)
1	Ground Floor	1	0	0	0	964.86	130.06
2	Typical Floor (1st-7th, 9th-12th, 14th)	12	288	29.07	8,372.16	11,578.32	11424
3	8th Refuge Floor	1	22	29.07	639.54	964.86	887.14
4	13th Refuge Floor	1	22	29.07	639.54	964.86	887.14
	Total	15	332		9,651.24	14,472.9	13,328.34

AMENITY BUILDING				
Sr.No.	Floor	No. of Floors	Total Construction Area (Sq. Mts.)	Total BUA (sq. m)
1	Ground Floor	1	553.46	543.25
2	First Floor	1	553.46	543.25
3	Second Floor	1	553.46	543.25
	Total	3	1,660.38	1,629.74
FITNESS CENTER IN RG-4				
Sr.No.	Floor	No. of Floors	Total Construction Area (Sq. Mts.)	
1	Ground Floor	1	135.06	
2	First Floor	1	67.00	
	Total	2	202.06	
CRECHE IN RG-1				
Sr.No.	Floor	No. of Floors	Total Construction Area (Sq. Mts.)	
1	Ground Floor	1	121.69	
2	First Floor	1	60.74	
	Total	2	182.83	
CRECHE IN RG-2				
Sr.No.	Floor	No. of Floors	Total Construction Area (Sq. Mts.)	
1	Ground Floor	1	30.00	

	Total	1					30.00
<u>TOTAL AREA STATEMENT</u>							
Building Type	No. of Bldg.	Total RERA carpet Area (Sq. mts)		Total Construction (Sq. mts)		Total BUA (Sq. mts)	
		Per Bldg.	Total	Per Bldg.	Total	Per Bldg.	TOTAL
EWS TYPE-1	3	8,023.32	24,069.96	12,195.15	36,585.45	11,202.31	33,606.93
EWS TYPE-2	3	9,651.24	28,953.72	14,472.90	43,418.70	13,328.34	39,985.02
AMENITY BUILDING	1	-	-	1,660.38	1,660.38	1,629.74	1,629.74
FITNESS CENTER RG-4	1	-	-	202.06	202.06	0.00	0.00
CRECHE IN RG-1	1	-	-	182.43	182.43	0.00	0.00
CRECHE IN RG-2	1	-	-	30.00	30.00	0.00	0.00
Total	10	17,674.56	53,023.68	28,742.92	82,079.02	26,160.39	75,221.69

(Above construction, area is inclusive of architectural projections, Staircase)

The Construction area mentioned is for Residential and Commercial structures excluding the construction area of underground water tank, overhead water tank, STP, WTP, security cabin, pump room, transformer/DG room etc.

1. Above Built-up Area is not to be taken as measurement for payment.
2. The Tenderer shall fully satisfy himself regarding of Built-Up Areas, if any discrepancy is noticed between drawings and built-up area statement as above, by the tenderer, it shall be brought to the notice of the Client before or at the time of the Pre-Bid conference.
3. All items of work, part of work or work itself shown either on the drawings or mentioned in the tender document or vice a versa are to be executed by the contractor. Nonappearance of any of the item either in the drawings or in the tender or vice a versa shall not vitiate the scope of the work or contract.
4. Time limit for completion of the work shall be **15 months** including monsoon from the date of issuance of the work order.
5. The successful Tenderer shall construct all the buildings of said work / project shall be executed as per Tender drawing and specification.
6. The said work shall be executed under supervision of the PMC/Client as per terms and conditions of the contract.
7. The tender shall submit their **Lump sum offer (Excluding all Taxes & GST)** for construction of the said work / project including all infrastructure amenities as mentioned in 1.1. (III) above and as per the scope of work mentioned in the document.

8. The work shall be carried out as per the tender drawings, document, specifications and instructions including obtaining necessary permissions, No Objection Certificates (NOC), Completion Certificates, Licenses, from various competent authorities.
9. The Contractor shall initially pay all fees/ deposits (refundable/ non-refundable) / charges/ supervision charges/ any other charges, etc. if any, required to be paid to various local/public/statutory authorities (such as Municipal Corporation, Maharashtra State Electricity Board and or its subsidiaries, Irrigation Dept. etc.) on behalf of the Client, as a requirement and up to completion of the said work/Project including all contractual obligations/scope etc. within stipulated time period and to the satisfaction of Client. The Client shall there after reimburse eligible payments on production of original receipts. The Client shall also require recommendation of the PMC for reimbursement of such payment and for their verification if necessary. No claim for extension in time limit shall be considered against the delay in payment in any case. The contractor shall ensure to pay the said charges due on time.

1.2 Eligibility Criteria:

- I. To be eligible for participation in the tendering the Tenderer shall have to pay Rs. 5,900/- (Rupees Five thousand nine hundred only), (Including GST) as the tender form fee and **Rs 91,46,897/- (Ninety-One Lacs Forty-Six Thousand Eight Hundred and Ninety-seven only)** as Earnest Money Deposit should be paid through online gateway mode and Produce Evidence of the Same.

To qualify for opening of the Financial Bid, the bidder shall submit the documentation demonstrating his qualification of the Technical and Financial Eligibility criteria as mentioned below.

- a) Work Experience: The bidder must have experience of having successfully completed works during the last 5 years ending previous day of last date of submission of tender:

Three similar completed works, each costing not less than the amount equal to 40% of the estimated cost put to tender viz., Rs.73.18 Cr

OR

Two similar completed works, each costing not less than the amount equal to 50% of the estimated cost put to tender viz., Rs. 91.47 Cr

OR

One similar completed works, each costing not less than the amount equal to 60% of the estimated cost put to tender viz., Rs. 109.76 Cr.

Note:

- i) The value of executed work shall be brought to the current costing level by enhancing the actual value of work at simple rate of 10% per annum, calculated from the date of completion of the projects to the last date of the month prior to the date of Tender.)

The experience certificate for the work shall be certified by the Officer of rank not less than Executive Engineer from the concerned Govt / Semi Government Department.

- ii) In case of private works, work completion certificates shall be certified by the client and project architect along with copy of completion certificates/occupation certificates issued by the concerned statutory authority and TDS certificate substantiating the cost of the project.
- b) Turnover: Average annual financial turnover of the bidder for the construction works should be at least 50% of the estimated cost put to tender during in any one year in the immediate last 3 consecutive financial years viz., Rs. 91.47 Cr
- c) Solvency Certificate: Solvency of the amount equal to 10% of the Estimated Cost put to tender (ECPT) viz., Rs 18.29 Cr
- d) Bidding Capacity as mentioned in this para shall be more than estimated cost of project. Bidding Capacity = $[A \times N \times 2] - B$, Where,

A = Maximum value of construction works executed in any one year during the last five years considering the completed as well as works in progress by enhancing the actual value of works at simple rate of 7% per annum.

N = Number of years prescribed for completion of work for which criteria has been invited (1.25 for this tender)

B = Value of existing commitments and on-going works and work issued to be Completed during the period of completion of work for which criteria have been invited.

Note:

- I. Maximum Annual Turnover for Building and Infrastructure Construction Works only will be considered from Form-V and checked with Audited Balance sheet certified & Signed by the Chartered Accountant. (Along with UDIN)
- II. The maximum turnover for Building and Infrastructure Construction Works shall consist of only construction cost without
- III. land cost, development cost etc. (in case of Flat / shop / offices sales) with certified by Chartered Accountant. (UDIN on certificate is required)
- IV. The existing Commitments i.e., work in hand will be taken in the Form -. I (A) and checked with Form - V, Column no. 13, showing the value of existing commitment and ongoing works as well as the stipulated period of completion, remaining for each of the works listed shall be signed by the bidder/ tenderer and counter signed by the authority / agency/ Executive Engineer or equivalent.

- V. The Tenderer shall upload Audited Certified Balance Sheet under the hand and seal of Chartered Accountant for five years i.e., 2017-18, 2018-19, 2019-20, 2020-21, 2021-2022 for verification with UDIN no. Payment receipt will be verified as per 26AS Form.
- VI. Bid capacity shall be calculated as above. Details of the payment received (Receipt) in Building and Infrastructure Construction Works only during last seven financial years shall be given in Form No. V, which should be signed and certified with seal by Chartered Accountant with UDIN no.

In addition, following documents should be uploaded: -

- a) Undertaking in the prescribed format as given in Form No. VII Chapter 16, declaring the truthfulness of the documents and Form No. VI Chapter No. 16. Declarations of the Contractors conversant with by laws and site on stamp paper of Rs. 100/- each shall be submitted for eligibility/qualification criteria.
- b) Income Tax Returns for last five financial years i.e., 2017-18, 2018-19, 2019-20, 2020-21, 2021-2022.
- c) Valid empanelment Registration issued by MHDC.
- d) Valid registration colour copy of company identification PAN, GST, ESIC, EPF & Professional Tax.
- e) The Bidder / Contractor shall upload TDS (26AS) certificate/ document during last 6 years i.e., F.Y. 2016-17, F.Y. 2017-18, F.Y. 2018-19, F.Y. 2019-20, F.Y. 2020-21, F.Y. 2021-22 and from current financial year F.Y. 2021-22.
- f) The power of attorney in case of a company or Partnership firm.
- g) All above documents should be submitted with proper page no. & indexing.

Notes:

- i. Certificates as mentioned in qualification criteria are required to be obtained from the officer not below the rank of Executive Engineer or equivalent officer. Original certificates shall be scan and uploaded.
- ii. **Form No. I to VIII** enclosed in **chapter no. 16** shall be filled in and signed properly. If these forms are found incomplete, incorrect, kept blank or wrongly filed, tenderer's envelope no.2 (financial bid) shall not be opened.
- iii. The abstract of work done in Building and Infrastructure Construction Works during last seven years shall be given in **Form no. V** should be signed by Chartered Accountant along with **UDIN & (Clause no. 1.2 Qualifying Criteria)**.
- iv. The Contractor or his identified sub-contractor should possess required valid electrical license for executing the building electrification works and should have executed similar electrical works. The details shall be furnished before undertaking work of electrification.

- v. List of documents for qualification criteria is attached in clause no. 1.6.8 for reference.
- 1.2.1 Even though the tenderers meet the above qualifying criteria, they are liable to be disqualified and their financial bid shall not be opened, if they have:
- a. Made misleading or false representations in the forms, statements and attachments submitted in proof of the qualification requirements and/or
 - b. Record of poor performance such as abandoning the works, not properly completing the contract, inordinate delays in completion, litigation history or financial failures, etc.
 - c. Participated in the previous bidding for the same work and had quoted unreasonably high bid price and could not furnish rational justification to the employer.
 - d. Adverse past performance of tenderer regarding quality, workmanship, timely completion and directly/ indirectly subletting of works.
 - e. Any company/ entity blacklisted by the competent authority is prohibited in participating in this tender.
 - f. The forms and all documents shall be uploaded such that the same shall be readable. (Non- readable document will not be considered & financial bid of bidder will not be opened).

1.3 Instructions to Tenderers

- i) The tender document can be downloaded from the aforesaid e-tendering portal by the prospective tenderer for free of cost. But to participate in the e-Tender process the tenderer has to pay through online gateway mode, This Tender fee is non-refundable.
- ii) Client shall not be responsible for any delay in submission of tender.
- iii) Address for Communication/queries:
Managing Director & CEO, Maharashtra Housing Development Corporation Ltd., Office No. 32, 3rd floor, Sakhar Bhavan, Ramnath Goenka Marg, Nariman Point, Mumbai, Maharashtra 400021. Ph. No.: (022) 4044 1000.
- iv) The client reserves the right to reject any or all the tenders without assigning any reasons whatsoever.
- v) Canvassing or lobbying in connection with the tender in any form, if noticed, tender by concern tenderer shall be liable for rejection.
- vi) The tenderer if firm or company in their forwarding letter, shall mention the name and details of all the partners/directors of the firms or the company (as the case shall be) and the name of the partner/director who holds the power of attorney with attested copy of registered power of attorney, authorizing him

- to conduct transaction on behalf of the firms or company.
- vii) The Corporation shall not release any foreign exchange.
 - viii) The tender notice along with all the tender documents including common set of deviation (CSD) shall form part of the contract agreement.
 - ix) The Tenderer shall give a declaration (as per format in Chapter 16, clause 16.1 & 16.2), that he is conversant with local conditions/ laws and availability of labour and materials specifications, tender drawings and he has quoted his offer with the consideration to all these factors.
 - x) The Client reserves right to revise or amend the Tender documents fully or part thereof, prior to the date notified or amended for the receipt to tender. Such deviations /amendments if any shall form part of contract agreement and communicated in the form of corrigendum, common set of deviation (CSD) or by a letter as shall be considered suitable and shall form the part of the Contract Agreement.
 - xi) The Client reserves right to revise layout by relocating the buildings shown in the tender documents (Layout) within the same site and same Built – up- area for which no additional payment shall be made for executing work as per revised layout plan.
 - xii) The tenders containing deviations from the contractual terms and conditions, specifications or other requirement and / or conditional tenders, shall be rejected.
 - xiii) The acceptance of tender will be communicated to the successful tenderer through email or by registered letter or fax by the Client.
 - xiv) If the tender offer is at variance with the estimated project cost by more than (-) 10% (If below more than 10%), then the Tenderer shall furnish additional Security Deposit as per clause 1.8 (B) below. If such additional security deposit is not furnished by the Tenderer as per terms and conditions mentioned in the acceptance letter, his **Earnest Money Deposit** shall be forfeited. Additional security deposit shall be submitted along with initial security deposit for the period of completion of work /project.
 - xv) The tender being uploaded by a firm; the tender shall be signed by each partner thereof or it shall be signed on behalf of firm by a person holding a registered Power of Attorney authorizing him to do so.
 - xvi) In the event of the Tender being submitted by a Company, the Tender shall be signed by a director or it shall be signed on behalf of Company by an authorized representative holding a registered Power of Attorney along with the resolution of the Board authorizing him to do so.
 - xvii) In the event of the Tender being submitted by a sole proprietor, the Tender

shall be signed by the sole proprietor.

- xviii) The online opening of tender shall be as per clause 1.6. In the event of a tender being accepted, the Contractor(s) shall, for the purpose of identification, sign each page of the contract documents.
- xix) The amount of earnest money shall be refunded by the Client to the unsuccessful Tenderers except Lowest three tenderers.
- xx) All corrections and additions or pasted slips in the uploaded documents shall be initiated by the authorized person of the Tenderer.

Tendering programme as mentioned on e-tendering portal:

Sr. No.	Particulars	Date & Time
1.	Publish Date	06.06.2023
2.	Document Download / Sale Start Date	06.06.2023
3.	Document Download / Sale End Date	27.06.2023
4.	Site visit	Till Bid Submission End Date
5.	Pre-Bid Meeting	14.06.2023
6.	Bid Submission Start Date	06.06.2023
7.	Bid Submission End Date	27.06.2023
8.	Eligibility-cum-Technical Bid Opening Date	28.06.2023
9.	Financial Bid Opening Date	It will be communicated at the time of opening of Eligibility-cum-Technical Bid

Notes:

All tenderers to note that, MHDC, reserves the right to extend/change the aforesaid dates.

Pre-Tender conference will be held on 14.06.2023 at 15.00 hours in the Office of the MHDC. Mumbai in the presence of The Managing Director & CEO, Director (Technical), The Executive Engineer, Accounts Officer, Maharashtra Housing Development Corporation and PMC.

A. Instructions for Participating in e-Tenders

- I. Tenderers shall have valid Digital Signature Certificate (DSC) obtained from any Certifying Authorities. In case of requirement of DSC, interested Bidders shall log on to <https://mahatenders.gov.in> and follow the procedure mentioned in the document 'Procedure for application of Digital Certificate'.
- II. For any assistance on the use of Electronic Tendering System, the Users may call the following numbers:
Helpdesk No. - 0120-4001 002, 0120-4200 462, 0120-4001 005, 0120-6277 787.
e-Tender Portal: - <https://mahatenders.gov.in>

III. Tenderers shall install the Mandatory Components available on the Home Page of <https://mahatenders.gov.in> and make the necessary Browser Settings provided under section 'Internet Explorer Settings'

IV. Guidelines to Bidders on the operations of Electronic Tendering System of Maharashtra Housing Development Corporation Ltd.
<https://mahatenders.gov.in/nicgep/app>

A. Pre-requisites to participate in the Tenders processed by Client:

1. Registration of contractors on Electronic Tendering System on <https://mahatenders.gov.in>.

The Tenderer interested in participating in the Tenders of Client processed using the Electronic Tendering System shall be required to enrol on the Electronic Tendering System to obtain User ID and password on the <https://mahatenders.gov.in> Once the ID is generated the Bidder can login into the Tendering portal of Government of Maharashtra. The process of Enrolment is explained in Step.

Registration of New Bidders/Vendor.: Registration link:
<https://mahatenders.gov.in/nicgep/app?page=LoginDetailsandservice=page>

(Or click on "online bidder enrolment" Link at Right side panel)

After submission of application for enrolment on the System, the application information shall be verified by the Authorized Representative of the Service Provider. If the information is found to be complete, the enrolment submitted by the Vendor shall be approved.

The Tenderers may obtain the necessary information on the process of enrolment either from Helpdesk Support Team: may visit the site <https://mahatenders.gov.in>

2. Obtaining a Digital Certificate:
Refer to "Bidders Manual Kit" on home page at mahatenders.gov.in or visit the below given direct link
<https://mahatenders.gov.in/nicgep/app;jsessionid=58E6D1F989D1A5C19DCCD3760DB4F790.mhspeg1?page=BiddersManualKitandservice=page>

B. Instructions to the Contractors/Bidders for the e-submission of the bids online through this tender site: <https://mahatenders.gov.in>

1. Tenderer shall register themselves on <https://mahatenders.gov.in> portal by clicking "Online Bidder Enrolment" and then map Digital Signature certificate.
2. Tenderer shall then login to the site giving user id/ password chosen during registration.
3. The e-token registered shall be used by the Tenderer only and shall not use by any other person.

4. The Tenderers can update well in advance, the documents such as certificates, purchase order details etc., under “My Documents” option and these can be selected as per tender requirements and then attached along with bid documents during bid submission.
5. After downloading/ getting the tender schedules, the Tenderer should go through them carefully and then submit the documents as asked, otherwise, the bid will be rejected.
6. If there are any clarifications, these shall be obtained online through the tender site, or through the contact details. Tenderer shall take into account the corrigendum’s published before submitting the bid online.
7. Tenderer, in advance, shall complete the bid documents to be submitted as indicated in the tender schedule and the same shall be in PDF/XLS/RAR/DWF formats. If there is more than one document, they can be clubbed together.
8. Tenderer shall keep ready transaction receipt of EMD as specified in the tender.
9. The Tenderer shall read the terms and conditions and shall accept the same to proceed further to submit the bids.
10. Tenderer shall read the “Announcements” and “Downloads” section of <https://mahatenders.gov.in> before submitting the Bid.
11. The Tenderer shall submit the tender document online well in advance before the prescribed time to avoid any delay or problem during the submission process.
12. After the bid submission, the acknowledgement number, generated by the e-tendering system shall be printed by the Tenderer and kept as a record of evidence for online submission of bid for that particular tender.
13. Tenderer shall pay EMD through online gateway mode. In case of non-realization of EMD payment Tenderer shall not be allowed to complete his Bid Preparation stage resulting in non-participation in the aforesaid e-Tender.
14. The Tender Inviting Authority (TIA) shall not be responsible for any sort of delay or the difficulties faced during online submission of the bid by the Tenderer.
15. The Tenderer shall submit the bid documents online through the site (<https://mahatenders.gov.in>) as indicated in the tender.
16. The tendering system will give a successful bid uploaded message after uploading all the bid documents and then a bid summary will be shown with the bid no, date and time of submission of the bid with all other relevant details. The documents submitted by the Tenderers will be digitally signed

using the e-token of the Tenderer. The bid summary shall be printed and kept as an acknowledgement of submission of the bid. The bid summary will be a proof of bid submission for a tender floated and shall also be an entry permit to participate in the bid opening date.

17. Bidder shall log in to the site well in advance for bid submission so that he can submit the bid in time i.e., on or before the bid submission end time. For any delay, due to any issue, bidder alone shall be responsible.
18. The bidder shall ensure that the bid documents submitted are free from virus. If the documents uploaded by the Bidder could not be opened, due to virus, during tender opening. The bid shall be treated as invalid.
19. The time settings fixed in the server and displayed at the top of the tender site, will be valid for all actions of requesting, bid submission, bid opening etc., in the e-tender system. The bidders shall follow these time settings during bid submission.
20. The bidder shall logout of the tendering system using the normal logout option available at the top right-hand corner and not by selecting the (X) option in the browser.
21. The bidder shall upload the technical document in. RAR format single file to upload in Eligibility-cum-Technical Bid and then Price Bid in .XLS format, single file, to upload in "Finance Bid".
22. For any other queries, the bidders can contact e-tender Helpdesk support team: - 0120-4001 002, 0120-4200 462, 0120-4001 005, 0120-6277 787.

1.4 Preparation of Bid/ Tendering Procedure

(A) Blank Tender Forms:

Blank tender forms can be downloaded online from the e-Tendering portal i.e., <https://mahatenders.gov.in/> as per the schedule defined in the tender notice.

(B) Language of Bid:

The language of bid shall be English unless otherwise specified.

(C) One Bid per Bidder:

Each bidder shall be submitting only one bid for one project. A bidder who submits or participates in more than one Bid (other than as a subcontractor or in cases of alternatives that have been permitted or requested) shall cause all the bids with the Bidder's participation to be disqualified.

(D) Cost of Biding:

The bidder shall bear all costs associates with the preparation and submission of his Bid, and he shall in no case be responsible and liable for those costs regardless of the conduct or outcome of their Bidding Process.

(E) Site Visit and verification of Information

Bidders are encouraged to submit their Bids after visiting the project site and ascertain for themselves the site conditions, location, surrounding, climate, availability of water, power, other utilities for construction.

(F) Pre-Tender Conference:

- i. A Pre-Tender Conference open to all prospective tenderers will be held at Mumbai on **14.06.2023 at 15:00 hrs.** in the Office of the Client in Presence of Competent Authorities wherein the prospective tenderers will have an opportunity to obtain clarifications regarding the work and the tender conditions. The prospective tenderers are free to ask any additional information or clarification in writing through email addressed at mhdc.mahahousing@maharashtra.gov.in mentioning name of the work up to **13.06.2023 at 16.00** before pre-bid conference. As a result of pre- bid conference modifications to bid document if any here in after referred as “Common set of deviations” (CSD) shall be uploaded on the e-Tendering portal and shall form part of tender documents and which shall be applicable to all tenderers. Any doubts, clarification about discrepancies in specification and drawing or insufficient information should be brought to notice / reported at the time of pre-bid meeting by the bidders. No claims will be entertained during progress of work.
- ii. The tender submitted by tenderer shall be based on the Bid document with CSD issued by the MHDC and his tender shall be unconditional.
- iii. All tenderers to note that, MMHDC reserves the right to extend / change the aforesaid dates.

(G) Documents comprising Bid:

E-Tender process shall be carried out in two stages i.e., Eligibility-cum-Technical Bid and Financial Bid.

i. Technical Bid:

Eligibility-cum-Technical Bid shall contain the following uploaded documents.

1. EMD amounting to **Rs 91,46,897/- (Ninety-One Lacs Forty-Six Thousand Eight Hundred and Ninety-seven only)** and Tender form fee including GST, Rs 5,900/- should be paid through online gateway mode.
2. Valid empanelment Registration issued by MHDC.
3. Tender fee receipt (scanned copy has to be uploaded on the web site & hard copy has to be submitted in an Eligibility-cum-Technical bid on or before last date of submission.)
4. Documents to be submitted
 - a) Undertaking in the prescribed format as given in chapter 16, Clause 16.10 (Form No. IX), declaring the truthfulness of the documents, and Form no VI in chapter 16 clause no. 16.07 declaration of conversant with bye laws and site on stamp paper of RS. 100/- each **shall be submitted for eligibility/qualification criteria.**
 - b) Income Tax Returns for last five financial years.
 - c) Valid registration colour copy of company identification PAN, GST, ESIC, EPF & Professional Tax.
 - d) The Bidder / Contractor shall upload TDS (26AS) certificate/ document during last 5 years.
5. **FORM NO. I A - Details of work** in hand with the value of work unfinished on the date of submission of tender. The certificates from the Executive Engineer or equivalent under whom the works are in progress shall be enclosed **FORM NO. I B- Details of work tendered for but where tendering process is still in process** shall submit in Form - I B
6. Details of works of similar type and magnitude carried out by the Tenderer. (Form no. III). Each work shall be support by the certificate from the Executive Engineer and in case of private works, the work done certificate under hand and seal of the Owner/Architect/Project Management Consultant shall be submitted along with commencement certificate/Letter of approval/Completion/Occupation certificate of Municipal Corporation, Municipal Council or Town planning Authority, any Govt. Authority concern.
7. List of immediately available with the tenderer for use on this work and List of Machinery proposed to be deployed on this work, but not immediately available and the manner in which it is proposed to be procured. **(Form No. II)**
8. Details of Technical Personnel with name, qualifications and experience in similar works, and a period of his working with Bidder 's organization, who will be working on this project as technical key persons for supervision, to receive technical instructions issued by the PMC / Client from time to time. **(Form No. IV)**

9. Details of payment received in Building Work and Infrastructure Construction work during last seven years certified by Chartered Accountant (**Form No. V along with UDIN**).
10. Scanned copy of original power of attorney for signing of tender document.
11. Certified balance sheets under seal of Chartered Accountant for last five financial years.
12. Any other additional documents / profile etc. in support of technical qualification.

Note: -

- a. The statement showing the value of existing commitments and ongoing work as well as the stipulated period of completion remaining for each of the works listed should be countersigned by the officer not below the rank of the Executive Engineer and Bid Capacity statement of the Contractor should be got certified from the Chartered Accountant.
- b. Works executed for government organizations/ Private organizations shall be considered. The necessary certificates are required to be obtained from the officer not below the rank of Executive Engineer, for government work. In case of works executed for private clients, private work shall be supported by certificate from concern architect, occupation certificate & TDS certificate issued by statutory authority. The work done for bidders own group company or subsidiary or sister companies shall not be considered. Works executed on BOT or Joint Venture basis shall not be considered.
- c. The value of executed work shall be brought to the current costing level by enhancing the actual value of work at simple rate of 10% per annum, calculated from date of completion to last date of receipt of tender.

ii. Financial Bid:

- a) For financial bid, download the Price Bid (**BOQ Format**) (**.xls file**) given in commercial bid section of website fill the details & quote price in it & upload the file. He should not quote his offer anywhere directly or indirectly in Envelope No.1. The contractor shall quote for the work as per details given in the main tender offer should be included of all prevailing taxes (Cess...etc) (Exclusive of GST). His tender shall be unconditional. **Online “Envelope No.2”** shall contain only the main tender. A tender submitted without this would be considered as invalid.
- b) If Price Bid file is blank then bid of such bidder will not be considered.

iii. Documents not submitted but still part of Contract:

- a) Common Set of Deviation (CSD)
- b) Corrigendum
- c) Letter of Acceptance (LOA)
- d) Work Order
- e) Standard Specifications of P.W.D. (National Building Code, Red Book, Manual, Etc.)
- f) Good Engineering Practice Manual

(H) Public Works Manual Bid Prices:

- i. The tenderer shall fill his quotes in the manner as prescribed in Financial Bid (lump sum quote).
- ii. Price quoted shall be inclusive of all Taxes except GST.
- iii. Applicability of Prices (frozen):
- iv. Currencies: All the prices shall be INR (Indian Rupees)

(I) Bid validity:

The bid of the Tenderer shall remain valid for **120 days** (One Hundred Twenty Days Only) after the deadline date for submission of e-Tender or withdrawn by Tenderer in writing after 120 days whichever is later. However, in exceptional cases, with intimation to the eligible tenderers, the Client reserves the right to extend such validity maximum up to 60 more days unilaterally and the same shall be binding on the tenderer.

(J) Conditional Proposals:

The Conditional Proposals shall be summarily rejected.

(K) Format of Signing:

An Undertaking as prescribed in standard formats in chapter 16, clause 16.1 and 16.2 of these documents shall be uploaded for eligibility/qualification criteria.

(L) All Pages to be Signed:

All the pages shall be duly signed by the contractor or his authorized representative (Authorization certificate is mandatory).

1.5 Manner of Submission of Tender and its Accompaniments:

Submission of Tender:

The tenderer shall ensure that The Managing Director & C.E.O, Maharashtra Housing Development Corporation Ltd. Mumbai, receives his e-tender before the expiry of the date and time. In the unloadable document type of templates, the Tenderers are required to select the relevant document/ compressed file (containing multiple documents) already uploaded in the briefcase.

Notes:

- a) The Tenderers shall upload a single document or a compressed file containing multiple documents against each unloadable option.
- b) The Hashes are the thumbprint of electronic data and are based on one – way algorithm. The Hashes establish the unique identity of Bid Data.
- c) The bid hash values are digitally signed using valid Class 1A Digital Certificate issued any Certifying Authority. The Contractors are required to obtain Digital Certificate in advance.
- d) After the hash value of bid data is generated, the Tenderers cannot make any change/ addition in its bid data. The Tenderers shall modify bids before the deadline for bid preparation and has submission as per Time Schedule mentioned in the e- Tender documents.

1.6 Opening of Tenders:

On the date specified in the tender notice, following procedure will be adopted for opening of the tender:

1. Eligibility-cum-Technical Bid

Eligibility-cum-Technical Bid of the tender will be opened online, in the presence of the Tenderer(s) who has/have submitted tenders or their authorized representatives who chose to attend, to verify its contents as per requirements. If the various documents contained in this bid do not satisfy the requirements of the Client, a note shall be recorded accordingly by the tender opening authority and the said tenderer's Financial Bid shall not be opened.

2. Financial Bid

Financial Bid shall be opened, in the presence of the Tenderer(s) who has/have submitted tenders or their authorized representatives who chose to attend on notified

date, only if contents of Eligibility-cum-Technical Bid are found to be acceptable to the Client. The lump sum amount quoted shall then be read out.

3. The Client reserves the absolute right to accept any offer not necessarily the lowest, based up on its final decision regarding tenderer's capacity, competency, financial status, experience, machineries available, experienced technical staff with the Tenderer etc. The past experience/ track record/ performance of tenderer in MHDC regarding quality, workmanship, punctuality, leakages – dampness– seepage in toilet blocks, terrace floor, underground - overhead water tanks, peripheral walls of building, and directly/ indirectly subletting of works, will be minutely reviewed during scrutiny of tender/awarding works, and even if the tender emerges as the lowest among the tenderers, he will be treated as ineligible for award of the work in view of **clause 1.2.1** above.

4. **Rejection of Bids:**

Notwithstanding anything contained in this Bid Document, the Client reserves the right to reject any Bid or to annual Bidding process and reject all bids at any time without any liability or any obligation for such acceptance, rejection or annulment and without assigning any reason therefore.

5. **Confidentiality:**

Information relating to examination, clarification, evaluation, comparison of bids and recommendations for award of the contract shall not be disclosed to any bidder/person who is not officially concerned with the Bidding process until the award to the successful bidder has been announced. Any effort by a bidder to influence the Client or his representative in bid process or award decision may result in rejection of bid.

6. **Correspondence with the Bidder –**

Save and except as provided in this document the Client shall not entertain any correspondence with any bidder in relation to acceptance or rejection of any bid.

7. **Technical bid of the tender**

It will be opened online, in the presence of the Tenderer(s) who has/ have submitted tenders or their authorized representatives who chose to attend, to verify its contents as per requirements. If the various documents contained in this bid do not satisfy the requirements of the Client, note shall be recorded accordingly by the tender opening authority and the said tender's Financial Bid shall not be opened.

8. Checklist for documents to be submitted for eligibility criteria.

- i. EMD Receipt.
 - ii. 1) List of similar work completed in last seven years is in Form III.
 - iii. 2) In Case of Govt. / Govt. undertaking bodies works certificate of completion of work stating cost of tender, cost of completion, start date, completion date.
 - iv. 3) In case of the private works, the work done certificate under the hand and seal of Owner / Architect / Project Management Consultant shall be submitted along with Commencement certificate / letter of approval or completion / occupancy certificate of Municipal Corporation, Municipal Council or Town Planning Authority, any Government Authority concern.”.
- i) **The average Annual Turn Over** of the Contractor for last 3 financial years from all Classes of civil engineering Construction work shall be 75% of Cost of work Years also shall be certified by chartered accountant.

For Bid capacity

- I) Maximum Annual Turnover
 - The information in Form no V – statement showing payment received (i.e., Receipt) in Building Works and Infrastructure Construction Works of last seven years for registered / unregistered contractor.
 - Form V shall be certified by the chartered accountant along with UDIN on each page if found not certified will not be considered.
- II) Form no. I(A) – Details of list of Works in Hand
- III) Certificate issued by competent Authority not below the rank of” Executive Engineer’ in case of government project.
- IV) In case of private work certificate shall be issued by client/ Architect
- ii) 1) Audited Certified Balance Sheet under the hand and seal of Chartered Accountant for five years.
- 2) Details of the works carried out in last Seven financial year should be given in form no. V and shall be signed by the tenderer or holder of his power of attorney and shall be certified by the chartered accountant.
- iii) Other Certificates
 - i. Permanent Account Number (PAN) card
 - ii. Goods and Service Tax (GST) certificate
 - iii. PF Certificate
 - iv. ESIC certificate
 - v. Company registration certificate.
- iv) Form no. 26 as of seven financial years should be uploaded.

- v) Bidder should submit Chartered Accountant Certified documents / Certificates / Clarification with unique document identification number (UDIN) of Institute of Chartered Accountant of India (ICAI). Document / Certified documents / Certificates / Clarification without (UDIN) will not be considered.

Note – If documents shown in appendix I are not submitted with the bid, then financial bid of bidder will not be opened.

*Incomplete, Incorrect and misleading information filled up form will not be taken in consideration for bid qualification.

1.7 Award of Work:

- 1.7.1 Responsive Price Bid shall be evaluated on the basis of total amount which consists of quoted Lump sum Amount for the project. Tenderer who has quoted lowest total amount shall be the Preferred Tenderer. The tenderer shall quote the lump sum Amount in the format provided in Price Bid form.
- 1.7.2 MHDC may choose either to accept the Tender of the Preferred Tenderer or to negotiate with him or reject the Tender.
- 1.7.3 If more than one Tenderer quotes same total amount, then such Tenderers (whose total amount are same) shall be called for submitting revised price Bid.
- 1.7.4 Upon acceptance of the Tender of the Preferred Tenderer with or without negotiations, MHDC shall declare the Preferred Tenderer as the successful Tenderer and enter into an agreement.
- 1.7.5 Notification of Award and Signing of Agreement:
 - i. The Preferred Tenderer whose Bid has been accepted be notified of the award by the Client prior to expiration of the Tender validity period by cable, telex or facsimile confirmed by registered letter, email. This letter (hereinafter and in the conditions of Contract called the “Letter of Acceptance”) shall state the sum that the Client shall pay the Contractor in consideration of the Survey, Soil Investigation, Structural Design, execution, completion, and DLP of the Works by the Contractor as prescribed by the Contract (hereinafter and in the Contract called the “Contract Price”).
 - ii. The Letter of Acceptance of award shall constitute the formation of the Contract, subject only to the furnishing of a Security Deposit in accordance with Clause 1.8. (B).
 - iii. The Agreement shall incorporate all agreements between the Client and the successful Tenderer.

- iv. Within **twenty-eight (28) days** of receipt of the Contract Agreement, the successful Tenderer shall sign, date, and return it to the Client.
- v. On receipt of Letter of Acceptance of award from the Client, the successful Tenderer shall furnish the Security Deposit in accordance with the conditions of contract, subject to Clause 1.8. (B).

1.7.6 Corrupt or Fraudulent Practices of fake document and information: -

The proposal for award if it determines that the Tenderer recommended for award has engaged in corrupt, fraudulent practices, fake documents & information in competing for the contract in question shall be rejected and will be declared ineligible, either indefinitely or for a stated period as decided by the Client. E.M.D. of such tenderer will be forfeited and debarred for future tender of Maharashtra Housing Development Corporation Ltd.

1.7.7 Failure of the successful Tenderer to submit the Security Deposit including additional Security Deposit, if any or sign the Contract shall constitute sufficient grounds for the annulment of the award and forfeiture of the E.M.D. In that event the Client may award the Contract to the next lowest evaluated Tenderer whose offer is substantially responsive.

1.8 Bid Security:

(A) Refund of EMD:

- i. The EMD shall not carry any interest.
- ii. The earnest money deposit of the tenderer shall be forfeited if the tenderer withdraws the bid after tender submission during the bid validity period.
- iii. The amount of earnest money shall be refunded to the unsuccessful Tenderers except Lowest three tenderers, within a month from date of opening of financial bid. The tenderer shall submit the Bank details for refund of EMD. If details of Bank are not correct and the amount refunded is transferred to any other account, the Client shall not be held responsible.
- iv. In case of the successful tenderer, Earnest Money Deposit shall be refunded on paying the initial security deposit and completing the agreement formalities mentioned in the tender document or shall be transferred towards a part of security deposit to be paid.
- v. Earnest Money Deposit of second and third lowest tenderer shall be refunded on payment of initial security deposit and completing the agreement formalities mentioned in the tender document by the successful tenderer.

- vi. If successful tenderer does not pay the security deposit as per clause 1.8. (B) below within time limit and does not complete the agreement formalities and / or does not execute the contract within prescribed time limit then his/their earnest money deposit shall be forfeited by Client.

(B) Security Deposit & Additional Security Deposit:

Security Deposit

I	Total Security Deposit	1% (One Percent) of tendered cost offered by Tenderer and Accepted by the Client.
II	Initial Security Deposit	The successful tenderer shall have to pay 50% Security Deposit within 15 days of issue of letter of acceptance of tender in the form of Demand Draft/ FDR/ Bank Guarantee/RTGS/NEFT of Nationalized Bank/ Scheduled Bank payable at Mumbai, in the name of The Managing Director & C.E.O. of The Maharashtra Housing Development Corporation Ltd., Mumbai for the project period (24 months) + 5 year defect liability period in the enclosed form and complete the contract documents failing which his earnest money will be forfeited to the Corporation.
III	Remaining Security Deposit	The balance 50% Security Deposit will be recovered from the R.A. Bills at the rate of 5% (Five Percent) of the bill amount until the total amount of security deposit is recovered.

Additional Security Deposit

The successful tenderer shall pay additional security Deposit (Additional Performance Security) along with the initial security deposit as per II above in the form of Bank Guarantee, when the below percentage quoted is more than 10% of cost put to tender. The amount of additional Security deposit shall be worked out as per following formula.

a) If the offer submitted is below schedule "B" rates by more than 1% but less than 10% of the estimated cost put to tender	1% of the estimated cost put to tender
b) If below by more than 10% of the estimated cost put to Tender	1% of the estimated cost put to tender plus an amount equal to the percentage by which the offer is below 10% of the estimated cost put to tender. (e.g., if the

	offer is 14.00% below, the Performance Security will be $1\% + (14\% - 10\%) = 5.00\%$ of the estimated cost put to tender.
c) If offer is more than 15% below.	For balance amount, submission of double amount in the form of Demand Draft/ Fixed Deposit Receipt/ Bank Guarantee is mandatory for example 19% below rate Up to 10% below rate = 01% Up to 15% below rate = 05% (15% - 10%) Up to 19% below rate = 08% $(19\% - 15\% = 4\% \times 2) = 08\%$ } Total = 14% Note: - (i) For more than 10% below and up to 15% below,, b" is applicable. (ii) For more than 15% below c is applicable.

- i) If the successful tenderer does not pay initial security deposit and additional security deposit if applicable within 15 days, including Government holidays, issue of letter of acceptance of tender, then E.M.D. paid may be forfeited and his tender shall be liable to be rejected.
- ii) The Term Deposit Receipt of initial Security Deposit shall be kept valid for the **project period 24 months + 5 years defect liability period for the said work (i.e., 24 months + 5 years)**. Before expiry of the Bank Guarantee/ Term Deposit Receipt, the contractor shall renew it or submit new Bank Guarantee/ Term Deposit Receipt of required period well before the expiry date, if necessary, or otherwise the Guarantee/ Term Deposit Receipt will be encashed accordingly.
- iii) In case of non-renewal of Bank guarantee/ Term Deposit Receipt or non-submission of Bank guarantee / Term Deposit Receipt well in time, a penalty of Rs. 5000/- (Rupees Five Thousand per day (after the expiry of the date) shall be imposed till submission of renewed Bank Guarantee / Term Deposit Receipt or new Bank guarantee.
- iv) All compensation or other sums payable by the Contractor under the terms of this contract or any other contract or an any account shall be deducted from his Security Deposit or from any sums which may be due to him or may become due to him by Client on any account and in the event of the Security deposit being reduced by reason of any such above noted deductions the Contractor shall within ten days of receipt of notice of demand from the client make good the deficit and reinstate the security deposits.
- v) There shall be no liability on the Client to pay any interest on the Security Deposit paid by or recovered from the Contractor.

(C) Refund of Security Deposit:

If the DLP period is more than 2 years, 90% of the amount of total security deposit shall be refunded after 2 years from the date of virtual completion of this project. Balance 10% Amount shall be refunded after expiry of defect liability period of 5 years from the date of virtual completion of this project, as mentioned in clause 12.4 of the conditions of contract.

The amount of the Additional performance security shall be refunded within 3 months after completion of Defect Liability Period, as per the Maharashtra, Public Works Department, Mantralaya, Mumbai G.R. No.CAT-2017/CR-8/(II)/Bldg-2, Dt. 27/9/2018

2 Additional Instructions to Tenderer

2.1 Definitions:

2.1.1 In the tender documents the following words and expressions shall have the meanings stated;

- i. **“Client”** shall mean Maharashtra Housing Development Corporation, (hereinafter referred to as “MHDC or Mahahousing”) a Corporation fully owned by the Government of Maharashtra and incorporated as a limited company under the Indian Companies Act, 2013 and having its Registered Office at Sakhar Bhavan, Office No. 32, 3rd floor, Ramnath Goenka Marg, Nariman Point, Mumbai, Maharashtra 400021 represented through its Joint Chairman and Managing Director & CEO.
- ii. **“Board of Directors”** shall mean the Board of Directors of Maharashtra Housing Development Corporation Limited.
- iii. **“Managing Director”** shall mean Managing Director of Maharashtra Housing Development Corporation Limited, holding that office, and also his successors and shall include any other officer authorized by him.
- iv. “Managing Director”, “Director (Technical)”, “Executive Engineer” and **“Deputy Engineer”, “Junior Engineer”**, used in the tender documents shall mean officers of Administrative/ Engineering rank of the Maharashtra Housing Development Corporation Limited, (by whatever designation he shall be known) under whose control and supervision of the work lies.
- v. **“Government”** shall mean that the Government of Maharashtra.
- vi. **“PWD”** means the Public Works Department of the Govt. of Maharashtra.
- vii. **“Project Management Consultant (PMC)”** shall mean a person, firm so appointed by the Client for comprehensive planning, architectural services, detailed design scrutiny, approval to design and drawings, construction supervision and shall include any person/persons duly authorized by the Client. and Engineer of the Client having access to the site for supervision of construction work and any other activity pertaining to Planning Designing, construction, etc.
- viii. **“Engineer / Engineer-in-Charge”** shall mean the Person from the PMC/Client for the purposes of the contract or any person appointed by the Client from time to time and notified to the Contractor under the contract.

- ix. **“Representative of Client”** shall mean any person authorized by the Client as his representative for the purposes of this contract having access to the site from time to time.
- x. **“Engineer’s Representative”** shall mean as Assistant of the Engineer, notified in writing to the contractor by the Client/Engineer from time to time.
- xi. **“Contract”** shall mean these Conditions, Specifications, the Drawings, the Tender, the Letter of Acceptance, the Contract Agreement (if completed) and such further documents as may be **expressly incorporated** in the Letter of Acceptance or Contract Agreement
- xii. **“Contractor”** shall mean a person(s) named as Contractor in the letter of acceptance by the Client and legal successor in title to this person (s).
- xiii. **“Contractor’s Representative”** shall mean as Assistant of the Contractor notified in writing to the Client/PMC by the Contractor from time to time.
- xiv. **“Day”** shall mean a calendar day of 24 hours from midnight to midnight irrespective of the number of hours worked in any day.
- xv. **“Drawing”** shall mean the drawings enclosed in the bid document and any modifications approved in writing by the Engineer/Client and any drawings issued from time to time and approved in writing by the Engineer/Client.
- xvi. **“Plant”** shall mean every plant/equipment temporary and necessary or considered necessary by the Engineer to execute, construct, complete and maintain the works and used in altered, modified substituted and additional work ordered in the time and the manner herein provided and all temporary materials and special and other articles of appliances of every sort, kind and description whatsoever, intended or used therefore
- xvii. **“Schedule of rates”** or simply D.S.R/e-D.S.R/SSR, shall mean the Schedule of the rates of the District / State in which jurisdiction the work lies.
- xviii. **“Tenderer / Bidder”** shall mean the one who has submitted its offer or participated in bid process of this Tender.
- xix. **“Site”** shall mean the lands, plot, and /or other places on which the works are to be executed under the contract which shall be allotted by Client / Users or any other places as may be specified by the Client.
- xx. **“Structural Design Engineer (SDE)”** shall mean a person complying with the criteria stipulated in the Bid document and appointed by the PMC who shall carry out structural designs and prepare Good for Construction (GFC) drawings of all buildings in the said work/project.
- xxi. **“Good for Construction Drawings”** shall mean the drawings prepared by the PMC and approved in writing by the Client.

- xxii. **“Temporary works”** shall mean all temporary works of every kind required on site for the execution and completion of permanent works and remedying any defects, or maintenance of the works.
- xxiii. **“User agency”** shall mean any unit for whom work is carried out and who will be the end user of the Project (s).
- xxiv. **“Urgent works”** shall mean any measures which, in the opinion of the PMC/Client, become necessary during the progress of the works to obviate any risk or accident or failure or which become necessary for security of the work or the persons working, thereon.

2.2 Errors, Omissions, Discrepancies and Interpretations:

- a) In case of following errors and /or discrepancies a reference shall be made to Client, whose elucidation, elaboration or decision shall be considered as final.
 - i. Between actual scaled and written dimension or description on a drawing/s
 - ii. Between the written description and dimension in the drawings and corresponding in the specifications.
- b) In case of discrepancy between amount quoted in figures and in words, the figures in words shall be considered for acceptance of tender and calculation of amounts.
- c) The special provision if any in scope of work shall gain precedence over corresponding contradictory provision (if any) in the detailed item specifications and standard specifications.
- d) Where the context so requires, words importing the singular only also include the plural and vice-versa.
- e) If there are varying or confliction provisions made in any one document forming part of the contract, the Accepting Authority shall be the contractor.

2.3 Tenderer to Study the Site Conditions and Tender Document:

- i. The Tenderer shall have before bidding be deemed to have visited and thoroughly studied the site carefully, examined and understood the nature of the said work, site conditions, foundation strata, the terrain, the climatic condition, water, electricity, resources, drainage connection, or drainage outlet points including availability of materials and labours, tax structure, applicable law, transport and communication facilities, the availability of land for right of way, temporary office and accommodations, quarters, and all other facts and facilities necessary or relevant for the formulation of the tender. Also, the general and the special conditions, the specification schedules and drawings shall be deemed to

have been studied by the tenderer and have fully informed himself regarding the local conditions and carefully gone through the PWD, M.J. P's standard specifications, PWD hand book, manual account code and the schedule of rate and carried out his own investigations at his own cost to arrive at the offer quoted in the tender.

- ii. In this regard tenderer shall refer necessary information given in this document/ drawings/ specification. In no case shall the Client be liable for any, loss, claim or damage arising out of any discrepancy in such information.
- iii. In case there is any doubt / omission regarding the wording in general conditions or the special conditions or the scope of work or the specifications and drawing or any other matter concerning the contract, the Bidder shall in good time, submit them to the Client to reach him before schedule pre-bid conference date and time in writing in order that such doubts can be clarified before submission of the Tender.
- iv. The clarification given by the Client, in respect thereof, as well as, any further information which the Client shall chose to furnish to the tenderers in the form of Common Set of Deviation (CSD) shall form a part of the Tender. Any claim at a later date based on either incorrectness or inadequacy of the information / data made available by the Client shall not be entertained. The Client shall be fully free from blame and all liabilities in this regard.

2.4 Brief Scope of Work:

- 2.4.1 The work shall be carried out as per the tender drawings, document, specifications and instructions on site including supplementary and explanatory drawings issued during execution. along with obtaining necessary permissions, No Objection Certificates (NOC), Licenses, etc.as per clause 1.1(3)
- 2.4.2 The construction of all buildings shall be for Reinforced Cement Concrete (RCC). Tenderers are requested to submit the lump sum offer for the work construction of all buildings and for infrastructure as per the scope given in chapter 3,5 and 6 of Bid Document. The detailed scope of work shall be as given in Chapter 3, 5 and Schedule 'C' given in Chapter 6. Apart from Mivan technology, contractor can use alternative construction material/ technology for construction and development of the project as approved by Government of Maharashtra GR No. PMAY- 2018 / Q.No. 08 HAD – 2 dt. 01/08/2018 only. Construction using Conventional Technology is not allowed for this Project except ancillary buildings.

2.5 Assistance in Procuring Priorities Noc's, Permits, etc:

The Client, on a written request by the Contractor, shall provide support in form of letters to the concerned authorities if found necessary. However, for the non-availability of such support or delay in this regard from the Client, no claims shall be entertained by the Client.

2.6 Co-ordination with Agencies:

- i. When several agencies for different sub-works on the project, work simultaneously on the project, there shall be full co-ordination and co-operation between different agencies to ensure timely completion of the whole project smoothly. The schedule dates for completion specified in contract shall therefore, be strictly adhered to and the Contractor shall make his independent arrangement for water, power, housing, etc. However, the contractors are at liberty to make mutual agreement and make joint arrangement with other agency with the approval of the PMC. The contractor shall not cause any disruption, discontent or disturbance of work, labour or arrangement etc., of other agencies working in the project location. No extra payment shall be considered as charges for coordination with other agencies.
- ii. In case of any dispute, disagreements between the agencies, the Client's decision regarding the co-ordination, co-operation and facilities to be provided by the agencies shall be final and binding and such a decision(s) shall not vitiate contract neither absolve the contractor of his obligations under the contract nor consider for any claim.

2.7 Relation with Public Authorities:

The Contractor shall comply with all rules regulations by laws and directions given from time to time by any local or public authority in connection with this work and shall pay fees or charges which are levied on it and same will be reimbursed by the Client after submission of original receipt by contractor and certification by PMC.

2.8 Treasure-Trove:

In the event of discovery by the Contractor or his employees, during the progress of the work any treasure, fossils, mineral or any other article of value or interest, the contractor shall give immediate intimation thereof to the Client and forthwith hand-over to the Client such treasure of things which shall be the property of the Corporation.

2.9 Patented Device:

Whenever the Contractor desires to use any devices, materials or process covered by the letter of patent or copyright, the right for such use shall be secured by suitable legal arrangement and agreement with patent owner and the copy of their agreement shall be filed with the Client if so desired by the latter. The Contractor shall indemnify the Client against the claims by any party on account of any infringement or unauthorized use by the Contractor.

2.10 Explosives:

- i. The contractor shall at its own expenses construct and maintain proper magazines, if such are required for the storage of explosives for use in connection with the work and such magazine being situated, constructed and maintained in accordance with rules applicable, The Contractor shall at his own expenses, obtain such license or licenses as shall be necessary for storing and using explosive. Notwithstanding that the location, etc. or storage of explosive are approved by the licensing authority, the Client shall not be liable whatsoever in connection with storage and use of explosive on the site or any accident for use of such explosives.
- ii. Any hazardous materials brought on site by the Contractor, for the execution of the work, shall be the sole responsibility of the Contractor. The Contractor shall at his own expenses obtain such license as shall be necessary for storing and using such hazardous materials. The Corporation shall not be liable whatsoever in connection with storage and use of such hazardous materials on the site or any accident involving the use of such hazardous materials.
- iii. The Contractor shall indemnify the Client and its respective employees, Directors and officers and agents from and against any claims, demands, loss, damage or expenses (including counsel fees and legal cost) relating to bodily injury or death of any person or damage to real and / or tangible property directly caused by the use or storage of such explosives and hazardous materials by the Contractor.

2.11 Quarries

- i. Quarries are not available with the Corporation. The contractor (s) shall have to arrange the same itself. Getting permission from Concerned Department or any statutory body shall be the responsibility of Contractor.
- ii. The contractor shall maintain at his own cost the books, registers etc., required to be maintained under the relevant rules and regulations regarding quarrying and

as directed by the Client. These books shall be open for inspection at all times by the Client or his representative and the Contractor shall furnish the copies or extracts of books or registers as and when required.

- iii. Quarrying permission shall be directly obtained by the Contractor from the necessary assistance. Royalty charges or any other taxes shall be borne by the Contractor.

Note:

Regarding royalty charges refer **Clause No. 12.64 of chapter no.12** “Conditions of Contract”.

2.12 Accident:

- i. In the event of an accident involving serious injuries or damages to human life or death of any of his employees and/or contract labours the same shall be reported forthwith to the nearest Police Station and within twenty-four hours of the occurrence to the Client and to the Labour Commissioner.
- ii. In case of fatal accident or serious non-fatal accidents, Contractor shall pay compensation to injured worker/dependents of deceased workers as decided by Labour Commissioner as per Workmen Compensation Act. The amount of compensation likely to be payable shall be kept under deposit by withholding that amount from running bills payable to the Contractor. Such amount shall be withheld till the final bills payable to Contractor and till the final decision is received from the Labour Commissioner whichever is later.
- iii. In case of sudden collapse of any structure or part thereof, due to faulty execution of work or due to lack of supervision or negligence from the Contractor, he shall be entirely responsible for damage of lives of peoples, structure and any losses to the Client, etc.

2.13 Police Protection

- i. For the special protection of labour Camp of the Contractor and its equipment, etc., the Corporation will help the Contractor as far as possible to arrange for such protection with the concerned authorities if so, requested by the Contractor in writing. The full cost of such protections shall be borne by the Contractor.
- ii. The Contractor shall also prevent the encroachment /trespass, dumping of the waste /debris, unauthorized construction, whether permanent or temporary in nature, on the common /open areas in the sites. The Contractor shall file a complaint in the concerned police station in this respect and it shall be Contractor

responsibility to remove the same with the assistance of the Client on taking proper police protection.

2.14 Protection / Shifting of Utilities:

- i. Identified utilities, where quantum of work is pre-determined
During the execution of work, the Contractor shall be required to deal with utilities such as underground / overhead live or dead telephone cable, HT, LT electrical cables, poles, existing live / dead water supply or sewer's lines etc., It shall therefore, be the responsibility of the Contractor to protect/ shift them carefully at his own cost as per the provisions in the contract. During shifting/ protecting these utilities, any damage whatsoever to these utilities shall be made good by him at his own cost, as directed by the Client and relevant authorities, Statutory fees if any shall only be reimbursed by the Client.
- ii. Un-identified utilities.
During the execution of work, if the Contractor is required to deal with un-identified utilities such as underground live or dead telephone cable, electrical cables, water supply or sewer lines, etc. shall bring to the notice of client and concerned department about the same. The Contractor shall protect/ shift them carefully as directed by the Client/agency. During shifting/ protecting these utilities, if any damage whatsoever to these utilities shall be made good by him at his own cost, as directed by the Client and relevant authorities, - Statutory fees if any and cost of shifting/ protecting these utilities shall be reimbursed by the Client.
- iii. If any delay occurs on account of shifting of above stated utilities, and also cutting and shifting of trees and other structures obstructing the work, where permissions have been accorded by the relevant authorities for such removal or shifting, no compensation or extension on account of delay, shall be entertained.

2.15 Working Methods and Construction Programme

- i. The Contractor shall furnish to the PMC with the copy to the Client, within **15 days** from the date of work order, construction programme containing working layout plan bar chart PERT Chart, etc. in quadruplicate indicating the date of actual start, the monthly progress to be achieved and the anticipated completion date of each major item of work to be done by him, also indicating dates of procurement and setting up of plants and machinery. The programme shall be such as to be, practicable of achievement towards the completion of whole work in the time limit; the particular item on the due dates specified in the contract and

shall have the approval of the Client. No revised programme shall be operative without such acceptance of the Client in writing. (The PMC is further empowered to ask for detailed programme say week by week, for any item or items, as will be directed by him including working method) The contractor shall be responsible for maintaining the progress according to milestone laid down in Clause no. 12.12 of the contract.

The monthly progress report shall be submitted by the contractor which shall include Physical progress and Financial Progress of the project.

- ii. The contractor shall deploy sufficient plant, equipment and labour as shall be necessary to maintain the progress The Contractor shall have to make his own arrangements for transportation and procuring plants and machinery required for the work. The working and shift hours restricted to one shift a day for operation to be done under the Corporation supervision shall be such as shall be approved by the Client. Night work that requires supervision shall not be permitted except when specifically allowed by Client each time, if requested by the contractor. The contractor shall provide necessary lighting arrangements and safety measures, etc. for night work as directed by the Client without extra cost.
- iii. Electric supply and water supply for construction purposes shall be arranged by the Contractor at his own cost including necessary connection and consumption charges. The Client shall however, render all reasonable assistance by way of recommendation only without no obligation/ liability.
- iv. Further the contractor shall submit the progress report of work in prescribed forms, charts, statements, photographs on yammer, reports, etc. at periodical intervals as shall be specified by the PMC/Client.
- v. The Contractor shall maintain details regarding machinery, equipment, labour, materials, etc. as shall be specified by the PMC/Client and submit the same at periodical intervals as shall be specified by the Client
- vi. The contractor shall establish and operate SCADA from the date of work order till end of construction period.

2.16 The Client and his Representative:

- i. The Client shall from time to time, in writing delegate to PMC any powers and authorities vested in the Client and shall furnish to the Contractor a copy of all such delegations of powers and authorities, any written instructions of approval given by the representative of the Client to the Contractor within the terms of such delegation (but not otherwise) shall bind the Contractor and the Corporation as though it had been given by the Client.

- ii. Approval of any work by the PMC or any part thereof or materials shall not prejudice the power of the Client to disapprove such work or materials and the contractor shall comply with the directions of the Client at any stage of the work without any additional cost.
- iii. The Client shall have full authority regarding inspection of the work and all such matters.

2.17 Management of Works by the Contractor:

The contractor shall himself manage the work and engage authorized full time Project Engineer (possessing the qualification as mentioned herein) on the site capable of managing and executing the work and understanding the specifications and contract conditions. The Project Engineer shall take orders from the PMC/Client as he will be responsible for carrying out the work. The Project Engineer shall not be changed without prior permission from Client.

The Contractor shall deploy staff as given in table below;

Sr. No.	Minimum Qualification of Staff appointed by Contractor	Minimum Year of Experience	Duties / responsibilities
1	Architect- G.D. Arch / B. Arch.	5 years (as per site requirement)	• Verification of layout, D.P. requirements drawings, location, dimension, etc. • Checking of working drawing and getting approval of PMC/Client
2	Structural Engineer- MTech/ M.E. (Structure)/ M.S. or equivalent)	5/10/15 years (as per site requirements)	• Verification of structural detailing shown in drawing, actual placing of reinforcement at site, sizes of RCC member,
3	Project Engineer - B.E. Civil, B.Tech. (Civil)	10 years	• Total management of project and completion within scheduled time period, furnishing Monthly progress report with photographs etc.
4	Site Engineer B.E/ DCE.	5 years	• Quality and workmanship of work, progress of work as per Bar chart, Site Supervision.
5	Quality assurance and	5 years	• Quality of material assurance and

Sr. No.	Minimum Qualification of Staff appointed by Contractor	Minimum Year of Experience	Duties / responsibilities
	quality control Engg. - B.E./ DCE		testing of material, Site Supervision. etc.
6	Electrical Engineer- B.E./ DEE	5 years (as per site requirement)	• Verification of electrical load, material quality, testing, Site Supervision. etc.
7	Water supply and drainage Engineer-B.E.	5 years (as per site requirement)	• Verification diameter of pipes, material, quality, workmanship, necessity, Site Supervision. etc.
8	Management Information System (MIS) / Planning Engg., B. E	5 years (as per site requirement)	• Planning, Daily Progress Report (DPR), Monthly Progress Report (MPR), etc.
9	Quantity Surveyor (QS) / Billing Engg, B. E	5 years (as per site requirement)	• Quantity Surveyor (QS), Billing, Cash flow, Bar Bending Schedule (BBS)
10	Safety Engineer, Advance Diploma in Industrial Safety / Certificate course of NICMAR	5 years	• Health and Safety Engineer (HSE), day to day check for work as per safety standards/ manual
11	Supervisors, Safety Steward as per Project requirement	3 years	• For day-to-day supervision of works
12	Surveyor and Asst. Surveyor	3 years	• Survey related works
13	Quality Assurance (QA) / Quality Control (QC) Lab In Charge	5 years	• Testing and keeping records of tests at site.

Notes:

- a) The staff at Sr. no 3 to 13 shall be deployed for full time. The services of Architect and Structural Engineer shall be made available as per the Project requirements and shall be available for discussion/meetings with the Client / PMC.

- b) The number of each key person listed above shall be mutually agreed between Client/PMC & Contractor depending on the Project requirements and it should be adequate enough to handle the prevailing / current project progress.
- c) Contractor shall submit CVs of employees to Client regarding the above specified post or designation.
- d) The contractor shall not change any key person(s) without the prior written consent of the Client. Any violation on this account shall attract penal action as decided by the Client against the Contractor.
- e) If staff as mentioned above is not appointed on site of work, the Client has full right to stop the work till appointment of such staff and also recover compensation from the Contractor equal to amount of salary of staff as determined by the Client till such staff is appointed. No claim or extension shall be entertained in the event of the same.
- f) Orders given to the Contractor's Project Engineer / representative shall be considered to have the force as if these had been given to the Contractor himself. If the Contractor fails to appoint Project Engineer / representative as directed by the Client, the Client shall have full powers to suspend the execution of the work until the contractor appoints such a responsible Project Engineer / representative. Delay so caused to the works shall not entitle the Contractor for any compensation or extension in the due date for completion of the work in this behalf.
- g) The contractor shall inform the PMC/ Client in writing when any portion of the work is ready for inspection giving him sufficient notice in advance to enable him to inspect the same without affecting the further progress of the work. An approval of materials or workmanship or approval of part of the work during the progress of execution shall not bind the PMC/Client or in any way affect him even to reject the work which is alleged to be completed.
- h) The Contractor shall provide at his cost, necessary ladders and such arrangements as to provide necessary facilities and assistance for proper inspection of all parts of the work at his own cost.
- i) If any staff of contractor failed in his duties / responsibilities at site of works then without any prior notice he shall be removed from site and contractor shall deploy proper responsible technical personnel at site within 7 days. If not deployed by Contractor, then Client can either deposit the fees or deploy other engineer / engineers and the cost will be borne by contractor's bill.

2.18 Site Order Book:

A site order book shall be maintained on site and it shall be the property of Client and the Contractor should promptly sign orders given there in by the Client or his

representative and comply with them. The Contractor shall report the compliance of the orders within the time specified by the representative of client in the site order book. The blank site order book shall be maintained for this purpose. The site order books shall have triplicate pages. Original page will be retained. at the site and other copies will be with PMC and Client.

2.19 Line Out of the Project:

- i. The contractor without any additional cost shall provide all labour, instrument and material required for line out, surveying, inspection and give mark out as decided by the PMC/Client and as considered necessary for the proper execution of the work. The checking or inspection of any setting out of any line or level or work by Engineer or his representative shall not in any way relieve the contractor of his responsibility for correctness thereof. The contractor shall carefully protect and preserve all benchmarks, site rails, pegs and stones etc. used in setting out the work. Contractor shall be required to redo / rectify or adjust the line of master layout and or of individual buildings any number of times till the PMC/Client is convinced about all bench marks in regard to correctness. The Client reserves the right to relocate the buildings shown in the Tender documents (Layout) within the same site and same Built-Up Area.
- ii. The Contractor shall be permitted to erect sheds at his own risk and cost, if suitable vacant area of Client/ user is available for the purpose of his own structures for stores, offices etc. at places approved by the Client/ user. On completion of the work, the Contractor shall remove all the temporary structures erected by him and restore the site to its original condition.
- iii. The Contractor shall not use any land in the site either for cultivation or for any other purpose except that required for breaking or stacking or transporting stones, Labour sheds, Stores, for camping of machinery etc.
- iv. Contractor shall use proper survey instruments/ equipment's like Theodolite, Total station, etc. For lineout, layout marking, verticality, centre line, etc. at site of work.
- v. Prior to start of work, Layout, working plan shall be got approved from PMC/Client.
- vi. Layout, working plan shall inclusive indicative all locations of buildings, infrastructure, stacking of material, installation of machineries, cement godown, storage of cement place, steel platform, storage of centering materials, Labour tents, Water supply, sanitary arrangement, site office, laboratory, sample room for materials, PMC's office, Contractor's office, Client site office, structures to be

dismantled, structures to be retained, water bodies, source of water, drainage, source of electricity, substation, Roads STP, cementary, existing trees to be cut and replant, etc.

- vii. Prior to start the actual work at site, contractor shall be aware of all necessary requisites, i.e. NOC's, Permissions, Source of water supply, Source of electricity, dumping grounds for debris, collection of storm water, drainage, sewerage disposal, location, users requirement regarding shifting of existing facilities HT & LT Line, use of materials from dismantled structures or disposal, use of top fertile soil for plantation source of metal / sand / steel / cement / Bricks / Blocks, Facilities for testing, etc.
- viii. Any works at site shall not start without intimation to Client / PMC and User, works at site shall not be started in haphazard manner without approval to the work manual and layout, working plan from PMC/Client.

2.20 Testing and Rejection of Work:

If it is found necessary from safety point of view to test any part of the structure, the test shall be carried out by the Contractor at his own cost. Defective work not meeting the specification of the contract is liable to be rejected at any stage by the Client. The Contractor on no account can refuse to rectify the defects merely on reasons that further work has been carried out. No payment shall be made for rectifications and additional tests required for this purpose to test the quality of works.

2.21 Completion of Project:

The Project shall be considered as complete only if

- i. Water supply, drainage / sanitary disposal and electricity supply, plumbing, fire-fighting equipment, lift, etc. are proper and usable without defects.
- ii. Operational functionality of the infrastructure in the project to the satisfaction of the client/user.
- iii. Obtaining full Occupation Certificate (O.C.) from the competent authority.
- iv. The user has taken over possession of Project.
- v. Completion certificate by PMC

The work shall not be considered to have been completed in accordance with the terms of the contract until the Engineer-in-charge shall have certified in writing to that effect. No approval of material or workmanship or approval of part of work during the progress of execution shall bind the Engineer-in-charge or in any way prevent him from even rejecting the work which is claimed to be complete and to suspend the issue of his certificate of completion until such alteration and modifications or

reconstruction have been affected at the cost of the Contractor as shall enable him to certify that the work has been completed to his satisfaction.

After the work is completed, the Contractor shall give notice of such completion to the Engineer-in-charge and within 30 days of receipt of such a notice the Engineer-in-charge shall inspect the work and if there is no defect in the work shall furnish the Contractor with a certificate indicating the date of completion. However, if there are any defects which in the opinion of the Engineer-in-charge are rectifiable he shall inform the Contractor the defects noticed. The Contractor after rectification of such defects shall then inform the Engineer-in-charge and Engineer-in-charge on his part shall inspect the work and issue the necessary completion certificate within 30 days if, the defects are rectified to his satisfaction, and if not, he shall inform the Contractor indicating defects yet to be rectified. The time cycle as above, shall continue.

The issue of completion certificate shall not be linked up with the site clearance on completion of the work.

2.22 Handing Over of work:

All the works and materials before being finally taken over by MHDC, it will be the entire liability of the contractor to guard, maintain and make good any damages of any magnitude; Interim payments made for such work will not alter this position. The handing over by the contractor and taking over by the Executive Engineer or his authorized representative will be always in writing copies of which will go to the Executive Engineer or his authorized representative and the contractor. It is, however, understood that before taking over such work, MHDC will not put it into regular use as distinct from casual or incidental one, except as specifically mentioned elsewhere in this contract, or as mutually agreed to.

2.23 Disputes Resolution in Bidding Process:

- i. In the event of any disputes between the Tenderer and the Client regarding bidding process, the dispute shall be referred to the Standing Committee set up by the Client. The Committee will examine disputes and claims of the Tenderer as are referred to it. The decision of Standing Committee shall be final and binding on the Client and Tenderer.
- ii. The bidding process shall be governed by and construed in accordance with the laws of India and the Courts at Mumbai shall have exclusive jurisdiction over all disputes arising under, pursuant to and / or in connection with the bidding process.

3 Scope of Civil Work in the Project

3.1 Scope of Civil Work

3.1.1 The scope of work shall mainly comprise of

- i. Construction of all buildings as per Tender drawing and Specification
- ii. Construction of all other infrastructure as per Tender Drawing and All other allied activities to complete the project, and Detail scope and specification given **Chapter 6 (Schedule C).**

a. **Excavation, dewatering, Backfilling and plinth –**

Excavation for foundation is proposed in all types of strata such as soil, soft murum, hard murum and boulders, etc. Anti-termite treatment is to be provided at foundation level. Back filling of foundation and footings/Pile caps is to be done with departmental soil available in excavation and balance quantity shall be brought from outside by the contractor. If required 2nd class burnt brick masonry/AAC Block Masonry in C. M. 1:6 shall be provided for inner walls of plinth of the building as per the detailed drawings and requirement.

b. **Antitermite Treatment**

Anti-termite treatment is to be provided at foundation level. Back filling of foundation and footing is to be done with departmental, murum available in excavation and balance quantity shall be brought from outside by the contractor. 200 mm thick AAC block masonry in Chemical mortar and 230 mm thick fly ash ` Masonry/AAC Block masonry in C. M. 1:6 shall be provided at the plinth of the building and as shown on drawings.

c. **PCC Bed concrete-**

15-20 cm thick Rubble soling below layer of bed PCC

Plain Cement Concrete of M -15 (as per detailed drawings) grade is provided below RCC Pile Caps/footings / foundations/Ground Beams/Pedestals.

d. **RCC Footing/Pile caps and foundation-**

For the Main Buildings as well as Ancillary Buildings such as Creche, Club House, Amenity Building and Commercial Buildings pile foundation with cast-in-situ piles of M-30 (as per detailed drawings) grade with pile caps is proposed.

e. **RCC Columns, Beams, Lintels, Slab, Waist Slab, Chajja, Pardi, Stair Case –**

All other structural members are proposed and designed as M-25 and M-30 (as per detailed drawings) grade Reinforced Cement Concrete. TMT- FE-500 (TOR) reinforcement of various diameters shall be used for R.C.C work.

For UGWT, OHWT, ~~OWC~~, RWHT, ~~STP~~, ~~WTP~~ base slab, top slab, beams, walls are designed as RCC M-30 grade

The Ancillary buildings shall be considered constructed using Conventional Formwork and Main Buildings with Mivan Technology.

f. **Water Proofing –**

Terrace Water proofing- Cement based Water Proofing Treatment to Terraces (Indian water proofing or alike) with brick bats laid in required slope to drain the water for any span after cleaning the base surface. Applying a coat of cement slurry admixed with approved water proofing compound and laying the brick bats on bottom layer in C.M.1:5 admixed with approved water proofing compound filling up to half depth of brick bats, curing this layer for 3 days, applying cement slurry over this layer joints of brick bats with C.M.1:3 admixed with approved water proofing compound and finally top finishing with average 20 mm thick layers of same mortar added with jute fibre at 1 Kg per bag including finishing the surface smooth with cement slurry admixed with approved water proofing compound. Marking finished surface with false squares of 300mm x 300 mm. making the junctions at the parapet rounded and tapered top for required height, with drip mould at the junction of plaster and parapet and curing and covering 10 years Guarantee against leak proofness on Court fee stamp paper of Rs. 500/- including ponding test.

Waterproofing in W.C. and Bath- Brick bat coba in all position including providing and laying 12mm bedding in cement mortar 1:3 on virgin concrete slab with waterproofing compound @ 1Kilogram/per bag of cement laying brick bat coba of required thickness in cm 1:5 with waterproofing compound 1 Kilogram/bag of cement grouting and finishing the top layer with 20mm thick brick bedding in cm mortar 1:3 with waterproofing compound 1 Kilogram/per bag of cement and testing the treated portion for 48 hours by pond test and covering ten years' guarantee on requisite stamp paper including curing.

Waterproofing in Lift pit below PL, on Chajja, OHT - Acrylic polymer modified cement-based water proofing coating with fibre glass mesh mixing at the rate of powder to liquid (2:1) by weight covering 9 to 10 sqm /kg with two coat using approved chemicals for masonry and concrete surface by brush covering 7 years guarantee on Stamp Papers.

Waterproofing Balcony- Water Proof Bedding 25 mm thick in C.M. 1:3 including using approved water proofing compound in specified proportion as per manufacturers specifications for per bag of cement including levelling, curing and covering 10 years guarantee on court fee stamp paper of Rs.500/- including ponding test.

UG Tanks-Box type Shahabad waterproofing for UG tanks.

(Refer schedule “C”)

g. Walls-

- (i) Fly ash brickwork /AAC blockwork 230 mm thick for plinth, plinth protection entrance steps etc. as per schedule C and Drawings
- (ii) RCC Non-structural walls for Main Buildings partition walls proposed to be constructed with Mivan construction method.
- (iii) RCC Walls for Lifts, OHWT
- (iv) Autoclaved Aerated Concrete Block masonry to be provided for buildings proposed to be constructed with conventional construction method.
- (v) Fly ash brick masonry for compound wall.

h. Doors-

- (i) Main doors - Solid core Flush Door Shutter commercial in single leaf 32 mm thick without ventilator commercial type of exterior grade with wood Polymer composite (WPC) door frame for all doors. (Refer schedule “C”)
- (ii) Internal doors - Solid core Flush Door Shutter commercial in single leaf 32 mm thick without ventilator commercial type of exterior grade with wood Polymer composite (WPC) door frame for all doors. (Refer schedule “C”)
- (iii) Bath/Toilets- 18 to 20 mm thick Green Marble stone frame with Fiberglass fiber glass Reinforced polyester door shutter 30 mm thick. (Refer Schedule C).
- (iv) Double shutter Openable powder coated aluminium door/window for balcony.
- (v) 2 Hr fire rated door for staircase.

i. Windows- (Refer Schedule C).

- (i) All frames / sills in Green Marble.
- (ii) Powder Coated Aluminium sliding window of two tracks.
- (iii) Powder coated aluminium louvered windows / ventilator.
- (iv) Powder Coated Aluminium openable window.
- (v) 50mm thick Precast Reinforcement Jali.

j. Flooring, Skirting and Dado-

- (i) 60 X 60 mirror / glossy finish vitrified tiles for flooring in rooms.
- (ii) 30 X 30 Antiskid ceramic tiles for flooring in toilets.
- (iii) 60 X 60 mirror / glossy finish vitrified tiles for Skirting in rooms.
- (iv) 30 X 45 ceramic tiles for Dado and Skirting in toilet.
- (v) 30 X 30 chequered tiles for parking.
- (vi) Polished Shahabad stone flooring for Lobby and staircase.

k. Plaster and Pointing-

- (i) External - 15 mm thick in cm 1:4 sand faced plaster and second coat with 6 to 8 mm thick finishing in cm 1:4 all Ancillary buildings and compound walls, gate columns etc. (Refer Schedule C)
- (ii) Internal –12 mm thick plaster in CM 1:4 with neeru finish for all internal walls, corridors of Ancillary buildings. (Refer Schedule C)
- (iii) Waterproof Plaster in W.C. and bath sunken side portion 12 mm thick in cement mortar 1:3 with neat finishing, floating using waterproofing compound at the rate of 1 Kilogram. per bag of cement of approved make and manufacturer and curing.
- (iv) Two coats of wall care putty for RCC walls and slabs / ceiling.

l. Painting-

- (i) Colour-wash of approved colour and shade for EWS building ceilings.
- (ii) Priming coat and Two coats of OBD Paint on internal walls painting for EWS building.
- (iii) OBD Paint for LIG building ceilings.
- (iv) Two coats of waterproof cement paint for external surface.
- (v) Priming coat and Two coats of Plastic Emulsion Paint on internal walls painting for LIG buildings.
- (vi) Colour-wash of approved colour and shade for Lift shaft.
- (vii) Primer and oil paint for M.S. grills & structural work

m. Staircase –

- (i) Machine cut machine polished 20 mm to 25 mm thick Shahabad stone for treads and risers in staircase.
- (ii) MS Ladder for overhead water tank.
- (iii) MS Gate at Entry to the plot.
- (iv) M.S. hand rails for staircase. (Refer Schedule 'C')

n. Counter wash hand basin of 55 x 40 CM size.

o. Granite Kitchen platform with SS sink of size 600mm x 450mm.

p. Iron Works

- (i) MS Hand railing for Staircase and balcony.

q. Infrastructure Works –

- (i) RCC walls for Storm Water Drains and Utility Trench.
- (ii) RCC Under Ground Water Tank & Rain water Harvesting tank
- (iii) Shahabad box type waterproofing for UGT tanks.
- (iv) Acrylic polymer modified cement based waterproofing coating with fiberglass mesh for internal faces of UGT & RWH tank.

- (v) MS Grill gates.
- (vi) Bituminous open parking and internal roads.
- (vii) Lawns and trees for soft landscaping and horticulture.
- (viii) UCR / Block Masonry Compound wall.

3.1.2 The scope of work shall include

- a) Site preparation such as levelling of sites by cuttings and fillings including conveyance and disposal of any excess / unserviceable excavated material to disposal field. Cutting or replantation of trees, if any, removing / shifting electric lines, telephone lines, plumbing lines, drains, sewer lines, chambers, rain water harvesting, water bodies, etc. as per requirement and reconnecting the same to proposed service lines as necessary as per drawings. The salvage value of dismantled serviceable or unserviceable material shall be deemed to be considered in the rate quote for the project.
- b) Construction of site office setup (about 60 m²)
 - 1. Construction of site office about 60 m² or more (for Contractor, PMC, Client) as shown in drawing along with meeting hall and separate rooms with necessary infrastructure facilities including toilets for all staff. Big table along with ten chairs shall be provided in meeting hall and two small tables along with three chairs shall be provided to rooms. Ceiling fans shall be provided along with Fans, tubes, plugs, water purifier, and other small requirements as possible. Construction of Field laboratory / along with sample room about 50 m² for testing the materials with stipulated testing equipment as given in Chapter (3.3.3), etc. The Contractor shall maintain the site office; toilets, laboratory, sample room and other temporary structures erected by him in good and hygienic conditions till the completion of the project and shall dismantle the same after the successful handing over of the project.
 - 2. Contractor/firm shall provide Brand New Laptop with required software's (MSP/AutoCAD Mandatory) with inverter (2 Hours Battery Backup) and A4 size printer (all in one) etc. along with A.M.C. All licenses should be in the name of Maharashtra Housing Development Corporation Limited, Mumbai.
- c) Plate load test / pile load test according to BIS or Bore Logs shall be taken before or during execution of the foundation work in consultation with PMC / CLIENT.

- d) Excavation and disposal of all excess excavated material to municipal dumping ground or as directed by Client / PMC including loading, unloading and transportation etc., Soling, PCC, RCC, brickwork, plaster (Internal and External), paint (Internal and External), water proofing, flooring, skirting, dado, Doors, Windows, structural works, roofing, false ceiling, anti-termite treatment and other miscellaneous works required for the completion of all buildings as per the Tender drawings and specifications, etc. complete.
- e) Buildings shall be of conventional cast in situ RCC structure with conventional / Aluform or using Alternative Building Technologies as mentioned in Part B, Schedule "C" of Chapter 6.
- ~~f) Furniture work, if any, as per interior drawings and specifications etc. complete.~~
- g) Water supply scheme as per tender drawings and specifications etc. complete.
- h) Plumbing arrangements with fittings and fixtures as per the drawings and specifications, etc. complete.
- i) Firefighting works as per drawing and specifications, etc. complete.
- j) STP MBBR of 1450 CMD capacity (Including taxes, transportation, installation supervision etc.) including all required necessary Civil work, etc. complete plant including cost of operator, chemical & other consumable (without energy Bill) for one year etc. complete as per design, drawings and specifications approved by the Client. S.T.P. treated water shall be potable and shall comply with the requirement of additional specifications approved by Client.
- k) Sewerage disposal system with inspection chambers connected to STP with Tertiary Treatment Systems per drawing and specifications, etc. complete. Based on consultant's recommendation/ specification.
- l) Organic Waste Converter (OWC) plant of 2200 kg/day capacity as per drawings and specification etc. complete based on consultant's recommendation / specification.
- m) 27m³/hr Pre-Treatment Plant for Rain Water. Rain Water Harvesting system well connected with Rain water pipes from terraces of building, storm water drains
- n) RCC Overhead tanks Domestic & flushing tank as per tender drawings and specifications etc. complete.
- o) Fire alarm system with conventional fire alarm panel, heat & smoke detectors in Amenity Fitness Centre, Creche, fire bell, external & internal electrification, providing and fixing of electricity meters, lightening arresters for each building, solar PV system as per drawings, specifications, etc. complete.

- p) Electrical meter/ panel room along with all accessories, equipment's, all types of civil works of substation & adequate / appropriate size shed for DG set as per the requirement of the power Supply Company, etc. complete.
- q) Bituminous Internal Road and parking area, (Ref schedule "C" infra specifications) as per tender drawings and specifications, etc. complete.
- r) Open / Underground / Trough Gutter Storm water drain / ~~RCC Hume Pipe NP2 Class of various sizes~~ with necessary inspection chambers for crossing, as per PMC 's designs, drawings and specifications etc. complete.
- s) Supply & installation of photo-voltaic solar power generation with net metering system, for each building as shown in drawings including structural work, etc. complete as per requirement and the Client's designs, drawings and specifications.
- t) Landscape, hardscape and horticulture work including civil works as per Client's designs, drawings and specifications etc. complete.
- u) Civil works for Elevators for Passengers elevator as per Client's designs, drawings and specifications, etc. complete.

3.1.3 Photographs & Installation of CCTV Camera:

The Contractor shall furnish the photographs in various angles and shall record the date of the work completed on photographs, to the satisfaction of the PMC/Client, at every stage of Project the Contractor shall install CCTV camera, and video conference facilities, etc. at site of work along with internet connection to observe the live activity of works in conference hall of Client. The photographs of site shall be uploaded on yammer in first week of every month to verify the progress of work by Client.

3.1.4 Official Secrecy:

The Contractor(s) shall take whenever required, all necessary steps to ensure that the persons employed on any work in connection with this contract, are aware that the Indian Official Secrets Act, 1923 (XIX of 1923) applies to them and shall continue so to apply even after execution of such work under the contract.

3.2 Work Specifications:

- 3.2.1 The whole work shall be carried out strictly in accordance with the Tender drawings (unless otherwise directed by the Client), description of the items in Schedule-B of Chapter 6 and the detailed specifications in the Standard Specification Book of the Public Work Department and additional specification in chapter 13 of this document.

- 3.2.2 Detailed specifications given in the Schedule-C of Chapter 6 are the specifications for items to be executed.
- 3.2.3 During construction of works, the Contractor(s) shall make all necessary arrangements, at his own cost, for proper drainage of the site by providing pipe, channels, temporary crossings, etc. as may be required according to the directions of the PMC/Client. The Contractor shall submit to the PMC/Client, proposals for ensuring the drainage of the site by taking into consideration the final drainage plan proposed in the project and for carrying out the works in the manner approved by the PMC/Client.
- 3.2.4 The Contractor(s) shall keep the site free from any encroachments. He shall be required to take, at his own cost, measures to protect the site from any encroachment and keep the site free from all encumbrances till final completion of work and handing over the site to the Client and or concerned authorities (i.e., user). Due care to be taken while execution & all safety measures of residents, movement plan of residential pedestrian and vehicular traffic and Site related movement plan to be prepared and approved.

3.2.5 Discrepancies in Drawings or Specifications:

- a) The drawings and specifications are to be considered as mutually explanatory to each other, and mutually supportive of each other, with the detailed drawing being followed in preference to small-scale drawings, and figured dimensions in preference to scale, and the special conditions if any, in preference to general conditions.
- b) In the event any discrepancies appear, as to the materials or the due and proper execution of the work, or as to the measurement or quality and valuations of the work executed under this contract, or as beyond the scope of work, there upon, the same shall be referred to the Client and the Client's decision shall be binding upon the Contractor and the Contractor shall execute the work according to such decision (subject as aforesaid) and without extra charge and shall also perform all such works and things as may be necessary for the proper completion of works as implied by the drawings and specifically shown and described in the said drawings and specifications. USE OF ISI (Indian Standards Institute) SPECIFICATIONS:
- c) In case where no particular specifications are given for any articles to be used under the contract, the relevant specification as per the Indian Standards Institute or National Building Code, U.K., U.S.A. or any other international standards, as may be approved by the Client, shall apply.

3.2.6 Missing Item / Specification / Works:

The Contractor shall bring to the notice of the Client, any missing item of construction activity or specification which in its view, is absolutely necessary for the completion of work and/or if any material of particular make/brand as mentioned in the approved list of material is not available, then same shall be brought to the notice of the Client. After the due notice as above, the Client shall give decision and the Client's decision shall be binding upon the Contractor. The Contractor shall execute the work according as per the conditions of contract.

The Construction work to be done is described in drawings & specifications. As it is not possible to show each & every minor detail in drawings, and where such instant occurs (small portion of work details not given in drawings & specifications) the construction work of that component or part of work is to be done considering the good civil engineering practices without any extra claim.

The information about important provisions given in tender drawings, particulars of items is for guidance to Contractor, work to be done as per specifications and as shown in the detailed drawings.

3.2.7 Additional / Extra Work During Due Course of Construction:

The Contractor shall carry out the additional or Extra work other than those mentioned in the Contract if it is ordered by Client in writing in due course of construction or even after completion of work as found necessary by the Client. The payment against such item shall be based on the current SSR of the State in which work lies, along with actual rate analysis.

3.2.8 Deleted / Additional Item of Work During Due Course of Construction:

The work which are not carried out by contractor as specified in contract and for which the Rate has been quoted by the Contractor or if the Client has informed in writing to the contractor for deletion of particular item of work and directed for new item of work to be performed instead, due to change in scope of work or specification or by any reasons whatsoever, then Client shall adjust the amount to be paid against the new item, with the amount of the deleted item of work at the current schedule of rate and thereby pay the differential amount for new item of work based on the current DSR of the division in which work lies along with actual rate.

3.2.9 No claims will be entertained on basis of additional working drawings, Extra detailed drawings provided during the execution of work because, inclusion of all minor detailed drawings is difficult to include at the time of calling the tender.

3.3 Materials and Samples

- 3.3.1 The samples of each class of material shall be got approved from the Engineer prior to its actual use. All material which cannot be tested in the field laboratory shall be got tested from designated Laboratory such as Govt. Engineering Colleges, National Test House, structwel, MHADA Laboratory, etc., and at specified codal frequency, as directed, and the Contractor shall bear all prescribed charges for the same. Also, material which can be tested in the field laboratory at specified codal frequency, at least 30% samples shall be got tested from designated Laboratory, as directed and Contractor shall bear all charges for the same as per prescribed. The Client/ Engineer is empowered to reject or approve materials based on test report and his decision shall be final and binding on the Contractor and no disputes in this regard shall be entertained.

Testing Report of authorized laboratory shall be produced prior to actual use of material at site of work, sample material shall be certified by PMC and ensure that the same material shall be allow and use at site of work. for all the purpose of items in construction activities. PMC is sole responsible for the material at site of work and to ensure use of such materials for the construction.

- 3.3.2 All materials and workmanship shall be as prescribed in the contract and shall be binding subjected from time to time, to such tests as the Engineer may direct at the place of manufacture of fabrication, or on the site or at such other place or places as may be specified in the contract, The Contractor(s) shall provide required assistance, instruments, machines, labour and materials for examining, measuring and testing of the quality, weight or quantity of any material to be used and shall supply samples of such materials before incorporation in the works as may be selected and required any by the Engineer, without any additional cost.

3.3.3 Samples and Testing of Materials:

- i. All materials to be used on work shall be got approved before starting of the work from the PMC/Client and shall pass the test and/or analysis required by it, which will be,
 - a) As specified in the specification for the items concerned and /or
 - b) Bureau of Indian Standards specifications (whichever and wherever applicable) or
 - c) Such recognized specifications acceptable to the Client.
- ii. The contractor at his own cost shall establish fully equipped Field laboratory and sample room of about 50 m² at site to carry out the routine tests of materials, which are to be used on the work. The tests shall be carried out in the field

laboratory. At least 30% tests shall be carried out in nearest quality control laboratory authorized by state or central government or any other laboratory approved by the Client and shall submit the test reports to the PMC/Client. The contractor shall supply all materials and install equipment, testing machines as per table below. No bills shall be prepared without approved test report as per frequency of standards and certified by PMC.

EQUIPMENT FOR FIELD LABORATORY:

1.	Slump Cones with platform	3 Nos.
2.	Cube Molds	36 Nos.
3.	Cement Testing Equipment (Initial/Final setting, accelerated compressive test)	1 Set
4.	Oven including set of pans	1 No.
5.	Cube Testing Machine	2 No.
6.	Weigh Balance	2 Nos.
7.	Spring Balance	2 Nos.
8.	I.S. Sieves (For Fine and Coarse Aggregate)	2 Sets
9.	Vernier Caliper	2 Nos
10.	Micro Meter	2 No.
11.	Silt Measuring Jar	4 Nos.
12.	Total Station	1 No.
13.	Spectrometer	2 No.

- iii. The contractor shall, at his own risk and cost make, all arrangements and /or shall provide all such facilities as the PMC/Client shall require for collecting, preparing required number of samples of the material to be used in the works, for test or analysis at such time and to such places as shall be directed by PMC. The Contractor shall arrange testing as required by the codes and specifications at his own cost. Testing shall be done in the presence of authorized representative of the Client/PMC at the nearest approved laboratory. If additional testing other than as required by the specification is ordered by the client, the testing charges shall be borne by the Contractor.
- iv. The Contractor or his authorized representative shall remain present in designated laboratory authorized by the client while testing the samples furnished

by him. The results of all the tests carried out in such laboratory shall be binding on the Contractor.

- v. The contractor shall if and when required, submit at his cost the samples of materials to be tested or analysed and if so directed, shall not make use of or incorporate in the work any materials represented by the samples until the required tests results or analysis have been made available and the materials finally accepted by the PMC / Client.
- vi. NDT (Non-Destructive Testing): - Minimum of five (5%) of structural elements shall be tested by non-destructive test, such as ultra-sonic test, pulse velocity test IS 1331, Part I and rebound hammer test IS 1331 part I and rebound hammer test IS 1331-part II probe penetration and maturity test. (As per clause no. 13.4 of Chapter 13 of this document.)
- vii. The contractor shall not be eligible for any claim or compensation either arising out of any delay in the work or due to any corrective measure required to be taken on account of unsatisfactory test results.
- viii. Mix design of concrete shall be obtained from laboratory approved by Client/ PMC. The cement consumption for controlled concrete shall be as per mix design of concrete as approved by Client/ PMC.
- ix. If at any stage of the works, it is found necessary from a safety point of view to test any material and / part of the structures constructed by the Contractor, the test shall be carried out by the Contractor when so directed by the Engineer/Client at its own cost and risk.
- x. Following Test Abstract duly signed shall be submitted by Contractor / PMC.

Sr. No.	Item	Quantity (Cum/Nos)	Test required as per standard Specification	Test carried out (Nos)	Test		PMC's Remark
					Passed (Nos)	Failed (Nos)	
Contractor Signature			Signature of PMC				

- xi. The payment for the work done shall be subjected to submission of the Quality Control (QC) results compliant to the requirement.

3.3.4 All samples shall be supplied by the Contractor(s) at his own cost if the test thereof is clearly included in the contract.

3.3.5 The approved samples of materials such as tiles, door window fixtures, water supply and sanitary fixtures etc. shall be duly signed by PMC/Client and Contractor or his

Project Engineer and shall be kept in sample room for reference. All samples shall be under the control of PMC for the purpose of execution at site.

- 3.3.6 The cost of carrying out any test shall be borne by the Contractor(s) as and when required so as to ascertain quality of work executed/ to be executed.
- 3.3.7 All sanitary and water fittings and pipes etc. shall be tested and their size and pattern shall confirm to the latest Municipal Bye-laws. This part of the work shall be carried out through a licensed plumber according to latest standard practice and as per the Municipal Bye-laws applicable in the area.
- 3.3.8 The Contractor(s) shall inform in advance in writing the dates of casting of slab to the PMC / Client and shall fix the programme for the same with the approval of the Client. Any change in programme shall also be intimated, in writing, in time, to the Client / PMC.
- 3.3.9 The Contractor(s) shall give hydraulic and smoke tests for all pipes and fittings and glazed stoneware pipes, as directed without any extra cost.
- 3.3.10 The terrace slabs of all buildings, Porch slab, W.Cs. baths, sinks, toilets, top slab of staircase cabins, overhead water tanks, all floors etc. must be thoroughly leak proof. Ponding Test in this regard shall be certified by PMC and submitted to the Client prior to submission of bills along with photographs.
- 3.3.11 Proper rain water drainage arrangement shall be ensured and certificate in this regard shall be produced duly signed by the contractor and PMC to the client. Proper rainwater protection shall be ensuring to avoid leakages, seepages, dampness, etc. Certificate in this regard duly signed by Contractor and PMC shall be submitted to the Client.
- 3.3.12 The Contractor shall give additional bank guarantee (from Nationalized / Schedule Bank) before completion of work, for water proofing & anti-termite work equivalent to 1% each of estimated cost of these items valid from the date of completion of the whole project up to end of 10 years. In case of leakage /dampness is observed the contractor shall immediately start and complete the repair work and complete the same to satisfaction of the PMC/Client within 30 days of notice of any leakages. If the contractor fails to do so then the Bank Guarantee as above shall be encashed by the Client for rectifying the work. However, no refund shall be payable to the contractor, even if the cost incurred on rectification is less than Bank guarantee amount for waterproofing and anti- termite work. (Refer clause 9.6 (III) and (IV) of chapter 9 of this document)

3.3.13 For residential building project, sample flat should be completed in all respect within two months period after completion of third slab of any one building. If contractor fails to complete sample flat then Rs.1,00,000/- (Rupees One Lakh Only) per month will be recovered from the contractor.

3.4 Infrastructure at Site and up to Site:

3.4.1 Infrastructure to be provided by the Contractor (s) at his costs shall include the following works.

- a) To make available potable water and electricity for construction purpose. (Specify sources)
- b) Temporary Access roads, Cross Drainage Works, etc. if required as per by- laws of the local authority for the site.

3.4.2 Infrastructure to be provided as per Contract

- a) Water supply distribution network within the campus including water meter, valves, meter room, fencing, etc. complete as per approved standards and network designs. Separate fire hydrant line shall be laid for fire-fighting including fire hydrants as per approved drawings.
- b) Sewerage system as per approved standard and network design including connecting to main sewer line or to septic tanks, soak pits etc. as the case may be and as per approved drawings.
- c) Tapping the power supply at source from nearest pillar box source as per drawings and as available on site. Street lighting as per Electrical supply company standards and as approved by the competent authority.
- d) The Contractor(s) shall obtain permissions/approval from local Municipal Corporation, Municipal Council and other authorities like PWD, Z.P, MIDC, MMRDA, Director of Town Planning, MIDC, etc., as the case may be, for infrastructure up to the plot under reference and shall ensure timely provision of such infrastructures within the contract period.

3.5 Completion and Handing Over

3.5.1 Project shall not be considered as complete, unless all items in schedule "C" are completed along with additional items if instructed and all works as shown on drawings along with additional details issued at the time of execution, modifications in drawings if any and instructions given on site by the PMC/Client from time to time and confirmed in writing. Electric and water supply to the building is commissioned

i.e., to say that the work completes in all respects to make the building ready for use and occupation including all infrastructural amenities.

- 3.5.2 Before handing over the buildings and the site to the Client, the Contractor (s) shall obtain certificate (whichever is applicable) for – (a) Completion of buildings; (b) Building occupation; (c) Malaria Department; (d) Debris removal (e) Assessment; and (f) Fire safety, from the Government authorities and certificate from other authorities, like M.I.D.C., M.S.E.D. Co. Ltd. & Handing and taking over certificate of P.W.D. Civil & P.W.D. Electrical, etc.

3.6 Green Building (Deleted)

~~Green building refers to both a structure and the application of processes that are environmentally responsible and resource-efficient throughout a building's life cycle: from planning to design, construction, operation, maintenance, renovation, and demolition." This requires close co-operation of the contractor, the architects, the engineers and the client at all project stages as such following issues shall be taken care of:~~

- ~~i. Preserve and Protect Landscape: – If there are existing grown up trees on site then protection and preservation/ replantation of trees.~~
- ~~ii. Soil Conservation~~
- ~~iii. If top soil is fertile then shall be removed and stack separately and later on shall be use for landscaping/plantation work, etc. balance soil if any shall be disposed of as directed by PMC/Client.~~
- ~~iv. Provision of drinking water, healthy and clean living conditions and sanitation facilities shall be provided for workers.~~
- ~~v. Reduction in air pollution during construction: – Measures to be taken on site to reduce air pollution.~~
- ~~vi. Efficient water use during construction: – Efforts to be taken to reduce use of potable water during construction.~~
- ~~vii. Building shall be designed in compliance with NBC code in order to be user and liable friendly.~~
- ~~viii. Minimize use of ozone depleting substance.~~
- ~~ix. Waste water treatment.~~
- ~~x. Waste recycle and reuse~~
- ~~xi. Reduction in waste during construction.~~
- ~~xii. Resource recovery from waste.~~
- ~~xiii. Maximum use of Natural Resources to minimize the cost of operation and maintenance (i.e. sunlight, wind flow, rain water etc.)~~

3.7 Defective Work and Defect Liability Period (DLP):

The term "Defect" includes imperfection, inadequacy, lack in supervision, mistakes and errors in work, shortfalls, shortcoming and faults and flaw in execution, use of substandard and non- approved material, imperfect fixing of material, improper fixtures and fittings, unsatisfactory quality and workmanship, dampness and

leakages; works not in line and level, interrupted electrical supply, choke-up of drainage, defective water supply line(s), inspection chamber(s), defective storm water drain(s), incorrect camber and gradient of road, etc. and all the work which is not considered to be satisfactory by the PMC / Client.

The Contractor shall be liable to carry out the rectification or replacement of defective work / material within a period of 15 days from the date of notification of such defect by PMC/Client, falling within the above definition during defect liability period of 5 (Five) years from the date of virtual completion of work duly certified by PMC and accepted by Client. After completion of project, if user points out any defect orally or in writing to the Client, such defect shall be rectified by the Contractor within 15 days. If Contractor does not rectify or attend such defects then action shall be taken against the Contractor.

After rectification, appropriate certificate from PMC regarding rectification of defects is essential. The deemed date of completion shall be considered as basis for computation of defect liability period.

4 Design Criteria for Structural Designs of Buildings

General description regarding parameters incorporated in the design shall be considering following requirements.

4.1 General Requirements:

- i. The Contractor shall construct the buildings on the site and construct all other infrastructure facilities as per the scope given in the contract.
- ii. The Contractor shall be fully responsible for safety, strength and durability of the structures.
- iii. The Contractor shall carry out necessary investigations and surveys to develop intimate knowledge of the site condition, availability, approaches to the sites, etc. This shall also include taking of trial pits, trial bores, Plate load test, testing of soils, obtaining test results from approved government laboratory, etc.
- iv. All concrete used for structural RCC members shall be Design Mix concrete and minimum grade of such concrete shall be of M25 Grade.
- v. If the SBC is less than 15 T/m² then either pile foundation or Raft Foundation with necessary ground improvement techniques shall ordinarily be adopted. If the SBC is in between 10 to 15T/m² then choice of other suitable foundation shall be adopted only with prior approval of client.
- vi. It shall be economical.
- vii. It shall ensure soundness of design and strength and durability of the structure, based on designs on requirement of adequate strength and satisfactory serviceability.
- viii. It shall ensure speedy and easy construction, and lead to appreciable economy.
- ix. It shall maintain the aesthetics as per the Architect's Plans and Architectural beauty as a whole. Architectural Plans / drawings shall be consonant with structural soundness based on designs requirement of adequate strength and stability.
- x. The necessary certificate of structural stability shall be given by the PMC, before issue of completion certificate by the PMC.

4.2 Restriction on Type of Structures:

- 4.2.1 Structures, which require material for which foreign exchange is required.
- 4.2.2 Structures, which require floating columns.

- 4.2.3 Cantilevers of Span more than 3 meters.
- 4.2.4 Structural configuration with soft storey / weak storey.
- 4.2.5 Configuration inducing large torsion in Horizontal / Vertical planes. if any variation in above conditions, then prior permission of Client must be taken.

4.3 Requirements and Data:

- 4.3.1 The designs of building, Number of stores and other details shall be based on Architect's drawings and satisfying prevailing by-laws of Local Municipal Authority.
- 4.3.2 If there is any vertical expansion in future, the design shall cater for that expansion.
- 4.3.3 If there is any horizontal expansion planned in future, provision for the same shall be made by way of providing expansion joints, combined footings, etc. or other provisions required for the same. Foundation near such joints also shall be designed accordingly taking this factor into account.
- 4.3.4 The parameters as mentioned below, which will affect the designs, shall be considered, while carrying out the designs.

a) Location of building (Coastal / Non-Coastal).	Kamptee, Dist. Nagpur, Maharashtra. (Non-coastal)
b) Exposure to environments as defined in I.S.: 456-2000.	Moderate
c) Seismic Zones for which the designs are to be prepared (This shall be as per seismic zone given in I.S.:1893-2016.	II
d) Seismic co-efficient to be adopted (As applicable in "c" 'above) for static Method of analysis as per I.S.: 1893-2002.	Z=0.1
e) Importance factor as defined in I.S.:1893-2002.	1.2
f) Type of foundation proposed [Depending upon the height of the building and S.B.C. of foundation strata.]	Pile Foundation. 600 & 750 dia pile with capacity 120T & 190T respectively.
g) Presence of sulphate, chloride and other harmful chemicals in the sub-soil as per test	

results and measures proposed to obviate their effects.	
h) Bearing capacity of founding strata (shall be got Approved by Field Executive Engineer of concerned work.	600 & 750 dia pile with capacity 120T & 190T respectively. Pile socket length is 6D as per soil report.
i) Depth of founding strata below finished ground level.	Pile depth 10mt + 6D socket length from O.G.L. as per Soil report.
j) Values of K1, K2, K3 factors required to calculate design wind speed as per I.S. 875-2015 (Part – 3)	K1, K3, K4=1 & K2=1.12
k) Type of Anti-Corrosive treatment.	(To be approved by Client)

4.4 Verification of Survey Data:

The following items shall be included in the scope of work of the project:

- i. The Contractor shall verify the relevant data from the tender document such as detailed land records and topographical surveys of the site at suitable contour intervals, boundaries of the site. If earlier fixed boundary stones are moved / missing, and ensure existing public utility service lines, existing streets/accesses in relation to the site, adjoining services, i.e., water supply, sewerage, roads and street lights, cemetery, nallahs, etc. with respect to adjoining plots.
- ii. The Contractor shall carry out layout of buildings and roads in consultation with the PMC before commencement of any work and ensure that the roads and buildings are fitted within the plot and also match properly with existing roads, road levels, drainage invert levels, etc. The entire proposed work of layout and levels shall be got approved in advance from the PMC.
- iii. Contractor shall have to carry out necessary investigation and surveys to develop the knowledge of the site condition, availability of quarries and other building materials, topography, labour availability, water and power availability, approaches to the sites etc. This shall also mean taking of trial pits and trial bores, testing of soils etc. for foundation conditions. Construction Agency shall contact various local authorities and obtain all the necessary relevant information about the plots especially regarding existing infrastructure facilities like sewerage, water supply mains, electric supply line, their present locations, levels etc. The Construction agency however will have to depend on his own detailed surveys for obtaining all the data.

4.5 Obligatory Provisions:

- 4.5.1 The building shall ~~be of conventional cast in situ RCC structure or~~ using Alternative Building Technologies as mentioned in Part B, Schedule “C” of Chapter 6.
- 4.5.2 Because of the possibility of reversal of shears in beams, due to the earthquake forces shear reinforcement shall be provided in the form of vertical stirrups only as they will be effective both for upward and downward shear. Closely spaced stirrups are preferable.
- 4.5.3 Design of Structural Components shall conform to the Criteria laid down in the latest editions of the following IS Codes of Practice and Standard specifications published up to the last date of receipt of tender form subject to the departures stipulated.

1.	IS:432 (Part I)	Mild steel and medium tensile steel bars and hard drawn steel wire for concrete reinforcement.
2.	IS:456	Indian Standard code of Practice for plain and Reinforced Concrete Fourth Revision
3.	IS800	Indian Standard code of Practice for use of structural steel in General Building construction.
4.	IS:875 All parts	Indian Standard code of Practice for structural safety of Building: Loading standards.
5.	IS:1080	Indian Standard code of Practice for design and construction of shallow foundation in soils other than raft, ring and shell
6.	IS:1642	Code of practice for fire safety of buildings General
7.	IS:1643	Code of practice for fire safety of buildings General (Exposure Hazards).
8.	IS:1644	Code of practice for fire safety of buildings. General (Exit Requirements and personal Hazards.)
9.	IS:1786	High strength deformed bars and wires for concrete reinforcement
10.	IS:1892	Code of practice for subsurface investigation for Foundations
11.	IS:1893	Criteria for earthquake resistant design of structure

12.	IS:1904	Indian Standard code of Practice for structural safety of Building: Shallow foundation.
13.	IS:2502	Code of Practice for Bending and Fixing of Bars for Concrete Reinforcement
14.	IS:2911 (all parts)	Code of practice for Designs and construction of pile foundation (All parts)
15.	IS:2950	Indian Standard code of Practice for design and construction of Raft foundation.
16.	IS:3370 All parts	Code of practice for concrete structures for storage of liquids (All parts)
17.	IS:3414	Indian standard for Designs and Installation of joints in buildings.
18.	IS:3792	Guide for heat insulation of non-industrial Buildings
19.	IS:3935	Code of practice for composite construction
20.	IS:4326	Indian Standard code of Practice for Earthquake resistant design and construction of buildings.
21.	IS:6935	Method for determination of water level in a bore Hole
22.	IS:13620	Fusion bonded epoxy coated reinforcing bars.
23.	IS:13920-1993	Code of practice for ductile detailing of reinforced concrete structures subjected to seismic force.
24.	IS:14893	Non-destructive integrity testing of piles (NDT) – Guidelines
25.	IS:15916	Building Design and Erection Using Prefabricated Concrete - Code of Practice
26.	SP:7	National Building Code of India
27.	SP:16	Designs aids for reinforced concrete to I.S .456-2000
28.	SP:22	Explanatory Handbook on codes for earthquake Engineering
29.	SP:24	Explanatory Hand-Book on I.S.456-2000 Code of practice for plain and reinforced concrete
30.	SP:26	Explanatory handbook on Masonry code
31.	SP:34 (S&T)	Handbook on concrete reinforcement and detailing
32.	SP:34 (1978)	Handbook on concrete reinforcement and Detailing

33.	SP:22-1982	Explanatory Handbook on codes for earthquake engineering
34.	IS:2502-1963	Detailing of reinforcement
35.	IS:1904-1986	Structural safety of building shallow foundation.
36.	IS:1911-1967 IS:875-1964	Unit/Weight/loads
37.	IS:4326-1993 IS:1893-1984	Code of practice of design and Construction of earthquake resistant construction.
38.	SP:26-1984	Explanatory handbook on Masonry code by Bureau of Indian Standards
39.	IS:456-2000	Code of practice for plain reinforced concrete
40.	IS:1904-1968 IS:2911-1979	Element design
41.	IS:3370 (Part-I, II and III)	Water retaining structure (Part I, II, III)
Note: The above list given is representative only and not exhaustive.		

4.5.4 For special provisions, like shells, folded plates, pre-stressed concrete, pre-cast elements etc., relevant I.S. Codes shall be followed. For aspects which are not covered by these codes or any other I.S. Codes available, relevant A.C.I./British Standard Codes may be referred to. As for the stipulations which are not dealt with in any of these codes, the decision of the Client concerned shall be final and binding.

4.5.5 The diameter of steel, Nos. of bars, spacing of steel and stirrups, development length of steel, curtailment of steel, ductile detailing of steel etc. shall be strictly according to I.S. codes if it is mentioned wrongly in drawings.

4.6 Open Footings / Columns:

- i. The footing generally shall have minimum projection of 30 cm beyond column face.
- ii. Minimum diameter of reinforcement in footing shall be 12 mm except in severe exposure conditions. In case of severe exposure conditions diameter of bars shall be 16 mm.

- iii. Approval to the type of foundation shall be taken from the Client based on opinion of Geotech engineer, Soil Investigation agency and opinion of Structural Designers as the case may be.

4.7 Expansion Joints:

Expansion joints shall be designed as per relevant IS codes and strictly drawing shall be got approved from client.

4.8 Reinforcement Steel:

Steel shall be obtained from original steel manufactures company only such as SAIL, TATA, JSW, RINL, Rashtriya ISPAT Nigam Ltd., Essar, Arcelor Mittal original manufacturer company as approved by Client.

4.9 A Certification regarding Centering and Steel Checking:

- i. Certificate of Centering shall be given by Structural Design Engineer / PMC
 - ii. Certificate of steel checking shall be given by Structural Design Engineer / PMC.
- No concrete work shall be started without submission of such certificates.

4.10 Non-Destructive Testing:

In case of building if **future expansion** is proposed, structural safety of existing structures shall be verified by condition survey with all possible tests, including NDT (Non-Destructive Testing) and the results shall be got approved from the Superintending Engineer, PHC. This data shall be got approved from the approving authorities and then shall be used for design purpose.

4.11 Structural stability certificate

Shall be submitted duly signed by Structural Consultant & Project Management Consultant after verifying all structural item after completion of building project.

4.12 Specifications/Instructions for Alternative Building Technologies

For alternate building technologies proposed / to be used by contractor, details and certification required are mentioned at part B of chapter 6 (Schedule C).

Note: Bidder will be allowed to construct houses by using cast in situ, pre- cast technology, or any proven cost-effective technology as per GR No. PMAY- 2018 / Q.No. 08 HAD – 2 dt. 01/08/2018. While bidding, the bidder shall clearly mention the technology, which he is going to use for this project. However, for quoting rates and for financial evaluation all types of technologies be treated at par. The technology

with which bidder has become eligible will be treated at par. The technology with which bidder has become eligible will be followed off the execution of work, any alteration is allowed. For proposed buildings structures the agency will have to get approval of structural analysis and design from IIT/VJTI/Government engineering college, etc.

5 Scope of Electrical Work

5.1 Brief Scope of Electrical Work:

- 5.1.1 All electrical work shall be done through Licensed Electrical Contractor (JV or Subcontractor). Including preparing documents for submission to Electricity Supply Company and for obtaining necessary Electric Load, electric energy meter. Also shifting of the existing LT/HT line in the premises, if any, before the starting of the civil work, Inspection and approval on Completion from Electricity Supply Company /PWD and obtaining completion certificate from Electricity Supply Company.
- 5.1.2 The MOU shall be signed between Civil Contractor and Electrical Sub Contractor before start of electrification work. Copies of MOU shall be submitted to client/PMC. Appointment of License holder Electrical sub-Contractor shall be got approved from client.
- 5.1.3 Supplying, Erecting, Testing and Commissioning various items as per drawings and specifications, conditions and detailed scope of the works, categorized under Electrical portion and also carrying out allied works so as to complete work in all respect and put it into desired beneficial use
- 5.1.4 The electrical works includes all internal and external electrifications in all respect. The internal electrification works include point wiring, erection of mains, wires/cables, plugs, sockets, ceiling roses, holders, distribution switch boards, five-star ceiling fans, LED tube light fittings, distribution board with MCB, RCBO, MCCB all types of LED tube lights, wall LED lights and lamps, etc. The internal electrification works shall be carried out taking into consideration safety aspects and aesthetics of the interiors of the buildings.
- 5.1.5 Scope is included of Liasoning with MSEDCL for Load sanction, load release & other formalities etc. complete with Liason with MSEDCL authorities for testing.
- 5.1.6 Scope is included of all Statutory approvals, Charging Permission from the Electrical Inspectorate including application, documentation, preparation of required drawings, stage-wise inspection of installation (if any), charging permission etc. complete as required

5.2 General Conditions:

5.2.1 Statutory Provisions

Statutory provisions from following Acts, Rules, and Regulations relate to Electrical portion of works. Execution of work shall comply with related provisions from Act,

Rules, NBC, IS, IES and Regulations respective to the work. Observance of same will be binding and will be sole responsibility of the Contractor.

- Electricity Act 2003 Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations 2010 as amended up to date
- Central Electricity Authority (Technical Standards for Construction of Electric Plants and Electric Lines) Regulations 2010 as amended up to date
- Central Electricity Authority (Safety requirements for construction, operation and maintenance of electric plants and electric lines) Regulations 2011 as amended up to date
- Energy Conservation Act
- Energy Conservation Building Code (ECBC).
- ECBC Rules
- Maharashtra Fire Prevention and Life Safety Measures Act 2006
- Maharashtra Fire Prevention and Life Safety Measures Rules 2009
- The Maharashtra Lifts Act, 1939
- Bombay Lift Rules 1958

5.2.2 Standards / Norms

Execution and Maintenance related matters shall follow related provisions from following Standards, Norms, Approved Method of Construction and Specifications. (Latest amended version shall be adopted)

- Bureau of Indian Standards (BIS)
- International Electro-Technical Commission (IEC),
- National Electrical Code (NEC 2011),
- National Building Code (NBC 2016)
- Maharashtra Pollution Control Board (MPCB)

5.3 Scope of Electrical Work:

5.3.1 Supply and erection LT feeder pillars etc. complete as per drawing, design, specifications by PMC/Client/MSEDCL.

Description	Details
Watt per Sq. M As per MSEDCL Notice	75 W Per Sq. M
Load Details	
Transformer Sizes:	8 Nos x 630 kVA
DG Set:	1 Nos x 500 KVA
Total Connected Load:	8812 KW
Total Running Load (Demand Load)	5968 KW

- 5.3.2 Supply and erection of LT cables as per SLD from substation to LT feeder panels, LT Feeder Panels to Building Panel.
- 5.3.3 Meter Room and on words scope is in the building internal work.
- 5.3.4 Supply and erection of bus bar Amp. Rating as per SLD in feeder panel Complete.
- 5.3.5 Supply and erection of Street Light 9 mts Pole (including neutral link & SP MCB) with 70-watt LED fixtures, street Light Panel, 4 core X 16.0 sq.mm Aluminium cable for streetlight.
- 5.3.6 Provision of fire alarm system with,-booster pump with necessary fire hydrants, hose pipes, hose reels as per drawing, design, specifications by PMC/Client/ Fire CFO.
- 5.3.7 Provision of photo voltaic grid connected solar power generation per building complete as per drawing, design, specifications by PMC/Client
- 5.3.8 13 passengers' capacity, 1.0 m/s speed, complete as per drawing, design, specifications by PMC/Client 8 passengers' capacity, 1.0 m/s speed
- 5.3.9 Conventional CU earthing, Spike type Lightning conductor for Lightning Protection.
- 5.3.10 Street lighting and area lighting with street light poles, street light fittings, cables, timer control panels, pole foundations, trenches etc. complete as per drawing, design, specifications by PMC/Client.

5.4 Execution:

- 5.4.1 Work must be executed by an agency having license in respective category.
- 5.4.2 Prior to commencement of work, a bar chart shall be prepared showing all activities in detail. Each activity shall be correlated with the activity from civil work wherever its progress is dependent on such civil work activity e.g., concealing of pipes in ceiling depend on slab casting, concealing of pipes in wall (drops) depend on brickwork.
- 5.4.3 Attention is called towards Section 126 and 135 of Electricity Act 2003. It will be sole responsibility of contractor to get electrical supply for construction purpose.
- 5.4.4 Work shall be executed as per the drawings provided by PMC / CLIENT. Contractor shall submit actual working / shop drawings to PMC / CLIENT before commencing the work. Prior approval will be necessary. In case of any discrepancy in design with respect to Statutory Provisions, Standards, Norms same shall be brought to notice at the time of pre-bid meeting so that there shall not be any extra item. In case matter crops up during execution it shall be communicated well in advance and got resolved so that there shall not any delay in targeted completion date.

- 5.4.5 Contractor shall prepare mock-up if so desired by PMC / CLIENT.
- 5.4.6 Observing and Complying Safety norms during execution shall have high priority. Worker on site shall be provided with Personal Protective Equipment (PPE). Use of PPE during work shall be strictly observed.
- 5.4.7 Material shall be from the approved list only. Manufacturer's Data Sheets shall be submitted and got approved from PMC / CLIENT before procurement. Non- ISI material shall not be used.
- 5.4.8 Work must be compliant with related Standards, Norms, Method of Construction, Guidelines and PWD Specifications in respect of various CSR items. Contractor and his supervisory staff must be well conversant with the said provisions.
- 5.4.9 Portion of work going to be hidden shall be got checked from PMC / CLIENT prior to concealing. Wherever required, photographs, video clip shall be submitted. Record drawings of such work must be prepared and got countersigned from PMC / CLIENT.
- 5.4.10 The switch boards in any condition shall not come out of the wall or should not be very deep in the wall. The earth resistance readings shall be within the prescribed limits. Continuous 2 runs of earthing shall run up to the top floor from the earthing pit, all the Dbs. & Panels shall be connected to the earthing strips/wire as specified.
- 5.4.11 Laying of LT underground cables with proper termination shall be done, dismantling of existing LT overhead lines also dismantling of the existing street light poles shall be carried out, Installation of LT switchgears and control gears, and feeder pillars etc. shall be done.
- 5.4.12 The panel shall be prepared considering all the safety norms. The load is divided as per the SLD, the panel is to be manufactured according to the SLD only. All the wires and cables are to be connected by lugs and glands as per the sizes required at the point of termination to any equipment.
- 5.4.13 All the street light poles are to be erected in plum. All the LT cable laid underground are to be covered by the HDPE/DWC/half round cement pipe for overall length. The additional run of the cable should be properly kept so that there is no damage caused.
- 5.4.14 The danger boards, rubber matting, fire extinguishers are to be provided at the substations, floor power Ducts, pump room.
- 5.4.15 Supervision charges, meter charges and deposits or any other deposits, refundable or non-refundable etc. to the supply company shall be borne by the Contractor. However, the contractor will have to obtain demand note and deposit the cheque in an advance etc., and no extension may be granted in this regard for delay. The

amount deposited will be only reimbursed on submission of the original challans / receipts.

- 5.4.16 The work of internal and external electrical work shall be completed by contractor strictly according to tender drawings, the details as required will be furnished during the execution of work, if any. (Without prior approval of the Client no extra item shall be entertained to the contractor).

5.5 Completion and Handing Over:

- 5.5.1 After physical completion, all required inspections, trials, tests shall be jointly done by Contractor and PMC / CLIENT/MSEDCL.
- 5.5.2 Wherever necessary, Completion Certificate / NOC from respective statutory authority shall be taken. The documents of Test Results, Completion Certificates, NOCs shall be submitted
- 5.5.3 Contractor shall provide Documentation related to Guarantees, Warranties, Manuals, Trouble Shooting Charts, Training to maintenance staff.
- 5.5.4 Contractor shall also provide Record / As-Built Drawings and Inventory. The Contractor shall hand over fully functional electrical installation.

5.6 Work Specifications:

- 5.6.1 The whole work shall be carried out in accordance with the approved detailed drawings (unless otherwise directed by the Client), description of the items in Chapter 6 Schedule-C and the detailed specifications in the Standard Specification Book of the Public Work Department, (Electrical) Government of Maharashtra,
- 5.6.2 Detailed specifications given in the Chapter 6; Schedule-C are the specifications for items to be executed.
- 5.6.3 Number of points, fans, fixtures, boards, DBs, Panels, equipment etc. shall not be less than that shown in the drawings.
- 5.6.4 Wire shall of appropriate size as per standards and FRLS insulation and in no case be less than 1.5 sq.mm. Copper in cross section. In case of concealed wiring conduit sizes and layouts must be compliant with IS 732 / IEC 60364. Casing / trucking will not be allowed above false ceiling. Space factor for casing / trucking shall not exceed 50%. Wire colour coding must be observed.
- 5.6.5 Height and location of switch boards shall be as per standard norms and may be ascertained before erection. All switchboard accessories shall be modular type.

- 5.6.6 For each circuit there shall be separate neutral and earthing from DB. Size of neutral and earthing shall be as per the IEC standards. Limit of wattage and number of points must be observed in respect of each circuit. Each circuit shall be marked on DB as well as switch board.
- 5.6.7 All DBs shall be in accessible position. Use of RCCB/RCBO is mandatory.
- 5.6.8 Electrical panel shall have at least 1m clear space in front of it. GA drawings of Electrical panels shall be got approved prior to fabrication.
- 5.6.9 Earthing must follow IS 3043. Number of earth pits will depend on earth resistance to be kept within limit of 1 ohm. Distance between two earth pits shall not be less than 3m. Size of earth wire shall not be less than 1.5 sq.mm. Copper / Aluminium / GI. Adequate watering arrangement shall be provided for pits. Each electrode shall have test terminal in accessible position
- 5.6.10 All wiring to be connected to emergency / standby supply shall be distinct.
- 5.6.11 There shall be proper alignment of lighting poles. Depending on strata additional concrete / improved grade shall be used for foundations. Each pole shall be earthed and each circuit shall be provided with RCCB. LED Lighting fixture must be earthed. Underground cabling shall be as per IS 1255 and mechanically protected.

5.7 Permanent Electrical Supply Arrangement and Infrastructure:

- 5.7.1 Overhead / underground lines obstructing construction shall be removed / shifted in consultation with Supplier and Electrical Inspector prior to construction. Provisions from CEA regulations must be observed in respect of building clearances.
- 5.7.2 Drawings of sub stations / Gen. Stations shall be got approved from respective statutory authority / supplier prior to actual commencement. Record drawings of all OH/UG distribution network shall be prepared showing all details including details of cross section of underground cable.
- 5.7.3 Electrical Supply application and all related documentation shall be done by the contractor. No Liasoning fee will be payable. The process shall be taken up in hand in appropriate time so that commissioning will not be delayed due to want of supply. The estimate received from Supplier of electricity shall be got checked from PMC / CLIENT prior to making payment. Work executed on behalf of Supplier shall be as per specifications / norms recommended by Supplier. All related compliance getting WCR and release of supply will be in the scope of Contractor.
- 5.7.4 All norms / specifications applicable under Energy Conservation Building Code (ECBC), shall be observed.

- 5.7.5 Equipment like Generator must be factory tested in front of PMC / CLIENT before dispatch.
- 5.7.6 Stack height of D.G. set shall be compliant with MPCB norms.
- 5.7.7 Number of earth electrodes / grid shall be decided based on keeping minimum earth resistance not above 1 ohm in any season. Earth test terminals shall be accessible.
- 5.7.8 For distribution system multiple neutral earthing may be adopted in consultation with PMC / CLIENT. Work shall be compliant with related regulations.

5.8 Materials and Samples:

- 5.8.1 The samples of each class of material shall be tested got approved from the PMC/Client prior to its actual use. The material shall be got tested from designated Laboratory, as directed by PMC/Client and the Contractor shall bear all prescribed charges for the same. The Client/ PMC is empowered to reject or approve materials based on test report and his decision shall be final and binding on the Contractor and no disputes in this regard shall be entertained. PMC shall certify such appeared material and kept in sample room. No bill shall be forwarded without test report and certified.
- 5.8.2 All materials and workmanship shall be as prescribed in the contract and shall be binding subjected from time to time, to such tests as the Engineer-in-Charge may direct at the place of manufacture or fabrication, or on the site or at such other place or places as may be specified in the contract, The Contractor(s) shall provide required assistance, instruments, machines, labour and materials for examining, measuring and testing of the quality, weight or quantity of any material to be used and shall supply samples of such materials before incorporation in the works as may be selected and required any by the Engineer- in-Charge, without any additional cost.
- 5.8.3 All samples shall be supplied by the Contractor(s) at his own cost.
- 5.8.4 Samples approved by PMC/Client duly signed by the Electrical Engineer and Contractor shall be kept Fans, fittings, switchgears, wires, cables, switch and sockets, etc. in PMC's site office / Sample room.
- 5.8.5 The cost of carrying out any test shall be borne by the Contractor(s) as and when required so as to ascertain quality of work executed/ to be executed.
- 5.8.6 All Electrical wires, cables, fixtures, switch sockets, conduits, etc. shall be tested as per relevant IS standard.

CHAPTER NO. 6

6 Schedule– C

NOTE 1: These specifications shall be followed to understand further technical scope in addition to detail scope of work or technical brief Civil / Electrical Specifications. In case of Contradictions, particular specification of material and work as given in detail scope of work shall be followed either in lieu or otherwise as per instructions of Client.

NOTE 2: Description of item with Specification No., Page No. as per Standard specification book of P.W. Dept. of Govt. of Maharashtra Vol I (1979) & Vol. II (1981) edition & P.W.D. Civil/ P.W.D. Electrical/ M.J.P./ C.S.R Item No./ Page No. e-SSR 2022-23 of Maharashtra (CIVIL), (ELECTRICAL) / 2019-20 (MJP) and Landscaping works, parks and gardens DSR of Year 2012-13.

Name of Work: Construction of Affordable Housing Project of 1824 EWS Tenements Under Pradhan Mantri Awas Yojana (U) on Kh. No 59/2 & 60, Bhilgaon, Tal. Kamptee, Dist. Nagpur (Package – 2)

I. Building Work – Civil

Sr. No.	Description of Items
1.	Excavation for foundation in earth, soil of all types, sand, gravel and soft murum, including removing the excavated material up to a distance of 50 m. beyond the building area and stacking and spreading as directed, dewatering, preparing the bed for the foundation and necessary back filling, ramming, watering including shoring and strutting etc. complete. (Lift up to 1.5 m.) By Mechanical Means.
2.	Providing preconstructional antitermite treatment as per I.S. 6313 (Part-II) by treating the bottom surface and sides of excavation at the rate of 5 litres of emulsion concentrate of 1.0 percent of clorophyrifos per square meter of surface area covering 10 years guarantee on bond paper.
3.	Providing preconstructional antitermite treatment as per I.S. 6313 (Part-II) treatment by treating the backfill in immediate contact with foundation at the rate of 5 litres of emulsion concentrate of 1.0 percent of clorophyrifos per square metre of vertical surface area covering 10 years guarantee on bond paper.
4.	Providing Preconstructional Antitermite Treatment as per I.S. 6313 (Part-II)

Sr. No.	Description of Items
	by treating the top surface of plinth filling at the rate of 5 litres of emulsion concentrate at 1.0 percent of clorophyrifos per square metre of surface area covering ten years guarantee on bond paper.
5.	Filling in plinth and floors with approved excavated material in 15cm. To 20cm. layers including watering and compacting etc. complete.
6.	Filling in plinth and floors with contractors' material/ brought from outside and approved by Engineer in charge in layers of 15 cm to 20 cm including watering and compaction etc. complete.
7.	Providing dry/trap/granite/quartzite/gneiss rubble stone soling 15 cm to 20cm thick including hand packing and compacting etc. complete.
8.	Providing and laying Cast in situ/Ready Mix cement concrete in M-10 of trap/ granite/ quartzite/ gneiss metal for foundation and bedding including bailing out water, Steel centering, formwork, laying/pumping, compacting, roughening them if special finish is to be provided, finishing if required and curing complete, with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand SI Grade finely washed etc).
9.	Providing and laying Cast in situ/Ready Mix cement concrete in M15 of trap/granite/quartzite/gneiss metal for steps including steel centering, formwork, laying/pumping, compacting, roughening them if special finish is to be provided, finishing uneven and honeycombed surface and curing etc. complete. The Cement Mortar 1:3 plaster is considered for rendering uneven and honeycombed surface, only. Newly laid concrete shall be covered by gunny bag, plastic, tarpaulin etc. (Wooden centering will not be allowed.), with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Crushed sand VSI Grade).
10.	Providing M-30 R.C.C. cast in situ single rim bored piles each of load capacity and as per design as directed placed through steel shell sunk to the required depth through all strata excluding provision of reinforcement as per detailed drawings and design approved by the Engineer-in-charge including placing concrete by tremie arrangements, compaction of concrete including

Sr. No.	Description of Items
	chipping and dressing and withdrawal of shell etc. complete
11.	Providing and laying Cast in situ/Ready Mix cement concrete M-30 of trap / granite /quartzite/ gneiss metal for R.C.C. Pile Caps as per detailed designs and drawing or as directed including bailing out water manually, steel centering, formwork, cover blocks, laying/pumping, compaction finishing and curing etc. complete with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Crushed sand VSI Grade) complete Excluding steel Reinforcement.
12.	Carrying out low strain pile integrity test including all arrangements for test, equipment's/accessories, materials, labour, submission of test report etc but excluding the cost of installation of job pile (installation of pile shall be paid separately as per relevant items) all complete as per specification and as directed by the engineer-in-charge.
13.	Carrying out Initial Lateral Load Test on vertical (Initial/test) piles up to maximum test load of 2.0 times the safe lateral load capacity on single pile as specified in accordance with IS 2911 part-4 including all necessary arrangements such as jacks, measuring devices, etc. complete as required. (For this test if adjoining piles is used for drawing support for test then the arrangements shall be made to record deflection, submission of test report etc. of both the piles. However, testing both these piles will be considered as one test only. Providing and installing test piles measured separately)
14.	Conducting routine vertical load test for 1.5 times safe load carrying capacity by means of High Strain Dynamic Load Test including provision and erection of crane- hammer-drop mechanism, acceleration and displacement velocity transducers to record both force and velocity including excavation, dewatering, preparation of pile head including building up of pile, additional reinforcement if any etc. all complete as per approved method (ASTM D4945-89), submission of test report, specifications and instructions of the Engineer for piles of following diameter.
15.	Providing and laying Cast in situ/Ready Mix cement concrete M-30 of trap / granite/quartzite/ gneiss metal for R.C.C. columns as per detailed designs and drawing or as directed including steel centering, formwork, cover blocks,

Sr. No.	Description of Items
	laying/pumping, compaction finishing the formed surfaces with cement mortar 1:3 of sufficient minimum thickness to give a smooth and even surface or roughening if special finish is to be provided and curing etc. complete,(Excluding reinforcement and structural steel). with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Crushed sand VSI Grade)
16.	Providing and laying Cast in situ/Ready Mix cement concrete M-30 of trap / granite/quartzite/ gneiss metal for R.C.C. beams and lintels as per detailed designs and drawings or as directed including steel centering, formwork, cover blocks, laying/pumping, compaction and roughening the surface if special finish is to be provided and curing etc. complete. (Excluding reinforcement and structural steel). with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Crushed sand VSI Grade)
17.	Providing and casting in situ cement concrete M-30 of /trap / granite /quartzite/gneiss metal for R.C.C. Chajja/ Shear Walls / Secondary walls / columns / Beams / Waist Slabs / Slabs / etc. as per detailed designs and drawing or as directed including Aluform centering, formwork, cover blocks compacting and roughening the surface if special finish is to be provided and curing complete. (Excluding reinforcement). With 50% River Sand + 50% Crushed sand, chemicals (Plasticiser, Retarders etc.) Rate includes Concrete joints dressing immediate after deshuttering. Giving surface Smooth & ready to paint using mechanical means
18.	Providing and casting in situ cement concrete M-25 of /trap / granite /quartzite/gneiss metal for R.C.C. Chajja/ Shear Walls / Secondary walls / columns / Beams / Waist Slabs / Slabs / etc. as per detailed designs and drawing or as directed including Aluform centering, formwork, cover blocks compacting and roughening the surface if special finish is to be provided and curing complete. (Excluding reinforcement). With 50% River Sand + 50% Crushed sand, chemicals (Plasticiser, Retarders etc.) Rate includes Concrete joints dressing immediate after deshuttering. Giving surface Smooth & ready to paint using mechanical means -
19.	Providing and casting in situ cement concrete M-30 of /trap / granite

Sr. No.	Description of Items
	/quartzite/gneiss metal for R.C.C. Chajja/ Shear Walls / Secondary walls / columns / Beams / Waist Slabs / Slabs / etc. as per detailed designs and drawing or as directed including Aluform centering, formwork, cover blocks compacting and roughening the surface if special finish is to be provided and curing complete. (Excluding reinforcement). With 50% River Sand + 50% Crushed sand, chemicals (Plasticiser, Retarders etc.) Rate includes Concrete joints dressing immediate after deshuttering. Giving surface Smooth & ready to paint using mechanical means -
20.	Providing and casting in situ cement concrete M-25 of /trap / granite /quartzite/gneiss metal for R.C.C. Chajja/ Shear Walls / Secondary walls / columns / Beams / Waist Slabs / Slabs / etc. as per detailed designs and drawing or as directed including Aluform centering, formwork, cover blocks compacting and roughening the surface if special finish is to be provided and curing complete. (Excluding reinforcement). With 50% River Sand + 50% Crushed sand, chemicals (Plasticiser, Retarders etc.) Rate includes Concrete joints dressing immediate after deshuttering. Giving surface Smooth & ready to paint using mechanical means-
21.	Providing and casting in situ cement concrete M-30 of /trap / granite /quartzite/gneiss metal for R.C.C. Chajja / Shear Walls / Secondary walls / columns / Beams / Waist Slabs / Slabs / etc. as per detailed designs and drawing or as directed including Aluform centering, formwork, cover blocks compacting and roughening the surface if special finish is to be provided and curing complete. (Excluding reinforcement). With 50% River Sand + 50% Crushed sand, chemicals (Plasticiser, Retarders etc.) Rate includes Concrete joints dressing immediate after deshuttering. Giving surface Smooth & ready to paint using mechanical means -
22.	Providing and casting in situ cement concrete M-25 of /trap / granite /quartzite/gneiss metal for R.C.C. Chajja / Shear Walls / Secondary walls / columns / Beams / Waist Slabs / Slabs / etc. as per detailed designs and drawing or as directed including Aluform centering, formwork, cover blocks compacting and roughening the surface if special finish is to be provided and curing complete. (Excluding reinforcement). With 50% River Sand + 50% Crushed sand, chemicals (Plasticiser, Retarders etc.) Rate includes Concrete joints dressing immediate after deshuttering. Giving surface Smooth & ready to

Sr. No.	Description of Items
	paint using mechanical means-
23.	Providing & fixing of customized Aluminium formwork for monolithic construction RCC member with respective uses of 100 times using grade 5052 aluminium for panel sheet of minimum 4 mm thick and grade 6061 (type 6) aluminium for extruded section. The formwork includes beam i.e., beam side component prop head for soffit beam, beam soffit panel, beam soffit bulk head and deck component i.e., deck panel, deck prop, prop length, deck mid, soffit length, deck beam bar, and wall component, i.e., wall panel, rocker, kiker, and internal soffit corner, external soffit corner internal corner, external corners etc, the panels are held in position by a simple pin and wedge system that passes through holes in the outside rub of each panel. The tolerance of finished panel to be (1 mm) shall be confirmed to IS 14687:1999 pins and wedges to be made of high-grade mild steel, all complete as per direction of engineer in charge (Cost of RCC work shall be paid separately).
24.	Providing and fixing during laying in RCC Slab rectangular 2 mm thick M.S. sheet fan hook box of size 180x100 mm with required depth with holes or notches as required, bottom and top sides of 1.6 mm M.S. sheet fixed with 3.5 mm dia. round headed screws together with 12 mm dia. M.S. bar bent to shape with hooked ends for fan hook and painting to exposed surface.
25.	Providing expansion joints in reinforced cement concrete framed structure (interior columns) with plain pre-moulded filler with confirming to IS 1838,1961, 25 mm thick with 75 mm wide and 1 mm thick stainless-steel plate finishing etc. complete as directed.
26.	Providing expansion joints in R.C.C. framed structure (exterior column) with plain pre-moulded filler 25 mm thick with 75 mm wide and 1 mm thick stainless-steel plate and finishing etc. complete.
27.	Providing expansion joints in R.C.C. framed structure (at floor level) with plain pre-moulded filler 25 mm thick with 75 mm wide and 1 mm thick stainless-steel plat and finishing etc. complete.
28.	Providing and fixing in position TMT - FE – 500D bar reinforcement of various diameters for R.C.C. piles, Pile caps, footings, foundations, slabs, beams columns, shear walls, Secondary walls, canopies, staircase, newels, Chajja, lintels pardis, copings, fins, arches etc. as per detailed designs, drawings and

Sr. No.	Description of Items
	schedules. including cutting, bending, hooking the bars, binding with wires or tack welding and supporting as required complete.
29.	Providing Autoclaved Aerated Concrete Block masonry of Ecolite or equivalent make conforming to IS:2185 (Part 3) - 1984 in extra fine jointing mortar of fix block of UltraTech or equivalent in superstructure including striking joints, raking out joints and scaffolding etc. Complete. (The test shall be carried out conforming to IS:6441 (Part I) - 1972)
30.	Providing cement based waterproofing treatment to terraces (Indian waterproofing or alike) with brickbats laid in required slope to drain the water for any span after cleaning the base surface. Applying a coat of cement slurry admixed with approved waterproofing compound and laying the brickbats on bottom layer in C.M.1:5 admixed with approved waterproofing compound filling up to half depth of brickbats, curing this layer for 3days, applying cement slurry over this layer joints of brickbats with C.M.1:3 admixed with approved waterproofing compound and finally top finishing with average 20mm.thick layers of same mortar added with jute fiber at 1Kg per bag including finishing the surface smooth with cement slurry admixed with approved waterproofing compound. Marking finished surface with false squares of 300mmx300mm.making the junctions at the parapet rounded and tapered top for required height, with drip mould at the junction of plaster and parapet and curing and covering 10 years Guarantee against leak proofness on Court fee stamp paper of Rs. 500/- including ponding test etc. complete.
31.	Providing Waterproofing in W.C. and Bath including brick bat coba in all position including providing and laying 12mm bedding in cement mortar 1:3 on vergin concrete slab with waterproofing compound @ 1Kilogram/per bag of cement laying brick bat coba of required thickness in cm 1:5 with waterproofing compound 1 Kilogram/bag of cement grouting and finishing the top layer with 20mm thick brick bedding in cm mortar 1:3 with waterproofing compound 1 Kilogram/per bag of cement and testing the treated portion for 48 hours by pond test and covering ten years' guarantee on requisite stamp paper including curing etc. complete.
32.	Providing Water Proof Bedding for flooring of Balconies 25 mm thick in C.M. 1:3 including using approved water proofing compound in specified proportion as per manufacturers specifications for per bag of cement including

Sr. No.	Description of Items
	levelling, curing and covering 10 years guarantee on court fee stamp paper of Rs.500/- including ponding test etc. complete.
33.	Providing and applying water proofing treatment for Lift pits and RCC Chajja, UG tanks in Bldg., OHT using acrylic polymer modified cement-based water proofing coating with fibre glass mesh mixing at the rate of powder to liquid (2:1) by weight covering 9 to 10 sqm /kg with two coat using approved chemicals for masonry and concrete surface by brush covering 7 years guarantee on Stamp Papers etc. complete
34.	Providing Waterproof Plaster in W.C. and bath 12 mm thick for dado in cement mortar 1:3 with neat finishing, floating using waterproofing compound at the rate of 1 Kilogram. per bag of cement of approved make and manufacturer and curing etc. complete.
35.	Providing and laying vitrified mirror / glossy finish tiles decorative type having size 590mm to 605mm x 590mm to 605mm of 8 to10mm thickness and confirming to IS.15622-2006 (group BI a) of approved make, shade and pattern for flooring in required position laid on a bed of 1:4 cement mortar including neat cement float, filling joints, curing and clearing etc. complete.
36.	Providing and laying vitrified mirror / glossy finish tiles having size 590 mm to 605 mm x 590 mm to 605 mm of 8 to 10 mm thickness and confirming to IS. 15622-2006 (group Bla) of approved make, shade and pattern for dado and skirting in required position fixed in 1:4 cement mortar including neat cement float, filling joints, curing and clearing etc. complete
37.	Providing and laying Antiskid Ceramic tiles of approved quality of size 30 cm x 30 cm and confirming to IS 15622-2006 (Group-B IIA) for antiskid flooring in required position laid on a bed of 1:4 cement mortar including cement float, filling joint with cement slurry cleaning curing etc. complete.
38.	Providing and laying Ceramic tiles having size 30 cm. x 45 cm. confirming to corresponding I.S. for dado and skirting in required position with readymade adhesive mortar of approved quality on plaster of 1:2 cement mortar including joint filling with white/ colour cement slurry cleaning curing etc. complete.
39.	Providing and laying Polished Shahabad Stone flooring 25mm to 30mm thick and required width wide in plain/ diamond pattern on a bed of 1:6 C.M. including cement float, filling joints, with neat cement slurry, curing, polishing

Sr. No.	Description of Items
	and cleaning etc. complete.
40.	Providing and laying machine cut machine polished Shahabad Stone slabs 20 to 25mm thick for treads and risers of steps and staircases, with rounded nosing for the treads on a bed of 1:4 cement mortar including cement float, filling joints with neat cement slurry, curing, polishing and cleaning etc. complete.
41.	Providing and laying ceramic tiles of having size 30 cm x 30 cm confirming to I.S.15622/2006 (group D II-A) and 7 to 8 mm thick for flooring in required position laid on a bed of 1:4 cement mortar including cement float, filling joint with white/colour cement slurry cleaning curing etc. complete.
42.	Providing and fixing green marble of 18 to 20 mm thick for door frame/ dado/ window boxing etc. On C.M. 1:6 including filling joints with polymer base filler nosing / moulding the sharp edges wherever necessary, curing, etc. complete.
43.	Providing and constructing granite kitchen platform with fixing of stainless steel sink 600 mm x 450 mm size as per detailed drawing including vertical both side polished kadappah stone 25 to 30 mm thick supports with kadappah top 35 to 40 mm thick and polished granite 16 to 20 mm top with side strips of granite at front and both sides of platform raised with two vertical granite supports 15 cm height and top granite of 75 x 40 cm including cutting, opening for sink of required size in kadappah as well as granite etc. complete. (Platform top size 5.00 m x 0.60 m and height are 0.75 m)
44.	Providing and laying chequered tiles for Parking area of approved quality of size 30 cm x 30 cm, confirming to corresponding I.S. for flooring in required position laid on a bed of 1:4 cement mortar including cement float, filling joint with cement slurry cleaning curing etc. complete.
45.	Providing and applying colour-wash of approved colour and shade in one coat to new surface including scaffolding, brushing and brooming down (excluding base coats of white wash) etc. Complete.
46.	Providing and applying priming coat on concrete/ masonry/ Asbestos Cement plastered surfaces including scaffolding if necessary, preparing the surface by thoroughly cleaning oil, grease, dirt and other foreign matter and sand papering as required etc. complete
47.	Providing and applying washable oil-bound distemper of approved colour and

Sr. No.	Description of Items
	shade to old and new surfaces in two coats including scaffolding, preparing the surfaces. (Excluding the primer coat.) etc. complete.
48.	Providing and applying two coats of Water Proof cement paint of approved manufacture and of approved colour to the plastered surfaces including scaffolding if necessary, cleaning and preparing the surface, watering for two days etc. complete.
49.	Providing & fixing factory-made Single leaf 2 hour fire rated door as per IS 3614 Part 2 & BS 476 Part 20 & 22, tested to UL 10 C and conforming to NFPA 80, tested at NTL /CBRI/ARAI of Plain Steel finish type with a.) Frame of size 143x58 mm and thickness of 1.6 mm thick skin pass galvanized iron sheet Conform to IS 277) formed to single or double rebate profile b.) Shutter of 46 mm thickness and constructed from 1.2 mm thick skin pas galvanized iron sheet (Conform to IS 277) infill Rockwool / Mineral wool/ Honey comb craft paper c.) Vision glass of 5mm thick clear/wired glass Circular or Rectangular d.) Finish : Primed with zinc phosphate stoving primer and finished with Polyurethane aliphatic grade or epoxy paint as required of thickness 60 micron e.) Accessories : Min 8 Anchor fastener of HILTI make, min 3 SS Ball Bearing Butt Hinges, Mortise sash lock with lever handle, Mortise dead bolt, Mortise latch, Door closer, Panic bar and external trim lock and Flush bolts complete.
50.	Providing and fixing factory made door/Window frames of required colour made with wood Polymer composite (WPC) extruded internal ribbed sections conforming to American Society for testing and materials (ASTM) in regard with density, tensile strength, flexural strength, Izod impact, water absorption and screw withdrawal strength with wall thickness of 5mm (+/-1mm) and with frames having mitred joints at each corner, joined together by two nos. of 10 x 8 mm & 40 mm long PVC dovetail keys including hard wood piece support along the full vertical and horizontal length in rebate portion hole for fixing hinges and fixtures and the frames fixed to Walls/columns with 15 mm long dash fasteners/holdfasts (dash fasteners/holdfasts to be paid separately) etc. complete as per the direction of Engineer in charge, with prior approval of concerned Superintending Engineer. For Profile size of 100 x 50 mm with 32 x 12 mm rebate
51.	Providing and fixing solid core flush door shutter in single leaf 32 mm thick decorative type of exterior grade as per detailed drawings approved face

Sr. No.	Description of Items
	veneers 3 mm thick on both faces or as directed, all necessary beads, mouldings and lipping, wrought iron hold fasts, chromium plated fixtures and fastenings, with brass mortise lock, chromium plated handles on both sides, and finishing with French Polish etc. complete.
52.	Providing and fixing solid core Flush Door Shutter commercial in single leaf 32 mm thick without ventilator commercial type of exterior grade as per detailed drawings with wrought iron hold fast, stainless steel fixtures and fastenings and handles on both sides and finishing with oil painting 3 coats complete.
53.	Providing and fixing fiber glass reinforced polyester door shutter 30 mm thick as per IS 14856 (2000) (Reaffirmed 2006) without ventilator including chromium plated fixtures and fastening with chromium plated handles on both sides, etc. complete
54.	Providing and fixing in position. (As per I.S. 1868 / 1982) Aluminium sliding window of two tracks with rectangular pipe having overall dimension 63.50 x 38.10 x 1.02 mm at weight 0.547 kg / Rmt. and window frame bottom track section 61.85 x 31.75 x 1.20 mm at weight 0.695 kg / Rmt. Top and side track section 61.85 x 31.75 x 1.30 mm at weight 0.659 kg / Rmt. The shutter should be of bearing bottom 40 x 18 x 1.25 mm at weight 0.417 kg / Rmt. Interlocking section 40 x 18 x 1.10 mm at weight 0.469 kg / Rmt. And handle section 40 x 18 x 1.25 mm at weight 0.417 kg / Rmt. and top section 40 x 18 x 1.25 mm at weight 0.417 kg / Rmt. As per detailed drawings and as directed by Engineer in charge with all necessary Aluminium sections fixtures and fastenings such as roller bearing in nylon casting and self-locking catch fitted in vertical section of shutter including 5mm thick plain glass with all required screws and nuts etc, complete. With colour Anodising with box.
55.	Providing and fixing in position powder coated aluminium louvered windows / ventilator of various sizes with powder coating as per detailed drawing and specifications including aluminium frames 80 x 38 mm x 1.22 mm box type, 5 mm thick sheet glass louvers, of approved quality etc. complete.
56.	Providing and fixing mild steel grill railing 15 Kilogram/Sq. M with Ghana teak wood hand rail of size 75mm x 60 mm and sill of 75mm.x 25mm and newel posts for staircases, including fabricating, fixtures, erecting, painting the grill

Sr. No.	Description of Items
	work with one coat of anticorrosive paint and two coats of oil painting with approved colours and polishing the sill, hand rail and the newel post with French polish.
57.	Providing and fixing in position after hoisting, precast 1:2 cement mortar Reinforced Jali 50 mm thick with 3 mm diameter mild steel bar reinforcement, of approved design on both sides, fixing in cement mortar 1:3 and curing etc. complete.

II. Infrastructure works – Civil

Sr.No.	Description of Items
	Amenity Building
1.	Excavation for foundation in earth, Soil of all types, sand, gravel and soft murum, including removing the excavated material up to a distance of 50 m. beyond the building area & stacking and spreading as directed, dewatering, preparing the bed for the foundation and necessary back filling, ramming, watering including shoring and strutting etc. complete. (Lift up to 1.5 m.) By Mechanical Means
2.	Providing preconstructional antitermite treatment as per I.S. 6313 (Part-II) by treating the bottom surface and sides of excavation at the rate of 5 litres of emulsion concentrate of 1.0 percent of clorophyrifos per square meter of surface area covering 10 years guarantee on bond paper.
3.	Providing preconstructional antitermite treatment as per I.S. 6313 (Part-II) treatment by treating the backfill in immediate contact with foundation at the rate of 5 litres of emulsion concentrate of 1.0 percent of clorophyrifos per square metre of vertical surface area covering 10 years guarantee on bond paper.
4.	Providing Preconstructional Antitermite Treatment as per I.S. 6313 (Part-II) by treating the top surface of plinth filling at the rate of 5 litres of emulsion concentrate at 1.0 percent of clorophyrifos per square metre of surface area covering ten years guarantee on bond paper.
5.	Filling in Plinth and floors with approved excavated materials in 15 cm to 20 cm layers including watering and compaction complete.

Sr.No.	Description of Items
6.	Filling in plinth and floors with contractors' material/brought from outside and approved by Engineer in charge in layers of 15 cm to 20 cm including watering and compaction etc. complete.
7.	Providing dry/ trap/ granite/ quartzite/ gneiss rubble stone soling 15 cm to 20 cm thick including hand packing and compacting etc. complete.
8.	Providing and laying Cast in situ/Ready Mix cement concrete in M15 of trap/granite/quartzite/gneiss metal for steps including steel centering, formwork, laying/pumping, compacting, roughening them if special finish is to be provided, finishing uneven and honeycombed surface and curing etc. complete. The Cement Mortar 1:3 plaster is considered for rendering uneven and honeycombed surface, only. Newly laid concrete shall be covered by gunny bag, plastic, tarpaulin etc. (Wooden centering will not be allowed.), with fully automatic micro-processor-based PLC with SCADA enabled reversible Drum Type mixer/concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Crushed sand VSI Grade)
9.	Providing M-30 R.C.C. cast in situ single rim bored piles each of load capacity and as per design as directed placed through steel shell sunk to the required depth through all strata except rock excluding provision of reinforcement as per detailed drawings and design approved by the Engineer-in-charge including placing concrete by tremie arrangements, compaction of concrete including chipping and dressing and withdrawal of shell etc. complete
10.	Providing and laying Cast in situ/Ready Mix cement concrete M-30 of trap / granite /quartzite/ gneiss metal for R.C.C. Pile Caps as per detailed designs and drawing or as directed including bailing out water manually, steel centering, formwork, cover blocks, laying/pumping, compaction finishing and curing etc. complete with fully automatic micro-processor-based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Crushed sand VSI Grade) complete Excluding steel Reinforcement.
11.	Carrying out low strain pile integrity test including all arrangements for test, equipment's/accessories, materials, labour, submission of test report etc but excluding the cost of installation of job pile (installation of pile shall be paid separately as per relevant items) all complete as per specification and as directed by the engineer-in-charge.

Sr.No.	Description of Items
12.	Carrying out Initial Lateral Load Test on vertical (Initial/test) piles up to maximum test load of 2.0 times the safe lateral load capacity on single pile as specified in accordance with IS 2911 part-4 including all necessary arrangements such as jacks, measuring devices, etc. complete as required. (For this test if adjoining piles is used for drawing support for test, then the arrangements shall be made to record deflection, submission of test report etc. of both the piles. However, testing both these piles will be considered as one test only. Providing and installing test piles measured separately)
13.	Conducting routine vertical load test for 1.5 times safe load carrying capacity by means of High Strain Dynamic Load Test including provision and erection of crane- hammer-drop mechanism, acceleration and displacement velocity transducers to record both force and velocity including excavation, dewatering, preparation of pile head including building up of pile, additional reinforcement if any etc. all complete as per approved method (ASTM D4945-89), submission of test report, specifications and instructions of the Engineer for piles of following diameter.
14.	Providing and laying Cast in situ/Ready Mix cement concrete M-30 of trap / granite/quartzite/ gneiss metal for R.C.C. columns as per detailed designs and drawing or as directed including steel centering, formwork, cover blocks, laying/pumping, compaction finishing the formed surfaces with cement mortar 1:3 of sufficient minimum thickness to give a smooth and even surface or roughening if special finish is to be provided and curing etc. complete,(Excluding reinforcement and structural steel). with fully automatic micro-processor-based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Crushed sand VSI Grade)
15.	Providing and laying Cast in situ/Ready Mix cement concrete M-30 of trap / granite /quartzite/ gneiss metal for R.C.C. beams and lintels as per detailed designs and drawings or as directed including steel centering, formwork, cover blocks, laying/pumping, compaction and roughening the surface if special finish is to be provided and curing etc. complete. (Excluding reinforcement and structural steel). with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)

Sr.No.	Description of Items
16.	Providing and laying Cast in situ/Ready Mix cement concrete M-30 of trap/ granite / quartzite/ gneiss metal for R.C.C. slabs and landings as per detailed designs and drawings including steel centering, formwork, cover blocks, laying/pumping, compaction finishing the formed surfaces with cement mortar 1:3 of sufficient minimum thickness to give a smooth and even surface or roughening if special finish is to be provided and curing etc. complete, (Excluding reinforcement and structural steel). with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)
17.	Providing and fixing during laying in RCC Slab rectangular 2 mm thick M.S. sheet Fan hook box of size 180x100 mm with required depth with holes or notches as required, bottom and top sides of 1.6 mm M.S. sheet fixed with 3.5 mm dia. round headed screws together with 12 mm dia. M.S. bar bent to shape with hooked ends for fan hook and painting to exposed surface. As directed by Engineer in charge.
18.	Providing and fixing in position TMT FE-500bar Reinforcement of various diameter's for R.C.C. pile caps, footings, foundations, slabs, beams, columns, canopies, staircases, newels, Chajja, lintels, pardies coping, fins, arches, etc. as per detailed designs, drawings and schedules, including cutting, bending, hooking the bars, binding with wires or tack welding and supporting as required etc. complete.
19.	Providing cement-based Water Proofing Treatment to Terraces (Indian water proofing or alike) with brick bats laid in required slope to drain the water for any span after cleaning the base surface. Applying a coat of cement slurry admixed with approved water proofing compound and laying the brick bats on bottom layer in C.M.1:5 admixed with approved water proofing compound filling up to half depth of brick bats, curing this layer for 3 days, applying cement slurry over this layer joints of brick bats with C.M.1:3 admixed with approved water proofing compound and finally top finishing with average 20 mm. thick layers of same mortar added with jute fiber at 1 Kg per bag including finishing the surface smooth with cement slurry admixed with approved water proofing compound. Marking finished surface with false squares of 300mm x 300 mm. making the junctions at the parapet rounded and tapered top for required height, with drip mould at the junction of plaster and parapet and curing and

Sr.No.	Description of Items
	covering 10 years Guarantee against leak proofness on Court fee stamp paper of Rs. 500/- including ponding test etc. complete.
20.	Providing Waterproofing in W.C. and Bath including brick bat coba in all position including providing and laying 12mm bedding in cement mortar 1:3 on vergin concrete slab with waterproofing compound @ 1Kilogram/per bag of cement laying brick bat coba of required thickness in cm 1:5 with waterproofing compound 1 Kilogram/bag of cement grouting and finishing the top layer with 20mm thick brick bedding in cm mortar 1:3 with waterproofing compound 1 Kilogram/per bag of cement and testing the treated portion for 48 hours by pond test and covering ten years' guarantee on requisite stamp paper including curing etc. complete.
21.	Providing Water Proof Bedding for flooring of Bath and WC 25 mm thick in C.M. 1:3 including using approved water proofing compound in specified proportion as per manufacturers specifications for per bag of cement including levelling, curing and covering 10 years guarantee on court fee stamp paper of Rs.500/- including ponding test etc. complete.
22.	Providing and applying water proofing treatment using acrylic polymer modified cement-based water proofing coating with fibre glass mesh mixing at the rate of powder to liquid (2:1) by weight covering 9 to 10 sqm /kg with two coat using approved chemicals for masonry and concrete surface by brush covering 7 years guarantee on Stamp Papers etc. complete
23.	Providing Autoclaved Aerated Concrete Block masonry of Ecolite or equivalent make conforming to IS:2185 (Part 3) - 1984 in extra fine jointing mortar of fix block of UltraTech or equivalent in superstructure including striking joints, raking out joints and scaffolding etc. Complete. (The test shall be carried out conforming to IS:6441 (Part I) - 1972)
24.	Providing and applying Two coats of wall care Putty on plastered surface and Ceiling and Walls to prepare surface even and smooth of approved make, etc complete.
25.	Providing internal cement plaster 12mm thick in single coat in cement mortar 1:4 without neeru finish to concrete or brick surfaces, in all positions including scaffolding and curing etc. complete.
26.	Providing neeru finish to plastered surfaces in all positions including scaffolding and curing etc. complete.

Sr.No.	Description of Items
27.	Providing sand faced plaster externally in cement mortar using approved screened sand, in all positions including base coat of 15 mm thick in cement mortar 1:4 using waterproofing compound at 1 kg per cement bag curing the same for not less than 2 days and keeping the surface of the base coat rough to receive the sand faced treatment 6 to 8 mm thick in cement mortar 1:4 finishing the surface by taking out grains and curing for fourteen days scaffolding etc. Complete.
28.	Providing Waterproof Plaster in W.C. and bath 12 mm thick for dado in cement mortar 1:3 with neat finishing, floating using waterproofing compound at the rate of 1 Kilogram. per bag of cement of approved make and manufacturer and curing etc. complete.
29.	Providing and laying Vitrified mirror / glossy finish tiles decorative type having size 590 mm to 605 mm x 590 mm to 605 mm of 8 to 10 mm thickness and confirming to IS. 15622-2006 (group Bla) of approved make, shade and pattern for flooring in required position laid on a bed of 1:4 cement mortar including neat cement float, filling joints, curing and clearing etc. complete.
30.	Providing and laying vitrified mirror / glossy finish tiles having size 590 mm to 605 mm x 590 mm to 605 mm of 8 to 10 mm thickness and confirming to IS. 15622-2006 (group Bla) of approved make, shade and pattern for Dado and skirting in required position fixed in 1:4 cement mortar including neat cement float, filling joints, curing and clearing etc. complete.
31.	Providing and laying Antiskid Ceramic tiles of approved quality of size 30 cm x 30 cm and confirming to IS 15622-2006 (Group-B IIA) for antiskid flooring in required position laid on a bed of 1:4 cement mortar including cement float, filling joint with cement slurry cleaning curing etc. complete.
32.	Providing and laying Ceramic tiles having size 30 cm. x 45 cm. confirming to corresponding I.S. for dado and skirting in required position with readymade adhesive mortar of approved quality on plaster of 1:2 cement mortar including joint filling with white/ colour cement slurry cleaning curing etc. complete.
33.	Providing and laying machine cut machine polished machine cut Kota stone slabs 20 to 25 mm thick for treads and risers of steps and staircases, with rounded nosing for the treads on a bed of 1:4 cement mortar including cement float, filling joints with neat cement slurry, curing, polishing and cleaning etc. complete

Sr.No.	Description of Items
34.	Providing and fixing green marble of 18 to 20 mm thick for door frame/ dado/ window boxing etc. On C.M. 1:6 including filling joints with polymer base filler nosing / moulding the sharp edges wherever necessary, curing, etc. complete.
35.	Providing and applying colour-wash of approved colour and shade in one coat to new surface including scaffolding, brushing and brooming down (excluding base coats of white wash) etc. Complete.
36.	Providing and applying washable powder distemper of approved colour and shade to old and new surfaces in two coats including scaffolding, preparing the surfaces. (Excluding the primer coat) etc. complete.
37.	Providing and applying two coats of Water Proof cement paint of approved manufacture and of approved colour to the plastered surfaces including scaffolding if necessary, cleaning and preparing the surface, watering for two days etc. complete.
38.	Providing and fixing factory made door/Window frames of required colour made with wood Polymer composite (WPC) extruded internal ribbed sections conforming to American Society for testing and materials (ASTM) in regard with density, tensile strength, flexural strength, Izod impact, water absorption and screw withdrawal strength with wall thickness of 5mm (+/-1mm) and with frames having mitred joints at each corner, joined together by two nos. of 10 x 8 mm & 40 mm long PVC dovetail keys including hard wood piece support along the full vertical and horizontal length in rebate portion hole for fixing hinges and fixtures and the frames fixed to Walls/columns with 15 mm long dash fasteners/holdfasts (dash fasteners/holdfasts to be paid separately) etc. complete as per the direction of Engineer in charge, with prior approval of concerned Superintending Engineer. For Profile size of 100 x 50 mm with 32 x 12 mm rebate
39.	Providing and fixing solid core Flush Door Shutter commercial in single leaf 32 mm thick without ventilator commercial type of exterior grade as per detailed drawings with wrought iron hold fast, stainless steel fixtures and fastenings and handles on both sides and finishing with oil painting 3 coats complete.
40.	Providing and fixing fiber glass Reinforced polyester door shutter 30 mm thick as per IS 14856 (2000) (Reaffirmed 2006) without ventilator including chromium plated fixtures and fastening with chromium plated handles on both sides, etc complete.

Sr.No.	Description of Items
41.	Providing and fixing anodized (anodic film must not be less than 15 microns i.e. AC-15 as per IS, the anodising must be scaleted by keeping the anodized section in boiling de anodized water for a period of one hour) aluminium fixed/ sliding / pivoted / top hung / side hung / louvered type windows, doors, ventilators and partitions conforming to IS:1948, 1961 with hollow bottom section & standard single / two / three / four track frame section all round (top, bottom and sides) with shutter frame made up of standard handle section standard interlocking section and standard top and bottom section with infill panel of plain /frosted / tinted glass/ Novapan sheet / acrylic sheet / hardener laminated sheet as listed below with nylon guides, gliders glass fixing PVC weather strips, neoprene gasket, PVC weep holes etc complete with approved type of locking arrangement, handles and other fixtures including fixing, filling the gaps with silicon sealant, cleaning polishing the aluminium section with petrol and / or other agents as specified by the manufacturers etc. complete as directed by Engineer In Charge. 6 mm thick Plain / frosted / tinted Glass.
42.	Providing and fixing rolling shutter fabricated from steel laths of minimum thickness 0.9 mm with lock plate of 3.15 mm thickness reinforced with 35 x 35 x 5 mm angle section fitted with sliding bolts and handles for both sides, deep M.S. channel section of depth and thickness not less than 65 mm and 3.15 mm respectively with hold fast arrangements, M.S. Bracket plate 300 x 300 x 3.15 mm minimum size and shape with square bar, suspension shaft of minimum 32 mm diameter, hood cover of M.S. sheet not less than 0.9 mm thickness and of any size at top and safety devices including mechanical gear operation arrangement consisting of worm gear wheels and worms of high grade cast iron or mild steel and one coat of red lead primer etc. complete. (I.S. 62481979) (With mechanical gear)
43.	Providing and fixing mild steel grill railing 15 Kilogram/Sq. m with Ghana teak wood hand rail of size 75mm x 60 mm and sill of 75mm.x 25mm and newel posts for staircases, including fabricating, fixtures, erecting, painting the grill work with one coat of anticorrosive paint and two coats of oil painting with approved colours and polishing the sill, hand rail and the newel post with French polish.
44.	Providing and fixing in position. (As per I.S.1868 / 1982) Aluminium sliding window of two tracks with rectangular pipe having overall dimension ?63.50 x 38.10 x 1.02 mm at weight 0.547 Kilogram/Running metre. and window frame

Sr.No.	Description of Items
	bottom track section 61.85 x 31.75 x 1.20 mm at weight 0.695 Kilogram/Running metre. Top and side track section 61.85 x 31.75 x 1.30 mm at weight 0.659 Kilogram/Running metre. The shutter should be of bearing bottom 40 x 18 x 1.25 mm at weight 0.417 Kilogram/Running metre. Inter locking section 40 x 18 x 1.10 mm at weight 0.469 Kilogram/Running metre. And handle section 40 x 18 x 1.25 mm at weight 0.417 Kilogram/Running metre. and top section 40 x 18 x 1.25 mm at weight 0.417 Kilogram/Running metre. As per detailed drawings and as directed by Engineer in charge with all necessary Aluminium sections fixtures and fastenings such as roller bearing in nylon casting and self-locking catch fitted in vertical section of shutter including 5 mm thick plain glass with all required screws and nuts etc, complete. With powder coating without box.
45.	Providing and fixing in position powder coated aluminium Louvered windows / ventilator of various sizes with powder coating as per detailed drawing and specifications including aluminium frames 80 x 38 mm x 1.22 mm box type, 5 mm thick sheet glass louvers, of approved quality etc. complete.
46.	Providing and fixing in position after hoisting, precast 1:2 cement mortar Reinforced Jali 50 mm thick with 3 mm diameter mild steel bar reinforcement, of approved design on both sides, fixing in cement mortar 1:3 and curing etc. complete.
47.	Design, Fabrication & Installation of 4 mm thick exterior grade PVDF coated aluminium composite panels of ALUCOBOND having 0.5 mm thick aluminium PVDF coated sheet available in standard colours + 3 mm core material + 0.5 mm aluminium sheet chemically treated (back sheet) bent as per requirement, fitted on anodised aluminium necessary coated rectangular grid work, hardware, fixtures, brackets, anchor, fasteners of S.S. 304 grade etc. complete duly sealed with weathering silicon manufactured by DOW corning / G.E. silicon for circular columns and curved beams etc complete. All section for supporting ACP shall be of size 50 x 50 x 2.5 mm at a distance of 1.2 m Horizontally & Vertically or as per requirement of the design shop drawing has to be submitted by the contractor for the approval of structural Engineer. a) For Terrace b) For Entrance Canopy REBATE FOR SIZES Size :39"-Bruhan Mumbai Mahanagar palika shri Harilal Bhagwati sarvsadharan Rugnalay (For 11" = 505.94*11"/25" = Rs. 222.17 per Inch. Size :46"-MCGM SHRI HARILAL BHAGWATI GENERAL HOSPITAL (For 4" = 505.94*4"/25" = Rs. 80.95 per

Sr.No.	Description of Items
	Inch
	Fitness Centre / Club house / Creche
1.	Excavation for foundation in earth, Soil of all types, sand, gravel and soft murum, including removing the excavated material up to a distance of 50 m. beyond the building area & stacking and spreading as directed, dewatering, preparing the bed for the foundation and necessary back filling, ramming, watering including shoring and strutting etc. complete. (Lift up to 1.5 m.) By Mechanical Means.
2.	Providing preconstructional antitermite treatment as per I.S. 6313 (Part-II) by treating the bottom surface and sides of excavation at the rate of 5 litres of emulsion concentrate of 1.0 percent of clorophyrifos per square meter of surface area covering 10 years guarantee on bond paper.
3.	Providing preconstructional antitermite treatment as per I.S. 6313 (Part-II) treatment by treating the backfill in immediate contact with foundation at the rate of 5 litres of emulsion concentrate of 1.0 percent of clorophyrifos per square metre of vertical surface area covering 10 years guarantee on bond paper.
4.	Providing Preconstructional Antitermite Treatment as per I.S. 6313 (Part-II) by treating the top surface of plinth filling at the rate of 5 litres of emulsion concentrate at 1.0 percent of clorophyrifos per square metre of surface area covering ten years guarantee on bond paper.
5.	Filling in Plinth and floors with approved excavated materials in 15 cm to 20 cm layers including watering and compaction complete.
6.	Filling in plinth and floors with contractors' material/brought from outside and approved by Engineer in charge in layers of 15 cm to 20 cm including watering and compaction etc. complete.
7.	Providing dry/ trap/ granite/ quartzite/ gneiss rubble stone soling 15 cm to 20 cm thick including hand packing and compacting etc. complete.
8.	Providing and laying Cast in situ/Ready Mix cement concrete in M15 of trap/granite/quartzite/gneiss metal for steps including steel centering, formwork, laying/pumping, compacting, roughening them if special finish is to be provided, finishing uneven and honeycombed surface and curing etc. complete. The Cement Mortar 1:3 plaster is considered for rendering uneven

Sr.No.	Description of Items
	and honeycombed surface, only. Newly laid concrete shall be covered by gunny bag, plastic, tarpaulin etc. (Wooden centering will not be allowed.), with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Crushed sand VSI Grade)
9.	Providing M-30 R.C.C. cast in situ single rim bored piles each of load capacity and as per design as directed placed through steel shell sunk to the required depth through all strata except rock excluding provision of reinforcement as per detailed drawings and design approved by the Engineer-in-charge including placing concrete by tremie arrangements, compaction of concrete including chipping and dressing and withdrawal of shell etc. complete
10.	Providing and laying Cast in situ/Ready Mix cement concrete M-30 of trap / granite /quartzite/ gneiss metal for R.C.C. Pile Caps as per detailed designs and drawing or as directed including bailing out water manually, steel centering, formwork, cover blocks, laying/pumping, compaction finishing and curing etc. complete with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Crushed sand VSI Grade) complete Excluding steel Reinforcement.
11.	Carrying out low strain pile integrity test including all arrangements for test, equipment's/accessories, materials, labour, submission of test report etc but excluding the cost of installation of job pile (installation of pile shall be paid separately as per relevant items) all complete as per specification and as directed by the engineer-in-charge.
12.	Carrying out Initial Lateral Load Test on vertical (Initial/test) piles up to maximum test load of 2.0 times the safe lateral load capacity on single pile as specified in accordance with IS 2911 part-4 including all necessary arrangements such as jacks, measuring devices, etc. complete as required. (For this test if adjoining piles is used for drawing support for test, then the arrangements shall be made to record deflection, submission of test report etc. of both the piles. However, testing both these piles will be considered as one test only. Providing and installing test piles measured separately)
13.	Conducting routine vertical load test for 1.5 times safe load carrying capacity by means of High Strain Dynamic Load Test including provision and erection of

Sr.No.	Description of Items
	crane- hammer-drop mechanism, acceleration and displacement velocity transducers to record both force and velocity including excavation, dewatering, preparation of pile head including building up of pile, additional reinforcement if any etc. all complete as per approved method (ASTM D4945-89), submission of test report, specifications and instructions of the Engineer for piles of following diameter
14.	Providing and laying Cast in situ/Ready Mix cement concrete M-30 of trap / granite/quartzite/ gneiss metal for R.C.C. columns as per detailed designs and drawing or as directed including steel centering, formwork, cover blocks, laying/pumping, compaction finishing the formed surfaces with cement mortar 1:3 of sufficient minimum thickness to give a smooth and even surface or roughening if special finish is to be provided and curing etc. complete,(Excluding reinforcement and structural steel). with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Crushed sand VSI Grade)
15.	Providing and laying Cast in situ/Ready Mix cement concrete M-30 of trap / granite /quartzite/ gneiss metal for R.C.C. beams and lintels as per detailed designs and drawings or as directed including steel centering, formwork, cover blocks, laying/pumping, compaction and roughening the surface if special finish is to be provided and curing etc. complete. (Excluding reinforcement and structural steel). with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)
16.	Providing and laying Cast in situ/Ready Mix cement concrete M-30 of trap/ granite / quartzite/ gneiss metal for R.C.C. slabs and landings as per detailed designs and drawings including steel centering, formwork, cover blocks, laying/pumping, compaction finishing the formed surfaces with cement mortar 1:3 of sufficient minimum thickness to give a smooth and even surface or roughening if special finish is to be provided and curing etc. complete, (Excluding reinforcement and structural steel). with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)

Sr.No.	Description of Items
17.	Providing and fixing during laying in RCC Slab rectangular 2 mm thick M.S. sheet Fan hook box of size 180x100 mm with required depth with holes or notches as required, bottom and top sides of 1.6 mm M.S. sheet fixed with 3.5 mm dia. round headed screws together with 12 mm dia. M.S. bar bent to shape with hooked ends for fan hook and painting to exposed surface. As directed by Engineer in charge.
18.	Providing and fixing in position TMT FE-500bar Reinforcement of various diameter's for R.C.C. pile caps, footings, foundations, slabs, beams, columns, canopies, staircases, newels, Chajja, lintels, pardies coping, fins, arches, etc. as per detailed designs, drawings and schedules, including cutting, bending, hooking the bars, binding with wires or tack welding and supporting as required etc. complete.
19.	Providing cement-based Water Proofing Treatment to Terraces (Indian water proofing or alike) with brick bats laid in required slope to drain the water for any span after cleaning the base surface. Applying a coat of cement slurry admixed with approved water proofing compound and laying the brick bats on bottom layer in C.M.1:5 admixed with approved water proofing compound filling up to half depth of brick bats, curing this layer for 3 days, applying cement slurry over this layer joints of brick bats with C.M.1:3 admixed with approved water proofing compound and finally top finishing with average 20 mm. thick layers of same mortar added with jute fiber at 1 Kg per bag including finishing the surface smooth with cement slurry admixed with approved water proofing compound. Marking finished surface with false squares of 300mm x 300 mm. making the junctions at the parapet rounded and tapered top for required height, with drip mould at the junction of plaster and parapet and curing and covering 10 years Guarantee against leak proofness on Court fee stamp paper of Rs. 500/- including ponding test etc. complete.
20.	Providing Waterproofing in W.C. and Bath including brick bat coba in all position including providing and laying 12mm bedding in cement mortar 1:3 on vergin concrete slab with waterproofing compound @ 1Kilogram/per bag of cement laying brick bat coba of required thickness in cm 1:5 with waterproofing compound 1 Kilogram/bag of cement grouting and finishing the top layer with 20mm thick brick bedding in cm mortar 1:3 with waterproofing compound 1 Kilogram/per bag of cement and testing the treated portion for 48 hours by pond test and covering ten years' guarantee on requisite stamp paper including

Sr.No.	Description of Items
	curing etc. complete.
21.	Providing Autoclaved Aerated Concrete Block masonry of Ecolite or equivalent make conforming to IS:2185 (Part 3) - 1984 in extra fine jointing mortar of fix block of UltraTech or equivalent in superstructure including striking joints, raking out joints and scaffolding etc. Complete. (The test shall be carried out conforming to IS:6441 (Part I) - 1972)
22.	Providing and applying Two coats of wall care Putty on plastered surface and Ceiling and Walls to prepare surface even and smooth of approved make, etc complete.
23.	Providing internal cement plaster 12mm thick in single coat in cement mortar 1:4 without neeru finish to concrete or brick surfaces, in all positions including scaffolding and curing etc. complete.
24.	Providing neeru finish to plastered surfaces in all positions including scaffolding and curing etc. complete.
25.	Providing sand faced plaster externally in cement mortar using approved screened sand, in all positions including base coat of 15 mm thick in cement mortar 1:4 using waterproofing compound at 1 kg per cement bag curing the same for not less than 2 days and keeping the surface of the base coat rough to receive the sand faced treatment 6 to 8 mm thick in cement mortar 1:4 finishing the surface by taking out grains and curing for fourteen days scaffolding etc. Complete.
26.	Providing Waterproof Plaster in W.C. and bath 12 mm thick for dado in cement mortar 1:3 with neat finishing, floating using waterproofing compound at the rate of 1 Kilogram. per bag of cement of approved make and manufacturer and curing etc. complete.
27.	Providing and laying Vitrified mirror / glossy finish tiles decorative type having size 590 mm to 605 mm x 590 mm to 605 mm of 8 to 10 mm thickness and confirming to IS. 15622-2006 (group Bla) of approved make, shade and pattern for flooring in required position laid on a bed of 1:4 cement mortar including neat cement float, filling joints, curing and clearing etc. complete.
28.	Providing and laying vitrified mirror / glossy finish tiles having size 590 mm to 605 mm x 590 mm to 605 mm of 8 to 10 mm thickness and confirming to IS. 15622-2006 (group Bla) of approved make, shade and pattern for Dado and skirting in required position fixed in 1:4 cement mortar including neat cement

Sr.No.	Description of Items
	float, filling joints, curing and clearing etc. complete
29.	Providing and laying Antiskid Ceramic tiles of approved quality of size 30 cm x 30 cm and confirming to IS 15622-2006 (Group-B IIA) for antiskid flooring in required position laid on a bed of 1:4 cement mortar including cement float, filling joint with cement slurry cleaning curing etc. complete.
30.	Providing and laying Ceramic tiles having size 30 cm. x 45 cm. confirming to corresponding I.S. for dado and skirting in required position with readymade adhesive mortar of approved quality on plaster of 1:2 cement mortar including joint filling with white/ colour cement slurry cleaning curing etc. complete.
31.	Providing and laying machine cut machine polished machine cut Kota stone slabs 20 to 25 mm thick for treads and risers of steps and staircases, with rounded nosing for the treads on a bed of 1:4 cement mortar including cement float, filling joints with neat cement slurry, curing, polishing and cleaning etc. complete.
32.	Providing and applying colour-wash of approved colour and shade in one coat to new surface including scaffolding, brushing and brooming down (excluding base coats of white wash) etc. Complete.
33.	Providing and applying washable powder distemper of approved colour and shade to old and new surfaces in two coats including scaffolding, preparing the surfaces. (Excluding the primer coat) etc. complete.
34.	Providing and applying two coats of Water Proof cement paint of approved manufacture and of approved colour to the plastered surfaces including scaffolding if necessary, cleaning and preparing the surface, watering for two days etc. complete.
35.	Providing and fixing factory made door/Window frames of required colour made with wood Polymer composite (WPC) extruded internal ribbed sections conforming to American Society for testing and materials (ASTM) in regard with density, tensile strength, flexural strength, Izod impact, water absorption and screw withdrawal strength with wall thickness of 5mm (+/-1mm) and with frames having mitred joints at each corner, joined together by two nos. of 10 x 8 mm & 40 mm long PVC dovetail keys including hard wood piece support along the full vertical and horizontal length in rebate portion hole for fixing hinges and fixtures and the frames fixed to Walls/columns with 15 mm long dash fasteners/holdfasts (dash fasteners/holdfasts to be paid separately) etc.

Sr.No.	Description of Items
	complete as per the direction of Engineer in charge, with prior approval of concerned Superintending Engineer. For Profile size of 100 x 50 mm with 32 x 12 mm rebate
36.	Providing and fixing green marble of 18 to 20 mm thick for door frame/ dado/ window boxing etc. On C.M. 1:6 including filling joints with polymer base filler nosing / moulding the sharp edges wherever necessary, curing, etc. complete.
37.	Providing and fixing solid core Flush Door Shutter commercial in single leaf 32 mm thick without ventilator commercial type of exterior grade as per detailed drawings with wrought iron hold fast, stainless steel fixtures and fastenings and handles on both sides and finishing with oil painting 3 coats complete.
38.	Providing and fixing fiber glass Reinforced polyester door shutter 30 mm thick as per IS 14856 (2000) (Reaffirmed 2006) without ventilator including chromium plated fixtures and fastening with chromium plated handles on both sides, etc complete.
39.	Providing and fixing anodized (anodic film must not be less than 15 microns i.e. AC-15 as per IS, the anodising must be scaleted by keeping the anodized section in boiling de-anodized water for a period of one hour) aluminium fixed/ sliding / pivoted / top hung / side hung / louvered type windows, doors, ventilators and partitions conforming to IS:1948, 1961 with hollow bottom section & standard single / two / three / four track frame section all round (top, bottom and sides) with shutter frame made up of standard handle section standard interlocking section and standard top and bottom section with infill panel of plain /frosted / tinted glass/ Novapan sheet / acrylic sheet / hardener laminated sheet as listed below with nylon guides, gliders glass fixing PVC weather strips, neoprene gasket, PVC weep holes etc complete with approved type of locking arrangement, handles and other fixtures including fixing, filling the gaps with silicon sealant, cleaning polishing the aluminium section with petrol and / or other agents as specified by the manufacturers etc. complete as directed by Engineer In Charge.6 mm thick Plain / frosted / tinted Glass
40.	Providing and fixing mild steel grill railing 15 Kilogram/sq. m with Ghana teak wood hand rail of size 75mm x 60 mm and sill of 75mm.x 25mm and newel posts for staircases, including fabricating, fixtures, erecting, painting the grill work with one coat of anticorrosive paint and two coats of oil painting with

Sr.No.	Description of Items
	approved colours and polishing the sill, hand rail and the newel post with French polish.
41.	Providing and fixing in position. (As per I.S.1868 / 1982) Aluminium sliding window of two tracks with rectangular pipe having overall dimension 763.50 x 38.10 x 1.02 mm at weight 0.547 Kilogram/Running metre. and window frame bottom track section 61.85 x 31.75 x 1.20 mm at weight 0.695 Kilogram/Running metre. Top and side track section 61.85 x 31.75 x 1.30 mm at weight 0.659 Kilogram/Running metre. The shutter should be of bearing bottom 40 x 18 x 1.25 mm at weight 0.417 Kilogram/Running metre. Inter locking section 40 x 18 x 1.10 mm at weight 0.469 Kilogram/Running metre. And handle section 40 x 18 x 1.25 mm at weight 0.417 Kilogram/Running metre. and top section 40 x 18 x 1.25 mm at weight 0.417 Kilogram/Running metre. As per detailed drawings and as directed by Engineer in charge with all necessary Aluminium sections fixtures and fastenings such as roller bearing in nylon casting and self-locking catch fitted in vertical section of shutter including 5 mm thick plain glass with all required screws and nuts etc, complete. With powder coating without box
42.	Providing and fixing in position powder coated aluminium Louvered windows / ventilator of various sizes with powder coating as per detailed drawing and specifications including aluminium frames 80 x 38 mm x 1.22 mm box type, 5 mm thick sheet glass louvers, of approved quality etc. complete.
	Road work
1.	Excavation for roadway in earth, soil of all sorts, sand, gravel or soft murum including dressing section to the required grade, camber and side slopes and conveying the excavated materials with all lifts up to a lead of 50m. and spreading for embankment or stacking as directed.
2.	Providing, laying, spreading and compacting stone aggregates of specific sizes to water bound macadam specification including spreading in uniform thickness, hand packing to proper grade and camber, applying and brooming requisite type of screening/ binding Materials to fill up the interstices of coarse aggregate, watering and compacting with vibratory roller to the required density. By Mechanical Means - Grading II (Using Screening Type B (11.2 mm) Aggregate).
3.	Providing, laying, spreading and compacting stone aggregates of specific sizes

Sr.No.	Description of Items
	to water bound macadam specification including spreading in uniform thickness, hand packing to proper grade and camber, applying and brooming requisite type of screening/ binding Materials to fill up the interstices of coarse aggregate, watering and compacting with Vibratory roller. to the required density. By Mechanical Means - Grading I (Using Screening Type A (13.2) mm Aggregate)
4.	Spreading hard murum/ soft murum/ gravel or kankar for side width complete.
5.	Compacting the hard murum side widths including laying in layers on each side with vibratory roller including artificial watering etc. complete.
6.	Surface Dressing--Providing and laying surface dressing as wearing course in single coat using crushed stone aggregates of specified size on a layer of bituminous binder laid on prepared surface and rolling with 8-10 tonne smooth wheeled steel roller. -- c)10mm nominal chipping size (bitumen9.0kg/10Sq.M-- Bitumen of specified grade (VG-30 bulk bitumen rates are considered to arrive at rates)
7.	Providing and constructing 50 mm. thick Modified Penetration Macadam (MPM) road surface including all materials, preparing the existing road surface, spreading 40mm. stone metal layers 30% crusher broken metal + 70% Hand broken (by breaking rubble obtained by blasting) heating and spraying the bitumen of specified grade @ 1.75 kg/sqm, spreading 12mm.size chips compacting with static roller having weight 8 to 10 MT. to achieve the desired degree of compaction as per Technical Specification Clause 506 etc. complete. Including picking of existing WBM surface. (VG-30 bulk bitumen rates are considered to arrive at rates)
	Underground Water Tanks/WTP
8.	Excavation for foundation in earth, soil of all types, sand, gravel and soft murum, including removing the excavated material up to a distance of 50 m. beyond the building area and stacking and spreading as directed, dewatering, preparing the bed for the foundation and necessary back filling, ramming, watering including shoring and strutting etc. complete. (Lift up to 1.5 m.) By Mechanical Means
9.	Excavation for foundation in hard murum and boulders including removing the excavated material up to a distance 50 metres, beyond the building area and stacking and spreading as directed, dewatering, preparing the bed for the

Sr.No.	Description of Items
	foundation and necessary backfilling, ramming, watering including shoring and strutting etc. complete. (Lift up to 1.5m.) By Mechanical Means
10.	Providing preconstructional antitermite treatment as per I.S. 6313 (Part-II) by treating the bottom surface and sides of excavation at the rate of 5 litres of emulsion concentrate of 1.0 percent of clorophyrifos per square meter of surface area covering 10 years guarantee on bond paper.
11.	Filling in plinth and floors with approved excavated material in 15cm. To 20cm. layers including watering and compacting etc. complete.
12.	Filling in plinth and floors with contractors' material/brought from outside and approved by Engineer in charge in layers of 15 cm to 20 cm including watering and compaction etc. complete.
13.	Providing dry/trap/granite/quartzite/gneiss rubble stone soling 15 cm to 20cm thick including hand packing and compacting etc. complete.
14.	Providing and laying Cast in situ/Ready Mix cement concrete in M-10 of trap/ granite/ quartzite/ gneiss metal for foundation and bedding including bailing out water, Steel centering, formwork, laying/pumping, compacting, roughening them if special finish is to be provided, finishing if required and curing complete, with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand SI Grade finely washed etc)
15.	Providing and laying Cast in situ/Ready Mix cement concrete in M15 of trap/granite/quartzite/gneiss metal for steps including steel centering, formwork, laying/pumping, compacting, roughening them if special finish is to be provided, finishing uneven and honeycombed surface and curing etc. complete. The Cement Mortar 1:3 plaster is considered for rendering uneven and honeycombed surface, only. Newly laid concrete shall be covered by gunny bag, plastic, tarpaulin etc. (Wooden centering will not be allowed.), with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Crushed sand VSI Grade)
16.	Providing and laying Cast in situ / Ready Mix cement concrete M-25 of trap / granite / quartzite / gneiss metal for R.C.C. work in foundations like raft, strip foundations, grillage and footings of R.C.C. columns and steel stanchions etc

Sr.No.	Description of Items
	.including bailing out water, Steel centering, formwork, cover blocks, laying / pumping, compaction and curing roughening the surface if special finish is to be provided (Excluding reinforcement and structural steel) etc. complete, with fully automatic microprocessor based PLC with SCADA enabled reversible Drum Type mixer / concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)
17.	Providing and laying Cast in situ / Ready Mix cement concrete M-25 of trap / granite / quartzite / gneiss metal for R.C.C. columns as per detailed designs and drawings or as directed including steel centering, formwork, cover blocks, laying / pumping, compaction finishing the formed surfaces with cement mortar 1:3 of sufficient minimum thickness to give a smooth and even surface or roughening if special finish is to be provided and curing etc. complete, (Excluding reinforcement and structural steel). with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer / concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)
18.	Providing and laying Cast in situ / Ready Mix cement concrete in M-25 of trap / granite / quartzite / gneiss metal for R.C.C. beams and lintels as per detailed designs and drawings or as directed including steel centering, formwork, cover blocks, laying / pumping, compaction and roughening the surface if special finish is to be provided and curing etc. complete. (Excluding reinforcement and structural steel). with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer / concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)
19.	Providing and laying Cast in situ / Ready Mix cement concrete M-25 of trap / granite / quartzite / gneiss metal for R.C.C. slabs and landings as per detailed designs and drawings including steel centering, formwork, cover blocks, laying / pumping, compaction finishing the formed surfaces with cement mortar 1:3 of sufficient minimum thickness to give a smooth and even surface or roughening if special finishes to be provided and curing etc. Complete, (Excluding reinforcement and structural steel). with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer / concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)

Sr.No.	Description of Items
20.	Providing and laying Cast in situ/Ready Mix cement concrete in M-25 of trap/ granite/ quartzite/ gneiss metal for R.C.C. pardi of required thickness including steel centering, formwork, cover blocks, laying/pumping, compacting and roughening them if special finish is to be provided and curing complete. (Excluding reinforcement and structural steel). with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)
21.	Providing and fixing in position TMT FE-500bar Reinforcement of various dia Metres for R.C.C. piles and pile caps, footings, foundations, slabs, beams, columns, canopies, staircases, newels, chajja, lintels, pardies coping, fins, arches, etc. as per detailed designs, drawings and schedules, including cutting, bending, hooking the bars, binding with wires or tack welding and supporting as required etc. complete.
22.	Providing and fixing 20 to 25 mm thick rough Shahabad box type waterproofing treatment to vertical outside faces of Reinforced Cement concrete walls of basement or underground floor including filling the gap of 25 mm between rough Shahabad and R.C.C. walls with cement grout mixed with water proofing liquid Algae proof or other alike with one tile lift method, brushing the joints horizontally with cement slurry mixed with water proofing liquid for width 30 to 35 mm and sloping coping over topmost tile with cement mortar 1:3, butting the bottom most with Cement concrete 1:2:4 mixed with water proofing liquid, curing, with 7 years guarantee on court fee stamp of Rs.100/- with ponding test etc complete. (Excluding Cement concrete 1:3:6 base concrete).
23.	Providing and fixing 20 to 25 mm thick rough Shahabad box type waterproofing for basement or underground floor on a base of Cement concrete 1:3:6, including sub base of 2.50-centimetre-thick Cement mortar 1:3 with conceal, secrete (RMW), Algae proof or other alike waterproofing liquids, raking of joints up to 35 mm depth, filling the joints with 6 mm size crushed metal, hand grouting with cement slurry, curing and cleaning, with 7 years guarantee on court fee stamp of Rs. 100/- with ponding test etc complete. (Excluding Cement concrete 1:3:6 base concrete).
24.	Providing Internal cement plaster 12mm thick in single coat in cement mortar 1:4 without neeru finish to concrete or brick surfaces, in all positions including scaffolding and curing etc. complete.

Sr.No.	Description of Items
25.	Providing Waterproof Plaster 12 mm thick for dado in cement mortar 1:3 with neat finishing, floating using waterproofing compound at the rate of 1 Kilogram. per bag of cement of approved make and manufacturer and curing etc. complete.
26.	Providing and laying machine cut machine Polished machine cut Kota stone slabs 20 to 25 mm thick for treads and risers of steps and staircases, with rounded nosing for the treads on a bed of 1:4 cement mortar including cement float, filling joints with neat cement slurry, curing, polishing and cleaning etc. complete.
27.	Providing and fixing Circular Manhole Cover of 0.56m internal dia in SFRC (Reinforced Concrete with Steel Fibre) with grade designation HD20 as per IS 12592 :2002 air tight fixed to Manhole frame, including testing and inspection as per the guidelines of IS 12592. (Fibre reinforcement for shrinkage crack resistance & steel as main reinforcement as per IS 12592), including fixing with M-30 concrete and locking arrangement, as per specification no SP-SWD-3 etc. complete as specified and as directed by Engineer in Charge.
	STP
1.	Excavation for foundation in earth, Soil of all types, sand, gravel and soft murum, including removing the excavated material up to a distance of 50 m. beyond the building area & stacking and spreading as directed, dewatering, preparing the bed for the foundation and necessary back filling, ramming, watering including shoring and strutting etc. complete. (Lift up to 1.5 m.) By Mechanical Means
2.	Excavation for foundation in Hard murum and boulders including removing the excavated material up to a distance of 50 metres, beyond the building area and stacking and spreading as directed, dewatering, preparing the bed for the foundation and necessary back filling, ramming, watering including shoring and strutting etc. complete. (Lift from 1.5 m to 3.0 m.) By Mechanical Means
3.	Excavation for foundation in soft rock and old cement or lime masonry foundations including removing the excavated material up to a distance of 50 metres beyond the building area and stacking as directed, including dewatering, preparing the bed for the foundation and necessary back filling with available earth /murum, ramming, watering including shoring and strutting etc. complete (lift up to 1.5m) By Mechanical Means

Sr.No.	Description of Items
4.	Excavation for foundation in soft rock and old cement or lime masonry foundations including removing the excavated material up to a distance of 50 metres beyond the building area and stacking as directed, including dewatering, preparing the bed for the foundation and necessary back filling with available earth /murum, ramming, watering including shoring and strutting etc. complete (Lift from 1.5m to 3.0 m.) By Mechanical Means
5.	Providing and laying Cast in situ/Ready Mix cement concrete in M-10 of trap/ granite/quartzite/ gneiss metal for foundation and bedding including bailing out water, Steel centering, formwork, laying/pumping, compacting, roughening them if special finish is to be provided, finishing if required and curing complete, with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type Mixer/concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Crushed sand VSI Grade)
6.	Providing and laying Cast in situ/Ready Mix cement concrete M-25 of trap / granite /quartzite/ gneiss metal for R.C.C. work in foundations like raft, strip foundations, grillage and footings of R.C.C. columns and steel stanchions etc. including bailing out water, Steel centering, formwork ,cover blocks, laying/pumping, compaction and curing roughening the surface if special finish is to be provided (Excluding reinforcement and structural steel) etc. complete, with fully automatic microprocessor based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Crushed sand VSI Grade)
7.	Providing and laying Cast in situ/Ready Mix cement concrete M-25 of trap / granite /quartzite/ gneiss metal for R.C.C. columns as per detailed designs and drawings or as directed including steel centering, formwork, cover blocks, laying/pumping, compaction finishing the formed surfaces with cement mortar 1:3 of sufficient minimum thickness to give a smooth and even surface or roughening if special finish is to be provided and curing etc. complete,(Excluding reinforcement and structural steel).with fully automatic microprocessor based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)
8.	Providing and laying Cast in situ / Ready Mix cement concrete M-25 of trap / granite / quartzite / gneiss metal for R.C.C. slabs and landings as per detailed designs and drawings including steel centering, formwork, cover blocks, laying

Sr.No.	Description of Items
	/ pumping, compaction finishing the formed surfaces with cement mortar 1:3 of sufficient minimum thickness to give a smooth and even surface or roughening if special finishes to be provided and curing etc. Complete, (Excluding reinforcement and structural steel). with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer / concrete Batch mix plant (Panmixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)
9.	Providing and laying Cast in situ / Ready-mix cement concrete in M-25 of trap / granite / quartzite / gneiss metal for R.C.C. beams and lintels as per detailed designs and drawings or as directed including steel centering, formwork, cover blocks, laying / pumping, compaction and roughening the surface if special finish is to be provided and curing etc. Complete. (Excluding reinforcement and structural steel). with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer / concrete Batch mix plant (Panmixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)
10.	Providing and laying Cast in situ/Ready Mix cement concrete in M-25 of trap/ granite/ quartzite/ gneiss metal for R.C.C. pardi of required thickness including steel centering, formwork, cover blocks, laying/pumping, compacting and roughening them if special finish is to be provided and curing complete. (Excluding reinforcement and structural steel).with fully automatic microprocessor based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)
11.	Providing and fixing in position TMT FE-500bar Reinforcement of various diameter's for R.C.C. pile caps, footings, foundations, slabs, beams, columns, canopies, staircases, newels, Chajja, lintels, pardies coping, fins, arches, etc. as per detailed designs, drawings and schedules, including cutting, bending, hooking the bars, binding with wires or tack welding and supporting as required etc. complete.
12.	Providing Autoclaved Aerated Concrete Block masonry of Ecolite or equivalent make conforming to IS:2185 (Part 3) - 1984 in extra fine jointing mortar of fix block of UltraTech or equivalent in superstructure including striking joints, raking out joints and scaffolding etc. Complete. (The test shall be carried out conforming to IS:6441 (Part I) - 1972)

Sr.No.	Description of Items
13.	Providing internal cement plaster 12mm thick in single coat in cement mortar 1:4 without neeru finish to concrete or brick surfaces, in all positions including scaffolding and curing etc. complete.
14.	Providing sand faced plaster externally in cement mortar using approved screened sand, in all positions including base coat of 15 mm thick in cement mortar 1:4 using waterproofing compound at 1 kg per cement bag curing the same for not less than 2 days and keeping the surface of the base coat rough to receive the sand faced treatment 6 to 8 mm thick in cement mortar 1:4 finishing the surface by taking out grains and curing for fourteen days scaffolding etc. Complete.
	SWD (Storm Water Drain)
1.	Excavation for foundation in earth, Soil of all types, sand, gravel and soft murum, including removing the excavated material up to a distance of 50 m. beyond the building area & stacking and spreading as directed, dewatering, preparing the bed for the foundation and necessary back filling, ramming, watering including shoring and strutting etc. complete. (Lift up to 1.5 m.) By Mechanical Means
2.	Providing dry/ trap/ granite/ quartzite/ gneiss rubble stone soling 15 cm to 20 cm thick including hand packing and compacting etc. complete.
3.	Providing and laying Cast in situ/Ready Mix cement concrete in M-10 of trap/ granite/quartzite/ gneiss metal for foundation and bedding including bailing out water, Steel centering, formwork, laying/pumping, compacting, roughening them if special finish is to be provided, finishing if required and curing complete, with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type Mixer /concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Crushed sand VSI Grade)
4.	Providing and laying in situ / ready mix M-20 cement concrete of trap /granite /quartzite /gneiss metal for approach slab / Rigid pavement including compacting, finishing, curing etc. complete. (Excluding reinforcement) (with reversible drum type mixer/ concrete batch mix plant (pan mixer) with SCADA with fine aggregates of required specifications (Natural sand / VSI sand finely washed etc), cement 6.40 bags/cumt)
5.	Providing and casting in situ / ready mix PCC M15 grade of trap metal for coping to head walls / parapet including centering, form work, compaction and

Sr.No.	Description of Items
	curing etc. complete. (With reversible drum type mixer/concrete batch mix plant (pan mixer) with SCADA with fine aggregates of required specifications (Natural sand / VSI sand finely washed etc)
6.	Providing and fixing in position TMT FE-500bar Reinforcement of various diameter's for R.C.C. pile caps, footings, foundations, slabs, beams, columns, canopies, staircases, newels, Chajja, lintels, pardies coping, fins, arches, etc. as per detailed designs, drawings and schedules, including cutting, bending, hooking the bars, binding with wires or tack welding and supporting as required etc. complete.
7.	Providing and fixing of 40mm thick FRP heavy duty grating of approved make for near the swimming pool of required shape and sizes. The rate shall include cost of providing 50mm x 50mm x 5mm thick MS angle, hold fast with application of two coats of primer and enamel paint of approved colour, etc. complete
	Compound Wall/Security Cabin
1.	Excavation for foundation in earth, Soil of all types, sand, gravel and soft murum, including removing the excavated material up to a distance of 50 m. beyond the building area & stacking and spreading as directed, dewatering, preparing the bed for the foundation and necessary back filling, ramming, watering including shoring and strutting etc. complete. (Lift up to 1.5 m.) By Mechanical Means
2.	Filling in plinth and floors with contractors' material/brought from outside and approved by Engineer in charge in layers of 15 cm to 20 cm including watering and compaction etc. complete.
3.	Providing and laying Cast in situ/Ready Mix cement concrete in M15 of trap/granite/quartzite/gneiss metal for steps including steel centering, formwork, laying/pumping, compacting, roughening them if special finish is to be provided, finishing uneven and honeycombed surface and curing etc. complete. The Cement Mortar 1:3 plaster is considered for rendering uneven and honeycombed surface, only. Newly laid concrete shall be covered by gunny bag, plastic, tarpaulin etc. (Wooden centering will not be allowed.), with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Crushed sand VSI Grade)

Sr.No.	Description of Items
4.	Providing and laying Cast in situ/Ready Mix cement concrete M-25 of trap / granite /quartzite/ gneiss metal for R.C.C. work in foundations like raft, strip foundations, grillage and footings of R.C.C. columns and steel stanchions etc. including bailing out water, Steel centering, formwork ,cover blocks, laying/pumping, compaction and curing roughening the surface if special finish is to be provided (Excluding reinforcement and structural steel) etc. complete, with fully automatic microprocessor based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Crushed sand VSI Grade)
5.	Providing and laying Cast in situ/Ready Mix cement concrete M-25 of trap / granite /quartzite/ gneiss metal for R.C.C. columns as per detailed designs and drawings or as directed including steel centering, formwork, cover blocks, laying/pumping, compaction finishing the formed surfaces with cement mortar 1:3 of sufficient minimum thickness to give a smooth and even surface or roughening if special finish is to be provided and curing etc. complete,(Excluding reinforcement and structural steel).with fully automatic microprocessor based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)
6.	Providing and laying Cast in situ / Ready Mix cement concrete in M-25 of trap / granite / quartzite / gneiss metal for R.C.C. beams and lintels as per detailed designs and drawings or as directed including steel centering, formwork, cover blocks, laying / pumping, compaction and roughening the surface if special finish is to be provided and curing etc. complete. (Excluding reinforcement and structural steel). with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer / concrete Batch mix plant (Panmixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)
7.	Providing fly ash brick masonry with conventional/ I.S. type bricks in cement mortar 1:6 in superstructure including striking joints, raking out joints, watering and scaffolding etc. Complete
8.	Providing and laying Cast in situ/Ready Mix cement concrete in M-20 of trap/ granite/ quartzite/gneiss metal for R.C.C. coping to plinth or parapet and sill of doors and windows moulded as per detailed drawings or chamfered as approved by the Engineer including steel centering, formwork, cover blocks,

Sr.No.	Description of Items
	laying/pumping, compacting, curing, finishing and roughening them if special finish is to be provided and curing complete. (Excluding reinforcement and structural steel).with fully automatic microprocessor based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Crushed sand VSI Grade)
9.	Providing and laying Cast in situ / Ready Mix cement concrete M-25 of trap / granite / quartzite / gneiss metal for R.C.C. slabs and landings as per detailed designs and drawings including steel centering, formwork, cover blocks, laying / pumping, compaction finishing the formed surfaces with cement mortar 1:3 of sufficient minimum thickness to give a smooth and even surface or roughening if special finishes to be provided and curing etc. Complete, (Excluding reinforcement and structural steel). with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer / concrete Batch mix plant (Panmixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)
10.	Providing and fixing in position TMT FE-500bar Reinforcement of various diameter's for R.C.C. pile caps, footings, foundations, slabs, beams, columns, canopies, staircases, newels, Chajja, lintels, pardies coping, fins, arches, etc. as per detailed designs, drawings and schedules, including cutting, bending, hooking the bars, binding with wires or tack welding and supporting as required etc. complete.
11.	Providing sand faced plaster externally in cement mortar using approved screened sand, in all positions including base coat of 15 mm thick in cement mortar 1:4 using waterproofing compound at 1 kg per cement bag curing the same for not less than 2 days and keeping the surface of the base coat rough to receive the sand faced treatment 6 to 8 mm thick in cement mortar 1:4 finishing the surface by taking out grains and curing for fourteen days scaffolding etc. Complete.
12.	Providing and fixing mild steel grill gate as per the detailed drawing including hinges, locking arrangements and painting in three coats of oil paint of approved colour and shade etc. complete. (Weight 18 Kilogram. to 20 Kilogram. Per Sqmt.) As directed by Engineer- in-charge.
13.	Providing and laying Vitrified mirror / glossy finish tiles decorative type having size 590 mm to 605 mm x 590 mm to 605 mm of 8 to 10 mm thickness and

Sr.No.	Description of Items
	confirming to IS. 15622-2006 (group Bla) of approved make, shade and pattern for flooring in required position laid on a bed of 1:4 cement mortar including neat cement float, filling joints, curing and clearing etc. complete.
14.	Providing and applying colour-wash of approved colour and shade in one coat to new surface including scaffolding, brushing and brooming down (excluding base coats of white wash) etc. Complete.
15.	Providing and applying two coats of Water Proof cement paint of approved manufacture and of approved colour to the plastered surfaces including scaffolding if necessary, cleaning and preparing the surface, watering for two days etc. complete.
16.	Providing and fixing solid core Flush Door Shutter commercial in single leaf 32 mm thick without ventilator commercial type of exterior grade as per detailed drawings with wrought iron hold fast, stainless steel fixtures and fastenings and handles on both sides and finishing with oil painting 3 coats complete.
17.	Providing and fixing in position. (As per I.S.1868 / 1982) Aluminium sliding window of two tracks with rectangular pipe having overall dimension 763.50 x 38.10 x 1.02 mm at weight 0.547 Kilogram/Running metre. and window frame bottom track section 61.85 x 31.75 x 1.20 mm at weight 0.695 Kilogram/Running metre. Top and side track section 61.85 x 31.75 x 1.30 mm at weight 0.659 Kilogram/Running metre. The shutter should be of bearing bottom 40 x 18 x 1.25 mm at weight 0.417 Kilogram/Running metre. Inter locking section 40 x 18 x 1.10 mm at weight 0.469 Kilogram/Running metre. And handle section 40 x 18 x 1.25 mm at weight 0.417 Kilogram/Running metre. and top section 40 x 18 x 1.25 mm at weight 0.417 Kilogram/Running metre. As per detailed drawings and as directed by Engineer in charge with all necessary Aluminium sections fixtures and fastenings such as roller bearing in nylon casting and self-locking catch fitted in vertical section of shutter including 5 mm thick plain glass with all required screws and nuts etc, complete. With powder coating without box.
	Parks and Gardens
1.	Preparing Lawns - (30Cm Depth) - Excavating the ground to a required depth of 30 CMS, removing & conveying unwanted stuff to a required distance as directed. Filling fresh garden soil / silt & manure in excavated area in 3:1 proportion to a height of 30cms. Mixing garden soil and manure thoroughly,

Sr.No.	Description of Items
	well, levelling, watering on previous night. Planting required variety of Lawn grass as directed and maintaining till well established by watering, weeding, clipping, rolling etc. complete
2.	Planting large flowering/Shady trees (Ht.1 mtrs.): -Digging pit with size 0.9m x 0.9m x 0.9m, removing and conveying unwanted stuff to a required distance as directed. Filling the pit with approved excavated material up to a height of 0.30m. and remaining height of 0.60m. with fresh garden soil & manure in 3:1 proportion. Mixing soil & manure thoroughly well, levelling, as directed, watering previous night, planting one- to two-year-old minimum 1 Mtrs height, healthy trees sapling as per specifications. Maintaining till well established by watering, weeding, stirring of soil replacing of casualties etc. complete.

III. Building Internal Plumbing & Sanitation

Sr. No.	Description of Items
	INTERNAL PLUMBING
1.	Providing and fixing White glazed with bottle trap earthenware Wash Hand Basin of 55x40 cm size including cold water pillar taps, brackets, rubber plugs and brass chain, stop tap, chromium plate bottle trap and necessary pipe connections including UPAC waste pipe and trap up to the outside face of the wall, making good the damaged surface, testing etc. complete.
2.	Providing, laying and fixing P.V.C. pipe of 40mm dia. with fittings such as bends, tees, reducers, clamps, etc. including necessary excavation, trench filling etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowki or as directed etc. complete
3.	Providing, laying and fixing P.V.C. pipe of 50mm dia. with fittings such as bends, tees, reducers, clamps, etc. including necessary excavation, trench filling etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete.
4.	Providing and fixing 75 mm dia stabilizer pipe/ P.V.C. soil vent/waste pipe and with necessary fixtures and fitting such as bends, tees, single junctions, slotted vent, clamps etc. complete
5.	Providing and fixing 100 mm dia stabilizer pipe/ P.V.C. soil vent/waste pipe and with necessary fixtures and fitting such as bends, tees, single junctions, slotted

Sr. No.	Description of Items
	vent, clamps etc. complete.
6.	Providing and fixing on walls/ ceiling/ floor 15 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete
7.	Providing and fixing on walls/ceiling/floor 20 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete
8.	Providing and fixing on walls/ ceiling/ floor 25 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete.
9.	Providing and fixing on walls/ ceiling/ floor 32 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete
10.	Providing and fixing on walls/ ceiling/ floor 40 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete
11.	Providing and laying in trenches 50mm dia. heavy grade having embossed as ISI Mark galvanized iron pipes of 6.33 Kilogram/meter necessary fitting remaking good the demolished portion with filling trenches and with primer of anti-corrosive oil paint,2coats complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete.
12.	Supplying and erecting ISI mark GI pipe 65 mm. dia 'C' class position with accessories complete. Specification no. CW-PLB/GP.
13.	Providing and laying in trenches 100mm dia. heavy grade having embossed as ISI Mark galvanized iron pipes of 14.74 Kilogram/meter necessary fitting remaking good the demolished portion with filling trenches and with primer of anti-corrosive oil paint,2coats complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete

Sr. No.	Description of Items
14.	Supplying & erecting cast iron double flange butterfly valve of size 65 mm dia. complete as per specification no. FF-VL/BFV
15.	Supplying & erecting cast iron double flange butterfly valve of size 75/80 mm dia. complete as per specification no. FF-VL/BFV
16.	Supplying and erecting CI flange 25 mm dia with necessary nuts bolts and washers with rubber seal in position complete.
17.	Supplying and erecting CI flange 50 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
18.	Supplying and erecting CI flange 65 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
19.	Supplying and erecting CI flange 75/80 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
20.	Supplying and erecting CI flange 100 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
21.	Providing and fixing C.P. Angular stop clock with wall flange of approved make continental including necessary sockets /union nut etc. complete.
22.	Providing and fixing C.P. Two-way BIB cock of approved make continental including necessary sockets/union nut etc. complete.
23.	Providing and fixing C.P. BIB cock with wall flange of approved make including necessary sockets/ union nut etc. complete.
24.	Supplying installation testing and commissioning of Hubless centrifugally cast-iron pipes as per ISI 15905 equivalent to NECO inclusive of all necessary special fitting like bends, tees, and caps, 'Y' bends for cleaning offsets, junctions, cowls, etc. lad under floor/fixed on walls and in pipe shafts etc. complete. The quoted rate should include lead and lifting charges to all levels, necessary scaffolding and supports like hanger supports grip bolts and CV sheet 304 grade clamps etc. with prior approval of concerned Superintending Engineer.
25.	c) 75 mm dia
26.	d) 100 mm dia
27.	Providing and fixing P.V.C. Rain water pipes of 110mm outer diameter and having wall thickness of 2.2 to 2.7 mm confirming to I.S. 135921992 including proper rainwater receiving recess with P.V.C. plug, bend, necessary fittings, such as, offsets, shoes, including fixing the pipe on wall using approved wooden cleats projecting 25mm to 40mm from face of wall a fixing with clips of

Sr. No.	Description of Items
	approved quality and number, filing the joint using rubber gasket with solvent cement and properly resting the shoe of pipes on C.C. or masonry blocks, including necessary scaffolding and maintenance for 3 yrs. for any leakages or dislocations of pipes. All the P.V.C. fittings and additional 2-piece socket clips shall be got approved from engineer in charge etc. complete.
28.	Providing and fixing 8 cm rigid PVC Nahani trap including PVC grating, bend, connecting piece of UPVC pipe up to the outside face of wall, making the good damaged surface and testing etc. complete (Prior approval of sample and brand by Ex. Engr. is necessary before use)
29.	Providing and fixing white vitreous China pedestal type water closet (European type W.C. pan) with seat and lid, 10 litre low level white P.V.C. flushing cistern, including flush pipe, with manually controlled device (handle lever), conforming to IS: 7231, with all fittings and fixtures complete, including cutting and making good the walls and floors wherever required:
30.	Providing and fixing in position UPVC ultra violet stabilized 110 mm outer diameter cowl dome confirming to I.S.-4985 including making joints with solvent cement etc. complete
31.	Providing and fixing 45cm wide mild steel ladder of 40mm x6mm mild steel flat stringers and steps of 18mm dia. Mild steel bar for water tank including fixing it in (1:2:4) M-15 cement concrete block 60cm x30cm x 30cm and painting the ladder complete. (Anti-Corrosive Paint)
32.	Providing and fixing Ductile iron Man Hole Covers/ Strom Water Grating and Grating with frame of Various sizes, weight and types and load bearing capacity as per EN-124, equivalent NECO with prior approval of concerned Superintending Engineer.
33.	C.I. Grating 150x150mm (At Terrace For rain water)
34.	C.I. Grating 100x100mm (At Terrace For rain water)
35.	Providing and fixing gun metal gate valve with C.I. wheel of approved quality (screwed end)
36.	1) 20MM DIA
37.	2) 25MM DIA
38.	3) 32MM DIA
39.	4) 40MM DIA
40.	Supplying & erecting cast iron double flange butterfly valve of size 50 mm dia. complete as per specification no. FFVL/BFV

Sr. No.	Description of Items
41.	Supplying & erecting cast iron double flange butterfly valve of size 65 mm dia. complete as per specification no. FFVL/BFV
42.	Supplying & erecting cast iron double flange butterfly valve of size 75/80 mm dia. complete as per specification no. FFVL/BFV
43.	Providing, laying and fixing, jointing Eco- drain 110 mm SN 8 NuDrain UPVC pipes or of equivalent make, manufacture as per EN 13476 or equivalent as per I.S.15328 with fittings such a bends, tees, coupler, etc., jointing with rubber lubricant including necessary excavation, trench refilling with selective excavated material etc. complete.
44.	Providing, laying and fixing, jointing Eco- drain 160 mm SN 4 NuDrain UPVC pipes or of equivalent make, manufacture as per EN 13476 or equivalent as per I.S.15328 with fittings such a bends, tees, coupler, etc., jointing with rubber lubricant including necessary excavation, trench refilling with selective excavated material etc. complete.
45.	Providing and constructing Brick Masonry Inspection Chamber 60cm x 45cm With R.C.C. raft in B.C. soil area including cost of reinforcement, 1:2:4 cement concrete channels half round G.S.W. pipes, Brick Masonry, plastering from inside and airtight 75 mm thick R.C.C. cover medium duty 100Kilogrametc. complete
46.	Providing and constructing Brick Masonry Inspection Chamber 90cmx45cm With R.C.C. raft in B.C. soil area including cost of reinforcement,1:2:4 cement concrete channels half round G.S.W. pipes, Brick Masonry, plastering from inside and airtight with 75 mm thick RCC cover medium duty 140Kilogrametc. complete
47.	Providing and fixing 15cm x 10cm salt glazed stoneware gully trap in cement concrete 1:4:8 outside the building including cast iron grating in the sink, connecting glazed stoneware pipe, brick masonry chamber with cast iron lid and cast-iron grating for the gully trap.
48.	Pressure Reducing Valve (PRV)
49.	1) 32MM DIA
50.	2) 40MM DIA
51.	Strainer for 40mm
52.	Strainer for 50mm
53.	Strainer for 65mm
54.	Strainer for 80mm

Sr. No.	Description of Items
55.	Supplying & erecting Float sensor switch for water level controller working on 250V & rated current 8A, with 2 mtrs. wire & High quality of wire & plastic body.
56.	Providing and fixing C.P. sink cock with raised J" shaped swinging casted spout of approved make including necessary sockets/ union nut etc. complete."
57.	Supplying and erecting non return valve 40 mm dia in position made of gun metal complete.
58.	Supplying and erecting non return valve 50 mm dia in position made of gun metal complete.
59.	Supplying and erecting non return valve 65 mm dia in position made of gun metal complete.
60.	Supplying and erecting non return valve 75/80 mm dia in position made of gun metal complete.
61.	Brass Mosquito Proof Grating Inverted J Type 100 MM Dia CI Cowl Ventilators
62.	Providing and fixing 50mm dia meter water meter with non-return valve including strainer, sockets/union nut and including water meter box making locking arrangement and lock. [without chamber]
63.	PRV For 15mm dia
64.	PRV For 20mm dia

	Creche Internal Plumbing
1.	Providing and fixing White glazed with bottle trap earthenware Wash Hand Basin of 55x40 cm size including cold water piller taps, brackets, rubber plugs and brass chain, stop tap, chromium plate bottle trap and necessary pipe connections including UPAC waste pipe and trap up to the outside face of the wall, making good the damaged surface, testing etc. complete.
2.	Providing, laying and fixing P.V.C. pipe of 40mm dia. with fittings such as bends, tees, reducers, clamps, etc. including necessary excavation, trench filling etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowki or as directed etc. complete
3.	Providing, laying and fixing P.V.C. pipe of 50mm dia. with fittings such as bends, tees, reducers, clamps, etc. including necessary excavation, trench filling etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete.
4.	Providing and fixing 75 mm dia stabilizer pipe/ P.V.C. soil vent/waste pipe and

	with necessary fixtures and fitting such as bends, tees, single junctions, slotted vent, clamps etc. complete
5.	Providing and fixing 100 mm dia stabilizer pipe/ P.V.C. soil vent/waste pipe and with necessary fixtures and fitting such as bends, tees, single junctions, slotted vent, clamps etc. complete.
6.	Providing and fixing on walls/ ceiling/ floor 15 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete
7.	Providing and fixing on walls/ceiling/floor 20 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete
8.	Providing and fixing on walls/ ceiling/ floor 25 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete.
9.	Providing and fixing on walls/ ceiling/ floor 32 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete
10.	Providing and fixing on walls/ ceiling/ floor 40 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete
11.	Providing and laying in trenches 50mm dia. heavy grade having embossed as ISI Mark galvanized iron pipes of 6.33 Kilogram/meter necessary fitting remaking good the demolished portion with filling trenches and with primer of anti-corrosive oil paint,2coats complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete.
12.	Supplying and erecting ISI mark GI pipe 65 mm. dia 'C' class position with accessories complete. Specification no. CW-PLB/GP.
13.	Providing and laying in trenches 100mm dia. heavy grade having embossed as ISI Mark galvanized iron pipes of 14.74 Kilogram/meter necessary fitting remaking good the demolished portion with filling trenches and with primer of anti-corrosive oil paint,2coats complete. Including removing existing pipe line if

	necessary and conveying and stacking the same in PWD chowky or as directed etc. complete
14.	Supplying & erecting cast iron double flange butterfly valve of size 65 mm dia. complete as per specification no. FF-VL/BFV
15.	Supplying & erecting cast iron double flange butterfly valve of size 75/80 mm dia. complete as per specification no. FF-VL/BFV
16.	Supplying and erecting CI flange 25 mm dia with necessary nuts bolts and washers with rubber seal in position complete.
17.	Supplying and erecting CI flange 50 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
18.	Supplying and erecting CI flange 65 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
19.	Supplying and erecting CI flange 75/80 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
20.	Supplying and erecting CI flange 100 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
21.	Providing and fixing C.P. Angular stop clock with wall flange of approved make continental including necessary sockets /union nut etc. complete.
22.	Providing and fixing C.P. Two-way BIB cock of approved make continental including necessary sockets/union nut etc. complete.
23.	Providing and fixing C.P. BIB cock with wall flange of approved make including necessary sockets/ union nut etc. complete.
24.	Supplying installation testing and commissioning of Hubless centrifugally cast-iron pipes as per ISI 15905 equivalent to NECO inclusive of all necessary special fitting like bends, tees, and caps, 'Y' bends for cleaning offsets, junctions, cowls, etc. lad under floor/fixed on walls and in pipe shafts etc. complete. The quoted rate should include lead and lifting charges to all levels, necessary scaffolding and supports like hanger supports grip bolts and CV sheet 304 grade clamps etc. with prior approval of concerned Superintending Engineer.
25.	c) 75 mm dia
26.	d) 100 mm dia
27.	Providing and fixing P.V.C. Rain water pipes of 110mm outer diameter and having wall thickness of 2.2 to 2.7 mm confirming to I.S. 135921992 including proper rainwater receiving recess with P.V.C. plug, bend, necessary fittings, such as, offsets, shoes, including fixing the pipe on wall using approved wooden cleats projecting 25mm to 40mm from face of wall a fixing with clips of

	approved quality and number, filing the joint using rubber gasket with solvent cement and properly resting the shoe of pipes on C.C. or masonry blocks, including necessary scaffolding and maintenance for 3 yrs. for any leakages or dislocations of pipes. All the P.V.C. fittings and additional 2-piece socket clips shall be got approved from engineer in charge etc. complete.
28.	Providing and fixing 8 cm rigid PVC Nahani trap including PVC grating, bend, connecting piece of UPVC pipe up to the outside face of wall, making the good damaged surface and testing etc. complete (Prior approval of sample and brand by Ex. Engr. is necessary before use)
29.	Providing and fixing white vitreous China pedestal type water closet (European type W.C. pan) with seat and lid, 10 litre low level white P.V.C. flushing cistern, including flush pipe, with manually controlled device (handle lever), conforming to IS: 7231, with all fittings and fixtures complete, including cutting and making good the walls and floors wherever required:
30.	Providing and fixing in position UPVC ultra violet stabilized 110 mm outer diameter cowl dome confirming to I.S.-4985 including making joints with solvent cement etc. complete
31.	Providing and fixing 45cm wide mild steel ladder of 40mm x6mm mild steel flat stringers and steps of 18mm dia. Mild steel bar for water tank including fixing it in (1:2:4) M-15 cement concrete block 60cm x30cm x 30cm and painting the ladder complete. (Anti-Corrosive Paint)
32.	Providing and fixing Ductile iron Man Hole Covers/ Strom Water Grating and Grating with frame of Various sizes, weight and types and load bearing capacity as per EN-124, equivalent NECO with prior approval of concerned Superintending Engineer.
33.	C.I. Grating 150x150mm (At Terrace For rain water)
34.	C.I. Grating 100x100mm (At Terrace For rain water)
35.	Providing and fixing gun metal gate valve with C.I. wheel of approved quality (screwed end)
36.	1) 20MM DIA
37.	2) 25MM DIA
38.	3) 32MM DIA
39.	4) 40MM DIA
40.	Supplying & erecting cast iron double flange butterfly valve of size 50 mm dia. complete as per specification no. FFVL/BFV
41.	Supplying & erecting cast iron double flange butterfly valve of size 65 mm dia. complete as per specification no. FFVL/BFV

42.	Supplying & erecting cast iron double flange butterfly valve of size 75/80 mm dia. complete as per specification no. FFVL/BFV
43.	Providing, laying and fixing, jointing Eco- drain 110 mm SN 8 NuDrain UPVC pipes or of equivalent make, manufacture as per EN 13476 or equivalent as per I.S.15328 with fittings such a bends, tees, coupler, etc., jointing with rubber lubricant including necessary excavation, trench refilling with selective excavated material etc. complete.
44.	Providing, laying and fixing, jointing Eco- drain 160 mm SN 4 NuDrain UPVC pipes or of equivalent make, manufacture as per EN 13476 or equivalent as per I.S.15328 with fittings such a bends, tees, coupler, etc., jointing with rubber lubricant including necessary excavation, trench refilling with selective excavated material etc. complete.
45.	Providing and constructing Brick Masonry Inspection Chamber 60cm x 45cm With R.C.C. raft in B.C. soil area including cost of reinforcement, 1:2:4 cement concrete channels half round G.S.W. pipes, Brick Masonry, plastering from inside and airtight 75 mm thick R.C.C. cover medium duty 100Kilogrametc. complete
46.	Providing and constructing Brick Masonry Inspection Chamber 90cmx45cm With R.C.C. raft in B.C. soil area including cost of reinforcement,1:2:4 cement concrete channels half round G.S.W. pipes, Brick Masonry, plastering from inside and airtight with 75 mm thick RCC cover medium duty 140Kilogrametc. complete
47.	Providing and fixing 15cm x 10cm salt glazed stoneware gully trap in cement concrete 1:4:8 outside the building including cast iron grating in the sink, connecting glazed stoneware pipe, brick masonry chamber with cast iron lid and cast-iron grating for the gully trap.
48.	Strainer for 40mm
49.	Strainer for 50mm
50.	Strainer for 65mm
51.	Strainer for 80mm
52.	Supplying & erecting Float sensor switch for water level controller working on 250V & rated current 8A, with 2 mtrs. wire & High quality of wire & plastic body.
53.	Providing and fixing C.P. sink cock with raised J" shaped swinging casted spout of approved make including necessary sockets/ union nut etc. complete."
54.	Supplying and erecting non return valve 40 mm dia in position made of gun metal complete.

55.	Supplying and erecting non return valve 50 mm dia in position made of gun metal complete.
56.	Supplying and erecting non return valve 65 mm dia in position made of gun metal complete.
57.	Supplying and erecting non return valve 75/80 mm dia in position made of gun metal complete.
58.	Brass Mosquito Proof Grating Inverted J Type 100 MM Dia CI Cowl Ventilators
59.	Providing and fixing 50mm dia meter water meter with non-return valve including strainer, sockets/union nut and including water meter box making locking arrangement and lock. [without chamber]
60.	PRV For 15mm dia
61.	PRV For 20mm dia
Fitness center Internal Plumbing	
1.	Providing and fixing White glazed with bottle trap earthenware Wash Hand Basin of 55x40 cm size including cold water pillar taps, brackets, rubber plugs and brass chain, stop tap, chromium plate bottle trap and necessary pipe connections including UPAC waste pipe and trap up to the outside face of the wall, making good the damaged surface, testing etc. complete.
2.	Providing, laying and fixing P.V.C. pipe of 40mm dia. with fittings such as bends, tees, reducers, clamps, etc. including necessary excavation, trench filling etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowki or as directed etc. complete
3.	Providing, laying and fixing P.V.C. pipe of 50mm dia. with fittings such as bends, tees, reducers, clamps, etc. including necessary excavation, trench filling etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete.
4.	Providing and fixing 75 mm dia stabilizer pipe/ P.V.C. soil vent/waste pipe and with necessary fixtures and fitting such as bends, tees, single junctions, slotted vent, clamps etc. complete
5.	Providing and fixing 100 mm dia stabilizer pipe/ P.V.C. soil vent/waste pipe and with necessary fixtures and fitting such as bends, tees, single junctions, slotted vent, clamps etc. complete.
6.	Providing and fixing on walls/ ceiling/ floor 15 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete
7.	Providing and fixing on walls/ceiling/floor 20 mm dia. CPVC pipe with

	necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete
8.	Providing and fixing on walls/ ceiling/ floor 25 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete.
9.	Providing and fixing on walls/ ceiling/ floor 32 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete
10.	Providing and fixing on walls/ ceiling/ floor 40 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete
11.	Providing and laying in trenches 50mm dia. heavy grade having embossed as ISI Mark galvanized iron pipes of 6.33 Kilogram/meter necessary fitting remaking good the demolished portion with filling trenches and with primer of anti-corrosive oil paint,2coats complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete.
12.	Supplying and erecting ISI mark GI pipe 65 mm. dia 'C' class position with accessories complete. Specification no. CW-PLB/GP.
13.	Providing and laying in trenches 100mm dia. heavy grade having embossed as ISI Mark galvanized iron pipes of 14.74 Kilogram/meter necessary fitting remaking good the demolished portion with filling trenches and with primer of anti-corrosive oil paint,2coats complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete
14.	Supplying & erecting cast iron double flange butterfly valve of size 65 mm dia. complete as per specification no. FF-VL/BFV
15.	Supplying & erecting cast iron double flange butterfly valve of size 75/80 mm dia. complete as per specification no. FF-VL/BFV
16.	Supplying and erecting CI flange 25 mm dia with necessary nuts bolts and washers with rubber seal in position complete.
17.	Supplying and erecting CI flange 50 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.

18.	Supplying and erecting CI flange 65 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
19.	Supplying and erecting CI flange 75/80 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
20.	Supplying and erecting CI flange 100 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
21.	Providing and fixing C.P. Angular stop cock with wall flange of approved make continental including necessary sockets /union nut etc. complete.
22.	Providing and fixing C.P. Two-way BIB cock of approved make continental including necessary sockets/union nut etc. complete.
23.	Providing and fixing C.P. BIB cock with wall flange of approved make including necessary sockets/ union nut etc. complete.
24.	Supplying installation testing and commissioning of Hubless centrifugally cast-iron pipes as per ISI 15905 equivalent to NECO inclusive of all necessary special fitting like bends, tees, and caps, 'Y' bends for cleaning offsets, junctions, cowls, etc. lad under floor/fixed on walls and in pipe shafts etc. complete. The quoted rate should include lead and lifting charges to all levels, necessary scaffolding and supports like hanger supports grip bolts and CV sheet 304 grade clamps etc. with prior approval of concerned Superintending Engineer.
25.	c) 75 mm dia
26.	d) 100 mm dia
27.	Providing and fixing P.V.C. Rain water pipes of 110mm outer diameter and having wall thickness of 2.2 to 2.7 mm confirming to I.S. 135921992 including proper rainwater receiving recess with P.V.C. plug, bend, necessary fittings, such as, offsets, shoes, including fixing the pipe on wall using approved wooden cleats projecting 25mm to 40mm from face of wall a fixing with clips of approved quality and number, filing the joint using rubber gasket with solvent cement and properly resting the shoe of pipes on C.C. or masonry blocks, including necessary scaffolding and maintenance for 3 yrs. for any leakages or dislocations of pipes. All the P.V.C. fittings and additional 2-piece socket clips shall be got approved from engineer in charge etc. complete.
28.	Providing and fixing 8 cm rigid PVC Nahani trap including PVC grating, bend, connecting piece of UPVC pipe up to the outside face of wall, making the good damaged surface and testing etc. complete (Prior approval of sample and brand by Ex. Engr. is necessary before use)
29.	Providing and fixing white vitreous China pedestal type water closet (European

	type W.C. pan) with seat and lid, 10 litre low level white P.V.C. flushing cistern, including flush pipe, with manually controlled device (handle lever), conforming to IS: 7231, with all fittings and fixtures complete, including cutting and making good the walls and floors wherever required:
30.	Providing and fixing in position UPVC ultra violet stabilized 110 mm outer diameter cowl dome confirming to I.S.-4985 including making joints with solvent cement etc. complete
31.	Providing and fixing 45cm wide mild steel ladder of 40mm x6mm mild steel flat stringers and steps of 18mm dia. Mild steel bar for water tank including fixing it in (1:2:4) M-15 cement concrete block 60cm x30cm x 30cm and painting the ladder complete. (Anti-Corrosive Paint)
32.	Providing and fixing Ductile iron Man Hole Covers/ Strom Water Grating and Grating with frame of Various sizes, weight and types and load bearing capacity as per EN-124, equivalent NECO with prior approval of concerned Superintending Engineer.
33.	C.I. Grating 150x150mm (At Terrace For rain water)
34.	C.I. Grating 100x100mm (At Terrace For rain water)
35.	Providing and fixing gun metal gate valve with C.I. wheel of approved quality (screwed end)
36.	1) 20MM DIA
37.	2) 25MM DIA
38.	3) 32MM DIA
39.	4) 40MM DIA
40.	Supplying & erecting cast iron double flange butterfly valve of size 50 mm dia. complete as per specification no. FFVL/BFV
41.	Supplying & erecting cast iron double flange butterfly valve of size 65 mm dia. complete as per specification no. FFVL/BFV
42.	Supplying & erecting cast iron double flange butterfly valve of size 75/80 mm dia. complete as per specification no. FFVL/BFV
43.	Providing, laying and fixing, jointing Eco- drain 110 mm SN 8 NuDrain UPVC pipes or of equivalent make, manufacture as per EN 13476 or equivalent as per I.S.15328 with fittings such a bends, tees, coupler, etc., jointing with rubber lubricant including necessary excavation, trench refilling with selective excavated material etc. complete.
44.	Providing, laying and fixing, jointing Eco- drain 160 mm SN 4 NuDrain UPVC pipes or of equivalent make, manufacture as per EN 13476 or equivalent as per I.S.15328 with fittings such a bends, tees, coupler, etc., jointing with rubber

	lubricant including necessary excavation, trench refilling with selective excavated material etc. complete.
45.	Providing and constructing Brick Masonry Inspection Chamber 60cm x 45cm With R.C.C. raft in B.C. soil area including cost of reinforcement, 1:2:4 cement concrete channels half round G.S.W. pipes, Brick Masonry, plastering from inside and airtight 75 mm thick R.C.C. cover medium duty 100Kilogrametc. complete
46.	Providing and constructing Brick Masonry Inspection Chamber 90cmx45cm With R.C.C. raft in B.C. soil area including cost of reinforcement,1:2:4 cement concrete channels half round G.S.W. pipes, Brick Masonry, plastering from inside and airtight with 75 mm thick RCC cover medium duty 140Kilogrametc. complete
47.	Providing and fixing 15cm x 10cm salt glazed stoneware gully trap in cement concrete 1:4:8 outside the building including cast iron grating in the sink, connecting glazed stoneware pipe, brick masonry chamber with cast iron lid and cast-iron grating for the gully trap.
48.	Strainer for 40mm
49.	Strainer for 50mm
50.	Strainer for 65mm
51.	Strainer for 80mm
52.	Supplying & erecting Float sensor switch for water level controller working on 250V & rated current 8A, with 2 mtrs. wire & High quality of wire & plastic body.
53.	Providing and fixing C.P. sink cock with raised J" shaped swinging casted spout of approved make including necessary sockets/ union nut etc. complete."
54.	Supplying and erecting non return valve 40 mm dia in position made of gun metal complete.
55.	Supplying and erecting non return valve 50 mm dia in position made of gun metal complete.
56.	Supplying and erecting non return valve 65 mm dia in position made of gun metal complete.
57.	Supplying and erecting non return valve 75/80 mm dia in position made of gun metal complete.
58.	Brass Mosquito Proof Grating Inverted J Type 100 MM Dia CI Cowl Ventilators
59.	Providing and fixing 50mm dia meter water meter with non-return valve including strainer, sockets/union nut and including water meter box making locking arrangement and lock. [without chamber]

60.	PRV For 15mm dia
61.	PRV For 20mm dia
Amenity Building Internal Plumbing	
1.	Providing and fixing White glazed with bottle trap earthenware Wash Hand Basin of 55x40 cm size including cold water pillar taps, brackets, rubber plugs and brass chain, stop tap, chromium plate bottle trap and necessary pipe connections including UPAC waste pipe and trap up to the outside face of the wall, making good the damaged surface, testing etc. complete.
2.	Providing, laying and fixing P.V.C. pipe of 40mm dia. with fittings such as bends, tees, reducers, clamps, etc. including necessary excavation, trench filling etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowki or as directed etc. complete
3.	Providing, laying and fixing P.V.C. pipe of 50mm dia. with fittings such as bends, tees, reducers, clamps, etc. including necessary excavation, trench filling etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete.
4.	Providing and fixing 75 mm dia stabilizer pipe/ P.V.C. soil vent/waste pipe and with necessary fixtures and fitting such as bends, tees, single junctions, slotted vent, clamps etc. complete
5.	Providing and fixing 100 mm dia stabilizer pipe/ P.V.C. soil vent/waste pipe and with necessary fixtures and fitting such as bends, tees, single junctions, slotted vent, clamps etc. complete.
6.	Providing and fixing on walls/ ceiling/ floor 15 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete
7.	Providing and fixing on walls/ceiling/floor 20 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete
8.	Providing and fixing on walls/ ceiling/ floor 25 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete.
9.	Providing and fixing on walls/ ceiling/ floor 32 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking

	the same in PWD chowky or as directed etc. complete
10.	Providing and fixing on walls/ ceiling/ floor 40 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete
11.	Providing and laying in trenches 50mm dia. heavy grade having embossed as ISI Mark galvanized iron pipes of 6.33 Kilogram/meter necessary fitting remaking good the demolished portion with filling trenches and with primer of anti-corrosive oil paint,2coats complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete.
12.	Supplying and erecting ISI mark GI pipe 65 mm. dia 'C' class position with accessories complete. Specification no. CW-PLB/GP.
13.	Providing and laying in trenches 100mm dia. heavy grade having embossed as ISI Mark galvanized iron pipes of 14.74 Kilogram/meter necessary fitting remaking good the demolished portion with filling trenches and with primer of anti-corrosive oil paint,2coats complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete
14.	Supplying & erecting cast iron double flange butterfly valve of size 65 mm dia. complete as per specification no. FF-VL/BFV
15.	Supplying & erecting cast iron double flange butterfly valve of size 75/80 mm dia. complete as per specification no. FF-VL/BFV
16.	Supplying and erecting CI flange 25 mm dia with necessary nuts bolts and washers with rubber seal in position complete.
17.	Supplying and erecting CI flange 50 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
18.	Supplying and erecting CI flange 65 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
19.	Supplying and erecting CI flange 75/80 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
20.	Supplying and erecting CI flange 100 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
21.	Providing and fixing C.P. Angular stop clock with wall flange of approved make continental including necessary sockets /union nut etc. complete.
22.	Providing and fixing C.P. Two-way BIB cock of approved make continental including necessary sockets/union nut etc. complete.

23.	Providing and fixing C.P. BIB cock with wall flange of approved make including necessary sockets/ union nut etc. complete.
24.	Supplying installation testing and commissioning of Hubless centrifugally cast-iron pipes as per ISI 15905 equivalent to NECO inclusive of all necessary special fitting like bends, tees, and caps, 'Y' bends for cleaning offsets, junctions, cowls, etc. lad under floor/fixed on walls and in pipe shafts etc. complete. The quoted rate should include lead and lifting charges to all levels, necessary scaffolding and supports like hanger supports grip bolts and CV sheet 304 grade clamps etc. with prior approval of concerned Superintending Engineer.
25.	c) 75 mm dia
26.	d) 100 mm dia
27.	Providing and fixing P.V.C. Rain water pipes of 110mm outer diameter and having wall thickness of 2.2 to 2.7 mm confirming to I.S. 135921992 including proper rainwater receiving recess with P.V.C. plug, bend, necessary fittings, such as, offsets, shoes, including fixing the pipe on wall using approved wooden cleats projecting 25mm to 40mm from face of wall a fixing with clips of approved quality and number, filing the joint using rubber gasket with solvent cement and properly resting the shoe of pipes on C.C. or masonry blocks, including necessary scaffolding and maintenance for 3 yrs. for any leakages or dislocations of pipes. All the P.V.C. fittings and additional 2-piece socket clips shall be got approved from engineer in charge etc. complete.
28.	Providing and fixing 8 cm rigid PVC Nahani trap including PVC grating, bend, connecting piece of UPVC pipe up to the outside face of wall, making the good damaged surface and testing etc. complete (Prior approval of sample and brand by Ex. Engr. is necessary before use)
29.	Providing and fixing white vitreous China pedestal type water closet (European type W.C. pan) with seat and lid, 10 litre low level white P.V.C. flushing cistern, including flush pipe, with manually controlled device (handle lever), conforming to IS: 7231, with all fittings and fixtures complete, including cutting and making good the walls and floors wherever required:
30.	Providing and fixing in position UPVC ultra violet stabilized 110 mm outer diameter cowl dome confirming to I.S.-4985 including making joints with solvent cement etc. complete
31.	Providing and fixing 45cm wide mild steel ladder of 40mm x6mm mild steel flat stringers and steps of 18mm dia. Mild steel bar for water tank including fixing it in (1:2:4) M-15 cement concrete block 60cm x30cm x 30cm and painting the

	ladder complete. (Anti-Corrosive Paint)
32.	Providing and fixing Ductile iron Man Hole Covers/ Storm Water Grating and Grating with frame of Various sizes, weight and types and load bearing capacity as per EN-124, equivalent NECO with prior approval of concerned Superintending Engineer.
33.	C.I. Grating 150x150mm (At Terrace For rain water)
34.	C.I. Grating 100x100mm (At Terrace For rain water)
35.	Providing and fixing gun metal gate valve with C.I. wheel of approved quality (screwed end)
36.	1) 20MM DIA
37.	2) 25MM DIA
38.	3) 32MM DIA
39.	4) 40MM DIA
40.	Supplying & erecting cast iron double flange butterfly valve of size 50 mm dia. complete as per specification no. FFVL/BFV
41.	Supplying & erecting cast iron double flange butterfly valve of size 65 mm dia. complete as per specification no. FFVL/BFV
42.	Supplying & erecting cast iron double flange butterfly valve of size 75/80 mm dia. complete as per specification no. FFVL/BFV
43.	Providing, laying and fixing, jointing Eco- drain 110 mm SN 8 NuDrain UPVC pipes or of equivalent make, manufacture as per EN 13476 or equivalent as per I.S.15328 with fittings such a bends, tees, coupler, etc., jointing with rubber lubricant including necessary excavation, trench refilling with selective excavated material etc. complete.
44.	Providing, laying and fixing, jointing Eco- drain 160 mm SN 4 NuDrain UPVC pipes or of equivalent make, manufacture as per EN 13476 or equivalent as per I.S.15328 with fittings such a bends, tees, coupler, etc., jointing with rubber lubricant including necessary excavation, trench refilling with selective excavated material etc. complete.
45.	Providing and constructing Brick Masonry Inspection Chamber 60cm x 45cm With R.C.C. raft in B.C. soil area including cost of reinforcement, 1:2:4 cement concrete channels half round G.S.W. pipes, Brick Masonry, plastering from inside and airtight 75 mm thick R.C.C. cover medium duty 100Kilogrametc. complete
46.	Providing and constructing Brick Masonry Inspection Chamber 90cmx45cm With R.C.C. raft in B.C. soil area including cost of reinforcement,1:2:4 cement concrete channels half round G.S.W. pipes, Brick Masonry, plastering from

	inside and airtight with 75 mm thick RCC cover medium duty 140Kilogrametc. complete
47.	Providing and fixing 15cm x 10cm salt glazed stoneware gully trap in cement concrete 1:4:8 outside the building including cast iron grating in the sink, connecting glazed stoneware pipe, brick masonry chamber with cast iron lid and cast-iron grating for the gully trap.
48.	Strainer for 40mm
49.	Strainer for 50mm
50.	Strainer for 65mm
51.	Strainer for 80mm
52.	Supplying & erecting Float sensor switch for water level controller working on 250V & rated current 8A, with 2 mtrs. wire & High quality of wire & plastic body.
53.	Providing and fixing C.P. sink cock with raised J" shaped swinging casted spout of approved make including necessary sockets/ union nut etc. complete."
54.	Supplying and erecting non return valve 40 mm dia in position made of gun metal complete.
55.	Supplying and erecting non return valve 50 mm dia in position made of gun metal complete.
56.	Supplying and erecting non return valve 65 mm dia in position made of gun metal complete.
57.	Supplying and erecting non return valve 75/80 mm dia in position made of gun metal complete.
58.	Brass Mosquito Proof Grating Inverted J Type 100 MM Dia CI Cowl Ventilators
59.	Providing and fixing 50mm dia meter water meter with non-return valve including strainer, sockets/union nut and including water meter box making locking arrangement and lock. [without chamber]
60.	PRV For 15mm dia
61.	PRV For 20mm dia

IV. Building Plumbing & Sanitation and External works

Sr. No.	Description of Items
1.	Providing, laying and fixing, jointing Eco- drain 160 mm SN 4 NuDrain UPVC pipes or of equivalent make, manufacture as per EN 13476 or equivalent as per I.S.15328 with fittings such a bends, tees, coupler, etc, jointing with rubber

	lubricant including necessary excavation, trench refilling with selective excavated material etc. complete.
2.	Providing, laying and fixing, jointing Eco- drain 200 mm SN 4 NuDrain UPVC pipes or of equivalent make, manufacture as per EN 13476 or equivalent as per I.S.15328 with fittings such a bends, tees, coupler etc, jointing with rubber lubricant including necessary excavation, trench refilling with selective excavated material etc. complete.
3.	Providing, laying and fixing, jointing Eco- drain 250 mm SN 4 NuDrain UPVC pipes or of equivalent make, manufacture as per EN 13476 or equivalent as per I.S.15328 with fittings such a bends, tees, coupler etc, jointing with rubber lubricant including necessary excavation, trench refilling with selective excavated material etc. complete.
4.	Providing, laying and fixing, jointing Eco- drain 315 mm SN 4 NuDrain UPVC pipes or of equivalent make, manufacture as per EN 13476 or equivalent as per I.S.15328 with fittings such a bends, tees, coupler etc, jointing with rubber lubricant including necessary excavation, trench refilling with selective excavated material etc. complete.
5.	Providing and constructing Brick Masonry Inspection Chamber 90cmx45cm With R.C.C. raft in B.C. soil area including cost of reinforcement,1:2:4 cement concrete channels half round G.S.W. pipes, Brick Masonry, plastering from inside and airtight with 75 mm thick R.C.C. cover medium duty 140Kilogrametc. complete.
6.	Providing, constructing and fixing 600 mm dia. Inspection chamber made up of Polypropylene / polyethylene material Having 315 mm diameter inlet(s)/outlets and with eccentric reducers as per requirement to connect 160 mm and 200 mm pipes having thickness of 5 mm and 600 mm in height, with in built shaft achieve the invert depth as per site condition of wall thickness of 5 mm having R.C.C. cover and frame fixing in 150 mm thick in c.c.1: 2:4, having crushed sand bed of 150 mm thick of size 750 mm dia. Including excavation and refilling the sides of chamber by sand crushed cushioning 150 mm thickness, connecting all required UPVC fittings with rubber lubricant etc. complete. As Directed by Engineer In charge
7.	Providing and fixing Ductile iron Man Hole Covers/ Strom Water Grating and Grating with frame of Various sizes, weight and types and load bearing capacity as per EN-124, equivalent NECO with prior approval of concerned

	Superintending Engineer.
8.	Providing and fixing in position sewer trap in last chamber.
9.	Supplying & erecting Float sensor switch for water level controller working on 250V & rated current 8A, with 2 mtrs. wire & High quality of wire & plastic body.
10.	Providing and laying in trenches 40mm dia. heavy grade having embossed as ISI Mark galvanised iron pipes of 4.49 Kilogram/metre necessary fitting remaking good the demolished portion with filling trenches and with primer of anti-corrosive oil paint,2coats complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowki or as directed etc. complete.
11.	Providing and laying in trenches 50mm dia. heavy grade having embossed as ISI Mark galvanised iron pipes of 6.33 Kilogram/metre necessary fitting remaking good the demolished portion with filling trenches and with primer of anti-corrosive oil paint,2coats complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowki or as directed etc. complete.
12.	Supplying and erecting ISI mark GI pipe 65 mm. dia 'C' class at position with accessories complete. Specification no. CW-PLB/GP.
13.	Providing and laying in trenches 80mm dia. heavy grade having embossed as ISI Mark galvanised iron pipes of 10.15 Kilogram/metre necessary fitting remaking good the demolished portion with filling trenches and with primer of anti-corrosive oil paint,2coats complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowki or as directed etc. complete.
14.	Providing and laying in trenches 100mm dia. heavy grade having embossed as ISI Mark galvanised iron pipes of 14.74 Kilogram/metre necessary fitting remaking good the demolished portion with filling trenches and with primer of anti-corrosive oil paint,2coats complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowki or as directed etc. complete
15.	Supplying and erecting G.I. pipe above ground of 'C' class ERW of size 150 mm dia with necessary fittings complete as per specification no. FF-PP

16.	Supplying and erecting G.I. pipe above ground of 'C' class ERW of size 200 mm dia with necessary fittings complete as per specification no. FF-PP
17.	Supplying & erecting cast iron double flange butterfly valve of size 50 mm dia. complete as per specification no. FF-VL/BFV
18.	Supplying & erecting cast iron double flange butterfly valve of size 65 mm dia. complete as per specification no. FF-VL/BFV
19.	Supplying & erecting cast iron double flange butterfly valve of size 80 mm dia. complete as per specification no. FF-VL/BFV
20.	Supplying & erecting cast iron double flange butterfly valve of size 100 mm dia. complete as per specification no. FF-VL/BFV
21.	Supplying & erecting cast iron double flange butterfly valve of size 150 mm dia. complete as per specification no. FF-VL/BFV
22.	Supplying & erecting cast iron double flange butterfly valve of size 200 mm dia. Complete as per specification no. FF-VL/BFV
23.	Supplying and erecting CI flange 25 mm dia with necessary nuts bolts and washers with rubber seal in position complete.
24.	Supplying and erecting CI flange 100 mm dia with necessary nuts bolts and washers with rubber seal in position complete.
25.	Supplying and erecting CI flange 150 mm dia with necessary nuts bolts and washers with rubber seal in position complete.
26.	Supplying and erecting CI flange 50 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
27.	Supplying and erecting CI flange 65 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
28.	Supplying and erecting CI flange 75/80 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
29.	Supplying and erecting CI flange 200 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
30.	Providing and fixing 50mm dia meter water meter with non-return valve including strainer, sockets/union nut and including water meter box making locking arrangement sand lock. [Without chamber].
31.	Providing and fixing 45 cm wide mild steel ladder of 40mm x 6 mm mild steel

	flat stringers and steps of 18 mm dia. mild steel bar for water tank including fixing it in (1:2:4) M-15 cement concrete block 60cm x 30cm x 30cm and painting the ladder complete. (Anti-Corrosive Paint)
32.	Providing and fixing White glazed with bottle trap earthenware Wash Hand Basin of 55x40 cm size including cold water pillar taps, brackets, rubber plugs and brass chain, stop tap, chromium plate bottle trap and necessary pipe connections including UPAC waste pipe and trap up to the outside face of the wall, making good the damaged surface, testing etc. complete.
33.	Providing, laying and fixing P.V.C. pipe of 40mm dia. with fittings such as bends, tees, reducers, clamps, etc. including necessary excavation, trench filling etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowki or as directed etc. complete.
34.	Providing, laying and fixing P.V.C. pipe of 50mm dia. with fittings such as bends, tees, reducers, clamps, etc. including necessary excavation, trench filling etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowki or as directed etc. complete.
35.	Providing and fixing 75 mm dia stabiliser pipe/ P.V.C. soil vent/waste pipe and with necessary fixtures and fitting such as bends, tees, single junctions, slotted vent, clamps etc. complete
36.	Providing and fixing 100 mm dia stabiliser pipe/ P.V.C. soil vent/waste pipe and with necessary fixtures and fitting such as bends, tees, single junctions, slotted vent, clamps etc. complete.
37.	Providing and fixing on walls/ ceiling/ floor 15 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowki or as directed etc. complete.
38.	Providing and fixing on walls/ceiling/floor 20 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowki or as directed etc. complete
39.	Providing and fixing on walls/ ceiling/ floor 25 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowki or as directed etc. complete.
40.	Providing and fixing on walls/ ceiling/ floor 32 mm dia. CPVC pipe with

	necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowki or as directed etc. complete.
41.	Providing and fixing on walls/ ceiling/ floor 40 mm dia. CPVC pipe with necessary fittings, remaking good the demolished portion etc. complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowki or as directed etc. complete
42.	Providing and laying in trenches 50mm dia. heavy grade having embossed as ISI Mark galvanised iron pipes of 6.33 Kilogram/metre necessary fitting remaking good the demolished portion with filling trenches and with primer of anti-corrosive oil paint,2coats complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete.
43.	Supplying and erecting ISI mark GI pipe 65 mm. dia 'C' class position with accessories complete. Specification no. CW-PLB/GP.
44.	Providing and laying in trenches 80mm dia. heavy grade having embossed as ISI Mark galvanised iron pipes of 10.15 Kilogram/metre necessary fitting remaking good the demolished portion with filling trenches and with primer of anti-corrosive oil paint,2coats complete. Including removing existing pipe line if necessary and conveying and stacking the same in PWD chowky or as directed etc. complete
45.	Supplying and erecting CI flange 25 mm dia with necessary nuts bolts and washers with rubber seal in position complete.
46.	Supplying and erecting CI flange 50 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
47.	Supplying and erecting CI flange 65 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
48.	Supplying and erecting CI flange 75/80 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
49.	Supplying and erecting CI flange 100 mm. dia with necessary nuts bolts and washers with rubber seal in position complete.
50.	Providing and fixing C.P. Angular stop clock with wall flange of approved make continental including necessary sockets /union nut etc. complete.
51.	Providing and fixing C.P. Two-way BIB cock of approved make continental

	including necessary sockets/union nut etc. complete.
52.	Providing and fixing C.P. BIB cock with wall flange of approved make including necessary sockets/ union nut etc. complete.
53.	Supplying installation testing and commissioning of Hubless centrifugally cast-iron pipes as per ISI 15905 equivalent to NECO inclusive of all necessary special fitting like bends, tees, and caps, 'Y' bends for cleaning offsets, junctions, cowls, etc. laid under floor/fixed on walls and in pipe shafts etc. complete. The quoted rate should include lead and lifting charges to all levels, necessary scaffolding and supports like hanger supports grip bolts and CV sheet 304 grade clamps etc. with prior approval of concerned Superintending Engineer.
54.	c) 75 mm dia d) 100 mm dia
55.	Providing and fixing P.V.C. Rain water pipes of 110mm outer diameter and having wall thickness of 2.2 to 2.7 mm confirming to I.S. 13592:1992 including proper rainwater receiving recess with P.V.C. plug, bend, necessary fittings, such as, offsets, shoes, including fixing the pipe on wall using approved wooden cleats projecting 25mm to 40mm from face of wall a fixing with clips of approved quality and number, filling the joint using rubber gasket with solvent cement and properly resting the shoe of pipes on C.C. or masonry blocks, including necessary scaffolding and maintenance for 3 yrs. for any leakages or dislocations of pipes. All the P.V.C. fittings and additional 2-piece socket clips shall be got approved from engineer in charge etc. complete.
56.	Providing and fixing 8 cm rigid PVC Nahani trap including PVC grating, bend, connecting piece of UPVC pipe up to the outside face of wall, making the good damaged surface and testing etc. complete (Prior approval of sample and brand by Ex. Engr. is necessary before use)
57.	Providing and fixing white vitreous China pedestal type water closet (European type W.C. pan) with seat and lid, 10 litre low level white P.V.C. flushing cistern, including flush pipe, with manually controlled device (handle lever), conforming to IS: 7231, with all fittings and fixtures complete, including cutting and making good the walls and floors wherever required:
58.	Providing and fixing in position UPVC ultra violet stabilized 110 mm outer diameter cowl dome confirming to I.S.-4985 including making joints with solvent cement etc. complete

59.	Providing and fixing 45cm wide mild steel ladder of 40mm x6mm mild steel flat stringers and steps of 18mm dia. Mild steel bar for water tank including fixing it in (1:2:4) M-15 cement concrete block 60cm x30cm x 30cm and painting the ladder complete. (Anticorrosive Paint)
60.	C.I. Grating 150x150mm (At Terrace For rain water)
61.	C.I. Grating 100x100mm (At Terrace For rain water)
62.	Providing and fixing gun metal gate valve with C.I. wheel of approved quality (screwed end)
63.	1) 20MM DIA 2) 25MM DIA 3) 32MM DIA 4) 40MM DIA
64.	Supplying & erecting cast iron double flange butterfly valve of size 50 mm dia. complete as per specification no. FFVL/BFV
65.	Supplying & erecting cast iron double flange butterfly valve of size 65 mm dia. complete as per specification no. FFVL/BFV
66.	Providing, laying and fixing, jointing Eco- drain 110 mm SN 8 NuDrain UPVC pipes or of equivalent make, manufacture as per EN 13476 or equivalent as per I.S.15328 with fittings such a bends, tees, coupler, etc, jointing with rubber lubricant including necessary excavation, trench refilling with selective excavated material etc. complete.
67.	Providing and constructing Brick Masonry Inspection Chamber 60cm x45cm With R.C.C. raft in B.C. soil area including cost of reinforcement,1:2:4 cement concrete channels half round G.S.W. pipes, Brick Masonry, plastering from inside and airtight 75mm thick R.C.C. cover medium duty 100 Kilogram etc. complete.
68.	Providing and constructing Brick Masonry Inspection Chamber 90cm x45cm With R.C.C. raft in B.C. soil area including cost of reinforcement,1:2:4 cement concrete channels half round G.S.W. pipes, Brick Masonry, plastering from inside and airtight 75mm thick R.C.C. cover medium duty 140 Kilogram etc. complete.
69.	Providing and fixing 15cm x 10cm salt glazed stoneware gully trap in cement concrete 1:4:8 outside the building including cast iron grating in the sink, connecting glazed stoneware pipe, brick masonry chamber with cast iron lid

	and cast-iron grating for the gully trap.
70.	Strainer for 40mm
71.	Strainer for 50mm
72.	Strainer for 65mm
73.	Strainer for 80mm
74.	Supplying & erecting Float sensor switch for water level controller working on 250V & rated current 8A, with 2 mtrs. wire & High quality of wire & plastic body.
75.	Providing and fixing C.P. sink cock with raised J" shaped swinging casted spout of approved make including necessary sockets/ union nut etc. complete."
76.	Supplying and erecting non return valve 40 mm dia in position made of gun metal complete.
77.	Supplying and erecting non return valve 50 mm dia in position made of gun metal complete.
78.	Supplying and erecting non return valve 65 mm dia in position made of gun metal complete.
79.	Supplying and erecting non return valve 75/80 mm dia in position made of gun metal complete.
80.	Brass Mosquito Proof Grating Inverted J Type 100 MM Dia CI Cowl Ventilators
81.	Supplying & erecting cast iron double flange butterfly valve of size 75/80 mm dia. complete as per specification no. FFVL/BFV
82.	Strainer for 80mm
83.	Supplying & erecting cast iron double flange butterfly valve of size 65 mm dia. complete as per specification no. FFVL/BFV
84.	Supplying, Fixing & Testing Pressure reducing Valve approved make with strainer and cleaning assembly and with necessary connectors.
85.	PRV for 15mm Dia
86.	PRV for 20mm Dia

V. Internal Electrical works

Sr. No.	Description
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Sr. No.	Description
1.	Supplying and erecting 25 mm dia. corrugated flexible polypropylene conduit
2.	Supplying and erecting PVC trunking (PVC casing-n capping) of size 20 mm with accessories on wall/ceiling as per specification No: WG-MA/CON.
3.	Supplying and laying HMS PVC conduit FRLS grade 20 mm dia. With necessary accessories in RCC work/false ceiling/false flooring as per specification No. WG-MA/CC
4.	Supplying and erecting mains with 1x1.5 sq.mm FRLSH copper PVC insulated wire laid in provided conduit/trunking/inside pole/Bus bars or any other places as per specification No: WG-MA/BW
5.	Supplying and erecting mains with 2x2.5 sq.mm FRLSH copper PVC insulated wire laid in provided conduit/trunking/inside pole/Bus bars or any other places. as per specification No: WG-MA/BW Note - 3) Rates for the earth wire shall be calculated as one-half rate of mains of bunch of two wire.
6.	Supplying and erecting mains with 2x4 sq.mm FRLSH copper PVC insulated wire laid in provided conduit/trunking/inside pole/Bus bars or any other places. as per specification No: WG-MA/BW
7.	Supplying and erecting flush, shock proof plug socket 6 pin 6A and 16A with switch 16A combined unit erected in recess in 16-gauge M.S sheet box duly painted and connected with earth terminal and leads
8.	Supplying and erecting ISI mark modular type T.V. Socket single outlet, duly erected on provided plate and box with wiring connections complete.
9.	Point wiring in for light/fan/bell in 20 mm metal conduit with 1.5 sq.mm (2+1E) FRLSH grade copper wire, flush type switch, earthing and required accessories as per specification No: WG-PW/SW
10.	Point wiring in PVC trunking (casing-capping) with 1.5 sq.mm (2+1E) FRLSH grade copper wire, flush type switch, earthing and required accessories as per specification No: WG-PW/SW
11.	Point wiring for independent plug in PVC trunking (casing-capping) with 1.5 sq.mm FRLSH grade copper wire, flush type switch, earthing and required accessories as per specification No: WG-PW/SW.
12.	Secondary point wiring for additional light/ bell point in PVC trunking (casing-capping) with 1.5 sq.mm (2+1E) FRLSH grade copper wire with required accessories as per specification No: WG-PW/SW.
13.	Wiring for plug on board with flush Switch socket, copper wiring and earthing as per specification No: WG-PW/SW

Sr. No.	Description
14.	Supplying and erecting bulk head LED fitting max. 10W with high transitivity diffuser with system lumens output of min. 1100 lumens, min. efficacy of 110 lumen/W, CRI>80, CCT up to 6500K, Beam Angle of 120 deg., THD0.95, operating range of 140-260V, in built surge protection of 2.5 kV, Life class of 50,000 Hrs. at L70B50, including driver, IP66, IK09 rated on provided PVC Block / wooden board with 3 years warranty.
15.	Supplying and erecting anodized aluminium corridor / passage light LED fitting (4 feet) Max. 22W with high transitivity diffuser with system lumens output of Min.2200 lumens, min. efficacy of 100 lumen/W, CRI>80, CCT up to 6000K, Beam Angle of 110 deg., Ripple0.95, operating range of 200-270V, surge protection of 2 kV, Life class of 50,000 Hrs. at L70B50, including driver, with end caps on provided PVC Block / wooden board with 3 years warranty.
16.	Supplying & erecting Aviation obstruction light consisting of yellow painted diecast aluminium alloy housing with Integral LED aviation light comprising of aluminium housing with polycarbonate enclosure complete.
17.	Supplying and erecting electronic type step regulator for ceiling fan 600/ 900/ 1200/ 1400 mm. sweep, complete erected on provided switch board.
18.	Supplying, installing, testing and commissioning of 3 kWp ON-GRID Solar PV System / Plant consisting solar PV poly crystalline panel (335 Wp X 09 Nos.) and Solar ON-GRID Inverter of similar capacity (THD<3%) with interconnecting wires / cables up to solar ON GRID inverter, proper base plate, canopy box to house Inverter and necessary accessories i.e. ACDB/ DCDB , MCB etc. wiring up to solar Inverter in casing and capping, display board, sign board, two distinct earthing, lightning arrester, Liasoning charges, remote monitoring system with maintaining card / chip for 5 years including net charges for RMS system, complete up to net meter installation and export to the grid with 5-year ON-SITE warranty and fully comprehensive maintenance contract as per specification no. ESD-RT-ONG.
19.	Supplying, installing, testing and commissioning of 5 kWp ON-GRID Solar PV System / Plant consisting Solar PV Poly Crystalline Panel (335 Wp X 15 Nos.) and Solar ON-GRID Inverter of similar capacity (THD<3%) with interconnecting wires / cables up to solar ON GRID inverter, proper base plate, canopy box to house Inverter and necessary accessories i.e. ACDB/ DCDB , MCB etc. wiring up to solar Inverter in casing and capping, display board, sign board, two distinct earthing, lightning arrester, Liasoning charges, remote monitoring

Sr. No.	Description
	system with maintaining card/chip for 5 years including net charges for RMS system, complete up to net meter installation and export to the grid with 5-year ON-SITE warranty and fully comprehensive maintenance contract as per Specification no. ESD-RT-ONG
20.	Supplying, installing, testing and commissioning of 10 kWp ON-GRID Solar PV System / Plant consisting Solar PV Poly Crystalline Panel (335 Wp X 30 Nos.) and Solar ON-GRID Inverter of similar capacity (THD<3%) with interconnecting wires / cables up to solar ON GRID inverter, proper base plate, canopy box to house Inverter and necessary accessories i.e. ACDB/ DCDB , MCB etc. wiring up to solar Inverter in casing and capping, display board, sign board, two distinct earthing, lightning arrester, Liasoning charges, remote monitoring system with maintaining card/chip for 5 years including net charges for RMS System, complete up to net meter installation and export to the grid with 5-year ON-SITE warranty and fully comprehensive maintenance contract as per Specification no. ESD-RT-ONG.
21.	Supplying and erecting ON LOAD four pole automatic transfer switch, 415V, 100-125A, 50 Hz with enclosure, flexibility of neutral connection, line and load reversibility, changeover with phase barriers, source separator, terminal covers along with staggered termination for cable connection and ATS controller with provision of time delay protection of OV, UV, phase sequence changing, single phasing and event logging, connected with provided leads complete complying to IS/IEC 60947-3 Part-1/6.
22.	Supplying, erecting & marking SPMCB 6A to 32A, C-series with rated short – circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB.
23.	Supplying, erecting & marking SPMCB 6A to 32A, B-series with rated short – circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB.
24.	Supplying, erecting & marking DPMCB 6A to 32A, C-series with rated short - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB
25.	Supplying, erecting & marking DPMCB 6 A to 32 A, B-series with rated short – circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB.
26.	Supplying, erecting & marking TPMCB 6A to 32A, with rated short – circuit

Sr. No.	Description
	breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB.
27.	Supplying, erecting and marking double pole isolator only switch version of miniature circuit breaker of 40A with required wiring connections and lugs etc. in provided distribution board.
28.	Supplying, erecting and marking triple pole isolators only switch version of miniature circuit breaker of 100A with required wiring connections and lugs etc. in provided distribution board.
29.	Supplying and erecting single pole and neutral distribution board (SPNDB), with 2 ways for incoming and 6 ways (6 poles) for outgoing SP MCBs, with door, 1.2mm thickness surface / flush mounted, IP 43 Protection on iron / GI frame as per specification no. SW-SWR/MCBDB.
30.	Supplying and erecting single pole and neutral distribution board (SPNDB), with 2 ways for incoming and 10 ways (10 poles) for outgoing SP MCBs, with door, 1.2mm thickness surface / flush mounted, IP 43 Protection on iron / GI frame as per specification no. SW-SWR/MCBDB.
31.	Supplying & erecting CRCA sheet metal one-way enclosures of 1.2mm thickness suitable for DP MCB /TP MCB/FP MCB/RCCB/RCBO complete erected on angle iron/GI frame.
32.	Supplying & erecting triple pole and neutral distribution board (TPNDB), 3 pole/4 pole MCCB as incomer & outgoing SP MCB (24 poles) or TP MCB of 8 ways (24 poles), with door, 1.2mm thickness, surface/flush mounted, on iron/GI frame (vertical busbar type) as per specification no. SW-SWR/MCBDB1
33.	Providing & erecting 3 Pole MCCB, 415 V, 100A, rated short-circuit breaking capacity 25 kA (Ics=100% of Icu), adjustable thermal (overload) setting and fixed magnetic setting with provided leads, provision for installation of shunt/UV/trip alarm contact. MCCB with phase barriers on both sides, insulation withstand capacity 800V, no line-load bias in provided enclosure/panel as per specification no. SW-SWR/MCCB
34.	Providing & erecting 4 Pole MCCB, 415 V, 100A, rated short-circuit breaking capacity 25 kA (Ics=100% of Icu), adjustable thermal (overload) setting and fixed magnetic setting with provided leads, provision for installation of shunt/UV/trip alarm contact. MCCB with phase barriers on both sides, insulation withstand capacity 800V, no line-load bias in provided enclosure/panel as per specification no. SW-SWR/MCCB Note: 5309 x 20%=

Sr. No.	Description
	1061.8 ie 5309 + 1061.8 = 6370.8
35.	Providing & erecting 3 Pole MCCB 415V, 400A, rated short-circuit breaking capacity 36 kA (Ics=100% of Icu), adjustable thermal (overload) setting and adjustable magnetic setting with provided leads, provision for installation of shunt/ UV/ trip alarm contact and MCCB should have phase barriers both sides, with insulation withstand capacity 800V, no line-load bias in provided enclosure /panel as per specification no. SW-SWR/MCCB
36.	Supplying, erecting & commissioning 2 pole RCCB 16/25A, electromagnetic type with 30/100/300 mA sensitivity with earth leakage trip indication complete as per specification no. SW-RCCB/RCCB.
37.	Supplying, erecting & commissioning 2 pole RCCB 40A, electromagnetic type with 30/100/300 mA sensitivity with earth leakage trip indication complete as per specification no. SW-RCCB/RCCB.
38.	Supplying, fixing and commissioning 4 pole RCBO 6/10/16/20/25A, with overcurrent, rated short-circuit breaking capacity (10 kA) and earth leakage protection, with 30/100/300 mA sensitivity and short circuit and earth leakage trip indications per specification no SW-RCCB/RCBO.
39.	Providing & erecting floor/wall mounting, MCCB panel board, with door, suitable for four pole incoming 400 A, 6 ways four pole outgoing, up to 100 A MCCBs on iron frame, as per specification no SW-SWR/MCCBPB. (Excluding MCCBs)
40.	Providing & erecting incoming and outgoing cable alleys suitable for MCCB panel board with door, 250A incoming and 8 ways outgoing or 400A incoming and 6-way outgoing made from CRCA sheet 1.6 mm thick duly powder coated and having vertical insulated partition for shielding between the shared face of cableway and DB.
41.	Supplying and erecting marine (Waterproof) plywood 18/19 mm thick fixed to wall or on provided panel board with necessary materials such as screws, wall fasteners, supports, nut bolts etc. complete with compliance to IS 720:2010
42.	Supplying and erecting GI sheet 1.6 mm (16 SWG) caution board of size 200 x 150 mm duly painted with red oxide and enamel paint in Marathi, Hindi and English erected at position for medium voltage (415 Volt) installation with compliance to IS 2551-1982
43.	Supplying and erecting bus bar chamber triple pole with neutral with four aluminium bars, 500V, 100A having 15mm (Width) X 3 mm (Thickness), with

Sr. No.	Description
	99.9 % aluminium purity as per E91E Grade & Current Density 0.8 A per sq. mm. compliance to IS 5082 and Enclosure fabricated with 16 SWG CRCA sheet and epoxy paint, with suitable number of ways and with SMC/DMC Insulators and accessories of suitable rating complete as per specification no. CPBB
44.	Supplying and erecting static energy Trivector meter 440 V 50 Hz. 20- 100A ISI mark meter AC 3 phase 4 wires LCD/LED display (Measuring and display Trivector, V, I, kW, PF, Hz, Last month history of kWh & MD in kW) with special features like TOD, Tamper logging, Optical Communication Port & Display in absence of Power complete erected on polished teak wood board duly wired with required size of weather proof PVC wires duly tested for class 1.0 accuracy by electric supply company.
45.	Supplying and erecting AC three phase LT CT multifunction net meter DLMS compliance of accuracy class 0.5, 3 x 240V, 50Hz, with backlit LCD display, optical port & RS 232 port (measures import kWh, kVAh, export kWh, kVAh, Net kWh, kVAh, V, I, kW, 6 months history of energy, load survey, TOD, tamper detection & logging, power ON/OFF events, instantaneous parameters of rating 1/5A with display in absence of power) with wiring connections and mounting hardware on provided panels complete with calibration certificate from manufacturer.
46.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 3½ core 35 sq. mm. aluminium conductor complete erected with glands & lugs, on wall/ trusses/pole or laid in provided trench/ pipe as per specification no. CB-LT/AL
47.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 3 core 2.5 sq. mm. copper conductor complete erected with glands & lugs, on wall/ trusses/ pole or laid in provided trench/ pipe as per specification no. CB-LT/CU
48.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 4 core 6 sq. mm. copper conductor complete erected with glands & lugs, on wall/ trusses/ pole or laid in provided trench/ pipe as per specification no. CB-LT/CU
49.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 4 core 16 sq. mm. copper conductor complete erected with glands & lugs, on wall/ trusses/ pole or laid in provided

Sr. No.	Description
	trench/ pipe as per specification no. CB-LT/CU
50.	Supplying and laying (including excavation of suitable width & depth up to 90 cm) 90 mm outside dia. Double wall corrugated pipes (DWC) of HDPE for enclosing cable below ground/road surface, to required depth complete.
51.	Providing earthing with galvanized iron earth plate size 60 x 60 x 0.6 cm complete with all materials, testing & recording the results as per specification no. EA-EP
52.	Providing earthing with copper earth plate size 60 x 60 x 0.315 cm with funnel with a wire mesh for watering and brick masonry block C.I. cover with minimum 25 kg Carbon based environment friendly back fill Ground Enhancing compound complete with all materials, testing & recording the results as per specification no ESE -LA
53.	Supplying and erecting GI wire of high purity of required sizes used for earthing or any other purposes on wall with necessary G. I. Clamps fixed on wall/cable/ conduit with screws complete.
54.	Supplying and erecting aluminium strip of required sizes used for earthing on wall and/or any other purpose with necessary aluminium clamps fixed on wall with insulating and non-corrosive bond with required joints. As per specification no. EA-EP.
55.	Supplying and erecting GI strip of required size used for earthing on wall and/or any other purpose with necessary GI clamps fixed on wall painted with insulating and non-corrosive bond with required joints as per specification no. EA-EP.
56.	Supplying & erecting conventional spike type air termination suitable to carry lightning stroke made up of heavy gauge 40 mm dia copper pipe of standard length with 5 Nos. copper spikes fixed on copper ball as air terminals duly threaded in copper pipe erected on provided foundation complete.
57.	Providing printed instruction chart for treating persons suffering from electric shock, printed in English & Marathi and duly laminated complete.
58.	Supplying and fixing PVC synthetic elastomer electrically insulating mat with class A insulation conforming to IS: 15652 – 2006 having 2 mm thickness up to 3.3 kV
59.	SITC of Electric Traction Passenger Lift with · Rated capacity: - 8 Passenger/544Kg ·Floors: - G+14 floor (15 Stops/15 Landings) · Travel: - 45 mtrs

Sr. No.	Description
	· Location of Lift Machine: - MR · Rated speed: - 1.0 mps VS · Car/Landing door clear opening of 800 mm wide x 2000 mm high · Clear Car size ofmm wide xmm deep x 2250mm high · Lift shaft available having clear size ofmm wide xmm deep,mm Pit depth,mm Overhead. Door's type: - COPO/TOPO Doors with frame made from SS 304 grade solid(non-cladded) sheet of 1.5mm thick in hairline finish for car and all landing doors with SS door architraves/frames The lift doors shall have minimum 1 hour fire rating (with submission of necessary valid test certificate issued by NABL accredited or Independent test laboratory).Lift car enclosure made from SS 304 grade solid (non-cladded) sheet of 1.5mm, thick with hairline finish with frame made from MS girders, bracing of adequate size with minimum safety factor of 5, with Toe Guard Apron, with necessary false ceiling with adequate LED lights, blower/fan for ventilation & SS chequered plate flooring, handrails, mirror, emergency light etc. The lift car interior design shall be done as per the directions of engineer in charge. COP with SS face plate having metallic push buttons with Braille code & luminous indicator around button with FPI, scrolling UP/DN LED indicator & with / without attendant key switch, OWD with audio-visual alarm, VAS in Marathi, Hindi & English with intercom system with telephone instrument in Lift car, LMR & FCC/ground floor. LOP with SS face plate having recess/surface push button box for all landings with scrolling UP/DN LED indicator having metallic push buttons with Braille code & luminous indicator around button with CPI, Lift car arrival & next travel direction audio-visual indication at all landings. · Lift controller based on microprocessor/ PLC with VVVF Drive having closed loop control system, with IBMS compatible having necessary port. The controller shall have necessary protections such as overcurrent, overvoltage, over speed, overheat for all devices including lift motor and travel direction protection, protection for phase loss at input or output etc. various operation modes such as maintenance, change direction by attendant, full load bypass, VIP call operation, self-levelling operation, call cancel facility, auto return to home landing, fireman evacuation operation, earthquake operation, door open/close operation from COP etc. The system can record information for minimum 30 latest faults, with real-time clock management and handheld keypad with LCD screen for view and setting of parameters. The control panel

Sr. No.	Description
	duly wired with proper size & strength, copper wire for power & control circuit, with provision for addition of floor/control card & allied accessories control panel having enclosure of 1.5mm CRCA sheet with powder coating with IP54 Protection class. ·ARD complete with necessary SMF VRLA batteries. ·Fireman controller having fireman switch at fire Landing. ·CCTV surveillance system comprises of 2 no's minimum 2.0 MP FHD IP based vandal proof Dome camera in lift car & in LMR/inside lift shaft top aimed on Lift machinery & controller with NVR kept in LMR/FCC with HDR data backup min. 90 days with min. 18" FHD TV monitor, to be kept in FCC/LMR as directed by Engineer In Charge. ·Lift Machine of Gearless PMSM of suitable kW with duty cycle of minimum 120 starts/hr (with submission of necessary valid test certificate issued by NABL accredited or independent test laboratory), with Traction pulley, OSG, electromagnetic brakes, entire assembly mounted on adequate size girders duly fixed on LMR floor/ shaft walls complete with main/diverter traction sheaves, suspension wire ropes/belts of adequate size & strength. ·Other mechanical parts such as 'T' section adequate size guide rails for car & counter weight with brackets fasteners, counter weight frame with necessary blocks, buffers with necessary support arrangement, MS pit ladder etc. erected with necessary steel work. ·Minor civil work such as alteration work if any necessary for erection of landing door frames and its accessories e.g., sill, header, hole pass etc complete with plaster finish, civil work for erection buffers, erection of lift machinery, adequate size core cuts if required & scaffolding for erecting guide rails, providing and fixing steel girders/RCC work having adequate strength for mounting and hoisting lift machine etc. complete as per specification no. LFT. ·General: - Job includes entire procedure of obtaining all necessary erection permissions & "License to Work the Lift" from Electrical Inspector (Lifts) with submission to the Engineer in Charge. ·The above rate includes Fully Comprehensive AMC for one year from the date of commissioning.
60.	SITC of Electric Traction Passenger Lift with ·Rated capacity: - 13 Passenger/884Kg · Floors: - G+14 floor (15 Stops/15 Landings)

Sr. No.	Description
	·Travel: - 45 mtrs. ·Location of Lift Machine: - MR · Rated speed: - 1.0mps VS ·Car/Landing door clear opening of 900/1000 mm wide x 2000 mm high ·Clear Car size ofmm wide xmm deep x 2250mm high ·Lift shaft available having clear size ofmm wide xmm deep,mm Pit depth,mm Overhead. ·Door's type: - COPO/TOPO Doors with frame made from SS 304 grade solid(non-cladded) sheet of 1.5mm thick in hairline finish for car and all landing doors with SS door architraves/frames the lift doors shall have minimum 1 hour fire rating (with submission of necessary valid test certificate issued by NABL accredited or independent test laboratory). ·Lift car enclosure made from SS 304 grade solid (non-cladded) sheet of 1.5mm, thick with hairline finish with frame made from MS girders, bracing of adequate size with minimum safety factor of 5, with Toe Guard Apron, with necessary false ceiling with adequate LED lights, blower/fan for ventilation & SS chequered plate flooring, handrails, mirror, emergency light etc. The lift car interior design shall be done as per the directions of engineer in charge. ·COP with SS face plate having metallic push buttons with Braille code & luminous indicator around button with FPI, scrolling UP/DN LED indicator & with / without attendant key switch, OWD with audio-visual alarm, VAS in Marathi, Hindi & English with intercom system with telephone instrument in Lift car, LMR & FCC/ground floor. ·LOP with SS face plate having recess/surface push button box for all landings with scrolling UP/DN LED indicator having metallic push buttons with Braille code & luminous indicator around button with CPI, Lift car arrival & next travel direction audio-visual indication at all landings. ·Lift controller based on microprocessor/ PLC with VVVF Drive having closed loop control system, with IBMS compatible having necessary port. The controller shall have necessary protections such as overcurrent, overvoltage, over speed, overheat for all devices including lift motor and travel direction protection, protection for phase loss at input or output etc. various operation modes such as maintenance, change direction by attendant, full load bypass, VIP call operation, self-levelling operation, call cancel facility, auto return to home landing, fireman evacuation operation, earthquake operation, door open/close operation from COP etc. The system can record information for

Sr. No.	Description
	minimum 30 latest faults, with real-time clock management and handheld keypad with LCD screen for view and setting of parameters, The control panel duly wired with proper size & strength, copper wire for power & control circuit, with provision for addition of floor/control card & allied accessories control panel having enclosure of 1.5mm CRCA sheet with powder coating with IP54 Protection class. ·ARD complete with necessary SMF VRLA batteries. ·Fireman controller having fireman switch at fire Landing. ·CCTV surveillance system comprises of 2 no's minimum 2.0MP FHD IP based vandal proof Dome camera in lift car & in LMR/inside lift shaft top aimed on Lift machinery & controller with NVR kept in LMR/FCC with HDR data backup for min. 90 days with min. 18" FHD TV monitor, to be kept in FCC/LMR as directed by Engineer In Charge. ·Lift Machine of Gearless PMSM of suitable kW with duty cycle of minimum 120 starts/hr (with submission of necessary valid test certificate issued by NABL accredited or independent test laboratory), with Traction pulley, OSG, electromagnetic brakes, entire assembly mounted on adequate size girders duly fixed on LMR floor/ shaft walls complete with main/diverter traction sheaves, suspension wire ropes/belts of adequate size & strength. ·Other mechanical parts such as 'T' section adequate size guide rails for car & counter weight with brackets fasteners, counter weight frame with necessary blocks, buffers with necessary support arrangement, MS pit ladder etc. erected with necessary steel work. ·Minor civil work such as alteration work if any necessary for erection of landing door frames and it's accessories e.g. sill, header, hole pass etc complete with plaster finish, civil work for erection buffers, erection of lift machinery, adequate size core cuts if required & scaffolding for erecting guide rails, providing and fixing steel girders/RCC work having adequate strength for mounting and hoisting lift machine etc. complete as per specification no. LFT. ·General: - Job includes entire procedure of obtaining all necessary erection permissions & "License to Work the Lift" from Electrical Inspector (Lifts) with submission to the Engineer in Charge. ·The above rate includes Fully Comprehensive AMC for one year from the date of commissioning.
61.	For Lift having 800mm wide x 2000mm high clear entrance in 1.5mm thick SS

Sr. No.	Description
	304 grade solid (non-cladded) landing door [Two panel], this includes all necessary accessories like LOP's with UP/DN buttons-arrows-indicators, extension of guide rails, shaft wiring with trunking, traveling cables, main hoisting ropes/belts, & OSG rope landing doors with all accessories etc. necessary for the normal safe functioning of lift installation complete.
62.	Supplying & erecting flat flexible 3 core 2.5 sq.mm PVC sheathed submersible type copper cable suitably clamped at fixed intervals with column pipe assembly complete.
63.	Supplying and erecting flat flexible 3 core 4 sq. mm PVC sheathed submersible type copper cable suitably clamped at fixed intervals with column pipe assembly complete.
64.	Supplying & erecting automatic control & DOL starter panel for 3 Ph, 415 volt, A.C. Submersible/centrifugal pump set for 3.0 HP to 7.5 HP
65.	Supplying and erecting minimum Three & above star rated submersible pump set suitable for erection on open well of 2.25 KW/3 HP with 415 V 50 c/s. A.C. supply, having delivery head from 27 to 9 m and discharge from 300 to 660 litres. Per minute & pipe of Suction 65mm /Delivery 50 mm diameter erected with necessary H type clamps as per specification No. WP-OSP (FOR TRANFER OF RAIN WATER)
66.	Supplying & erecting Fully Automatic Star/delta starter suitable for 12.5 H. P to 15 H.P. for 3 phase squirrel cage motor totally enclosed with overload and no volt relay complete erected on provided angle iron frame. BOOSTER PUMP STARTER
67.	Supplying & erecting D.O.L Starter suitable for 415 V, 3 phase 50 cycles with no volt coil and overload element with necessary materials and connected to supply up to 7.5 H.P. (FOR TRANFER OF RAIN WATER)
68.	Supplying, installation, testing and commissioning of booster fire pump [BP] suitable for water discharge of 900 LPM at 35 m head driven by electric motor 415 volts, 3 phase 50 Hz, AC supply of 7.5 kW or of suitable kW capacity for manual/automatic operation and consisting of following: (a) Horizontal/vertical type, single/multi stage, centrifugal casing pump of cast iron body & bronze/CI impeller with stainless steel shaft (SS410 grade), mechanical seal conforming to IS 1520. (b) Squirrel cage induction motor, TEFC, synchronous speed 3000 RPM, suitable for operation on with IP 55 protection for enclosure, horizontal foot

Sr. No.	Description
	mounted type with Class- 'F' insulation, conforming to IS-325. © M.S. fabricated common base plate, coupling, coupling guard, foundation bolts etc. as required. (d)Erected on provided suitable size cement concrete foundation duly plastered with anti-vibration pads with perfect aligning, proper levelling complete pump set with accessories duly painted with two coats of synthetic enamel paint of fire red colour over a coat of primer (ISC code 536 as per IS 2932 of 2003) complete, as per specification no. FF-MFP/BP
69.	Supplying and erecting submersible pump set suitable for erection on open well of 6.0 HP with 415 V 50 c/s A.C. supply, having delivery head from 50m and discharge from 350 to 500ltrs. per minute) & pipe of Suction 50mm /Delivery 50 mm diameter erected with necessary H type clamps for Domestic for EWS Type 1,2 and 3.
70.	Supplying and erecting submersible pump set suitable for erection on open well of 3.0 HP with 415 V 50 c/s A.C. supply, having delivery head from 50m and discharge from 175 to 300ltrs. per minute) & pipe of Suction 50mm /Delivery 50 mm diameter erected with necessary H type clamps for Flushing for EWS Type 1,2 and 3.

VI. Building Internal Electrical Item Wise Specification

Sr. No.	Description of Items
A	Building Work
1.	Common Panel Supplying & erecting triple pole and neutral distribution board (TPNDB), 3 pole/4 pole MCCB as incomer & outgoing SP MCB (24 poles) or TP MCB of 8 ways (24 poles), with door, 1.2mm thickness, surface/flush mounted, on iron/GI frame (vertical busbar type) as per specification no. SW-SWR/MCDB1
2.	Providing & erecting 3 Pole MCCB, 415 V, 100A, rated short-circuit breaking capacity 25 kA (Ics=100% of Icu), adjustable thermal (overload) setting and fixed magnetic setting with provided leads, provision for installation of shunt/UV/trip alarm contact. MCCB with phase barriers on both sides, insulation withstand capacity 800V, no line-load bias in provided enclosure/panel as per specification no. SW-SWR/MCCB

Sr. No.	Description of Items
3.	Supplying and erecting ON LOAD four pole automatic transfer switch, 415V, 100-125A, 50 Hz with enclosure, flexibility of neutral connection, line and load reversibility, changeover with phase barriers, source separator, terminal covers along with staggered termination for cable connection and ATS controller with provision of time delay protection of OV, UV, phase sequence changing, single phasing and event logging, connected with provided leads complete complying to IS/IEC 60947-3 Part-1/6.
4.	Supplying, erecting & marking TPMCB 6A to 32A, with rated short - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB
5.	Supplying, erecting & marking SPMCB 6A to 32A, B-series with rated short - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB
6.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 3½ core 35 sq. mm. aluminium conductor complete erected with glands & lugs, on wall/ trusses/pole or laid in provided trench/ pipe as per specification no. CB-LT/AL
7.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 4 core 6 sq. mm. copper conductor complete erected with glands & lugs, on wall/ trusses/ pole or laid in provided trench/ pipe as per specification no. CB-LT/CU
8.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 3 core 2.5 sq. mm. copper conductor complete erected with glands & lugs, on wall/ trusses/ pole or laid in provided trench/ pipe as per specification no. CB-LT/CU
B	METERING PANEL BOARD
1.	Providing & erecting floor/wall mounting, MCCB panel board, with door, suitable for four pole incoming 400 A, 6 ways four pole outgoing, up to 100 A MCCBs on iron frame, as per specification no SW-SWR/MCCBPB. (Excluding MCCBs)
2.	Providing & erecting 3 Pole MCCB 415V, 400A, rated short-circuit breaking capacity 36 kA (Ics=100% of Icu), adjustable thermal (overload) setting and adjustable magnetic setting with provided leads, provision for installation of shunt/ UV/ trip alarm contact and MCCB should have phase barriers both

Sr. No.	Description of Items
	sides, with insulation withstand capacity 800V, no line-load bias in provided enclosure /panel as per specification no. SW-SWR/MCCB
3.	Providing & erecting 4 Pole MCCB, 415 V, 100A, rated short-circuit breaking capacity 25 kA ($I_{cs}=100\%$ of I_{cu}), adjustable thermal (overload) setting and fixed magnetic setting with provided leads, provision for installation of shunt/UV/trip alarm contact. MCCB with phase barriers on both sides, insulation withstand capacity 800V, no line-load bias in provided enclosure/panel as per specification no. SW-SWR/MCCB Note: $5309 \times 20\% = 1061.8$ i.e. $5309 + 1061.8 = 6370.8$
4.	Supplying and erecting bus bar chamber triple pole with neutral with four aluminium bars, 500V, 100A having 15mm (Width) X 3 mm (Thickness), with 99.9 % aluminium purity as per E91E Grade & Current Density 0.8 A per sq. mm. compliance to IS 5082 and Enclosure fabricated with 16 SWG CRCA sheet and epoxy paint, with suitable number of ways and with SMC/DMC Insulators and accessories of suitable rating complete as per specification no. CP-BB
5.	Supplying, erecting & marking DPMCB 6A to 32A, C-series with rated short - circuit breaking capacity (I_{cn}) 10kA in provided distribution board as per specification no. SW-SWR/MCB
6.	Supplying erecting and marking double pole isolator only switch version of miniature circuit breaker of 40A with required wiring connections and lugs etc. in provided distribution board complete.
7.	Supplying & erecting CRCA sheet metal one-way enclosures of 1.2mm thickness suitable for DP MCB /TP MCB/FP MCB / RCCB / RCBO complete erected on angle iron/GI frame.
8.	Supplying and erecting mains with 2x4 sq.mm FRLSH copper PVC insulated wire laid in provided conduit/trunking/inside pole/Bus bars or any other places. as per specification No: WG-MA/BW
9.	Supplying and erecting mains with 1x2.5 sq.mm FRLSH copper PVC insulated wire laid in provided conduit/trunking/inside pole/Bus bars or any other places. as per specification No: WG-MA/BW Note - 3) Rates for the earth wire shall be calculated as one-half rate of mains of bunch of two wire.

Sr. No.	Description of Items
10.	Supplying and erecting 25 mm dia. corrugated flexible polypropylene conduit.
11.	Providing & erecting incoming and outgoing cable alleys suitable for MCCB panel board with door, 250A incoming and 8 ways outgoing or 400A incoming and 6-way outgoing made from CRCA sheet 1.6 mm thick duly powder coated and having vertical insulated partition for shielding between the shared face of cableway and DB.
12.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 3½ core 35 sq. mm. aluminium conductor complete erected with glands & lugs, on wall/ trusses/pole or laid in provided trench/ pipe as per specification no. CB-LT/AL
13.	Supplying and erecting bus bar chamber triple pole with neutral with four aluminium bars, 500V, 100A having 15mm (Width) X 3 mm (Thickness), with 99.9 % aluminium purity as per E91E Grade & Current Density 0.8 A per sq. mm. compliance to IS 5082 and Enclosure fabricated with 16 SWG CRCA sheet and epoxy paint, with suitable number of ways and with SMC/DMC Insulators and accessories of suitable rating complete as per specification no. CPRR
14.	Supplying and erecting marine (Waterproof) plywood 18/19 mm thick fixed to wall or on provided panel board with necessary materials such as screws, wall
15.	Supplying and erecting static energy Trivector meter 440 V 50 Hz. 20-100A ISI mark meter AC 3 phase 4 wires LCD/LED display (Measuring and display Trivector, V, I, kW, PF, Hz, Last month history of kWh & MD in kW) with special features like TOD, Tamper logging, Optical Communication Port & Display in absence of Power complete erected on polished teak wood board duly wired with required size of weather proof PVC wires duly tested for class 1.0 accuracy by electric supply company.
C	LDB WITH ACCESSORIES
1.	Supplying, erecting & marking SPMCB 6A to 32A, B-series with rated short - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB
2.	Supplying, erecting & marking SPMCB 6A to 32A, C-series with rated short - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB

Sr. No.	Description of Items
3.	Supplying, erecting & commissioning 2 pole RCCB 16/25A, electromagnetic type with 30/100/300 mA sensitivity with earth leakage trip indication complete as per specification no. SW-RCCB/RCCB
4.	Supplying, erecting & commissioning 2 pole RCCB 40A, electromagnetic type with 30/100/300 mA sensitivity with earth leakage trip indication complete as per specification no. SW-RCCB/RCCB
5.	Supplying and erecting single pole and neutral distribution board (SPNDB), with 2 ways for incoming and 6 ways (6 poles) for outgoing SP MCBs, with door, 1.2mm thickness surface / flush mounted, IP 43 Protection on iron / GI frame as per specification no. SW-SWR/MCBDB
6.	Supplying and erecting single pole and neutral distribution board (SPNDB), with 2 ways for incoming and 10 ways (10 poles) for outgoing SP MCBs, with door, 1.2mm thickness surface / flush mounted, IP 43 Protection on iron / GI frame as per specification no. SW-SWR/MCBDB
D	INTERNAL ELECTRIFICATION
1.	Supplying and erecting mains with 2x4 sq.mm FRLSH copper PVC insulated wire laid in provided conduit/trunking/inside pole/Bus bars or any other places. as per specification No: WG-MA/BW
2.	Supplying and erecting mains with 1x2.5 sq.mm FRLSH copper PVC insulated wire laid in provided conduit/trunking/inside pole/Bus bars or any other places. as per specification No: WG-MA/BW Note - 3) Rates for the earth wire shall be calculated as one-half rate of mains of bunch of two wire.
3.	Supplying and laying HMS PVC conduit FRLS grade 20 mm dia. With necessary accessories in RCC work/false ceiling/false flooring as per specification No. WG-MA/CC.
4.	LIGHT Point wiring in PVC trunking (casing-capping) with 1.5 sq.mm (2+1E) FRLSH grade copper wire, flush type switch, earthing and required accessories as per specification No: WG-PW/SW
5.	Secondary point wiring for additional light/ bell point in PVC trunking (casing-capping) with 1.5 sq.mm (2+1E) FRLSH grade copper wire with required accessories as per specification No: WG-PW/SW
6.	FAN Point wiring in PVC trunking (casing-capping) with 1.5 sq.mm (2+1E)

Sr. No.	Description of Items
	FRLSH grade copper wire, flush type switch, earthing and required accessories as per specification No: WG-PW/SW
7.	EX.FAN Point wiring in PVC trunking (casing-capping) with 1.5 sq.mm (2+1E) FRLSH grade copper wire, flush type switch, earthing and required accessories as per specification No: WGPW/ SW
8.	BELL Point wiring in PVC trunking (casing-capping) with 1.5 sq.mm (2+1E) FRLSH grade copper wire, flush type switch, earthing and required accessories as per specification No: WGPW/ SW
9.	Wiring for plug on board with flush Switch socket, copper wiring and earthing as per specification No: WG-PW/SW
10.	Supplying and erecting flush, shock proof plug socket 6 pin 6A and 16A with switch 16A combined unit erected in recess in 16-gauge M.S sheet box duly painted and connected with earth terminal and leads
11.	Point wiring for independent plug in PVC trunking (casing capping) with 1.5 sq.mm FRLSH grade copper wire, flush type switch, earthing and required accessories as per specification No: WG-PW/SW
12.	Supplying and erecting electronic type step regulator for ceiling fan 600/ 900/ 1200/ 1400 mm. sweep, complete erected on provided switch board.
13.	Supplying and erecting PVC trunking (PVC casing-n-capping) of size 20 mm with accessories on wall/ceiling as per specification No: WG-MA/CON.
14.	Supplying and erecting mains with 2x2.5 sq.mm FRLSH copper PVC insulated wire laid in provided conduit/trunking/inside pole/Bus bars or any other places. as per specification No: WG-MA/BW
15.	Supplying and erecting mains with 1x1.5 sq.mm FRLSH copper PVC insulated wire laid in provided conduit/trunking/inside pole/Bus bars or any other places as per specification No: WG-MA/BW 3) Rates for the earth wire shall be calculated as one-half rate of mains of bunch of two wire.
16.	Supplying and erecting ISI mark modular type T.V. Socket single outlet, duly erected on provided plate and box with wiring connections complete.
17.	Supplying and erecting anodized aluminium corridor / passage light LED fitting (4 feet) Max. 22W with high transitivity diffuser with system lumens output of Min.2200 lumens, min. efficacy of 100 lumen/W, CRI>80, CCT up to 6000K, Beam Angle of 110 deg., Ripple<5%, THD<10%, p.f. >0.95, operating range of

Sr. No.	Description of Items
	200-270V, surge protection of 2 kV, Life class of 50,000 Hrs. at L70B50, including driver, with end caps on provided PVC Block / wooden board with 3 years warranty.
18.	Supplying & erecting conventional spike type air termination suitable to carry lightning stroke made up of heavy gauge 40 mm dia copper pipe of standard length with 5 Nos. copper spikes fixed on copper ball as air terminals duly threaded in copper pipe erected on provided foundation complete.
19.	Providing earthing with copper earth plate size 60 x 60 x 0.315 cm with funnel with a wire mesh for watering and brick masonry block C.I. cover with minimum 25 kg Carbon based environment friendly back fill Ground Enhancing compound complete with all materials, testing & recording the results as per specification no ESE -LA
20.	Supplying and erecting aluminium strip of required sizes used for earthing on wall and/or any other purpose with necessary aluminium clamps fixed on wall with insulating and non-corrosive bond with required joints. As per specification no. EA-EP.
21.	For Solar System Supplying, erecting & terminating FR XLPE insulated, galvanized steel formed wire armoured (strip) cable 1100 V, 4 core 16 sq. mm. copper conductor complete erected with glands & lugs, on wall/ trusses/ pole or laid in provided trench/ pipe as per specification no. CB-LT/CU.
22.	Supplying and erecting AC three phase LT CT multifunction net meter DLMS compliance of accuracy class 0.5, 3 x 240V, 50Hz, with backlit LCD display, optical port & RS 232 port (measures import kWh, kVAh, export kWh, kVAh, Net kWh, kVAh, V, I, kW, 6 months history of energy, load survey, TOD, tamper detection & logging, power ON/OFF events, instantaneous parameters of rating 1/5A with display in absence of power) with wiring connections and mounting hardware on provided panels complete with calibration certificate from manufacturer.
23.	Supplying, erecting and marking triple pole isolators only switch version of miniature circuit breaker of 100A with required wiring connections and lugs etc. in provided distribution board.
E	EARTHING & STRIP
1.	Providing earthing with galvanized iron earth plate size 60 x60 x 0.6 cm complete with all materials, testing & recording the results as per specification

Sr. No.	Description of Items
	no. EA-EP
2.	Supplying and erecting GI strip of required size used for earthing on wall and/or any other purpose with necessary GI clamps fixed on wall painted with insulating and non-corrosive bond with required joints as per specification no. EA-EP
3.	Supplying and erecting GI wire of high purity of required sizes used for earthing or any other purposes on wall with necessary G. I. Clamps fixed on wall/cable/conduit with screws complete.
4.	Supplying & erecting Aviation obstruction light consisting of yellow painted diecast aluminium alloy housing with Integral LED aviation light comprising of aluminium housing with polycarbonate enclosure complete.
5.	Supplying and fixing PVC synthetic elastomer electrically insulating mat with class A insulation conforming to IS: 15652 – 2006 having 2 mm thickness up to 3.3 kV.
6.	Providing printed instruction chart for treating persons suffering from electric shock, printed in English & Marathi and duly laminated complete.
7.	Supplying and erecting GI sheet 1.6 mm (16 SWG) caution board of size 200 x 150 mm duly painted with red oxide and enamel paint in Marathi, Hindi and English erected at position for medium voltage (415 Volt) installation with compliance to IS 2551-1982
8.	Supplying and laying (including excavation of suitable width & depth up to 90 cm) 90 mm outside dia. double wall corrugated pipes (DWC) of HDPE for enclosing cable below ground/road surface, to required depth complete.
F	LIFT WORK
1.	Supplying and erecting single pole and neutral distribution board (SPNDB), with 2 ways for incoming and 6 ways (6 poles) for outgoing SP MCBs, with door, 1.2mm thickness surface / flush mounted, IP 43 Protection on iron / GI frame as per specification no. SW-SWR/MCBDB
2.	Supplying, erecting & marking DPMCB 6 A to 32 A, B-series with rated short - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB
3.	Supplying, erecting & marking SPMCB 6A to 32A, C-series (for motor/power) in

Sr. No.	Description of Items
	provided distribution board as per specification no. SW-SWR/MCB
4.	Point wiring in for light/fan/bell in 20 mm metal conduit with 1.5 sq.mm (2+1E) FRLSH grade copper wire, flush type switch, earthing and required accessories as per specification No: WG-PW/SW
5.	Supplying and erecting bulk head LED fitting max. 10W with high transitivity diffuser with system lumens output of min. 1100 lumens, min. efficacy of 110 lumen/W, CRI>80, CCT up to 6500K, Beam Angle of 120 deg., THD<10%, p.f. >0.95, operating range of 140-260V, in built surge protection of 2.5 kV, Life class of 50,000 Hrs. at L70B50, including driver, IP66, IK09 rated on provided PVC Block / wooden board with 3 years warranty.
6.	SITC of Electric Traction Passenger Lift with
	Rated capacity: - 13 Passenger/884 Kg
	Floors: - G+14 floor (15 Stops)
	Travel: - 43.7 to 47.6 mtrs
	Location of Lift Machine: - MR
	Rated speed: - 1.0mps VS
	Door's type: - COPO/TOPO Doors with frame having clear opening of 800 mm wide x 2000 mm high made from SS 304 grade sheet of 1.5mm, thick in hairline finish for car and all landing doors with SS door architraves/frames
	Lift car enclosure made from SS 304 grade sheet of 1.5mm, thick with hairline finish with frame made from MS girders, bracing of adequate size with minimum safety factor of 5, with Toe Guard Apron, with necessary false ceiling with adequate LED lights, blower/fan for ventilation & SS chequered plate flooring, handrails, mirror, emergency light etc.
	COP with SS face plate having metallic push buttons with Braille Code & luminous indicator around button with FPI, scrolling UP/DN LED indicator & with/without attendant key switch, OWD with audio-visual alarm, VAS in Marathi, Hindi & English with intercom system with telephone instrument in Lift car, LMR & CC/ground floor
	Clear Car size of 1100 mm wide x 2000 mm deep x 2100 mm high
	LOP with SS face plate having recess/surface push button box for all landings with scrolling UP/DN LED indicator having metallic push buttons with Braille

Sr. No.	Description of Items
	Code & luminous indicator around button with CPI, Lift car arrival & next travel direction audio-visual indication at all landings
	Lift controller based on microprocessor/ PLC with VVVF Drive having closed loop control system, with IBMS compatible having necessary port, control panel duly wired with proper size & strength copper wire for power & control circuit, with provision for addition of floor/control card & allied accessories control panel having enclosure of 1.5mm CRCA sheet with powder coating with IP54 Protection class
	ARD complete with necessary SMF VRLA batteries
	Fireman controller having fireman switch at fire Landing,
	CCTV surveillance system comprises of 2nos minimum 2.0MP FHD IP based vandal proof Dome camera in lift car & in LMR/inside lift shaft top aimed on Lift machinery & controller with NVR kept in LMR/FCC with HDR data backup for 60 days with 18" FHD TV monitor, to be kept in FCC/LMR as directed by Engineer In Charge.
	Lift Machine of Gearless PMSM of suitable kW with Traction pulley, OSG, electromagnetic brakes, entire assembly mounted on adequate size girders duly fixed on LMR floor/ shaft walls complete with main/diverter traction sheaves, suspension wire ropes/belts of adequate size & strength
	Other mechanical parts such 's"T' section adequate size guide rails for car & counter weight with brackets fasteners, counter weight frame with necessary blocks, buffers with necessary support arrangement, MS pit ladder etc. erected with necessary steel work
	Minor civil work for alteration if any and erection of door frames and accessories, erection buffers, erection of lift machinery, adequate size core cuts if required & scaffolding for erecting guide rails fixing of girders for mounting lift machine etc. complete as per specification no. LFT.
	Lift shaft available having clear size of 1950 mm wide x 3000 mm deep.1350 mm Pit depth, 4200 mm Overhead.
	For Lift having 800mm wide x 2000mm high clear entrance in 1.5mm thick SS 304 grade landing door [Two panel], this includes all necessary accessories like 'OP's with UP/DN buttons-rows indicators, extension of guide rails, shaft wiring with trunking, traveling cables, main hoisting ropes/belts, & OSG rope

Sr. No.	Description of Items
	landing doors with all accessories etc. necessary for the normal safe functioning of lift installation complete.
7.	SITC of Electric Traction Passenger Lift with
	Rated capacity: - 8 Passenger/544Kg
	Floors: - G+14 floor (15 Stops)
	Travel: - 43.7 to 47.6 mtrs
	Location of Lift Machine: - MR
	Rated speed: - 1.0mps VS
	Door's type: - COPO/TOPO Doors with frame having clear opening of 800 mm wide x 2000 mm high made from SS 304 grade sheet of 1.5mm, thick in hairline finish for car and all landing doors with SS door architraves/frames
	Lift car enclosure made from SS 304 grade sheet of 1.5mm, thick with hairline finish with frame made from MS girders, bracing of adequate size with minimum safety factor of 5, with Toe Guard Apron, with necessary false ceiling with adequate LED lights, blower/fan for ventilation & SS chequered plate flooring, handrails, mirror, emergency light etc.
	COP with SS face plate having metallic push buttons with Braille Code & luminous indicator around button with FPI, scrolling UP/DN LED indicator & with/without attendant key switch, OWD with audio-visual alarm, VAS in Marathi, Hindi & English with intercom system with telephone instrument in Lift car, LMR & CC/ground floor
	Clear Car size of 1100 mm wide x 2000 mm deep x 2100 mm high
	LOP with SS face plate having recess/surface push button box for all landings with scrolling UP/DN LED indicator having metallic push buttons with Braille Code & luminous indicator around button with CPI, Lift car arrival & next travel direction audio-visual indication at all landings
	Lift controller based on microprocessor/ PLC with VVVF Drive having closed loop control system, with IBMS compatible having necessary port, control panel duly wired with proper size & strength copper wire for power & control circuit, with provision for addition of floor/control card & allied accessories control panel having enclosure of 1.5mm CRCA sheet with powder coating with IP54 Protection class

Sr. No.	Description of Items
	ARD complete with necessary SMF VRLA batteries
	Fireman controller having fireman switch at fire Landing,
	CCTV surveillance system comprises of 2nos minimum 2.0MP FHD IP based vandal proof Dome camera in lift car & in LMR/inside lift shaft top aimed on Lift machinery & controller with NVR kept in LMR/FCC with HDR data backup for 60 days with 18" FHD TV monitor, to be kept in FCC/LMR as directed by Engineer in Charge.
	Lift Machine of Gearless PMSM of suitable kW with Traction pulley, OSG, electromagnetic brakes, entire assembly mounted on adequate size girders duly fixed on LMR floor/ shaft walls complete with main/diverter traction sheaves, suspension wire ropes/belts of adequate size & strength
	Other mechanical parts such 's"T' section adequate size guide rails for car & counter weight with brackets fasteners, counter weight frame with necessary blocks, buffers with necessary support arrangement, MS pit ladder etc. erected with necessary steel work
	Minor civil work for alteration if any and erection of door frames and accessories, erection buffers, erection of lift machinery, adequate size core cuts if required & scaffolding for erecting guide rails fixing of girders for mounting lift machine etc. complete as per specification no. LFT.
	Lift shaft available having clear size of 1950 mm wide x 3000 mm deep.1350 mm Pit depth, 4200 mm Overhead.
	For Lift having 800mm wide x 2000mm high clear entrance in 1.5mm thick SS 304 grade landing door [Two panel], this includes all necessary accessories like 'OP's with UP/DN buttons-rows indicators, extension of guide rails, shaft wiring with trunking, traveling cables, main hoisting ropes/belts, & OSG rope landing doors with all accessories etc. necessary for the normal safe functioning of lift installation complete
G	WATER PUMPS
1.	Supplying and erecting submersible pump set suitable for erection on open well of 6 HP & 7.5 HP with 415 V 50 c/s.A.C. supply, having delivery head from 50m and discharge from 200 to 400ltrs. per minute) & pipe of Suction 50mm /Delivery 50 mm diameter erected with necessary H type clamps for Domestic for EWS Type

Sr. No.	Description of Items
2.	Supplying and erecting submersible pump set suitable for erection on open well of 5 HP & 3 HP with 415 V 50 c/s.A.C. supply, having delivery head from 50m and discharge from 150 to 300ltrs. per minute) & pipe of Suction 50mm /Delivery 50 mm diameter erected with necessary H type clamps for Flushing for EWS Type
3.	Supplying & erecting flat flexible 3 core 2.5 sq.mm PVC sheathed submersible type copper cable suitably clamped at fixed intervals with column pipe assembly complete.
4.	Supplying and erecting flat flexible 3 core 4 sq. mm PVC sheathed submersible type copper cable suitably clamped at fixed intervals with column pipe assembly complete.
5.	Providing & erecting 4 Pole MCCB, 415 V, 100A, rated short-circuit breaking capacity 25 kA ($I_{cs}=100\%$ of I_{cu}), adjustable thermal (overload) setting and fixed magnetic setting with provided leads, provision for installation of shunt/UV/trip alarm contact. MCCB with phase barriers on both sides, insulation withstand capacity 800V, no line-load bias in provided enclosure/panel as per specification no. SW-SWR/MCCB Note: $5309 \times 20\% = 1061.8$ i.e., $5309 + 1061.8 = 6370.8$
6.	Supplying, fixing and commissioning 4 pole RCBO 6/10/16/20/25A, with overcurrent, rated short-circuit breaking capacity (10 kA) and earth leakage protection, with 30/100/300 mA sensitivity and short circuit and earth leakage trip indications per specification no SW-RCCB/RCBO
7.	Supplying & erecting automatic control & DOL starter panel for 3 Ph, 415 volt, A.C. Submersible/centrifugal pump set for 3.0 HP to 7.5 HP
8.	Supplying and erecting minimum Three & above star rated submersible pump set suitable for erection on open well of 2.25 KW/3 HP with 415 V 50 c/s. A.C. supply, having delivery head from 27 to 9 m and discharge from 300 to 660 litres. per minute & pipe of Suction 65mm /Delivery 50 mm diameter erected with necessary H type clamps as per specification No. WP-OSP (FOR TRANSFER OF RAIN WATER)
9.	BOOSTER PUMP STARTER Supplying & erecting Fully Automatic Star/delta starter suitable for 12.5 H. P to 15 H.P. for 3 phase squirrel cage motor totally enclosed with overload and no volt relay complete erected on provided angle iron frame.

Sr. No.	Description of Items
10.	Supplying & erecting D.O.L Starter suitable for 415 V, 3 phase 50 cycles with no volt coil and overload element with necessary materials and connected to supply up to 7.5 H.P. (FOR TRANSFER OF RAIN WATER)
11.	Supplying, installation, testing and commissioning of booster fire pump [BP] suitable for water discharge of 900 LPM at 35 m head driven by electric motor 415 volts, 3 phase 50 Hz, AC supply of 7.5 kW or of suitable kW capacity for manual/automatic operation and consisting of following: (a) Horizontal/vertical type, single/multi stage, centrifugal casing pump of cast iron body & bronze/CI impeller with stainless steel shaft (SS410 grade), mechanical seal conforming to IS 1520. (b) Squirrel cage induction motor, TEFC, synchronous speed 3000 RPM, suitable for operation on with IP 55 protection for enclosure, horizontal foot mounted type with Class 'F' insulation, conforming to IS-325. (c) M.S. fabricated common base plate, coupling, coupling guard, foundation bolts etc. as required. (d) Erected on provided suitable size cement concrete foundation duly plastered with anti-vibration pads with perfect aligning, proper levelling complete pump set with accessories duly painted with two coats of synthetic enamel paint of fire red colour over a coat of primer (ISC code 536 as per IS 2932 of 2003) complete, as per specification no. FF-MFP/BP
H	FIRE ALARM SYSTEM
1.	Supplying, erecting testing and commissioning manual call point (Pill box) with break glass, push button (resettable type) in metal enclosure complete as per specification no. FF-FAAS/MCP
2.	Supplying, erecting, testing and commissioning hooters having high (100dB @1m) and low (94dB @ 1m) volume setting, group addressing facility allowing multiple sounders to be activated with CRCA enclosure complete as per specification no. FF-FAAS/HTR
3.	Supplying, installing, testing and commissioning of 2 Zones Microprocessor based conventional fire alarm control panel (FACP) with standard accessories, 16x2 Character LCD Display, provision for zone wise contact and beep sound alarm, suitable to operate on 120-220 V AC, -0 - 49 Deg C, 93 ± 2 Percentage RH (non- condensing) at 32 ± 2 Deg C complete as per specification no. FF-FAAS/FACP
4.	Supplying, installing, testing and commissioning FR, XLPE armoured cable 2

Sr. No.	Description of Items
	core 1.5 sq.mm. copper conductor complete erected on wall/ ceiling complete as per specification no. CBLT/CU
I	PUBLIC ADDRESS SYSTEM
1.	Supplying, erecting, testing and commissioning 120 W amplifier suitable to operate on 230 Volts A.C. / 12 Volts D.C. supply complete as per specification no. FF-PA/AFR
2.	Supplying, erecting, testing and commissioning 15 to 30 Watts sound column or ceiling mounted complete as per specification no. FF-A/SOC
3.	Supplying and fixing 6U wall mount rack (Dimension-DxWxH – 500x600x367mm) as per specification No. WG-NAS/RAK
4.	Supplying, erecting, testing and commissioning rust free microphone floor stand of chrome plating/powder coating or S.S. material, 500 mm to 1500 mm height and sturdy mechanism ensuring trouble free movement complete.
5.	Supplying, erecting, testing and commissioning stand mounted / handheld type wired microphone having 2 mV/Pa sensitivity, impedance level limit 300 ohms and –0 - 16000 Hz suitable to operate in -10 deg C to +55 deg C with min. 7 metre shielded cable and 3 Pin Professional XLR Connector, stereo jack complete as per specification no. FF-PA/MIC
6.	Supplying, erecting, testing and commissioning of 0.5 sq. mm speaker wire complete.
7.	Supplying and erecting HMS PVC conduit FRLS grade 20 mm dia. With necessary accessories in wall/floor with chiselling appropriately as per specification No: WG-MA/CC.
8.	Supplying, installing, testing and commissioning of 12 kWp ON-GRID Solar PV System / Plant consisting solar PV poly crystalline panel (335 Wp) and Solar ON-GRID Inverter of similar capacity (THD<3%) with interconnecting wires / cables up to solar ON GRID inverter, proper base plate, canopy box to house Inverter and necessary accessories i.e. ACDB/ DCDB , MCB etc. wiring up to solar Inverter in casing and capping, display board, sign board, two distinct earthing, lightning arrester, Liasoning charges, remote monitoring system with maintaining card / chip for 5 years including net charges for RMS system, complete up to net meter installation and export to the grid with 5 year ON-SITE warranty and fully comprehensive maintenance contract as per

Sr. No.	Description of Items
	specification no. ESD-RT-ONG For EWS building Type-1
9.	Supplying, installing, testing and commissioning of 15 kWp ON-GRID Solar PV System / Plant consisting solar PV poly crystalline panel (335 Wp) and Solar ON-GRID Inverter of similar capacity (THD<3%) with interconnecting wires / cables up to solar ON GRID inverter, proper base plate, canopy box to house Inverter and necessary accessories i.e. ACDB/ DCDB , MCB etc. wiring up to solar Inverter in casing and capping, display board, sign board, two distinct earthing, lightning arrester, Liasoning charges, remote monitoring system with maintaining card / chip for 5 years including net charges for RMS system, complete up to net meter installation and export to the grid with 5 year ON-SITE warranty and fully comprehensive maintenance contract as per specification no. ESD-RT-ONG For EWS building Type-2
10.	Supplying, installing, testing and commissioning of 14 kWp ON-GRID Solar PV System / Plant consisting solar PV poly crystalline panel (335 Wp) and Solar ON-GRID Inverter of similar capacity (THD<3%) with interconnecting wires / cables up to solar ON GRID inverter, proper base plate, canopy box to house Inverter and necessary accessories i.e. ACDB/ DCDB , MCB etc. wiring up to solar Inverter in casing and capping, display board, sign board, two distinct earthing, lightning arrester, Liasoning charges, remote monitoring system with maintaining card / chip for 5 years including net charges for RMS system, complete up to net meter installation and export to the grid with 5 year ON-SITE warranty and fully comprehensive maintenance contract as per specification no. ESD-RT-ONG For EWS building Type-3.

VII. Infra Electrical Item Wise Specification

Sr. No.	Description of Items
A	LT Feeder Pillar, Panels & Accessories
1.	Supplying and erecting Feeder pillar-1 shall be outdoor Type (IP-52) with canopy, compartmental cubicle type freestanding with appropriate cable entries, front operating, front maintained wherever required complete with base frame etc. erected in 14 gauge CRCA sheet and surface treated with phosphating seven tank process and duly powder coated with MSEDCL

Sr. No.	Description of Items
	approved colour, feeder pillar should be with supporting angles, self-locks, gasket and slanting top to be erected on provided foundation. Incomer and outgoing as per SLD
2.	Supplying and erecting Feeder pillar-2 shall be outdoor Type (IP-52) with canopy, compartmental cubicle type freestanding with appropriate cable entries, front operating, front maintained wherever required complete with base frame etc. erected in 14 gauge CRCA sheet and surface treated with phosphating seven tank process and duly powder coated with MSEDCL approved colour, feeder pillar should be with supporting angles, self-locks, gasket and slanting top to be erected on provided Foundation Incomer and outgoing as per SLD
3.	Supplying and erecting Feeder pillar-3, Feeder pillar shall be outdoor Type (IP-52) with canopy, compartmental cubicle type freestanding with appropriate cable entries, front operating, front maintained wherever required complete with base frame etc. erected in 14 gauge CRCA sheet and surface treated with phosphating seven tank process and duly powder coated with MSEDCL approved colour, feeder pillar should be with supporting angles, self-locks, gasket and slanting top to be erected on provided foundation. Incomer and outgoing as per SLD
4.	Supplying and erecting Feeder pillar-4, Feeder pillar shall be outdoor Type (IP-52) with canopy, compartmental cubicle type freestanding with appropriate cable entries, front operating, front maintained wherever required complete with base frame etc. erected in 14 gauge CRCA sheet and surface treated with phosphating seven tank process and duly powder coated with MSEDCL approved colour, feeder pillar should be with supporting angles, self-locks, gasket and slanting top to be erected on provided foundation. Incomer and outgoing as per SLD
5.	Supplying and erecting Feeder pillar-5, Feeder pillar shall be outdoor Type (IP-52) with canopy, compartmental cubicle type freestanding with appropriate cable entries, front operating, front maintained wherever required complete with base frame etc. erected in 14 gauge CRCA sheet and surface treated with phosphating seven tank process and duly powder coated with MSEDCL approved colour, feeder pillar should be with supporting angles, self-locks, gasket and slanting top to be erected on provided foundation. Incomer and outgoing as per SLD

Sr. No.	Description of Items
6.	Supplying and erecting Feeder pillar-6, Feeder pillar shall be outdoor Type (IP-52) with canopy, compartmental cubicle type freestanding with appropriate cable entries, front operating, front maintained wherever required complete with base frame etc. erected in 14 gauge CRCA sheet and surface treated with phosphating seven tank process and duly powder coated with MSEDCL approved colour, feeder pillar should be with supporting angles, self-locks, gasket and slanting top to be erected on provided foundation. Incomer and outgoing as per SLD
7.	Supplying and erecting Feeder pillar-7 shall be outdoor Type (IP-52) with canopy, compartmental cubicle type freestanding with appropriate cable entries, front operating, front maintained wherever required complete with base frame etc. erected in 14 gauge CRCA sheet and surface treated with phosphating seven tank process and duly powder coated with MSEDCL approved colour, feeder pillar should be with supporting angles, self-locks, gasket and slanting top to be erected on provided foundation. Incomer and outgoing as per SLD
8.	Supplying and erecting Substation Main feeder pillar-8, Feeder pillar shall be outdoor Type (IP-52) with canopy, compartmental cubicle type freestanding with appropriate cable entries, front operating, front maintained wherever required complete with base frame etc. erected in 14 gauge CRCA sheet and surface treated with phosphating seven tank process and duly powder coated with MSEDCL approved colour, feeder pillar should be with supporting angles, self-locks, gasket and slanting top to be erected on provided foundation. Incomer and outgoing as per S 'D
9.	'Supply, Installation, testing & commissioning of MV panel of DG distribution panel, panel shall be outdoor Type (IP-52) with canopy, compartmental cubicle type freestanding with appropriate cable entries, front operating, front maintained wherever required complete with base frame etc. erected in 14 gauge CRCA sheet and surface treated with phosphating seven tank process and duly powder coated with MSEDCL approved colour, feeder pillar should be with supporting angles, self-locks, gasket and slanting top to be erected on provided foundation Feeder Pillar Details: Name: DG distribution panel Duty: Outdoor with canopy (IP 52)

Sr. No.	Description of Items
	Location: Near DG Incomer and outgoing as per S'D
10.	Providing & erecting 4 Pole MCCB, 415V, 800A, rated short-circuit breaking capacity 50 kA ($I_{cs}=100\%$ of I_{cu}), adjustable thermal (overload) setting and adjustable magnetic setting with provided leads, provision for installation of shunt/UV/trip alarm contact and MCCB should have phase barriers both sides, with insulation withstand capacity 800V, no line-load bias in provided enclosure/panel as per specification no. SW-SWR/MCCB
11.	Providing & erecting 4 Pole MCCB 415V, 630A, rated short-circuit breaking capacity 36 kA ($I_{cs}=100\%$ of I_{cu}), adjustable thermal (overload) setting and adjustable magnetic setting with provided leads, provision for installation of shunt/UV/trip alarm contact and MCCB should have phase barriers both sides, with insulation withstand capacity 800V, no line-load bias in provided enclosure/panel as per specification no. SW-SWR/MCCB
12.	Providing & erecting 4 Pole MCCB 415V, 400A, rated short-circuit breaking capacity 36 kA ($I_{cs}=100\%$ of I_{cu}), adjustable thermal (overload) setting and adjustable magnetic setting with provided leads, provision for installation of shunt/ UV/ trip alarm contact and MCCB should have phase barriers both sides, with insulation withstand capacity 800V, no line-load bias in provided enclosure /Panel as per specification no. SW-SWR/MCCB
13.	Providing & erecting 4 Pole MCCB, 415V, 200A, rated short-circuit breaking capacity 25 kA ($I_{cs}=100\%$ of I_{cu}), adjustable thermal (overload) setting and adjustable magnetic setting with provided leads, provision for installation of shunt/UV/trip alarm contact and MCCB should have phase barriers both sides, with insulation withstand capacity 800V, no line-load bias in provided enclosure/panel as per specification no. SW-SWR/MCCB
14.	Providing & erecting 4 Pole MCCB, 415 V, 100A, rated short-circuit breaking capacity 25 kA ($I_{cs}=100\%$ of I_{cu}), adjustable thermal (overload) setting and fixed magnetic setting with provided leads, provision for installation of shunt/UV/trip alarm contact. MCCB with phase barriers on both sides, insulation withstand capacity 800V, no line-load bias in provided enclosure/panel as per specification no. SW-SWR/MCCB
15.	Supplying and erecting terminal spreaders suitable for 4 pole, above 630A up

Sr. No.	Description of Items
	to 800A MCCB (Set of 8)
16.	Supplying and erecting terminal spreaders suitable for 4 pole, above 400A up to 630A MCCB (Set of 8)
17.	Supplying and erecting terminal spreaders suitable for 4 pole, above 100A up to 250A MCCB (Set of 8)
18.	Supplying and erecting extended rotary handle for MCCB with adjustable telescopic rod suitable above 630A up to 800A MCCB complete.
19.	Supplying and erecting extended rotary handle for MCCB with adjustable telescopic rod suitable for above 250A up to 630A MCCB complete.
20.	Supplying and erecting extended rotary handle for MCCB with adjustable telescopic rod suitable for up to 250A MCCB complete.
21.	Supplying, installing, testing and commissioning air circuit breaker manual fixed type, non-draw out type, 650V, 1000A rated at 50° C without deration, 12kV Impulse, 4 Pole (100% Neutral Protection), 50 kA (Ics=Icu=Icw for 1s),with standard accessories with ready to close, individual fault trip, alarm and trip indication, mechanical and electrical anti-pumping, soft/ hard rating plug (including hard rating plug CT) with Micro-processor based LCD display release with LSIG and temperature rise protection & with 4 C/O contacts for On-Off-Trip indication, min. 5 event and trip history on breaker display on provided iron frame as per specification no. SW-SWR/ACB
22.	Supplying and erecting bus bar chamber triple pole with neutral with four aluminium bars, 500V 630A having 50mm (Width) X 10 mm (Thickness), with 99.9 % aluminium purity as per E91E Grade & Current Density 0.8 A per sq. mm. compliance to IS 5082 and enclosure fabricated with 16 SWG CRCA sheet and epoxy paint, with suitable number of ways and with SMC/DMC Insulators and accessories of suitable rating complete as per specification no. CPBB
23.	Supplying and erecting bus bar chamber triple pole with neutral with four aluminium bars, 500V 800A having 100mm (Width) X 5 mm (Thickness), with 99.9 % aluminium purity as per E91E Grade & Current Density 0.8 A per sq. mm. compliance to IS 5082 and enclosure fabricated with 16 SWG CRCA sheet and epoxy paint, with suitable number of ways and with SMC/DMC Insulators and accessories of suitable rating complete as per specification no.

Sr. No.	Description of Items
	CPBB
24.	Supplying and erecting LT CT multifunction DLMS compliance energy meter of accuracy of class 0.5, 3x240V,50 Hz with optical & RS 232 port, backlit LCD, measure & display Trivector energy, load survey, TOD, tamper detection & logging, power ON /OFF events, instantaneous parameters of rating 1/5 A with display in absence of power complete with zero adjustment & test certificate from manufacturer erected on provided M.S. box and connected to CTs.
25.	Supplying and erecting LT current transformer with bar primary 50/5 to 1000/5 ratio with 15 VA burden erected in provided CRCA box duly secured with insulating materials connected to the meter with test certificates.
26.	Supplying & erecting LED type indicating pilot lamp with LED colours red / yellow / green, 230 / 250V on provided box / panel with necessary material, wiring & connections etc.
27.	Supplying & erecting LED type indicating pilot lamp with LED colour blue, 230 / 250 V on provided box / panel with necessary material, wiring & connections etc.
28.	Supplying and erecting metal work in CRCA sheet including Iron work for supports with fabrication of boxes, panel boards, etc. including cutting, bending, drilling, welding, riveting, treated with anti-rust treatment and duly powder coated or painted with one coat of red lead paint and 2 coats of enamel paint complete.
29.	Supplying, erecting & marking SPMCB 0.5A to 5A, C-series with rated short - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB
B	LT Cable & Accessories
1.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 3½core 400 sq. mm. aluminium conductor complete erected with glands & lugs, on wall/ trusses/pole or laid in provided trench/ pipe as per specification no. CB-LT/AL
2.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 3½ core 300 sq. mm. aluminium conductor complete erected with glands & lugs, on wall/ trusses/pole or laid in provided trench/ pipe as per specification no. CB-LT/AL

Sr. No.	Description of Items
3.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 3½core 240 sq. mm. aluminium conductor complete erected with glands & lugs, on wall/ trusses/pole or laid in provided trench/ pipe as per specification no. CB-LT/AL
4.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 3½ core 185 sq. mm. aluminium conductor complete erected with glands & lugs, on wall/ trusses/pole or laid in provided trench/ pipe as per specification no. CB-LT/AL
5.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 3½ core 95 sq. mm. aluminium conductor complete erected with glands & lugs, on wall/ trusses/pole or laid in provided trench/ pipe as per specification no. CB-LT/AL
6.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 3½ core 70 sq. mm. aluminium conductor complete erected with glands & lugs, on wall/ trusses/pole or laid in provided trench/ pipe as per specification no. CB-LT/AL
7.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 3½ core 35 sq. mm. aluminium conductor complete erected with glands & lugs, on wall/ trusses/pole or laid in provided trench/ pipe as per specification no. CB-LT/AL
8.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 3½ core 25 sq. mm. aluminium conductor complete erected with glands & lugs, on wall/ trusses/pole or laid in provided trench/ pipe as per specification no. CB-LT/AL
C	Civil Work (For Electrical Works)
1.	Supplying & laying (including excavation) reinforced cement concrete pipe of IS 458:2003 NP-2 class of 150 mm internal diameter in proper line, level and slope including providing and fixing collars in cement mortar 1:2 and curing etc complete.
2.	Supplying & laying (including excavation) reinforced cement concrete pipe of IS 458:2003 NP-2 class of 250 mm internal diameter in proper line, level and slope including providing and fixing collars in cement mortar 1:2 and curing etc. complete.

Sr. No.	Description of Items
3.	Making cement concrete foundation including excavation (60 x 60) cm x deep 1/6 of pole length in 1:3:6 cement concrete 20 x 25 mm stone metal, 45 cm x 45 cm x 45 cm /45 cm dia. x 45 cm. height plinth duly plastered, with necessary curing and finishing complete. (For above 9 m poles)
4.	Making trench in soft soil having 0.75 m depth and minimum 0.3 m width for laying provided cables up to voltage level of 1.1kV complete as per specification No. CW-EXN-CTR
5.	Making trench in Hard murum/Tar Road having 0.75 m depth and minimum 0.3mtr width as per IS for laying provided cables up to voltage level 1.1Kv cable complete as per specification No. CW-EXN-CTR
D	Earthing & Accessories
1.	Providing earthing with copper earth plate size 30 x 30 x 0.315 cm complete with all materials, testing & recording the results as per specification no. EA-EP
2.	Providing earthing with galvanized iron earth plate size 60 x 60 x 0.6 cm with funnel with a wire mesh for watering and brick masonry block C.I. cover complete with all materials, testing & recording the results as per specification no. EA-EP
3.	Supplying and erecting copper strip of required size used for earthing on wall and/or any other purpose with necessary copper clamps fixed on wall painted with bituminous paint in an approved manner with joints required. As per specification no. EA-EP.
4.	Supplying and erecting GI strip of required size used for earthing on wall and/or any other purpose with necessary GI clamps fixed on wall painted with bituminous paint in an approved manner with joints required. As per specification no EA-EP.
E	DG Set & Accessories
1.	Supplying, erecting, testing and commissioning of diesel generating set with alternator of 500 kVA output continuous rating, 3 Phase, 415 V, 50c/s 0.8 p. f. A.C supply, a totally enclosed air cooled / liquid cooled multi-cylinder diesel engine developing suitable BHP at 1500 rpm with 10% overload for 1 hour in 12 hours, along with standard accessories, self-excited, self-regulated, screen protected alternator with static excitation system running at 1500 RPM as per IS 4722-2001 with voltage regulation +/- 5 %, with performance class G2 and

Sr. No.	Description of Items
	maximum fuel consumption 81.90 Ltr/hr @75% loading. Both the engine and alternator direct coupled on a common fabricated steel base frame and mounted on anti-vibrating pads with standard control panel comprising meters, switchgears, indicators connected with suitable wires/cables, the complete set enclosed in composite Acoustic enclosure as fully assembled integral unit made of 16 SWG CRCA Sheet, sound absorbing material to restrict sound level up to 75 dB at 1.0 m, provided with first filling of oil, diesel not less than 690 Litre etc. on provided M20 Grade CC Foundation and obtaining necessary approval from Electrical Inspector as per specification no. GEN-DG.
2.	Supplying, erecting, testing and commissioning of microprocessor-based AMF panel suitable for diesel generating set of above 400 kVA up to 500 kVA capacity Three phase, 415 Volts, 50Hz A.C. with all standard features, safeties etc as per specification no. GEN-AMF.
F	Street Light
1.	Making M- 20 grade reinforced cement concrete foundation by considering the safe soil bearing capacity at site as 10 T/Sq.M at 1.5 m depth including supply of steel, concrete, excavation and fixing provided nut bolts with the help of template, duly plastered as per design complete (for 9m & 10m high octagonal /conical GI pole).
2.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 4 core 16 sq. mm. aluminium conductor complete erected with glands & lugs, on wall/ trusses/pole or laid in provided trench/ pipe as per specification no. CB-LT/AL
3.	Supplying and erecting steel tubular swaged pole 165.1 x 139.7 x 114.3 mm dia with 4.5 x 4.5 x 3.65 mm thickness (5+2+2 m) length respectively and total 9 m long with pole cap, base plate in provided foundation (weight 147 kg) as per specification no. OH-PL/STP
4.	Supplying and erecting Street light bracket for erection of Single fitting made from 40 mm. dia 'B' class G.I. pipe 1.0 m in length along with pole cap of 300 mm length 100 mm dia duly welded as per specification no. FG-BKT/BPC
5.	Supplying and erecting integrated LED street light fitting 70-75W IP65 & IK08 class having single piece pressure die-cast aluminium housing, having system lumens output of Min. 7700 Lumens, min. efficacy of 110 lumen/W, CRI>70, CCT up to 6500K, THD<10%, p.f. >0.95, operating range of 140-270V, inbuilt

Sr. No.	Description of Items
	surge protection of 10 kV, Life class of 50,000 Hrs. at L70B50, including driver complete with 3 Years warranty as per specification No FG-ODF/FLS2.
6.	Supplying & erecting three phase, 415V, street light control panel up to the max load of 12kW, TPN MCB 63A, powder coated CRCA 14 SWG sheet, outdoor type, having IP54,IK10 protection, suitable rating contactor, 24 hrs. astronomical time switch with minimum 5 years battery back-up, auto/manual selector switch, ON/OFF push buttons, indicator lamps, control wiring, metering device, etc. for automatic operation, with overcurrent, short circuit, earth fault protection on provided iron frame / CC foundation.
7.	Supplying & erecting FRP box of size 150 x 125 x 100 mm, 2.7 mm thick complete on pole/wall as per specification no. CB-SB
8.	Supplying, erecting & marking FPMCB 40A to 63A, with rated –Short - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB
H	Creche Electrical Work
1.	Supplying & erecting triple pole and neutral distribution board (TPNDB), SP/TP MCBs total 4 ways /12 Poles, with door, 1.2mm thickness surface/flush mounted, IP 43 Protection on iron/GI frame (vertical busbar type) as per specification no. SW-SWR/MCBDB
2.	Supplying, erecting & marking FPMCB 6A to 32A, with rated short - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB
3.	Supplying and erecting ON LOAD four pole automatic transfer switch, 415V, 100-125A, 50 Hz with enclosure, flexibility of neutral connection, line and load reversibility, changeover with phase barriers, source separator, terminal covers along with staggered termination for cable connection and ATS controller with provision of time delay protection of OV, UV, phase sequence changing, single phasing and event logging, connected with provided leads complete complying to IS/IEC 60947-3 Part-1/6 .
4.	Supplying, erecting & marking TPMCB 6A to 32A, with rated short - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB
5.	Supplying, erecting & marking SPMCB 6A to 32A, B-series with rated short -

Sr. No.	Description of Items
	circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB
6.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 3½ core 25 sq. mm. aluminium conductor complete erected with glands & lugs, on wall/ trusses/pole or laid in provided trench/ pipe as per specification no. CB-LT/AL
7.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 4 core 6 sq. mm. copper conductor complete erected with glands & lugs, on wall/ trusses/ pole or laid in provided trench/ pipe as per specification no. CB-LT/CU
I	METERING PANEL BOARD
1.	Supplying and erecting marine (Waterproof) plywood 18/19 mm thick fixed to wall or on provided panel board with necessary materials such as screws, wall fasteners, supports, nut bolts etc. complete with compliance to IS 720:2010
2.	Supplying and erecting static energy Trivector meter 440 V 50 Hz. 20-100A ISI mark meter AC 3 phase 4 wires LCD/LED display (Measuring and display Trivector, V, I, kW, PF, Hz, Last month history of kWh & MD in kW) with special features like TOD, Tamper logging, Optical Communication Port & Display in absence of Power complete erected on polished teak wood board duly wired with required size of weather proof PVC wires duly tested for class 1.0 accuracy by electric supply company.
J	INTERNAL ELECTRIFICATION
1.	LIGHT Point wiring in PVC trunking (casing-capping) with 1.5 sq.mm (2+1E) FRLSH grade copper wire, flush type switch, earthing and required accessories as per specification No: WG-PW/SW
2.	Secondary point wiring for additional light/ bell point in PVC trunking (casing-capping) with 1.5 sq.mm (2+1E) FRLSH grade copper wire with required accessories as per specification No: WG-PW/SW
3.	FAN Point wiring in PVC trunking (casing-capping) with 1.5 sq.mm (2+1E) FRLSH grade copper wire, flush type switch, earthing and required accessories as per specification No: WG-PW/SW
4.	EX.FAN Point wiring in PVC trunking (casing-capping) with 1.5 sq.mm (2+1E) FRLSH grade copper wire, flush type switch, earthing and required accessories

Sr. No.	Description of Items
	as per specification No: WGPW/ SW
5.	Wiring for plug on board with flush Switch socket, copper wiring and earthing as per specification No: WG-PW/SW
6.	Supplying and erecting electronic type step regulator for ceiling fan 600/ 900/ 1200/ 1400 mm. sweep, complete erected on provided switch board.
7.	Supplying and erecting PVC trunking (PVC casing-n-capping) of size 20 mm with accessories on wall/ceiling as per specification No: WG-MA/CON.
8.	Supplying and erecting mains with 2x2.5 sq.mm FRLSH copper PVC insulated wire laid in provided conduit/trunking/inside pole/Bus bars or any other places. as per specification No: WG-MA/BW
9.	Supplying and erecting mains with 1x1.5 sq.mm FRLSH copper PVC insulated wire laid in provided conduit/trunking/inside pole/Bus bars or any other places as per specification No: WG-MA/BW 3) Rates for the earth wire shall be calculated as one-half rate of mains of bunch of two wire.
K	INTERNAL LIGHTING FIXTURE
1.	Supplying and erecting anodized aluminium corridor / passage light LED fitting (4 feet) Max. 22W with high transitivity diffuser with system lumens output of Min.2200 lumens, min. efficacy of 100 lumen/W, CRI>80, CCT upto 6000K, Beam Angle of 110 deg., Ripple<5%, THD<10%, p.f. >0.95, operating range of 200-270V, surge protection of 2 kV, Life class of 50,000 Hrs. at L70B50, including driver, with end caps on provided PVC Block / wooden board with 3 years warranty.
2.	Supplying & erecting 10-12W, 230V integral type LED Lamp suitable for B-22/E-27 base
3.	Supplying and erecting bulk head LED fitting max. 10W with high transitivity diffuser with system lumens output of min. 1100 lumens, min. efficacy of 110 lumen/W, CRI>80, CCT up to 6500K, Beam Angle of 120 deg., THD<10%, p.f. >0.95, operating range of 140-260V, in built surge protection of 2.5 kV, Life class of 50,000 Hrs. at L70B50, including driver, IP66, IK09 rated on provided PVC Block / wooden board with 3 years warranty.
L	LIGHTNING ARRESTER

Sr. No.	Description of Items
1.	Supplying & erecting conventional spike type air termination suitable to carry lightning stroke made up of heavy gauge 40 mm dia copper pipe of standard length with 5 Nos. copper spikes fixed on copper ball as air terminals duly threaded in copper pipe erected on provided foundation complete.
2.	Providing earthing with copper earth plate size 60 x 60 x 0.315 cm with funnel with a wire mesh for watering and brick masonry block C.I. cover with minimum 25 kg Carbon based environment friendly back fill Ground Enhancing compound complete with all materials, testing & recording the results as per specification no ESE -LA
3.	Supplying and erecting aluminium strip of required sizes used for earthing on wall and/or any other purpose with necessary aluminium clamps fixed on wall with insulating and non-corrosive bond with required joints. As per specification no. EA-EP.
M	SOLAR SYSTEM
1.	Supplying, installing, testing and commissioning of 3 kWp ON-GRID Solar PV System / Plant consisting solar PV poly crystalline panel (335 Wp X 09 Nos.) and Solar ON-GRID Inverter of similar capacity (THD<3%) with interconnecting wires / cables up to solar ON GRID inverter, proper base plate, canopy box to house Inverter and necessary accessories i.e. ACDB/ DCDB , MCB etc. wiring up to solar Inverter in casing and capping, display board, sign board, two distinct earthing, lightning arrester, Liasoning charges, remote monitoring system with maintaining card / chip for 5 years including net charges for RMS system, complete up to net meter installation and export to the grid with 5-year ON-SITE warranty and fully comprehensive maintenance contract as per specification no. ESD-RT-ONG
2.	Supplying and erecting AC three phase LT CT multifunction net meter DLMS compliance of accuracy class 0.5, 3 x 240V, 50Hz, with backlit LCD display, optical port & RS 232 port (measures import kWh, kVAh, export kWh, kVAh, Net kWh, kVAh, V, I, kW, 6 months history of energy, load survey, TOD, tamper detection & logging, power ON/OFF events, instantaneous parameters of rating 1/5A with display in absence of power) with wiring connections and mounting hardware on provided panels complete with calibration certificate from manufacturer.
3.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed

Sr. No.	Description of Items
	wire armoured (strip) cable 1100 V, 4 core 16 sq. mm. copper conductor complete erected with glands & lugs, on wall/ trusses/ pole or laid in provided trench/ pipe as per specification no. CB-LT/CU.
4.	Supplying, erecting and marking triple pole isolators only switch version of miniature circuit breaker of 100A with required wiring connections and lugs etc. in provided distribution board.
N	EARTHING & STRIP
1.	Providing earthing with galvanized iron earth plate size 60 x60 x 0.6 cm complete with all materials, testing & recording the results as per specification no. EA-EP
2.	Supplying and erecting GI strip of required size used for earthing on wall and/or any other purpose with necessary GI clamps fixed on wall painted with insulating and non-corrosive bond with required joints As per specification no. EA-EP
3.	Supplying and erecting GI wire of high purity of required sizes used for earthing or any other purposes on wall with necessary G. I. Clamps fixed on wall/cable/ conduit with screws complete.
O	AVIATION LIGHT
1.	Supplying & erecting Aviation obstruction light consisting of yellow painted diecast aluminium alloy housing with Integral LED aviation light comprising of aluminium housing with polycarbonate enclosure complete
2.	Supplying and fixing PVC synthetic elastomer electrically insulating mat with class A insulation conforming to IS: 15652 – 2006 having 2 mm thickness up to 3.3 kV
3.	Providing printed instruction chart for treating persons suffering from electric shock, printed in English & Marathi and duly laminated complete.
4.	Supplying and erecting GI sheet 1.6 mm (16 SWG) caution board of size 200 x 150 mm duly painted with red oxide and enamel paint in Marathi, Hindi and English erected at position for medium voltage (415 Volt) installation with compliance to IS 2551-1982
5.	Supplying and laying (including excavation of suitable width & depth up to 90 cm) 90 mm outside dia. double wall corrugated pipes (DWC) of HDPE for enclosing cable below ground/road surface, to required depth complete.

Sr. No.	Description of Items
P	WATER PUMP
1.	Supplying and erecting minimum Three & above star rated submersible pump set suitable for erection on open well of 2.25 KW/3 HP with 415 V 50 c/s. A.C. supply, having delivery head from 27 to 9 m and discharge from 300 to 660 litres. per minute & pipe of Suction 65mm /Delivery 50 mm diameter erected with necessary H type clamps as per specification No. WP-OSP DOMESTIC PUMP
2.	Supplying and erecting minimum Three & above star rated submersible pump set suitable for erection on open well of 2.25 KW/3 HP with 415 V 50 c/s. A.C. supply, having delivery head from 27 to 9 m and discharge from 300 to 660 litres. per minute & pipe of Suction 65mm /Delivery 50 mm diameter erected with necessary H type clamps as per specification No. WP-OSP FLUSHING PUMP
3.	Supplying & erecting flat flexible 3 core 2.5 sq.mm PVC sheathed submersible type copper cable suitably clamped at fixed intervals with column pipe assembly complete.
4.	Supplying, erecting & marking FPMCB 6A to 32A, with rated short - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB
5.	Supplying, fixing and commissioning 4 pole RCBO 6/10/16/20/25A, with overcurrent, rated short-circuit breaking capacity (10 kA) and earth leakage protection, with 30/100/300 mA sensitivity and short circuit and earth leakage trip indications per specification no SW-RCCB/RCBO
6.	Supplying and erecting minimum Three & above star rated submersible pump set suitable for erection on open well of 2.25 KW/3 HP with 415 V 50 c/s. A.C. supply, having delivery head from 27 to 9 m and discharge from 300 to 660 litres. per minute & pipe of Suction 65mm /Delivery 50 mm diameter erected with necessary H type clamps as per specification No. WP-OSP (FOR TRANSFER OF RAIN WATER)
7.	Supplying & erecting D.O.L Starter suitable for 415 V, 3 phase 50 cycles with no volt coil and overload element with necessary materials and connected to supply up to 7.5 H.P. (FOR TRANSFER OF RAIN WATER, DOMESTIC PUMP & FLUSHING PUMP)

Sr. No.	Description of Items
Q	FIRE ALARM SYSTEM
1.	Supplying, erecting testing and commissioning manual call point (Pill box) with break glass, push button (resettable type) in metal enclosure complete as per specification no. FF-FAAS/MCP
2.	Supplying, erecting, testing and commissioning hooters having high (100dB @1m) and low (94dB @ 1m) volume setting, group addressing facility allowing multiple sounders to be activated with CRCA enclosure complete as per specification no. FF-FAAS/HTR
3.	Supplying, erecting testing and commissioning optical type smoke detector on suitable back box with necessary connections complete as per specification no. FF-FAS/SD
4.	Supplying, installing, testing and commissioning of 2 Zones Microprocessor based conventional fire alarm control panel (FACP) with standard accessories, 16x2 Character LCD Display, provision for zone wise contact and beep sound alarm, suitable to operate on 120-220 V –C, 0 - 49 Deg C, 93 ± 2 Percentage RH (non- condensing) at 32 ± 2 Deg C complete as per specification no. FF-FAAS/FACP
5.	Supplying, installing, testing and commissioning FR, XLPE armoured cable 2 core 1.5 sq.mm. copper conductor complete erected on wall/ ceiling complete as per specification no. CBLT/CU
R	PUBLIC ADDRESS SYSTEM
1.	Supplying, erecting, testing and commissioning 120 W amplifier suitable to operate on 230 Volts A.C. / 12 Volts D.C. supply complete as per specification no. FF-PA/AFR
2.	Supplying, erecting, testing and commissioning 15 to 30 Watts sound column or ceiling mounted complete as per specification no. FF-A/SOC
3.	Supplying and fixing 6U wall mount rack (Dimension-DxWxH – 500x600x367mm) as per specification No. WG-NAS/RAK
4.	Supplying, erecting, testing and commissioning rust free microphone floor stand of chrome plating/powder coating or S.S. material, 500 mm to 1500 mm height and sturdy mechanism ensuring trouble free movement complete.
5.	Supplying, erecting, testing and commissioning stand mounted / handheld type wired microphone having 2 mV/Pa sensitivity, impedance level limit 300 ohms

Sr. No.	Description of Items
	a-d 50 - 16000 Hz suitable to operate in -10 deg C to +55 deg C with min. 7 metre shielded cable and 3 Pin Professional XLR Connector, stereo jack complete as per specification no. FF-PA/MIC
6.	Supplying, erecting, testing and commissioning of 0.5 sq. mm speaker wire complete.
7.	Supplying and erecting HMS PVC conduit FRLS grade 20 mm dia. With necessary accessories in wall/floor with chiselling appropriately as per specification No: WG-MA/CC.
T	Amenity Building Electrical Work
1.	Supplying & erecting triple pole and neutral distribution board (TPNDB), 3 pole/4 pole MCCB as incomer & outgoing SP MCB (24 poles) or TP MCB of 8 ways (24 poles), with door, 1.2mm thickness, surface/flush mounted, on iron/GI frame (vertical busbar type) as per specification no. SW-SWR/MCBDB1
2.	Providing & erecting 3 Pole MCCB, 415 V, 100A, rated short-circuit breaking capacity 25 kA (Ics=100% of Icu), adjustable thermal (overload) setting and fixed magnetic setting with provided leads, provision for installation of shunt/UV/trip alarm contact. MCCB with phase barriers on both sides, insulation withstand capacity 800V, no line-load bias in provided enclosure/panel as per specification no. SW-SWR/MCCB
3.	Supplying and erecting ON LOAD four pole automatic transfer switch, 415V, 100-125A, 50 Hz with enclosure, flexibility of neutral connection, line and load reversibility, changeover with phase barriers, source separator, terminal covers along with staggered termination for cable connection and ATS controller with provision of time delay protection of OV, UV, phase sequence changing, single phasing and event logging, connected with provided leads complete complying to IS/IEC 60947-3 Part-1/6.
4.	Supplying, erecting & marking TPMCB 6A to 32A, with rated short - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB
5.	Supplying, erecting & marking SPMCB 6A to 32A, B- series (for lighting) in provided distribution board as per specification no. SW-SWR/MCB
6.	Supplying, erecting & terminating FR XLPE insulated, galvanized steel formed wire armoured (strip) cable 1100 V, 3½ core 35 sq. mm. aluminum conductor

Sr. No.	Description of Items
	complete erected with glands & lugs, on wall/ trusses/pole or laid in provided trench/ pipe as per specification no. CB-LT/AL
7.	Supplying, erecting & terminating FR XLPE insulated, galvanized steel formed wire armoured (strip) cable 1100 V, 4 core 6 sq. mm. copper conductor complete erected with glands & lugs, on wall/ trusses/ pole or laid in provided trench/ pipe as per specification no. CB-LT/CU
8.	Supplying, erecting & terminating FR XLPE insulated, galvanized steel formed wire armoured (strip) cable 1100 V, 3 core 2.5 sq. mm. copper conductor complete erected with glands & lugs, on wall/ trusses/ pole or laid in provided trench/ pipe as per specification no. CB-LT/CU
U	METERING PANEL BOARD FOR SHOPS
1.	Providing & erecting 3 Pole MCCB of 200A,415V capacity with short circuit rating 25 kA (Ics=100% of Icu) adjustable thermal (overload) setting and fixed magnetic setting with provided leads, provision for installation of shunt/ UV/ trip alarm contact and MCCB should have phase barriers both sides, in provided enclosure on iron /GI frame as per specification no. SW-SWR/MCCB
2.	Providing & erecting 4 Pole MCCB, 415 V, 100A, rated short-circuit breaking capacity 25 kA (Ics=100% of Icu), adjustable thermal (overload) setting and fixed magnetic setting with provided leads, provision for installation of shunt/UV/trip alarm contact. MCCB with phase barriers on both sides, insulation withstand capacity 800V, no line-load bias in provided enclosure/panel as per specification no. SW-SWR/MCCB
3.	Supplying and erecting bus bar chamber triple pole with neutral with four aluminum bars, 500V 200A having 25mm (Width) X 5 mm (Thickness) , with 99.9 % aluminum purity as per E91E Grade & Current Density 0.8 A per sq. mm. compliance to IS 5082 and Enclosure fabricated with 16 SWG CRCA sheet and epoxy paint, with suitable number of ways and with SMC/DMC Insulators and accessories of suitable rating complete as per specification no. CPBB
4.	Supplying, erecting & marking DPMCB 6A to 32A, C-series with rated short - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB
5.	Supplying, erecting & marking DPMCB 40A to 63A, C-series with rated –hort - circuit breaking capacity (Icn) 10kA in provided distribution board as per

Sr. No.	Description of Items
	specification no. SW-SWR/MCB
6.	Supplying erecting and marking double pole isolator only switch version of miniature circuit breaker of 40A with required wiring connections and lugs etc. in provided distribution board complete.
7.	Supplying & erecting CRCA sheet metal one-way enclosures of 1.2mm thickness suitable for DP MCB /TP MCB/FP MCB / RCCB / RCBO complete erected on angle iron/GI frame.
8.	Supplying & erecting mains with 2x4 sq.mm FRLSH copper PVC insulated wire laid in provided conduit/trunking/inside pole/Bus bars or any other places. as per specification No: WGMA/BW
9.	Supplying & erecting mains with 1x2.5 sq.mm FRLSH copper PVC insulated wire laid in provided conduit/trunking/inside pole/Bus bars or any other places. as per specification No: WGMA/BW–Note - 3) Rates for the earth wire shall be calculated as one-half rate of mains of bunch of two wire.
10.	Supplying and erecting 25 mm dia. corrugated flexible polypropylene conduit
11.	Supplying and erecting marine (Waterproof) plywood 18/19 mm thick fixed to wall or on provided panel board with necessary materials such as screws, wall fasteners, supports, nut bolts etc. complete with compliance to IS 720:2010
12.	Supplying and erecting static energy Trivector meter 440 V 50 Hz. 20-100A ISI mark meter AC 3 phase 4 wires LCD/LED display (Measuring and display Trivector, V, I, kW, PF, Hz, Last month history of kWh & MD in kW) with special features like TOD, Tamper logging, Optical Communication Port & Display in absence of Power complete erected on polished teak wood board duly wired with required size of weather proof PVC wires duly tested for class 1.0 accuracy by electric supply company.
V	LDB WITH ACCESSORIES
1.	Supplying, erecting & marking SPMCB 6A to 32A, B-series with rated –hort - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB
2.	Supplying, erecting & marking SPMCB 6A to 32A, C-series with rated –hort - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB

Sr. No.	Description of Items
3.	Supplying, erecting & commissioning 2 pole RCCB 16/25A, electromagnetic type with 30/100/300 mA sensitivity with earth leakage trip indication complete as per specification no. SW-RCCB/RCCB
4.	Supplying, erecting & commissioning 2 pole RCCB 40A, electromagnetic type with 30/100/300 mA sensitivity with earth leakage trip indication complete as per specification no. SW-RCCB/RCCB
5.	Supplying and erecting single pole and neutral distribution board (SPNDB), with 2 ways for incoming and 6 ways (6 poles) for outgoing SP MCBs, with door, 1.2mm thickness surface / flush mounted, IP 43 Protection on iron / GI frame as per specification no. SW-SWR/MCBDB
6.	Supplying and erecting single pole and neutral distribution board (SPNDB), with 2 ways for incoming and 10 ways (10 poles) for outgoing SP MCBs, with door, 1.2mm thickness surface / flush mounted, IP 43 Protection on iron / GI frame as per specification no. SW-SWR/MCBDB
7.	Supplying, erecting & marking DPMCB 6A to 32A, C-series with rated –Short - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB
8.	Supplying & erecting CRCA sheet metal one-way enclosures of 1.2mm thickness suitable for DP MCB /TP MCB/FP MCB / RCCB / RCBO complete erected on angle iron/GI frame.
W	INTERNAL ELECTRIFICATION
1.	Supplying & erecting mains with 2x6 sq.mm FRLSH copper PVC insulated wire laid in provided conduit/trunking/inside pole/Bus bars or any other places. as per specification No: WG-MA/ BW
2.	Supplying & erecting mains with 1x4 sq.mm FRLSH copper PVC insulated wire laid in provided conduit/trunking/inside pole/Bus bars or any other places. as per specification No: WG-MA/ BW–Note - 3) Rates for the earth wire shall be calculated as one-half rate of mains of bunch of two wire.
3.	Supplying & erecting mains with 2x4 sq.mm FRLSH copper PVC insulated wire laid in provided conduit/trunking/inside pole/Bus bars or any other places. as per specification No: WGMA/BW
4.	Supplying & erecting mains with 1x2.5 sq.mm FRLSH copper PVC insulated wire laid in provided conduit/trunking/inside pole/Bus bars or any other places.

Sr. No.	Description of Items
	as per specification No: WGMA/BW–Note - 3) Rates for the earth wire shall be calculated as one-half rate of mains of bunch of two wire.
5.	Supplying and laying HMS PVC conduit FRLS grade 20 mm dia. With necessary accessories in RCC work/false ceiling/false flooring as per specification No. WG-MA/CC.
6.	LIGHT Point wiring in PVC trunking (casing-capping) with 1.5 sq.mm (2+1E) FRLSH grade copper wire, flush type switch, earthing and required accessories as per specification No: WG-PW/SW
7.	Secondary point wiring for additional light/ bell point in PVC trunking (casing-capping) with 1.5 sq.mm (2+1E) FRLSH grade copper wire with required accessories as per specification No: WG-PW/SW
8.	FAN Point wiring in PVC trunking (casing-capping) with 1.5 sq.mm (2+1E) FRLSH grade copper wire, flush type switch, earthing and required accessories as per specification No: WG-PW/SW
9.	EX.FAN Point wiring in PVC trunking (casing-capping) with 1.5 sq.mm (2+1E) FRLSH grade copper wire, flush type switch, earthing and required accessories as per specification No: WGPW/ SW
10.	Wiring for plug on board with flush Switch socket, copper wiring and earthing as per specification No: WG-PW/SW
11.	Supplying and erecting flush, shock proof plug socket 6 pin 6A and 16A with switch 16A combined unit erected in recess in 16-gauge M.S sheet box duly painted and connected with earth terminal and leads
12.	Supplying and erecting electronic type step regulator for ceiling fan 600/ 900/ 1200/ 1400 mm. sweep, complete erected on provided switch board.
13.	Supplying and erecting PVC trunking (PVC casing-n-capping) of size 20 mm with accessories on wall/ceiling as per specification No: WG-MA/CON.
14.	Supplying and erecting mains with 2x2.5 sq.mm FRLSH copper PVC insulated wire laid in provided conduit/trunking/inside pole/Bus bars or any other places. as per specification No: WG-MA/BW
15.	Supplying and erecting mains with 1x1.5 sq.mm FRLSH copper PVC insulated wire laid in provided conduit/trunking/inside pole/Bus bars or any other places as per specification No: WG-MA/BW 3) Rates for the earth wire shall be calculated as one-half rate of mains of

Sr. No.	Description of Items
	bunch of two wire.
X	INTERNAL LIGHTING FIXTURE
1.	Supplying and erecting anodized aluminum corridor / passage light LED fitting (4 feet) Max. 22W with high transitivity diffuser with system lumens output of Min.2200 lumens, min. efficacy of 100 lumen/W, CRI>80, CCT up to 6000K, Beam Angle of 110 deg., Ripple<5%, THD<10%, p.f. >0.95, operating range of 200-270V, surge protection of 2 kV, Life class of 50,000 Hrs. at L70B50, including driver, with end caps on provided PVC Block / wooden board with 3 years warranty.
2.	Supplying & erecting 10-12W, 230V integral type LED Lamp suitable for B-22/E-27 base
3.	Supplying and erecting bulk head LED fitting max. 10W with high transitivity diffuser with system lumens output of min. 1100 lumens, min. efficacy of 110 lumen/W, CRI>80, CCT up to 6500K, Beam Angle of 120 deg., THD<10%, p.f. >0.95, operating range of 140-260V, in built surge protection of 2.5 kV, Life class of 50,000 Hrs. at L70B50, including driver, IP66, IK09 rated on provided PVC Block / wooden board with 3 years warranty.
Y	LIGHTNING ARRESTER
1.	Supplying & erecting conventional spike type air termination suitable to carry lightning stroke made up of heavy gauge 40 mm dia copper pipe of standard length with 5 Nos. copper spikes fixed on copper ball as air terminals duly threaded in copper pipe erected on provided foundation complete.
2.	Supplying, installation and testing of maintenance free earthing comprising of Electrode of 17.2 mm diameter Low Carbon Steel with 250-micron molecular copper bonded earthing rod of Length 3m along with 25 kg Carbon based environment friendly back fill ground enhancing compound required to fill up the excavated earth with required quantity complete, and recording the results Specification no. EA-MOBI
3.	Supplying and erecting aluminum strip of required sizes used for earthing on wall and/or any other purpose with necessary aluminum clamps fixed on wall with insulating and non-corrosive bond with required joints. As per specification no. EA-EP.

Sr. No.	Description of Items
Z	SOLAR SYSTEM
1.	Supplying, installing, testing and commissioning of 3 kWp ON-GRID Solar PV System / Plant consisting Solar PV Poly Crystalline Panel (335 Wp X 9 Nos.) and Solar ON-GRID Inverter of similar capacity (THD<3%) with interconnecting wires / cables up to solar ON GRID inverter, proper base plate, canopy box to house Inverter and necessary accessories i.e. ACDB/ DCDB , MCB etc. wiring up to solar Inverter in casing and capping, display board, sign board, two distinct earthing, lightning arrester, Liasoning charges, remote monitoring system with maintaining card/chip for 5 years including net charges for RMS system, complete up to net meter installation and export to the grid with 5 year ON-SITE warranty and fully comprehensive maintenance contract as per Specification no. ESD-RT-ONG
2.	Supplying and erecting AC three phase LT CT multifunction net meter DLMS compliance of accuracy class 0.5, 3 x 240V, 50Hz, with backlit LCD display, optical port & RS 232 port (measures import kWh, kVAh, export kWh, kVAh, Net kWh, kVAh, V, I, kW, 6 months history of energy, load survey, TOD, tamper detection & logging, power ON/OFF events, instantaneous parameters of rating 1/5A with display in absence of power) with wiring connections and mounting hardware on provided panels complete with calibration certificate from manufacturer.
3.	Supplying, erecting & terminating FR XLPE insulated, galvanized steel formed wire armoured (strip) cable 1100 V, 4 core 16 sq. mm. copper conductor complete erected with glands & lugs, on wall/ trusses/ pole or laid in provided trench/ pipe as per specification no. CB-LT/CU.
4.	Supplying, erecting and marking triple pole isolators only switch version of miniature circuit breaker of 100A with required wiring connections and lugs etc. in provided distribution board.
AA	EARTHING & STRIP
1.	Providing earthing with galvanized iron earth plate size 60 x60 x 0.6 cm complete with all materials, testing & recording the results as per specification no. EA-EP
2.	Supplying and erecting GI strip of required size used for earthing on wall and/or any other purpose with necessary GI clamps fixed on wall painted with insulating and non-corrosive bond with required joints As per specification no.

Sr. No.	Description of Items
	EA-EP
3.	Supplying and erecting GI wire of high purity of required sizes used for earthing or any other purposes on wall with necessary G. I. Clamps fixed on wall/cable/ conduit with screws complete.
AB	AVIATION LIGHT
1.	Supplying & erecting Aviation obstruction light consisting of yellow painted diecast aluminum alloy housing with Integral LED aviation light comprising of aluminum housing with polycarbonate enclosure complete.
2.	Supplying and fixing PVC synthetic elastomer electrically insulating mat with class A insulation conforming to IS: 15652 – 2006 having 2 mm thickness up to 3.3 kV
3.	Providing printed instruction chart for treating persons suffering from electric shock, printed in English & Marathi and duly laminated complete.
4.	Supplying and erecting GI sheet 1.6 mm (16 SWG) caution board of size 200 x 150 mm duly painted with red oxide and enamel paint in Marathi, Hindi and English erected at position for medium voltage (415 Volt) installation with compliance to IS 2551-1982
5.	Supplying and laying (including excavation of suitable width & depth up to 90 cm) 90 mm outside dia. double wall corrugated pipes (DWC) of HDPE for enclosing cable below ground/road surface, to required depth complete.
AC	LIFT
1.	SITC of Electric Traction Passenger Lift with· Rated capacity :- 8 Passenger/544Kg· Floors :- G+1 floor (2 Stops/2 Landings)· Travel :- 3 to 4.2 mtrs· Location of Lift Machine:- MR· Rated speed :- 1.0 mps VS· Car/Landing door clear opening of 800 mm wide x 2000 mm high· Clear Car si©ofmm wide xmm deep x 2250mm high· Lift shaft available having clear©ze of©m wide xmm deep,.....mm Pit depth,mm Overhead· Doors type :- COPO/TOPO Doors with frame made from SS 304 grade solid(non-cladded) sheet of 1.5mm thick in hairline finish for car and all landing doors with SS door architraves/frames The lift doors shall have minimum 1 hour fire rating (with submission of necessary valid test certificate issued by NABL accredited or Independent test laboratory).· Lift car enclosure made from SS

Sr. No.	Description of Items
	304 grade solid (non-cladded) sheet of 1.5mm, thick with hairline finish with frame made from MS girders, bracing of adequate size with minimum safety factor of 5, with Toe Guard Apron, with necessary false ceiling with adequate LED lights, blower/fan for ventilation & SS Chequered plate flooring, handrails, mirror, emergency light etc. The lift car interior design shall be done as per the directions of engineer in charge.· COP with SS face plate having metallic push buttons with Braille code & luminous indicator around button with FPI, scrolling UP/DN LED indicator & with / without attendant key switch, OWD with audio-visual alarm, VAS in Marathi, Hindi & English with intercom system with telephone instrument in Lift car, LMR & FCC/ground floor.· LOP with SS face plate having recess/surface push button box for all landings with scrolling UP/DN LED indicator having metallic push buttons with Braille code & luminous indicator around button with CPI, Lift car arrival & next travel direction audio-visual indication at all landings.· Lift controller based on microprocessor/ PLC with VVVF Drive having closed loop control system, with IBMS compatible having necessary port. The controller shall have necessary protections such as overcurrent, overvoltage, over speed, overheat for all devices including lift motor and travel direction protection, protection for phase loss at input or output etc. various operation modes such as maintenance, change direction by attendant, full load bypass, VIP call operation, self levelling operation, call cancel facility, auto return to home landing, fireman evacuation operation, earthquake operation, door open/close operation from COP etc. The system can record information for minimum 30 latest faults, with real-time clock management and handheld keypad with LCD screen for view and setting of parameters. The control panel duly wired with proper size & strength, copper wire for power & control circuit, with provision for addition of floor/control card & allied accessories control panel having enclosure of 1.5mm CRCA sheet with powder coating with IP54 Protection class.· ARD complete with necessary SMF VRLA batteries.· Fireman controller having fireman switch at fire Landing.· CCTV surveillance system comprises of 2 no's minimum 2.0 MP FHD IP based vandal proof Dome camera in lift car & in LMR/inside lift shaft top aimed on Lift machinery & controller with NVR kept in LMR/FCC with HDR data backup min. 90 days with min. 18" FHD TV monitor, to be kept in FCC/LMR as directed by Engineer In Charge.· Lift Machine of Gearless PMSM of suitable kW with duty cycle of minimum 120 starts/hr (with submission of necessary valid test certificate issued by NABL accredited or Independent test

Sr. No.	Description of Items
	laboratory), with Traction pulley, OSG, electromagnetic brakes, entire assembly mounted on adequate size girders dulyfixed on LMR floor/ shaft walls complete with main/diverter traction sheaves, suspension wire ropes/belts of adequate size & strength. Other mechanical 'parts such as 'T' section adequate size guide rails for car & counter weight with brackets fasteners, counter weight frame with necessary blocks, buffers with necessary support arrangement, MS pit ladder etc. erected with necessary steel work. Minor civil work such as alteration work if any necessary for erection of landing door frames and it's accessories e.g. sill, header, hole pass etc. complete with plaster finish, civil work for erection buffers, erection of lift machinery, adequate size core cuts if required & scaffolding for erecting guide rails, providing and fixing steel girders/RCC work having adequate strength for mounting andhoisting lift machine etc. complete as per specification no. LFT..
2.	For Lift having 800mm wide x 2000mm high clear entrance in 1.5mm thick SS 304 grade solid(non-cladded) landing door [Two panel], this includes all necessary accessories like LOP's with UP/DN buttons-arrows-indicators, extension of guide rails, shaft wiring with trunking, travelling cables, main hoisting ropes/belts, & OSG rope, landing doors with all accessories etc. complete for the safe functioning of lift installation.
3.	Supplying and erecting single pole and neutral distribution board (SPNDB), with 2 ways for incoming and 6 ways (6 poles) for outgoing SP MCBs, with door, 1.2mm thickness surface / flush mounted, IP 43 Protection on iron / GI frame as per specification no. SW-SWR/MCBDB.
4.	Supplying, erecting & marking DPMCB 6 A to 32 A, B-series with rated short - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB.
5.	Supplying, erecting & marking SPMCB 6A to 32A, C-series (for motor/power) in provided distribution board as per specification no. SW-SWR/MCB.
6.	Point wiring in for light/fan/bell in 20 mm metal conduit with 1.5 sq.mm (2+1E) FRLSH grade copper wire, flush type switch, earthing and required accessories as per specification No: WG-PW/SW.
7.	Supplying and erecting bulk head LED fitting max. 10W with high transitivity diffuser with system lumens output of min. 1100 lumens, min. efficacy of 110 lumen/W, CRI>80, CCT up to 6500K, Beam Angle of 120 deg., THD<10%, p.f.

Sr. No.	Description of Items
	>0.95, operating range of 140-260V, in built surge protection of 2.5 kV, Life class of 50,000 Hrs. at L70B50, including driver, IP66, IK09 rated on provided PVC Block / wooden board with 3 years warranty.
AC	WATER PUMP
1.	Supplying and erecting minimum Three & above star rated submersible pump set suitable for erection on open well of 2.25 KW/3 HP with 415 V 50 c/s. A.C. supply, having delivery head from 27 to 9 m and discharge from 300 To 660 liters. per minute & pipe of Suction 65mm /Delivery 50 mm diameter erected with necessary H type clamps as per specification No. WP-OSP DOMESTIC PUMP
2.	Supplying and erecting minimum Three & above star rated submersible pump set suitable for erection on open well of 2.25 KW/3 HP with 415 V 50 c/s. A.C. supply, having delivery head from 27 to 9 m and discharge from 300 To 660 liters. per minute & pipe of Suction 65mm /Delivery 50 mm diameter erected with necessary H type clamps as per specification No. WP-OSP FLUSHING PUMP
3.	Supplying & erecting flat flexible 3 core 2.5 sq.mm PVC sheathed submersible type copper cable suitably clamped at fixed intervals with column pipe assembly complete.
4.	Supplying and erecting flat flexible 3 core 4 sq mm PVC sheathed submersible type copper cable suitably clamped at fixed intervals with column pipe assembly complete.
5.	Providing & erecting 4 Pole MCCB, 415 V, 100A, rated short-circuit breaking capacity 25 kA (Ics=100% of Icu), adjustable thermal (overload) setting and fixed magnetic setting with provided leads, provision for installation of shunt/UV/trip alarm contact. MCCB with phase barriers on both sides, insulation withstand capacity 800V, no line-load bias in provided enclosure/panel as per specification no. SW-SWR/MCCB
6.	Supplying, fixing and commissioning 4 pole RCBO 6/10/16/20/25A, with overcurrent, rated short-circuit breaking capacity (10 kA) and earth leakage protection, with 30/100/300 mA sensitivity and short circuit and earth leakage trip indications per specification no SW-RCCB/RCBO
7.	Supplying and erecting minimum Three & above star rated submersible pump set suitable for erection on open well of 2.25 KW/3 HP with 415 V 50 c/s. A.C.

Sr. No.	Description of Items
	supply, having delivery head from 27 to 9 m and discharge from 300 To 660 liters. per minute & pipe of Suction 65mm /Delivery 50 mm diameter erected with necessary H type clamps as per specification No. WP-OSP (FOR TRANFER OF RAIN WATER)
8.	Supplying & erecting Fully Automatic Star/delta starter suitable for 12.5 H. P to 15 H.P. for 3 phase squirrel cage motor totally enclosed with overload and no volt relay complete erected on provided angle iron frame. (STARTER FOR BOOSTER PUMP STARTER)
9.	Supplying & erecting D.O.L Starter suitable for 415 V, 3 phase 50 cycles with no volt coil and overload element with necessary materials and connected to supply up to 7.5 H.P. (FOR TRANFER OF RAIN WATER, DOMESTIC PUMP & FLUSHING PUMP)
10.	Supplying, installation, testing and commissioning of booster fire pump [BP] suitable for water discharge of 900 LPM at 35 m head driven by electric motor 415 volts, 3 phase 50 Hz, AC supply of 7.5 kW or of suitable kW capacity for manual/automatic operation and consisting of following: (a) Horizontal/vertical type, single/multi stage, centrifugal casing pump of cast iron body & bronze/CI impeller with stainless steel shaft (SS410 grade), mechanical seal conforming to IS 1520. (b) Squirrel cage induction motor, TEFC, synchronous speed 3000 RPM, suitable for operation on with IP 55 protection for enclosure, horizontal foot mounted t'p' with Class-'F' insulation, conforming to IS-325. (c) M.S. fabricated common base plate, coupling, coupling guard, foundation bolts etc. as required. (d)Erected on provided suitable size cement concrete foundation duly plastered with anti-vibration pads with perfect aligning, proper levelling complete pump set with accessories duly painted with two coats of synthetic enamel paint of fire red colour over a coat of primer (ISC code 536 as per IS 2932 of 2003) complete, as per specification no. FF-MFP/BP
AD	FIRE ALARM SYSTEM
1.	Supplying, erecting testing and commissioning manual call point (Pill box) with break glass, push button (resettable type) in metal enclosure complete as per specification no. FF-FAAS/MCP
2.	Supplying, erecting, testing and commissioning hooters having high (100dB @1m) and low (94dB @ 1m) volume setting, group addressing facility allowing

Sr. No.	Description of Items
	multiple sounders to be activated with CRCA enclosure complete as per specification no. FF-FAAS/HTR
3.	Supplying, erecting testing and commissioning heat detector with combination "ROR" and Fixed Temp." complete as per specification no. FF-FAS/HD
4.	Supplying, erecting testing and commissioning optical type smoke detector on suitable back box with necessary connections complete as per specification no. FF-FAS/SD
5.	Supplying, installing, testing and commissioning of 2 Zones Microprocessor based conventional fire alarm control panel (FACP) with standard accessories, 16x2 Character LCD Display, provision for zone wise contact and beep sound alarm, suitable to operate on 1–0-220 V AC, 0 - 49 Deg C, 93 ± 2 Percentage RH (non- condensing) at 32 ± 2 Deg C complete as per specification no. FF-FAAS/FACP
6.	Supplying, installing, testing and commissioning FR, XLPE armoured cable 2 core 1.5 sq.mm. copper conductor complete erected on wall/ ceiling complete as per specification no. CBLT/CU
AE	PUBLIC ADDRESS SYSTEM
1.	Supplying, erecting, testing and commissioning 120 W amplifier suitable to operate on 230 Volts A.C. / 12 Volts D.C. supply complete as per specification no. FF-PA/AFR
2.	Supplying, erecting, testing and commissioning 15 to 30 Watts sound column or ceiling mounted complete as per specification no. FF-A/SOC
3.	Supplying and fixing 6U wall mount rack (Dimension-D x W x H – 500x600x367mm) as per specification No. WG-NAS/RAK
4.	Supplying, erecting, testing and commissioning rust free microphone floor stand of chrome plating/powder coating or S.S. material, 500 mm to 1500 mm height and sturdy mechanism ensuring trouble free movement complete.
5.	Supplying, erecting, testing and commissioning stand mounted / handheld type wired microphone having 2 mV/Pa sensitivity, impedance level limit 3–0 ohms and 50 - 16000 Hz suitable to operate in -10 deg C to +55 deg C with min. 7 metre shielded cable and 3 Pin Professional XLR Connector, stereo jack complete as per specification no. FF-PA/MIC

Sr. No.	Description of Items
6.	Supplying, erecting, testing and commissioning of 0.5 sq. mm speaker wire complete.
7.	Supplying and erecting HMS PVC conduit FRLS grade 20 mm dia. With necessary accessories in wall/floor with chiselling appropriately as per specification No: WG-MA/CC.

AS	Fitness Centre/Creche Electrical Work
1.	Supplying & erecting triple pole and neutral distribution board (TPNDB), SP/TP MCBs total 4 ways /12 Poles, with door, 1.2mm thickness surface/flush mounted, IP 43 Protection on iron/GI frame (vertical busbar type) as per specification no. SW-SWR/MCBDB
2.	Supplying, erecting & marking FPMCB 6A to 32A, with rated short - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB
3.	Supplying and erecting ON LOAD four pole automatic transfer switch, 415V, 100-125A, 50 Hz with enclosure, flexibility of neutral connection, line and load reversibility, changeover with phase barriers, source separator, terminal covers along with staggered termination for cable connection and ATS controller with provision of time delay protection of OV, UV, phase sequence changing, single phasing and event logging, connected with provided leads complete complying to IS/IEC 60947-3 Part-1/6 .
4.	Supplying, erecting & marking TPMCB 6A to 32A, with rated short - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB
5.	Supplying, erecting & marking SPMCB 6A to 32A, B-series with rated short - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB
6.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 3½ core 25 sq. mm. aluminium conductor complete erected with glands & lugs, on wall/ trusses/pole or laid in provided trench/ pipe as per specification no. CB-LT/AL.
7.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 4 core 6 sq. mm. copper

	conductor complete erected with glands & lugs, on wall/ trusses/ pole or laid in provided trench/ pipe as per specification no. CB-LT/CU
AT	METERING PANEL BOARD
1.	Supplying and erecting marine (Waterproof) plywood 18/19 mm thick fixed to wall or on provided panel board with necessary materials such as screws, wall fasteners, supports, nut bolts etc. complete with compliance to IS 720:2010
2.	Supplying and erecting static energy Trivector meter 440 V 50 Hz. 20-100A ISI mark meter AC 3 phase 4 wires LCD/LED display (Measuring and display Trivector, V, I, kW, PF, Hz, Last month history of kWh & MD in kW) with special features like TOD, Tamper logging, Optical Communication Port & Display in absence of Power complete erected on polished teak wood board duly wired with required size of weather proof PVC wires duly tested for class 1.0 accuracy by electric supply company.
AU	INTERNAL ELECTRIFICATION
1.	LIGHT Point wiring in PVC trunking (casing-capping) with 1.5 sq.mm (2+1E) FRLSH grade copper wire, flush type switch, earthing and required accessories as per specification No: WG-PW/SW
2.	Secondary point wiring for additional light/ bell point in PVC trunking (casing-capping) with 1.5 sq.mm (2+1E) FRLSH grade copper wire with required accessories as per specification No: WG-PW/SW
3.	FAN Point wiring in PVC trunking (casing-capping) with 1.5 sq.mm (2+1E) FRLSH grade copper wire, flush type switch, earthing and required accessories as per specification No: WG-PW/SW
4.	EX.FAN Point wiring in PVC trunking (casing-capping) with 1.5 sq.mm (2+1E) FRLSH grade copper wire, flush type switch, earthing and required accessories as per specification No: WGPW/ SW
5.	Wiring for plug on board with flush Switch socket, copper wiring and earthing as per specification No:WG-PW/SW
6.	Supplying and erecting flush, shock proof plug socket 6 pin 6A and 16A with switch 16A combined unit erected in recess in 16 gauge M.S sheet box duly painted and connected with earth terminal and leads
7.	Supplying and erecting electronic type step regulator for ceiling fan 600/

	900/ 1200/ 1400 mm. sweep, complete erected on provided switch board.
8.	Supplying and erecting PVC trunking (PVC casing-n-capping) of size 20 mm with accessories on wall/ceiling as per specification No: WG-MA/CON.
9.	Supplying and erecting mains with 2x2.5 sq.mm FRLSH copper PVC insulated wire laid in provided conduit/trunking/inside pole/Bus bars or any other places. as per specification No: WG-MA/BW
10.	Supplying and erecting mains with 1x1.5 sq.mm FRLSH copper PVC insulated wire laid in provided conduit/trunking/inside pole/Bus bars or any other places as per specification No: WG-MA/BW 3) Rates for the earth wire shall be calculated as one half rate of mains of bunch of two wire.
AV	INTERNAL LIGHTING FIXTURE
1.	Supplying and erecting anodized aluminium corridor / passage light LED fitting (4 feet) Max. 22W with high transitivity diffuser with system lumens output of Min.2200 lumens , min. efficacy of 100 lumen/W, CRI>80, CCT upto 6000K, Beam Angle of 110 deg., Ripple<5%, THD<10%, p.f. >0.95, operating range of 200-270V, surge protection of 2 kV, Life class of 50,000 Hrs. at L70B50, including driver, with end caps on provided PVC Block / wooden board with 3 years warranty.
2.	Supplying & erecting 10-12W, 230V integral type LED Lamp suitable for B-22/E-27 base
3.	Supplying and erecting bulk head LED fitting max. 10W with high transitivity diffuser with system lumens output of min. 1100 lumens, min. efficacy of 110 lumen/W, CRI>80, CCT upto 6500K, Beam Angle of 120 deg., THD<10%, p.f. >0.95, operating range of 140-260V, in built surge protection of 2.5 kV, Life class of 50,000 Hrs. at L70B50, including driver, IP66, IK09 rated on provided PVC Block / wooden board with 3 years warranty.
AW	LIGHTNING ARRESTER
1.	Supplying & erecting conventional spike type air termination suitable to carry lightning stroke made up of heavy gauge 40 mm dia copper pipe of standard length with 5 Nos. copper spikes fixed on copper ball as air terminals duly threaded in copper pipe erected on provided foundation complete.

2.	Providing earthing with copper earth plate size 60 x 60 x 0.315 cm with funnel with a wire mesh for watering and brick masonry block C.I. cover with minimum 25 kg Carbon based environment friendly back fill Ground Enhancing compound complete with all materials, testing & recording the results as per specification no ESE -LA
3.	Supplying and erecting aluminium strip of required sizes used for earthing on wall and/or any other purpose with necessary aluminium clamps fixed on wall with insulating and non-corrosive bond with required joints. As per specification no. EA-EP.
AX	SOLAR SYSTEM
1.	Supplying, installing, testing and commissioning of 3 kWp ON-GRID Solar PV System / Plant consisting solar PV poly crystalline panel (335 Wp X 09 Nos.) and Solar ON-GRID Inverter of similar capacity (THD<3%) with interconnecting wires / cables up to solar ON GRID inverter, proper base plate, canopy box to house Inverter and necessary accessories i.e. ACDB/ DCDB , MCB etc. wiring up to solar Inverter in casing and capping, display board, sign board, two distinct earthing, lightning arrester, liaisoning charges, remote monitoring system with maintaining card / chip for 5 years including net charges for RMS system, complete up to net meter installation and export to the grid with 5 year ON-SITE warranty and fully comprehensive maintenance contract as per specification no. ESD-RT-ONG
2.	Supplying and erecting AC three phase LT CT multifunction net meter DLMS compliance of accuracy class 0.5, 3 x 240V, 50Hz, with backlit LCD display, optical port & RS 232 port (measures import kWh, kVAh, export kWh, kVAh, Net kWh, kVAh, V, I, kW, 6 months history of energy, load survey, TOD, tamper detection & logging, power ON/OFF events, instantaneous parameters of rating 1/5A with display in absence of power) with wiring connections and mounting hardware on provided panels complete with calibration certificate from manufacturer.
3.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 4 core 16 sq. mm. copper conductor complete erected with glands & lugs, on wall/ trusses/ pole or laid in provided trench/ pipe as per specification no. CB-LT/CU.
4.	Supplying, erecting and marking triple pole isolators only switch version of miniature circuit breaker of 100A with required wiring connections and lugs

	etc. in provided distribution board.
AY	EARTHING & STRIP
1.	Providing earthing with galvanized iron earth plate size 60 x60 x 0.6 cm complete with all materials, testing & recording the results as per specification no. EA-EP
2.	Supplying and erecting GI strip of required size used for earthing on wall and/or any other purpose with necessary GI clamps fixed on wall painted with insulating and non-corrosive bond with required joints As per specification no. EA-EP
3.	Supplying and erecting GI wire of high purity of required sizes used for earthing or any other purposes on wall with necessary G. I. Clamps fixed on wall/cable/ conduit with screws complete.
AZ	AVIATION LIGHT
1.	Supplying & erecting Aviation obstruction light consisting of yellow painted die cast aluminium alloy housing with Integral LED aviation light comprising of aluminium housing with polycarbonate enclosure complete.
2.	Supplying and fixing PVC synthetic elastomer electrically insulating mat with class A insulation conforming to IS: 15652 – 2006 having 2 mm thickness up to 3.3 kV
3.	Providing printed instruction chart for treating persons suffering from electric shock, printed in English & Marathi and duly laminated complete.
4.	Supplying and erecting GI sheet 1.6 mm (16 SWG) caution board of size 200 x 150 mm duly painted with red oxide and enamel paint in Marathi, Hindi and English erected at position for medium voltage (415 Volt) installation with compliance to IS 2551-1982
5.	Supplying and laying (including excavation of suitable width & depth up to 90 cm) 90 mm outside dia. double wall corrugated pipes (DWC) of HDPE for enclosing cable below ground/road surface, to required depth complete.
BA	WATER PUMP
1.	Supplying and erecting minimum Three & above star rated submersible pump set suitable for erection on open well of 2.25 KW/3 HP with 415 V 50 c/s. A.C. supply, having delivery head from 27 to 9 m and discharge from

	300 to 660 litres. per minute & pipe of Suction 65mm /Delivery 50 mm diameter erected with necessary H type clamps as per specification No. WP-OSP DOMESTIC PUMP
2.	Supplying and erecting minimum Three & above star rated submersible pump set suitable for erection on open well of 2.25 KW/3 HP with 415 V 50 c/s. A.C. supply, having delivery head from 27 to 9 m and discharge from 300 to 660 litres. per minute & pipe of Suction 65mm /Delivery 50 mm diameter erected with necessary H type clamps as per specification No. WP-OSP FLUSHING PUMP
3.	Supplying & erecting flat flexible 3 core 2.5 sq.mm PVC sheathed submersible type copper cable suitably clamped at fixed intervals with column pipe assembly complete.
4.	Supplying, erecting & marking FPMCB 6A to 32A, with rated short - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification no. SW-SWR/MCB
5.	Supplying, fixing and commissioning 4 pole RCBO 6/10/16/20/25A, with overcurrent, rated short-circuit breaking capacity (10 kA) and earth leakage protection, with 30/100/300 mA sensitivity and short circuit and earth leakage trip indications per specification no SW-RCCB/RCBO
6.	Supplying and erecting minimum Three & above star rated submersible pump set suitable for erection on open well of 2.25 KW/3 HP with 415 V 50 c/s. A.C. supply, having delivery head from 27 to 9 m and discharge from 300 to 660 litres. per minute & pipe of Suction 65mm /Delivery 50 mm diameter erected with necessary H type clamps as per specification No. WP-OSP (FOR TRANSFER OF RAIN WATER)
7.	Supplying & erecting D.O.L Starter suitable for 415 V, 3 phase 50 cycles with no volt coil and overload element with necessary materials and connected to supply up to 7.5 H.P. (FOR TRANSFER OF RAIN WATER, DOMESTIC PUMP & FLUSHING PUMP)
BB	FIRE ALARM SYSTEM
1.	Supplying, erecting testing and commissioning manual call point (Pill box) with break glass, push button (resettable type) in metal enclosure complete as per specification no. FF-FAAS/MCP
2.	Supplying, erecting, testing and commissioning hooters having high (100dB

	@1m) and low (94dB @ 1m) volume setting, group addressing facility allowing multiple sounders to be activated with CRCA enclosure complete as per specification no. FF-FAAS/HTR
3.	Supplying, erecting testing and commissioning optical type smoke detector on suitable back box with necessary connections complete as per specification no. FF-FAS/SD
4.	Supplying, installing, testing and commissioning of 2 Zones Microprocessor based conventional fire alarm control panel (FACP) with standard accessories, 16x2 Character LCD Display, provision for zone wise contact and beep sound alarm, suitable to operate on 120-220 V AC, 0 - 49 Deg C, 93 ± 2 Percentage RH (non- condensing) at 32 ± 2 Deg C complete as per specification no. FF-FAAS/FACP
5.	Supplying, installing, testing and commissioning FR, XLPE armoured cable 2 core 1.5 sq.mm. copper conductor complete erected on wall/ ceiling complete as per specification no. CBLT/CU
BC	PUBLIC ADDRESS SYSTEM
1.	Supplying, erecting, testing and commissioning 120 W amplifier suitable to operate on 230 Volts A.C. / 12 Volts D.C. supply complete as per specification no. FF-PA/AFR
2.	Supplying, erecting, testing and commissioning 15 to 30 Watts sound column or ceiling mounted complete as per specification no. FF-A/SOC
3.	Supplying and fixing 6U wall mount rack (Dimension-DxWxH – 500x600x367mm) as per specification No. WG-NAS/RAK
4.	Supplying, erecting, testing and commissioning rust free microphone floor stand of chrome plating/powder coating or S.S. material, 500 mm to 1500 mm height and sturdy mechanism ensuring trouble free movement complete.
5.	Supplying, erecting, testing and commissioning stand mounted / handheld type wired microphone having 2 mV/Pa sensitivity, impedance level limit 300 ohms and 50 - 16000 Hz suitable to operate in -10 deg C to +55 deg C with min. 7 metre shielded cable and 3 Pin Professional XLR Connector, stereo jack complete as per specification no. FF-PA/MIC
6.	Supplying, erecting, testing and commissioning of 0.5 sq mm speaker wire complete.

7.	Supplying and erecting HMS PVC conduit FRLS grade 20 mm dia. With necessary accessories in wall/floor with chiselling appropriately as per specification No: WG-MA/CC.
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VIII. Firefighting Item Wise Specification

Sr. No.	Description of Items
1.	Providing Round bottom fire bucket of 9 Litres capacity as per IS: 2546 made out of 24-gauge G.I. sheet with extra handle at bottom duly painted white inside and red outside with FIRE Mark, on provided stand/ wall hook complete.
2.	Supplying & erecting Carbon Dioxide (CO2) fire extinguisher of 4.5 kg. capacity cartridge type conforms to IS 2878 /15683 complete erected with necessary clamp made from 50 x 6 mm. M. S. flat with nut & bolts grouted in wall complete.
3.	Supplying and erecting ABC powder type fire extinguisher as per IS: 13849 /15683 of 6 kg capacity with necessary clamp for erection on wall.
4.	Supplying, installation, testing and commissioning of main fire pump and Sprinkler Pump(MFP /EP) suitable for water discharge of 2280 to 2850 LPM at 120 to 90 m head driven by electric motor 415 volts, 3 phase 50 Hz, AC supply of 75kW or of suitable kW capacity for manual/automatic operation and consisting of following :(a) Horizontal type, single stage, centrifugal/split casing pump of cast iron body & bronze impeller with stainless steel shaft(SS410 grade), mechanical seal conforming to IS 1520. (b) Squirrel cage induction motor, TEFC, synchronous speed 3000 RPM, suitable for operation with IP 55 protection for enclosure, horizontal foot mounted type with Class-'F' insulation, conforming to IS-325. (c) M.S. fabricated common base plate, coupling, coupling guard, foundation bolts etc. as required. (d) Erected on provided suitable size cement concrete foundation duly plastered with anti-vibration pads with perfect aligning, proper levelling complete pump set with accessories duly painted with two coats of synthetic enamel paint of fire red colour over a coat of primer (ISC code 536 as per IS 2932 of 2003) complete, as per specification no. FF-MFP/SSC/EP

5.	Supplying, installation, testing and commissioning of jockey fire pump and Sprinkler Pump (JP) suitable for water discharge of 180 to 240 LPM at 110 to 90 m head driven by electric motor 415 volts, 3 phase 50 Hz, AC supply of 11 kW or of suitable kW capacity for manual/automatic operation and consisting of following : (a) Horizontal/vertical type, multi stage, centrifugal casing pump of cast iron body & bronze impeller with stainless steel shaft (SS410 grade), mechanical seal conforming to IS 1520. (b) Squirrel cage induction motor, TEFC, synchronous speed 3000 RPM, suitable for operation with IP 55 protection for enclosure, horizontal foot mounted type with Class-'F' insulation, conforming to IS-325. (c) M.S. fabricated common base plate, coupling, coupling guard, foundation bolts etc. as required. (d) Erected on provided suitable size cement concrete foundation duly plastered with anti-vibration pads with perfect aligning, proper levelling complete pump set with accessories duly painted with two coats of synthetic enamel paint of fire red colour over a coat of primer (ISC code 536 as per IS 2932 of 2003) complete, as per specification no. FF-MFP/JP
6.	Supplying, installation, testing and commissioning of diesel engine driven main fire pump suitable for automatic operation and consisting of following, complete in all respects, as required : (Diesel Driven Pump) Horizontal type, multistage, centrifugal pump of cast of iron body and bronze impeller with stainless steel shaft, mechanical seal conforming to IS 1520.Suitable HP, 1500 RPM water cooled with radiator, diesel engine conforming to relevant IS standard complete with auto starting mechanism, 12 /24 volts electric starting equipment, diesel tank, exhaust pipe extended up to 10 m outside pump house duly insulated with 50 mm thick glass wool with 1.0 mm thick aluminium sheet cladding, residential silencer, instruments and protection as per standard specification, stop solenoid for auto stop in the event of fault with audio indications, painted with post office red colour etc. as required. M.S fabricated, common base plate, coupling, coupling guard, foundation bolts etc. as required. Suitable cement concrete foundation duly plastered and with anti-vibration pads. 2800 lpm at 88 m head.
7.	CONTROL PANELS Fabrication, supply, Installation testing & commissioning of Electrical control panel of cubical construction, floor mounted type, fabricated out of 2mm thick

	CRCA sheet, compartmentalised with hinged lockable doors, dust and vermin proof, powder coated of approved shade after 7 tank treatment process, cable alley, interconnection with suitable size copper conductor cable/solid copper strip, having switchgears and accessories, mountings and internal wiring, earth terminals, numbering etc. complete in all respect, suitable for main fire pump, pressurisation pump & diesel pump set complete as per CPWD specification with following in coming and Outgoings, suitable for operation on 415V, 3 phase, 50Hz Ac Supply with enclosure protection class IP 42 as required :
8.	INCOMING 250A, 50kA 4 Pole MCCB, Ics=100% Icu rating Digital Voltmeter 0-500V with selector switch Digital Ammeter (0-250 A) with selector switch & CTs etc. LED type RYB phase indicating lamps, ON, OFF, trip indicating lamps Set of Copper Bus Bar 300A OUTGOING (Note: All outgoing feeders for pumps should have digital Ammeter with selector switches, and LED type ON, OFF, trip indicating lamps)
9.	Main Fire Pump 125 A, 50kA TPN MCCB, Ics=100% Icu, with fully automatic Star/Delta starter suitable for 60 HP pump with overload protection, current sensing type single phase preventor complete with all accessories and internal wiring required for automatic operation, selector switch for local/remote, auto/manual/OFF operation.
10.	Jockey Pump 63 A, 50kA TPN MCCB, Ics=100% Icu, with suitable HP fully automatic Star/Delta starter with overload protection, current sensing type single phase preventor complete with all accessories and internal wiring required for automatic operation, selector switch for local/remote, auto/manual/OFF operation.
11.	DIESEL ENGINE CONTROL Control for diesel engine comprising - Automatic/Manual selector switch & 3 attempts starting device, timers and relays as required, push buttons, start/stop in manual mode Indicating lamp for high/ Low Lub. Oil pressure, High Water Temp and Engine on indication Battery charger suitable for 12V/24 V DC with boost and trickle selector switch, 0-30 V DC volt meter, and 0-20 A DC Ammeter All standard relays and accessories for automatic operation of diesel engine System Controller Designing, Supply, Installation, Testing and commissioning of system controller to control operation of main electric fire

	pump, diesel pump, Pressurization pump, Terrace pump in sequence as per specification consisting of relays, timers. Sensors, annunciation window for fault indication, complete as per specification
12.	Supplying and erecting G.I. pipe above ground of 'C' class ERW of size 25 mm dia with necessary fittings complete as per specification no. FF-PP
13.	Supplying and erecting ISI mark GI pipe 32 mm dia.'C' class at position with accessories complete as per specification no. CW-PLB/GP.
14.	Supplying and erecting ISI mark GI pipe 40 mm dia.'B' class at position with accessories complete as per specification no. CW-PLB/GP.
15.	Supplying and erecting G.I. pipe 'C' class ERW 50 mm dia with necessary fittings complete as per specification no. FF-PP
16.	Supplying and erecting G.I. pipe above ground of 'C' class ERW of size 65 mm dia with necessary fittings complete as per specification no. FF-PP
17.	Supplying and erecting G.I. pipe above ground of 'C' class ERW of size 75 mm dia with necessary fittings complete as per specification no. FF-PP
18.	Providing Floor mounting stand for keeping 4 Nos. of FIRE buckets 1500mm in length, 900mm in height frame made out of 30x30x4 mm angle iron with cross supports for legs, welded with 4 hooks and duly painted with one coat of red lead and two coats of silver paint complete.
19.	Supplying and erecting G.I. pipe above ground of 'C' class ERW of size 100 mm dia with necessary fittings complete as per specification no. FF-PP
20.	Supplying and erecting G.I. pipe above ground of 'C' class ERW of size 150 mm dia with necessary fittings complete as per specification no. FF-PP
21.	Supplying and erecting G.I. pipe above ground of 'C' class ERW of size 200 mm dia with necessary fittings complete as per specification no. FF-PP
22.	Providing coating of bitumen paint & 4mm thick wrapping for underground 100 mm ring main of firefighting system.
23.	Providing coating of bitumen paint & 4mm thick wrapping for underground 150 mm ring main of firefighting system.
24.	Providing coating of bitumen paint & 4mm thick wrapping for underground 200 mm ring main of firefighting system.
25.	Supplying and erecting 100 mm dia Cast Iron end line strainer of Y-type flanged end pattern, PN16 pressure rating, SS screen, end connection with

	Flanged / Screwed / Socket Weld / Butt Weld End etc with standard OAR (open area Ratio) for positive suction complete as per specification no. FF-VL/ELS
26.	Supplying and erecting 200 mm dia Cast Iron end line strainer of Y-type flanged end pattern, PN16 pressure rating, screen of SS end connection with Flanged / Screwed / Socket Weld / Butt Weld End etc with standard OAR (open area Ratio) for positive suction complete as per specification no. FF-VL/ELS
27.	Supplying and erecting 100 mm dia. cast iron double flange butterfly valve of size complete with PN16 pressure rating, as per specification no. FF-VL/BFV
28.	Supplying and erecting 150 mm dia. cast iron double flange sluice valve complete with PN16 pressure rating, as per specification no. FF-VL/SV
29.	Supplying and erecting 100 mm dia. cast iron double flange butterfly valve of size complete with PN16 pressure rating, as per specification no. FF-VL/BFV
30.	Supplying and erecting 200 mm dia. cast iron double flange butterfly valve of size complete with PN16 pressure rating, as per specification no. FF-VL/BFV
31.	Supplying and erecting 100 mm dia cast iron double flange NRV complete with PN16 pressure rating, as per specification no. FF-VL/NRV
32.	Supplying and erecting 150 mm dia cast iron double flange ball type NRV complete with PN16 pressure rating, as per specification no. FF-VL/NRV
33.	Supplying and erecting 25 mm dia gun metal gate valve complete with PN16 pressure rating, as per specification no. FF-VL/GV
34.	Supplying and erecting stainless steel single outlet hydrant valve fitted with necessary accessories complete as per specification no. FF-VL/HV
35.	Supplying and installing wall mounting swinging Hose reel drum fitted with 19 mm dia. 30m long high-pressure polypropylene (Polyhose) fitted with necessary accessories complete as per specification no. FF-FFA/HV
36.	Supplying firefighting-controlled percolation (C.P.) Hosepipe, 63mm dia, 15m in length, fitted with necessary accessories complete as per specification no. FF-FFA/CPH
37.	Supplying & erecting gun metal chrome plated nozzle, 19 mm dia complete as per specification no. FF-FFA/NZ
38.	Supplying and erecting fire brigade header (Siamese connection) of 150 mm

	dia for supplying water to wet riser system complete as per specification no. FF-FFA/SMC
39.	Supplying and installing air vessel of 300 mm dia 1.5m in height M.S. tank installed inside pump house / on terrace complete as per specification no. FF-FFA/ACT
40.	Supplying and erecting 20/25mm dia gun metal air release cock, with necessary G.I. coupling to be fitted on top of air vessel or on wet riser complete as per specification no. FFFFA/ARV
41.	Supplying and installing pressure gauge of 100 mm dia., 0-300 PSI or 0-14 kg per cm square fitted with 12/15 mm dia. pad cock valve, and G.I. pipe, elbow etc. complete as per specification no. FF-FFA/PG
42.	Supplying and installing pressure switch with 12/15 mm dia isolation valve, G.I. nipple, elbow, etc complete as per specification no. FF-FFA/PS
43.	Supplying and erecting one no. brass orifice plate having 6 mm. thickness, 140 mm outer dia and required/proper diameter of inner hole at every single outlet hydrant valve as per specification no. FF-FFA/OP
44.	Supplying and erecting M.S./CRCA cabinet for housing Floor Hydrant valve, hose pipe, hose reel and branch pipe (size 1000 x 736 x 736 mm.) made from 16 SWG sheet and angle iron 25 x 25 x 4 mm. having front doors with viewing glass (8"x6") and locking arrangement with necessary fixing material such as rubber bidding etc. duly painted in post box red colour.
45.	Providing/conducting complete fire audit including submission of fire audit report for the whole premises.
46.	Supplying, erecting, testing and commissioning of 15 mm (½") dia. NBCM Body chrome finished quartzoid bulb sprinkler having 68° C fixed temperature rating with deflector disc of conventional construction complete as per specification no. FF-SPR
47.	Supplying, installation, testing and commissioning of diesel engine driven main fire pump suitable for automatic operation and consisting of following, complete in all respects, as required : (Diesel Driven Pump) Horizontal type, multistage, centrifugal pump of cast of iron body and bronze impeller with stainless steel shaft, mechanical seal conforming to IS 1520. Suitable HP, 1500 RPM water cooled with radiator, diesel engine conforming to relevant IS standard complete with auto starting mechanism, 12 /24 volts electric starting equipment, diesel tank, exhaust pipe extended up to 10 m outside pump house duly insulated

	with 50 mm thick glass wool with 1.0 mm thick aluminium sheet cladding, residential silencer, instruments and protection as per standard specification, stop solenoid for auto stop in the event of fault with audio indications, painted with post office red colour etc. as required. M.S fabricated, common base plate, coupling, coupling guard, foundation bolts etc. as required. Suitable cement concrete foundation duly plastered and with anti-vibration pads.
48.	2850 lpm at 88 m Head
49.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 3½ core 95 sq. mm. aluminium conductor complete erected with glands & lugs, on wall/ trusses/pole or laid in provided trench/ pipe as per specification no. CB-LT/AL
50.	Supplying and erecting LT current transformer with bar primary 50/5 to 1000/5 ratio with 15 VA burden erected in provided CRCA box duly secured with insulating materials connected to the meter with test certificates.
51.	Supplying and erecting marine (Waterproof) plywood 18/19 mm thick fixed to wall or on provided panel board with necessary materials such as screws, wall fasteners, supports, nut bolts etc. complete with compliance to IS 720:2010
52.	Supplying and erecting porcelain base iron clad cut out HC type 415V, 100A erected on polished wooden block.
53.	Supplying and erecting GI sheet 1.6 mm (16 SWG) caution board of size 200 x 150 mm duly painted with red oxide and enamel paint in Marathi, Hindi and English erected at position for medium voltage (415 Volt) installation with compliance to IS 2551-1982
54.	Providing printed instruction chart for treating persons suffering from electric shock, printed in English & Marathi and duly laminated complete.
55.	Providing Round bottom fire bucket of 9 Litres capacity as per IS: 2546 made out of 24-gauge G.I. sheet with extra handle at bottom duly painted white inside and red outside with FIRE Mark, on provided stand/ wall hook complete.
56.	Supplying & erecting Carbon Dioxide (CO ₂) fire extinguisher of 4.5 kg. capacity cartridge type conform to IS 2878 /15683 complete erected with necessary clamp made from 50 x 6 mm. M. S. flat with nut & bolts grouted in wall complete.
57.	Supplying and erecting ABC powder type fire extinguisher as per IS: 13849 /15683 of 6 kg capacity with necessary clamp for erection on wall.

IX. External Water Supply, Drainage, Pumps and Panels

1.	Supplying and erecting submersible pump set suitable for erection on open well of 10 HP with 415 V 50 c/s. A. C. supply, having delivery head from 20m and discharge from 200 to 800ltrs. per minute (Max efficiency at 16 m head & 500 ltr discharge) & pipe of Suction 50mm /Delivery 50 mm diameter erected with necessary H type clamps as per specification no. WP-OSP (Rain water transfer Pump)
2.	Supplying and erecting submersible pump set suitable for erection on open well of 10 HP with 415 V 50 c/s. A. C. supply, having delivery head from 20m and discharge from 200 to 800ltrs. per minute (Max efficiency at 16 m head & 500 ltr discharge) & pipe of Suction 50mm /Delivery 50 mm diameter erected with necessary H type clamps as per specification no. WP-OSP (Flushing water Pump)
3.	Providing and fixing 45 cm wide mild steel ladder of 40mm x 6 mm mild steel flat stringers and steps of 18 mm dia. mild steel bar for water tank including fixing it in (1:2:4) M-15 cement concrete block 60cm x 30cm x 30cm and painting the ladder complete. (Anti-Corrosive Paint)
4.	Supplying, erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 3 core 2.5 sq. mm. copper conductor complete erected with glands & lugs, on wall/ trusses/ pole or laid in provided trench/ pipe as per specification no. CB-LT/CU
5.	Supplying & erecting automatic control panel for 3 Phase, 415-volt, A.C, Submersible/centrifugal pump set up to 7.5 HP consisting of DOL starter having relay range 9-14 AMP, S.P.P., Combined ammeter/ voltmeter, phase indicating lamp enclosed in CRCA powder coated Vibration proof enclosure with IP 54 protection. Control Panel should offer single phasing, phase reversal, phase imbalance etc.
6.	Supplying & erecting Dewatering Monoblock pump set 415V, 3 phase, 50C/s, A.C. supply of 3.73KW/ 5 H.P of 9 to 18 m head with 1125 to 500 LPM discharge, 15.5mm Solid Handling Capacity, non-clogging minimised abrasive wear, large impeller clearance & maintenance free pump set with waterproof cable duly sealed with epoxy material built in protection to prevent the motor from burn out due to abnormal operation, specially designed high efficiency motor with F class insulation & the pump mechanically sealed in the oil chamber,. etc. complete as per specification no. DWP- SMP (For

	Fire Pump Room- sump pump) 1 WORKING + 1 STAND BY
7.	Supplying & erecting D.O.L Starter suitable for 415 V, 3 phase 50 cycles with no volt coil and overload element with necessary materials and connected to supply up to 7.5 H.P.

X. Additional Item wise Specification of Civil Work

Sr. No.	Item Description	Additional Specification
1.	Excavation for foundation in earth, soil of all types, sand, gravel and soft murum, including removing the excavated material up to a distance of 50 m. beyond the building area and stacking and spreading as directed, dewatering, preparing the bed for the foundation and necessary back filling, ramming, watering including shoring and strutting etc. complete. (Lift up to 1.5 m.) By Mechanical Means	Contractor should take care while carrying out excavation work, when services like water, drainage, underground electricity communications lines are found, the contractor will have to carry out excavation in such portions as per directions of Engineer-in-Charge. No extra payment will be made to the contractor
2.	Excavation for foundation in Hard murum and boulders including removing the excavated material up to a distance of 50 metres, beyond the building area and stacking and spreading as directed, dewatering, preparing the bed for the foundation and necessary back filling, ramming, watering including shoring and strutting etc. complete. (Lift from 1.5 m to 3.0 m.) By Mechanical Means	Contractor should take care while carrying out excavation work, when services like water, drainage, underground electricity communications lines are found, the contractor will have to carry out excavation in such portions as per directions of Engineer-in-Charge. No extra payment will be made to the contractor
3.	Providing pre-constructional anti-termite treatment as per I.S. 6313 (Part-II) treatment by treating the	As shown in drawings & as directed by Engineer-in-Charge.

Sr. No.	Item Description	Additional Specification
	backfill in immediate contact with foundation at the rate of 5 litres of emulsion concentrate of 1.0 percent of clorophyrifos per square metre of vertical surface area covering 10 years guarantee on bond paper.	
4.	Providing preconstructional antitermite treatment as per I.S. 6313 (Part-II) by treating the top surface of plinth filling at the rate of 5 litres of emulsion concentrate at 1.0 percent of clorophyrifos per square metre of surface area covering ten years guarantee on bond paper.	As shown in drawings & as directed by Engineer-in-Charge.
5.	Filling in plinth and floors with approved excavated material in 15cm. to 20cm. layers including watering and compacting etc. complete.	As shown in drawings & as directed by Engineer-in-Charge. Mechanical compaction should be done
6.	Filling in plinth and floors with contractors' material/brought from outside and approved by Engineer in charge in layers of 15 cm to 20 cm including watering and compaction etc. complete.	As shown in drawings & as directed by Engineer-in-Charge. Mechanical compaction should be done
7.	Providing and laying Cast in situ/Ready Mix cement concrete in M-10 of trap/ granite/ quartzite/ gneiss metal for foundation and bedding including bailing out water, Steel centering, formwork, laying/pumping, compacting, roughening them if special finish is to be provided, finishing if required and	The ingredient of concrete shall be mixed in electric batch mix plant of 30 cum/hour capacity and transported by concrete pump as directed by Engineer-in-Charge. The concrete shall be vibrated by plate / needle / form type vibrators. The acceptance criteria shall be 28 days strength only & as per IS 456-2000. The provision of clause 38, conditions of contract are not applicable to

Sr. No.	Item Description	Additional Specification
	curing complete, with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)	this item.
8.	Providing and laying Cast in situ/Ready Mix cement concrete M-25 of trap / granite /quartzite/ gneiss metal for R.C.C. work in foundations like raft, strip foundations, grillage and footings of R.C.C. columns and steel stanchions etc. including bailing out water, Steel centering, formwork, cover blocks, laying/pumping, compaction and curing roughening the surface if special finish is to be provided (Excluding reinforcement and structural steel) etc. complete, with fully automatic microprocessor based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)	The ingredient of concrete shall be mixed in electric batch mix plant of 30 cum/hour capacity and transported by concrete pump as directed by Engineer- in-Charge. The concrete shall be vibrated by plate / needle / form type vibrators. The acceptance criteria shall be 28 days strength only & as per IS 456-2000. The provision of clause 38, conditions of contract are not applicable to this item.
9.	Providing and laying Cast in situ/Ready Mix cement concrete M-25 of trap / granite /quartzite/ gneiss metal for R.C.C. columns as per detailed designs and drawings or as directed including steel centering,	The ingredient of concrete shall be mixed in electric batch mix plant of 30 cum/hour capacity and transported by concrete pump as directed by Engineer- in-Charge. The concrete shall be vibrated by plate / needle / form type vibrators. The acceptance criteria

Sr. No.	Item Description	Additional Specification
	formwork, cover blocks compacting and roughening if special finish is to be provided and curing etc. complete. (Excluding reinforcement and structural steel) with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)	shall be 28 days strength only & as per IS 456-2000. The provision of clause 38, conditions of contract are not applicable to this item.
10	Providing and laying Cast in situ/Ready Mix cement concrete in M-25 of trap/ granite/quartzite/ gneiss metal for R.C.C. beams and lintels as per detailed designs and drawings or as directed including steel centering, formwork, cover blocks, laying/pumping, compaction and roughening the surface if special finish is to be provided and curing etc. complete. (Excluding reinforcement and structural steel) with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)	The ingredient of concrete shall be mixed in electric batch mix plant of 30 cum/hour capacity and transported by concrete pump as directed by Engineer- in-Charge. The concrete shall be vibrated by plate / needle / form type vibrators. The acceptance criteria shall be 28 days strength only & as per IS 456-2000. The provision of clause 38, conditions of contract are not applicable to this item.
11	Providing and laying Cast in situ/Ready Mix cement concrete M-25 of trap / granite /quartzite/ gneiss	The ingredient of concrete shall be mixed in electric batch mix plant of 30 cum/hour capacity and transported by concrete pump

Sr. No.	Item Description	Additional Specification
	metal for R.C.C. columns as per detailed designs and drawings or as directed including steel centering, formwork, cover blocks compacting and roughening if special finish is to be provided and curing etc. complete. (Excluding reinforcement and structural steel) with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)	as directed by Engineer- in-Charge. The concrete shall be vibrated by plate / needle / form type vibrators. The acceptance criteria shall be 28 days strength only & as per IS 456-2000. The provision of clause 38, conditions of contract are not applicable to this item.
12	Providing and laying Cast in situ/Ready Mix cement concrete in M-25 of trap/ granite/ quartzite/ gneiss metal for R.C.C. beams and lintels as per detailed designs and drawings or as directed including steel centering, formwork, cover blocks, laying/pumping, compaction and roughening the surface if special finish is to be provided and curing etc. complete. (Excluding reinforcement and structural steel) with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)	The ingredient of concrete shall be mixed in electric batch mix plant of 30 cum/hour capacity and transported by concrete pump as directed by Engineer- in-Charge. The concrete shall be vibrated by plate / needle / form type vibrators. The acceptance criteria shall be 28 days strength only & as per IS 456-2000. The provision of clause 38, conditions of contract are not applicable to this item.

Sr. No.	Item Description	Additional Specification
13	Providing and laying Cast in situ/Ready Mix cement concrete M-25 of trap/ granite / quartzite/ gneiss metal for R.C.C. slabs and landings as per detailed designs and drawings including steel centering, formwork, cover blocks, compaction, finishing the formed surfaces with cement mortar 1:3 of sufficient minimum thickness to give a smooth and even surface or roughening if special finish is to be provided and curing etc. complete. (Excluding reinforcement and structural steel) with fully automatic microprocessor-based PLC with SCADA enabled reversible Drum Type mixer/ concrete Batch mix plant (Pan mixer) etc. complete. With fine aggregate (Natural Sand / Crushed sand VSI Grade finely washed etc)	The ingredient of concrete shall be mixed in electric batch mix plant of 30 cum/hour capacity and transported by concrete pump as directed by Engineer-in-Charge. The concrete shall be vibrated by plate / needle / form type vibrators. The acceptance criteria shall be 28 days strength only & as per IS 456-2000. The provision of clause 38, conditions of contract are not applicable to this item.
14	Providing and fixing in position TMT-FE-500 bar reinforcement of various diameters for R.C.C. pile caps, footings, foundations, slabs, beams columns, canopies, staircase, newels, chajjas, lintels pardis, copings, fins, arches etc. as per detailed designs, drawings and schedules. including cutting, bending, hooking the bars, binding with wires or tack welding and supporting as required complete.	As shown in drawings & as directed by Engineer-in-Charge. The contractor will have to procure steel having ISI mark along with manufactures test certificate. Contractor will have to retest the steel from approved laboratory as directed by PMC. No extra payment will be made for carrying out such tests. Provision and application of Clause 38 conditions of contract will not be done for this item.
15	Providing and fixing during laying in	As shown in drawings & as directed by

Sr. No.	Item Description	Additional Specification
	RCC Slab rectangular 2 mm thick M.S. sheet fan hook box of size 180x100 mm with required depth with holes or notches as required, bottom and top sides of 1.6 mm M.S. sheet fixed with 3.5 mm dia. round headed screws together with 12 mm dia. M.S. bar bent to shape with hooked ends for fan hook and painting to exposed surface	Engineer-in-Charge
16	Providing cement-based water proofing treatment to terraces (Indian water proofing or alike) with brick bats laid in required slope to drain the water for any span after cleaning the base surface. Applying a coat of cement slurry admixed with approved water proofing compound and laying the brick bats on bottom layer in C.M.1:5 admixed with approved water proofing compound filling up to half depth of brick bats, curing this layer for 3 days, applying cement slurry over this layer joints of brick bats with C.M.1:3 admixed with approved water proofing compound and finally top finishing with average 20 mm. thick layers of same mortar added with jute fiber at 1 Kg per bag including finishing the surface smooth with cement slurry admixed with approved water proofing compound. Marking finished surface with false squares of 300mm x 300	As shown in drawings & as directed by Engineer-in-Charge

Sr. No.	Item Description	Additional Specification
	mm. making the junctions at the parapet rounded and tapered top for required height, with drip mould at the junction of plaster and parapet and curing and covering 10 years Guarantee against leak proofness on Court fee stamp paper of Rs. 500/- including ponding test etc. complete.	
17	Providing waterproofing in W.C. and bath including brick bat coba in all position including providing and laying 12mm bedding in cement mortar 1:3 on vergin concrete slab with waterproofing compound @ 1kg/per bag of cement laying brick bat coba of required thickness in cm 1:5 with waterproofing compound 1 kg/bag of cement grouting and finishing the top layer with 20mm thick brick bedding in cm mortar 1:3 with waterproofing compound 1 kg/per bag of cement and testing the treated portion for 48 hours by pond test and covering ten years' guarantee on requisite stamp paper including curing etc. complete.	As shown in drawings & as directed by Engineer-in-Charge
18	Providing & Fixing FRP water tank (refer OHT layout for additional details enclosed in Vol. II)	As directed by Engineer-in-Charge
19	Providing and applying Two coats of wall care Putty on plastered surface and Ceiling and Walls to prepare surface even and smooth of approved make, etc complete.	As shown in drawings & as directed by Engineer-in-Charge

Sr. No.	Item Description	Additional Specification
20	Providing and laying vitrified mirror / glossy finish tiles decorative type having size 590 mm to 605 mm x 590 mm to 605 mm of 8 to 10 mm thickness and confirming to IS. 15622-2006 (group Bla) of approved make, shade and pattern for flooring in required position laid on a bed of 1:4 cement mortar including neat cement float, filling joints, curing and clearing etc. complete. a) Flooring	As shown in the drawing and as directed by engineer-in-charge. The make of material shall be approved by Engineer-in-Charge
21	Providing and laying ceramic tiles having size 30 cm. x 45 cm. confirming to corresponding I.S. for dado and skirting in required position with readymade adhesive mortar of approved quality on plaster of 1:2 cement mortar including joint filling with white/ colour cement slurry cleaning curing etc. complete	As shown in the drawing and as directed by engineer-in-charge. The make of material shall be approved by Engineer-in-Charge
22	Providing and laying ceramic tiles of having size 30 cm x 30 cm confirming to I.S.15622/2006 (group D II-A) and 7 to 8 mm thick for flooring in required position laid on a bed of 1:4 cement mortar including cement float, filling joint with white/colour cement slurry cleaning curing etc. complete.	As shown in the drawing and as directed by engineer-in-charge. The make of material shall be approved by Engineer-in-Charge
23	Providing and laying Antiskid Ceramic tiles of approved quality of size 30 cm x 30 cm and confirming to IS 15622-2006 (Group-B IIA) for antiskid flooring in required position	As shown in the drawing and as directed by engineer-in-charge. The make of material shall be approved by Engineer-in-Charge

Sr. No.	Item Description	Additional Specification
	laid on a bed of 1:4 cement mortar including cement float, filling joint with cement slurry cleaning curing etc. complete.	
24	Providing and constructing granite kitchen platform with fixing of stainless steel sink 600 mm x 450 mm size as per detailed drawing including vertical both side polished kadappah stone 25 to 30 mm thick supports with kadappah top 35 to 40 mm thick and polished granite 16 to 20 mm top with side strips of granite at front and both sides of platform raised with two vertical granite supports 15 cm height and top granite of 75 x 40 cm including cutting, opening for sink of required size in kadappah as well as granite etc. complete. (Platform top size 5.00 m x 0.60 m and height is 0.75 m)	As shown in the drawing and as directed by engineer-in-charge. The make of material shall be approved by Engineer-in-Charge
25	Providing and fixing M40 grade thick vibrated pull cast or similar type concrete frame with chamfer conforming to I.S. 6524:1983 having 6 mm dia. bars 3 Nos. And stirrups @250 mm c/c and fixing in wall with 6 Nos of hold fast of 12 mm dia bars 500 mm long including primer and oil painting etc, complete) frame size 60 mm x 100 mm.	As shown in drawings & as directed by Engineer-in-Charge
26	Providing and fixing solid core flush door shutter in single leaf 32 mm	As shown in drawings & as directed by Engineer-in-Charge. The make of the door

Sr. No.	Item Description	Additional Specification
	thick decorative type of exterior grade as per detailed drawings approved face veneers 3 mm thick on both faces or as directed, all necessary beads, mouldings and lipping, wrought iron hold fasts, chromium plated fixtures and fastenings, with brass mortise lock, chromium plated handles on both sides, and finishing with French Polish etc. complete.	shall be approved by Engineer-in-Charge
27	Providing and fixing fiber glass reinforced polyster door shutter 30 mm thick as per IS 14856 (2000) (Reaffirmed 2006) without ventilator including chromium plated fixtures and fastening with chromium plated handles on both sides, etc complete	As shown in drawings & as directed by Engineer-in-Charge. The make of the door shall be approved by Engineer-in-Charge
28	Providing and fixing in position. (as per I.S.1868 / 1982) Aluminium sliding window of two tracks with rectangular pipe having overall dimension 63.50 x 38.10 x 1.02 mm at weight 0.547 kg/Rmt. and window frame bottom track section 61.85 x 31.75 x 1.20 mm at weight 0.695 kg/Rmt. Top and side track section 61.85 x 31.75 x 1.30 mm at weight 0.659 kg/Rmt. The shutter should be of bearing bottom 40 x 18 x 1.25 mm at weight 0.417 kg/Rmt. Inter locking section 40 x 18 x 1.10 mm at weight 0.469 kg/Rmt. And handle section 40 x 18 x 1.25 mm at weight 0.417	As shown in drawings & as directed by Engineer-in-Charge. The make of the window shall be approved by Engineer-in-Charge.

Sr. No.	Item Description	Additional Specification
	kg/Rmt. and top section 40 x 18 x 1.25 mm at weight 0.417 kg/Rmt. As per detailed drawings and as directed by Engineer in charge with all necessary Aluminium sections fixtures and fastenings such as roller bearing in nylon casting and self-locking catch fitted in vertical section of shutter including 5 mm thick plain glass with all required screws and nuts etc, complete. With powder coating without box.	
29	Providing and fixing in position powder coated aluminium louvered windows / ventilator of various sizes with powder coating as per detailed drawing and specifications including aluminium frames 80 x 38 mm x 1.22 mm box type, 5 mm thick sheet glass louvers, of approved quality etc. complete.	As shown in drawings & as directed by Engineer-in-Charge. The make of the window shall be approved by Engineer-in-Charge.
30	Providing and fixing in position after hoisting, precast 1:2 cement mortar reinforced jali 50 mm thick with 3 mm diameter mild steel bar reinforcement, of approved design on both sides, fixing in cement mortar 1:3 and curing etc. complete	As shown in drawings & as directed by Engineer-in-Charge. The make of the window shall be approved by Engineer-in-Charge.
31	Providing and applying washable powder distemper of approved colour and shade to old and new surfaces in two coats including scaffolding, preparing the surfaces. (Excluding the primer coat) etc.	As shown in drawings & as directed by Engineer-in-Charge

Sr. No.	Item Description	Additional Specification
	complete.	
32	Providing and applying white-wash in two coats on old / new plastered or masonry surfaces and asbestos cement sheets including scaffolding and preparing the surface by brushing and brooming down etc. complete.	As shown in drawings & as directed by Engineer-in-Charge
33	Providing and applying three coats of water proof cement paint of approved manufacture and of approved colour to new plastered surfaces including scaffolding if necessary, cleaning and preparing the surface, watering for two days etc. complete.	As shown in drawings & as directed by Engineer-in-Charge
34	Providing and fixing heavy duty inter locking concrete gray paving blocks of 60 mm thickness of having a strength of 300 kg/Sq.cm. of approved quality and shape on a bed of crushed sand of 25 to 30 mm thick including skirting joints and cleaning etc. complete.	As shown in drawings & as directed by Engineer-in-Charge

NOTE:

- 1) All above work shall be carried out as per wording of the item and as directed by the PMC/Client.
- 2) Client's decision regarding specification of item shall be final and binding.
- 3) Being a Lumpsum Tender Mode of Measurement are not applicable. 4) Items, which are not incorporated in the project, shall be ignored.
- 5) Items, which are not specified shall be binding on contractor to complete the total project.

PART B - Alternative Building Technologies

(Specifications /Instructions for Alternative Building Technologies)

Apart from Mivan technology, any contractor can use alternative construction material/ technology for construction and development of the project as approved by Government of Maharashtra GR No. शासन निर्णय क्र. प्र.आ.यो. - २०१८/प्र.क्र.८ गृनिधो-२ 01.08.2018

Bidders engaging such technologies/materials shall have to fulfil the following additional technical criteria:

Eligibility Criteria

The following additional certificates should be submitted in Eligibility-cum-Technical Bid:

A. Certification:

1. The alternate technology / materials must have the following certification:
 - a) It should be certified by any of the following institutions: Building Materials and Technology Promotion Council/ Housing and Urban Development Council/ Central Building Research Institute/ Indian Institute of Technology
 - b) For the purpose of testing and quality assurance, either (i) Building materials used InTechnology should have IS codes or (ii) the institutions specified 1(a) must have also certified construction manual and quality assurances plan including material testing.
 - c) The contractor/technology provider should produce all the certificates relevant to the above clauses.
2. The above clarification should include the following critical quality aspects:
 - Structural strength, serviceability, durability and stability
 - Structural life of minimum 50 years
 - Thermal and energy performance and/ or efficiency
 - Water penetration
 - Behavior in fire
 - Durability
 - Behavior against other natural hazards, like earthquake, flood, cyclone
 - Construction methodology
 - Quality assurance including material testing
 - The value of the above criteria should be at least as much as provided by the traditional building technology as provided in this tender document or more.

B. Other Certifications:

The alternative building material / technology has to be used at least once in India. Bidder will have to submit completion certificate for at least one building of G+4/ P+5 or meters of height in use in India given by client and building use permission issued by local government authority.

The developer/contractor should produce Certification of the following by a licensed structural Engineer:

- Structural safety: conforming with IS Codes, resistance to static and dynamic actions, both individually and in combination, impacts, intentional and unintentional abuse, accidental actions.
- Structural serviceability: resistance to loss of function, damage and avoidance of user comfort.
- Structural durability: performance retention of the structure (a structural warranty of minimum 50 years).

C. Details to be submitted by the Contractor in Part-B

While giving proposal for alternative technology the contractor shall submit the following.

- Product description
- Detailed technical information on how all the materials of the technology (including testing of the materials) meet IS Codes OR certification in this respect given by the above-mentioned institution.
- Construction methodology.
- Design assumptions, detailed calculations, and references to the necessary design standards and detailed design drawings.
- Design calculations
- The structural design calculations must clearly demonstrate structural integrity and stability, including connection details
- Design calculations should have proper sketches annotated in English.
- Quality Assurance Plan

The screening committee will evaluate the technology on the basis of documents/ certificates/design which will be submitted by the developer/contractor as mentioned in the details of submission.

D. Bidding Condition:

- i) The minimum specification for interior and exterior finishing items can be suitably modified in accordance with alternative building material / technology such that the impact / effect of final finish is either maintained or enhanced than of conventional finish.
- ii) There shall not be any additional weightage for alternative building material / technology. iii. The decision of screening committee shall be final with respect to the technical eligibility
- iii) of other alternative building material / technology.

E. Scope of Work:

The scope of work for the project is as mentioned in this document.

In addition, the general scope of work, the developer/contractor opting for alternative technology should also perform the following:

- The proposals and costs must be based for plans for a specified area strictly and overall scheme.
- The contractor shall arrange necessary training for the use and maintenance of the product, if needed.
- If needed, briefing session and exhibition of the construction technology shall be arranged prior opening of price bid.

Note: - Land Plan attached below is as per PMC record, Bidder has to Visit site prior to quote tender and also to carry out preliminary survey to ascertain land area and other data required for quoting tender.

CHAPTER NO. 7

7 List of Approved Material for Civil and Interior Works

A. LIST OF CIVIL AND INFRASTRUCTURAL MATERIAL:

Sr. No.	Name of material	Few Names of companies or as approved by Client
1	Cement	
	a) Ordinary Portland cement (O.P.C.) (For R.C.C work)	Ultra Tech / ACC / Birla Plus / Birla Gold / Vasavdatta / Ambuja / Bharti / JSW / Dalmia / Chettinad / Emmami / Vicat / Orient / JK Laxmi / Lafarge
2	b) Portland pozzolana cement / slag cement (PPC)/ PSC (For plaster / masonry / flooring work)	Birla A-1 / ACC King / Vasavdatta / Ultra Tech / Manikgarth / Birla Gold / Coromondal / Bharti / JSW / Dalmia / Chettinad / Emmami / Vicat / Orient
3	Steel	
a	Steel	Tiscon / Vizac / Jindal / Sail / Tata/ Essar / Arcelor Mittal
b	Corrosion Resistance Steel (CRS)	Tiscon / Vizac / Jindal / Sail / Tata/ Essar / Arcelor Mittal
c	Epoxy coated	Electrotherm India Ltd., Hari Om
4	Ready mix concrete (RMC)	Godrej /ACC / Ultra Tech / Lafarge
5	Polycarbonate Sheet 6mm thick	Lexan
6	Artificial sand	Artificial Sand prepared through vertical SWHAT shaft impactor machine shall be used
7	Natural sand	Use river sand having silt content not more than 4%
8	Aggregate 10 mm and 20 mm	N.A.- As using of ready-mix concrete.
9	Clay Burnt Bricks	N.A.
10	Fly Ash Bricks	Local and approved as approved by PMC and Client
11	AAC blocks	Siporex/ Ultra Tech, Magicrete/ Ecolite/ Aerocon / Flyocrete / Powerlite /Dolite /

		Harmony
12	Adhesive to AAC Block/tile	K-2 Joy chemicals, Gobain weber, ultratech, flyobond, blockol by Ambuja cement, NSA Gold, NSA Silver, saint Gobain, block bond.
13	Concrete cover block	Disha/Astra
14	Mechanical splice	Splice Tek
15	Waterproof solid core flush door	Anchor / Century / Green
16	FRP doors	House of door / Perma finish / Syntax / Everest / Fibroplast
17	Lamination sheets' (1.50 mm thick)	Royal touch / Green Lam / Merino / Century / Formica/ Sundeck/ Signature
18	Door Fixtures and Fastenings i Main internal door a) Stainless steel	Shalimar/ Vinus (NVII)/Union
19	Cold and Hot water mixer	Jaquar / Johnson / Ess-Ess/ Parryware/ Roca / Hindustan /Kohler/ Toto/ Duravit/ Kajaria / Kerovit/Cera/Duravit, kajaria – Kenvit, Jal, Prayag, Hindware Sanitory and Americal standard, Plumbing fixtures & Cera, Kohier, Toto fittings.
20	Con. Stop cock	
21	Bib cock	
23	Two-way Bib cock	
24	Long neck Bib cock (Kitchen)	
25	Pillar cock for WHB	
26	Angle cock for Geyser WHB and wall	
27	Flush valve open etc.	
28	Waste coupling	
29	Bottle trap of WHB	
30	Hand Shower (Health faucet)	
31	Towel rod (600 mm long)	
32	Towel ring	

33	S.S. Hook	
34	Health Faucet	
35	Shower Rose	
36	Mirror supported on stud	Modi / Saint Gobain
37	PVC Water Tank	Syntax
38	Conventional Section 18 gauge heavy	Jindal / Hindalco / Superfine.
39	Vitrified tiles (600mm x 600mm), Vitrified Antiskid tiles, Vitrified Dado tiles.	Kajaria/ Johnson/ Nitco /RAK/Asian/Orient/Bell/ Symphony / Somani / Simpolo
40	a) Primer (External and Internal)	Asian / Nerolac / Berger / Dulux / Snowcem
41	b) Oil bound Distemper to internal walls	
42	c) Synthetic Enamel Paint to M.S. grill	
43	d) Polymer based external paint	Apex from Asian Paints / Excel from Nerolac Paints / Shalimar / Berger / Pearl excel from Godavari
44	CPVC Pipe	Finolex / Supreme /Jain/Astral Ashirwad / Ajay / Kitec/Kelvin/ Kerovit by Kajaria/ HIL
45	U.P.V.C. Pipes	Finolex / Suprem /Jain/Astral / Ashirwad / Ajay / Kitec / Kerovit by Kajaria/ HIL
46	G.I. Pipes	Tata/ Jindal / APL Apollo
47	Pump	Kirloskar / CRI/Wilo/KSB/Compton grooves / Grundfos
48	Water resistant white cement based wall care putty	J K White/ Birla White/ Ultra Tech / Wall plast
49	Gypsum Plaster	India Gypsum / Commix / Buildon / Magicrete
50	Texture plaster 3 to 5 mm thick applying by trowel.	Ultra Tech / Asian / Nerolac/ Berger/ Nitco / Rinova / Godavari
51	Glass	Modi / Hindustan / Saint-Gobain/Ashai
52	Paver Blocks	Gubbi / Shirke/Disha / Concreteshow
53	APP/ SBS membrane water proofing	Tikidan / APEX / IWL/Asian/Pidilite
54	Restroom Cubicle system for Toilets	Merino

	of Amenity block	
55	Urinal partitions for Amenity block	Merino
56	Expansion Joint treatment – a) Vertical b) Horizontal	Sanfield make a) SNPG-600 b) SRFL -600/ Kantaflex / Kanta / Maruti techno / MYK Schomburg
57	Membrane Water Proofing	Tikidan / APEX / IWL / STP Ltd. / Asian
58	Window	Fenesta/Kommerling/Ashai/Wintech
59	Grating and Manholes	HP Composites

B. INTERIOR WORK

LIST OF INTERIOR MATERIAL:

Sr. No.	Description	Few Names of Companies or as Approved by Client
A	CARPENTRY WORKS	
1	Timber Wood	Supplier to be approved by PMC/ PMC/Client
2	Commercial Plywood / Marine Plywood – ISI make	KIT Ply, Green, National, Merino, Anchor, Century, duroply, virgo ply
3	Flexy Plywood	KIT Ply, Green, National, Anchor, Century
4	Plain/Laminated Particle Board	Green, Ecoboard, Novapan, Decoboard, Kalpataru
5	Plain/Laminated Medium Density Fiber Board	Nuwood, Durative, Green
6	Block Board	KIT Ply, Green, National, Anchor, Century
7	Flush Door	KIT Ply, Green, National, Anchor, Century, Kutti, Swastik
8	Plaster board	India Gypsum, Boral
9	Metal Ceiling Tiles and Grid	Techno Ceiling Products, Hunter Douglas
10	Soft Board	Jolly Board
11	Calcium Silicate Board	Ramco, Aerolite
12	Veneer 4 mm	Timex, Euro, Greenlam, Century, Jalaram

Sr. No.	Description	Few Names of Companies or as Approved by Client
13	Laminates 1mm	National, Royal Touch, Formica, Greenlam, Merino, Century mica, New mica
14	Decorative Laminate sheet	Sibu, Bloom
15	Wooden Fire Doors	Shakti, Aadhunik
B	HARDWARES	
1	Drawer channels	
a)	Local Make	Earl Bihari, Windor, Enox, Ozone
b)	Imported Make	Geze, Hafele, Hettich, Dorma
2	Screws	Nettlefold, GKW
3	Fittings and Locks-Handles, Tower Bolts, Hinges, etc.	
a)	Local Make	Earl Bihari, Enox, Neki, Magnum
b)	Imported Make	Geze, Hafele, Hettich, Dorma
4	Hardware for Glass doors and partitions	
a	Floor Spring for doors	Geze, Hafele, Hettich, Dorma
b	Patch fittings	Geze, Hafele, Hettich, Dorma
c	Floor Lock	Geze, Hafele, Hettich, Dorma
d	Top Pivot	Geze, Hafele, Hettich, Dorma
e	Pull handle	Geze, Hafele, Hettich, Dorma
5	Hardware for Flush Doors	
a	Ball bearing butt Hinges	Geze, Hafele, Hettich, Dorma
b	Dead lock	Geze, Hafele, Hettich, Dorma
c	Pull handle	Geze, Hafele, Hettich, Dorma
d	Concealed Door closer	Geze, Hafele, Hettich, Dorma
e	Door Stopper	Geze, Hafele, Hettich, Dorma
6	Anchor Fastener	Fischer, Hilti
7	Adhesive	Fevicol SH, Vamicol, Araldite of Ciba Geigy

Sr. No.	Description	Few Names of Companies or as Approved by Client
8	Wood Preservative	Wood guard, Term seal
C	WALL FINISHES	
1	Internal Paint	Dulux, Asian Paints
2	External Paint	Snowcem, Nitcocem, Asian Paints Apex Ultima
3	Polyurethane Paint	Thorax coating U K
4	Fire Retardant Paint	Shalimar Paints, Noble Paints
5	Textured Paint	Spectrum, Terraco, Renova
6	Wallpapers	Marshall, M B International, Arte, Ego
D	MISCELLANEOUS	
1	Wall Acoustical material	Armstrong, Anutone, Techno Ceiling Products
2	Insulation material Glass wool	TWIGA, Phenol herm, Kimmco, Pilkington, Owens Corning
3	Antistatic Vinyl flooring	Armstrong, Wonder floor, Nora
4	Artificial Leather	Pride, National
5	Float Glass/Back painted Glass	Saint Gobain, Form 5, Gleverbel
6	Modular demountable Glass partitions	Methis, JEB
7	Mirror	Modi, Asahi, Saint Gobain
8	Writing Board	White Mark, AlkoSign, Altop, Alkon
9	Lockers and Storage Compactor	Godrej, Kompress, Steelage
10	Hand Dryer	Ascon, Kimberly Clarke, Technocrats
11	Fabric protection	Scotchguard Birla 3M, Fabguard Dove Corp.
12	Frosted Film	Garware, 3M
13	Wooden Flooring	Pergo, Tarkett, Armstrong, Ego
14	SS Railing	Neki, Ozone, Enox
15	Conventional Sections	Jindal, Indal, Hindalco
16	Slotted angel racks	MEK, Godrej, Vishwakarma
17	Rubber gasket	Neoprene, EPDM

Sr. No.	Description	Few Names of Companies or as Approved by Client
18	Synthetic Resin	DuPont Corian
19	Auto glow Signage	Prolite, Cross Fire, Martin
20	Signage	Neki, 3M
21	HDL Toilet Cubicles	Merino, SPCMG
22	HDL Lockers	Merino, T Line
23	Conventional Skirting	Windor, Doyle Asia, Bottomline
24	Vinyl Graphics Films	Supplier Approved by Architect
26	Metal Fire Doors	MPP, Matrix, Shakti Mate
27	Roller Blinds	VISTA, Wall track, Hunter Douglas

Notes:

1. The names of companies / manufacturer are given for guidance. The materials shall be used after obtaining approval from PMC and Client. Approved samples shall be preserved in sample room at site of work to ensure the quality / make for actual execution at site of work, no deviation from approved sample is allowed unless approved by Client.
2. No extension of time shall be granted because of delay in getting approval from Client.
3. The PMC/Client reserves right to ask for invoices and other details, if found necessary.
4. PMC/Client reserves the right to reject material if the same is not approved or substandard material used by Contractor at any stage (i.e., even during Defect Liability Period).
5. The contractor shall provide test certificates from the designated laboratories.
6. All the material shall be of first quality and ISI marked.
7. Certification of PMC regarding presence during testing and sample approved is mandatory.
8. PMC certification of all construction, finishes, plumbing, electrification and site infrastructure material etc. is mandatory.
9. All requisite certificate shall be enclosed along with contractor (s) bill.
10. No deviation from approved sample is permissible without prior approval of client.
11. If materials of companies as mentioned in the table are not available then other company material only after approval of the client, shall be used.
12. If any dispute client's decision shall be final.

CHAPTER NO. 8

8 List of Approved Material for Electrical Works

Sr. No.	Name of material	Company name
1	2	3
1	LED Light	Philips/ Wipro / GE/Polycab/GM/HPL/Crompton / Luker/ Surya / Kent / Jaguar/Holonix
2	Ceiling Fans	Cromptons / Orient / Havells /Polycab/ GM/ Panasonic/ Usha/ RR Electric
3	Exhaust Fans	Cromptons / Havells / Orient/ Polycab/ GM/ Panasonic/ usha
4	Modular Switches, Accessories	Anchor /Havells/RR Electrical/Vinay/ Modular/ Cromptons/ GM/ wipro/ Polycab/ legrand/HPL/ Precission/ Panasonic/ ABB/ L&T
5	MCB's - 10 KA. D, B	Legrand / Siemens /Kent/RR Electrical/ Wipro/ L&T/C& S/Polycab/ GM/ Hager/HPL/ Anchor/In-do-asian Honeywell/ Kent/Havells
6	ELCB/RCCB - 10.0KA. D, B, ACB	Legrand/Wipro/Siemens/L&T/ GM C&S/polycab /HPL/Panasonic/Indo-Asian/Honeywell/RR Electrical/ Kent/ABB
7	HRC Switch Fuse Units	L&T / GE / C&S/ GM /HPL/Indo-Asian
8	CFL/LED bulb Holder	Anchor /Havells/Bajaj/GM/Wipro/Polycab/ HPL/ Precission/ Panasonic/ L&T/Modular
9	Wires: 1100V/660V Grade FRLS	Finolex/Polycab/GM/Anchor/L&T/ KEI /RR Kable/ Episilon/ Rotoplast/Kent
10	Casing capping and Accessories	Precision /Anchor/GM /Press fit / AKG
11	Distribution Boards	Legrand/Siemens/L&T/GM/C&S/Polycab/Wipro/ HPL/Panasonic/ Hager/ Honeywell
12	Load break switch	Legrand / Siemens /L&T/Hager/HPL
13	Cables Glands and Lugs	Finolex/GM/Polycab
14	LED Streetlight Fittings	Panasonic/HPL/Wipro/GM/Crompton/Jaquar/Kent/ Surya/RR Electric/ Halonix

Sr. No.	Name of material	Company name
1	2	3
15	LED Flood Light	Wipro/Philips/Havells/C&S/Polycab/ GM//Panasonic/HPL/Wipro/Crompton/Jaquar /Kent/Surya/ RR Electrical/Halonix
16	LED Pathway Light	Wipro / Philips / Havells/ C&S/ Polycab / GM/ Panasonic/HPL/Wipro/Crompton/Jaquar /Kent/Surya/ RR Electrical/Halonix
17	XLPE Cables and Accessories	Polycab / Havel's / Finolex/ KEI/Anchor/HP/ RR cable/Rotoplast/Kent
18	Aluminium Raceways	Jindal / Bentec/Honeywell
19	Industrial Sockets	Legrand / L&T/HPL
20	Changeover Switch	L&T/ HPL / Havells / C&S/Siemens/Wipro
21	Capacitors	L&T / BENTEC
22	Lightning Arrestor	Siemens / Crompton
23	BUSBAR	L&T/HPL/BENTEC
24	Main LT Panel / APFC Panel	Siemens/ L&T / Devta / SewTech
25	L.T. Feeder Pillar	Siemens/ L&T /Indian Electrical /AKG /SAMCON
26	Metering Cubicle	Siemens/ L&T /Indian Electrical
27	11KV isolator and D.O. fuse.	Topaz / Siemens
28	Distribution Transformer	Crompton, ABB, L&T, Siemens, MSEDCL/ Approved
29	Diesel Generator Set	Kirloskar /Cummins / Cromptons/Sterling Wilson/Powerica/Greaves Cotton
30	Lift	Schindler/OTIS / HMAV/ Thyssenkrupp/Omega /KONE
31	Water Pumps	Kirloskar / Crompton
32	Solar Water Heating Systems	Tata Solar / Jain Solar /Surya/polycab/bosch/ Hawells/Panasonic
33	Audio System	Bose/Senneheiser

Sr. No.	Name of material	Company name
1	2	3
34	Water Heater	Bajaj/Recold/Crompton
35	MCB, MCB, DB, MCCB, ACB	Leagrand/Seimens/L&T/ABB/C&S/ESCO,Indoasian/Benlo/HPL. / Indoasian
36	Floor Mounting MCCB Panels	ABB/Snider/Seimens/L&T/ C&S/Indoasian/ HPL
37	Pumps	Kirloskar/Crompton/KSB/ CRI
38	Octagonal / Conical / High mast Pole	Bajaj/Transrail/Surya/Volmont/Utkarsh
39	Decorative Poles	Wipro/Suncity/ K-Lite
40	Decorative Fitting	Wipro/Suncity/ K-Lite/Philips
41	Solar Power Generator / Solar PV System	Havells/Panasonic/Waze
42	Street Light Poles	Havells/Panasonic/Waze/

APPROVED MAKES FOR FIRE FF and FAS, FA SYSTEM		
Sr. No.	Item Description	Approved Makes
1	Main & Jockey Fire Pumps	Kirloskar/Armstrong/Xylem/ABB/Wilo (Mather+Platt)
2	Electric LT Motors	ABB/WEG/Siemens/ Kirloskar/Crompton
3	Standby Diesel Engine Driven Pumps	Kirloskar/Mather & Platt/ Armstrong/ Patterson/ Pentair/Clark
4	GI Pipes	Jindal / Tata Steel / APL Apollo
5	Fire Brigade Connections	Newage /HD Fire/Shah Bhogilal
6	Di Body Gate Valves	Viking/Victaulic/Tyco/Fivalco/Kirloskar/ Hd Fire
7	Pressure Gauges	Waree, Wika, Ashcroft, Baumer/Dwyer
8	Water Flow Meters	Gerand / Siemens/Krohne Marshall / Dwyer/Any UI Listed/Any Fm Approved Pumps
9	Fire Hose Boxes	Newage/Venus
10	Check Valves/ Nrv	Kirloskar/Audco/Tyco/C &R

APPROVED MAKES FOR FIRE FF and FAS, FA SYSTEM			
Sr. No.	Item Description	Approved Makes	
11	Y Type/Pot /Bucket Strainers	Appollo Valves/Alfa Valves/Bombay Chemical Equipment Co.	
12	Branch Pipes	Newage/Shah Bhogilal	
13	CI Body Gate Valves	Viking/Victaulic/Tyco/Fivalco/Kirloskar/ Hd Fire /Audco/Zoloto	
14	Globe Valves	Viking/Victaulic/Tyco/Fivalco/Kirloskar/Hd Fire /Audco/ Leader/ Zoloto	
15	Ball Valves	Claval/Tyco/Fivalco/Kirloskar/Hd Fire /Audco/Leader/ Zoloto	
16	Hose Reels	Newage/Minimax/Jyoti Fire	
17	Alarm Valves	Hd Fire/ Tyco/Viking/Reliable/ Zoloto	
18	Sprinkler Gong	Hd Fire/ Tyco/Viking/Reliable	
19	Flow Switches	System Sensor/Indfoss/Danfoss/Potter/Dwyer	
20	Fire, Smoke Detectors, Manual Call Point, Monitor & Control Modules	Honeywell / Symplex /Notifier USA (Present System At Actrec)	
21	Sprinkler Heads	Tyco / Reliable/Viking/HD Fire	
22	Pressure & Level Transmitter	Honeywell/Siemens/Emerson/Yokogawa	
23	Pressure Switches	System Sensor/Indfoss/Danfoss/Potter/Dwyer	
24	Electrical & Control Cables	Polycab/Cci/RR Kables/ Havells/ Metaplast /Associated	
25	Auto Transfer Switch	Siemens/L&T/Schneider/Asco/Emerson	
26	MCCB & MCB	Siemens/L&T/Schneider/Asco/Emerson/ ABB/ Legrand / Indoasian	
27	Fire Pumps MCC Electrical Panel	Elecmec/ Siemens/L&T/Indcon/ Havells	
28	Push Button Switches	Siemens/L&T/Schneider/Connectwell/Te chnik/ ABB/ Seimmery	
29	Cable Terminations	Connectwell/Wago/Elmex	
30	Voltmeter, Ammeter	Kaycee/Rishabh/Meco/Me	

APPROVED MAKES FOR FIRE FF and FAS, FA SYSTEM		
Sr. No.	Item Description	Approved Makes
31	Load Manager	Siemens/L&T/Schneider/Asco/Emerson/Selec/Conserve
32	GI Perforated Cable Trays	Elcon/Aditya Steel/Indal/Sai Cable Trays/Legrand
33	Copper Earthing Electrodes	Spa Infrastructure/Isg Global/Galaxy/SGI/ABB/ JEF
34	Fire Extinguishers	Safex/ Kanex/ Bharati Fire/ Minimax/ Intime
35	Chain Pulley System Motorized	Hercules/Crane /Titan Make or Eqvt.
36	Indicating Lamps on MCC Panel	Teknic Or Eqvt.
37	Dc Relay	Oen/Pla
38	Current Transformers	Automatic Electric/Me
39	Light Fittings	Philips/Bajaj/Crompton/Anchor/
40	Electric Fittings	Anchor/Roma/Havells/Hager/Schneider
41	Cable Glands	Baliga/ Flameproof Equipment/ Fcg/ Ex Protecta
42	Pressure Relief Valve on Network	Frenger Lesser/Tyco/ Claval
43	Pressure Relief Valve in Pump House	Tyco/ Claval

LIST OF APPROVED MAKES for HVAC

Sr. No.	Item	Manufacturer/ Brand Name
1.	AC UNITS (VRF/NON VRF)	Daikin/Toshiba/Mitsubishi/Blue Star
2.	Refrigerant Pipes & Fittings	Maxflow/Mandev/Heling
3.	Centrifugal fans	Kruger/Nicrota/Systemair
4.	Cabinet/Circular Fan	Kruger/Nicrota/Systemair/VTs
5.	Anchor Fasteners	Hilti/ 3M/Canon
6.	GI Sheets	Jindal/TATA/SAIL/ Universal/ ISPAT
7.	Factory Fabricated Ducts	Roltastar/ZECO/Asawa

Sr. No.	Item	Manufacturer/ Brand Name
8.	Damper	Dynacraft/Cosmos/Ravistar/Carryaire
9.	Fire Dampers	Dynacraft/George & Rao/Carryaire/ Cosmos
10.	Aluminium Grilles/Diffuser	Dyanacraft /Cosmos/ Ravistar/Carryaire
11.	Acoustic Insulation	Armasound
12.	Disc Valve	Dyanacraft /Cosmos/ Ravistar/Carryaire
13.	Nitrile Rubber Insulation	Aeroflex/Armaflex/Kflex
14.	Under-deck Insulation	Supreme/Owens Corning
15.	UPVC Pipe	Supreme/Prince/Airtech
16.	Drain Pipe/Refrigerant Pipe/ Insulation	Aeroflex/Armaflex/Kflex
17.	Switchgear, Push Buttons & Indicating Lamps	L&T/ABB/SIEMENS
18.	Motors	L&T/ABB/SIEMENS
19.	AC Controllers	Belimo/ABB/SEIMENS/Honeywell/Schiendler
20.	Actuators (Motorized)	Belimo/ABB/SEIMENS/Honeywell/Schiendler
21.	VFD	L&T/ABB/SIEMENS
22.	Anchor Fasteners	Hilti/3M/Canon

13. The names of companies / manufacturer are given for guidance. The materials shall be used after obtaining approval from PMC and Client. Approved samples shall be preserved in sample room at site of work to ensure the quality / make for actual execution at site of work, no deviation from approved sample is allowed unless approved by Client.

14. No extension of time shall be granted because of delay in getting approval from Client.

15. The PMC/Client reserves right to ask for invoices and other details, if found necessary.

16. PMC/Client reserves the right to reject material if the same is not approved or substandard material used by Contractor at any stage (i.e., even during Defect Liability Period).

17. The contractor shall provide test certificates from the designated laboratories.

18. All the material shall be of first quality and ISI marked.

19. Certification of PMC regarding presence during testing and sample approved is mandatory.

20. PMC certification of all construction, finishes, plumbing, electrification and site infrastructure material etc. is mandatory.
21. All requisite certificate shall be enclosed along with contractor (s) bill.
22. No deviation from approved sample is permissible without prior approval of client.
23. If materials of companies as mentioned in the table are not available then other company material only after approval of the client, shall be used.

LIST OF IS – CODE:

Sr. No.	Item	Manufacturer/ Brand Name
1.	IS: 277-1977	Galvanized steel sheet (plain & corrugated) (Amendment - 01)
2.	IS: 655-1963	Metal Air Duct (Amendment - 02)
3.	IS: 659-1964	Safety Code for Air-conditioning (Amendment - 01)
4.	IS: 660-1963	Safety Code for Mechanical refrigeration
5.	IS: 900-1992	Code of practice for Installation & Maintenance of Induction Motors
6.	IS: 2441-1984	Code of practice & measurement procedure for testing refrigerant compressors (Amendment - 01)
7.	IS: 4894-1987	Test code for Centrifugal Fans (Amendment - 01)
8.	IS: 5111-1993	Code of practice & measurement procedure for testing refrigerant compressors (Amendment - 01)
9.	IS: 7613-1975	Method of Testing panel type air filter for Air conditioning & ventilation purpose
10.	IS: 3588-1987	Specification for electrical axial flow fans
11.	SMACNA – 1995	Sheet metal – Ducting, Manufacturing & Installing (low- & high-pressure ducting)
12.	BS. EN 779: 1998	Air Filters.

9 Payment Schedule

9.1 Preamble

The Contractor shall make himself aware of the Client's procedure regarding measurements and payment of interim bills and the final bill for the Project /Work.

9.2 Initial Measurement of Work:

The PMC shall measure the initial ground levels and all hidden measurements in the presence of the contractor's representative if he chooses to remain present. The said initial levels and measurements shall be recorded, in the authorized field book or measurement book of the Client, by the authorized Engineer of PMC. The record of such levels and measurements shall be jointly signed by the PMC and the Contractor. The Contractor shall be entitled to have a true copy of the same at his cost. Any failure on the part of the Contractor to remain present at time of taking such levels and hidden measurements, etc. and have the same recorded by PMC before the commencement of the work, shall render him liable to accept the same. Likewise, the Contractor shall not execute / commence any item of work which shall render its subsequent measurements difficult or impossible without first getting the same jointly measured by himself and the authorized representative of the PMC/Client. If such item of works is executed by the Contractor without measurements by the PMC, then the measurements recorded by the PMC shall deemed to be final and binding on the Contractor. For such items/components the criteria shall be based on these measurements taken and recorded by the PMC. To verify the progress of work at site contractor shall submit monthly bill to PMC with intimation to the client. PMC shall clearly mention the targeted milestone as per tender and achieved milestone.

9.3 Payments:

- i) Interim payments shall be made as per payment schedule given in the contract for the corresponding completed stage of work, as per the certification by the PMC. All payments shall be made in Indian Rupees (INR), after adjustment for deductions of advance payments, security deposit, other recoveries in terms of the Contract and taxes at source, as applicable under the law.
- ii) This being a lump sum contract the payment shall be stage wise. In case of any ambiguity or doubts in respect of stages of work, the decision of the Client shall be final.
- iii) This stage wise payment shall be based with respect to stage of construction and

its corresponding percentage mention in payment schedule under clause 9.6.
Interim stage wise payment.

- iv) PMC shall measure initial ground levels and final levels in presence of contractor 's representative, for measuring the quantity of work done. The Contractor shall supply required labour and equipment, etc. for taking joint measurements. Such measurements shall be signed by the Contractor or his authorized representative and the PMC. The Contractor shall obtain copies of the said measurements and copies of the bills paid to him at his own cost and responsibility. Contractor / PMC shall create digitized copy of measurements for onward submission to the client. PMC shall submit Hard and soft copy to the Client.
- v) For the record PMC shall measure initial ground levels and final levels in presence of contractor's representative. The Contractor shall supply required labour and equipment, etc. for taking joint measurements. Such measurements shall be signed by the Contractor or his authorized representative and the PMC. The Contractor shall obtain copies of the said measurements and copies of the bills paid to him at his own cost and responsibility. Contractor / PMC shall create digitized copy of measurements for onward submission to the client. PMC shall submit Hard and soft copy to the Client.
- vi) The PMC shall record the hidden measurement / stages of work in the measurement book supplied by the Client with certification about quality and quantity of work executed. The Contractor and PMC shall sign in M.B. at appropriate place in standard format of the Client as a token of acceptance of the quality and quantity of the work done.
- vii) Deputy Engineer / Project Engineers of the Client shall exercise 25% check and Executive Engineer of Client shall exercise 5% check of work executed on site. The payment of Contractor 's bill is subjected to above checking.
- viii) Generally, such bill shall not exceed one / two in the month for which the qualitative and quantitative assessment of work is done. The Contractor shall submit running account bills to the PMC with intimation to the Client in an appropriate form along with all necessary statements / documents i.e. test reports, certificate, abstract of tests, progress charts, basic calculations of % of work claimed for payment. No interest shall be payable against delay in payment by the Client. The Contractor / PMC shall submit the parawise compliance of previous inspection notes, instructions issued by the Client / PMC. The Client shall not release the payment of bill without satisfactory parawise compliance of earlier inspection notes / instructions.
- ix) Contractor / PMC shall submit the part wise compliance of previous inspection notes, instructions issued by the Client/PMC. The Client shall not release the

payment of bill without satisfactory part wise compliance of earlier inspection notes/instructions.

- x) Payment of the Contractor(s) shall be made by RTGS only. For RTGS mode of payment the relevant details of bank account shall be provided by the Contractor to the PMC/Client. In the event of wrong details by the Contractor, the Client shall not be liable for any loss caused due to such incorrect RTGS transaction.
- xi) For the purpose of handover of facilities to the user, joint visit of representative of client / PMC / Contractor / user shall be mandatory prior to final completion of works to cure the defects if any observed at site of work.
- xii) Prior to final bill just after completion of second joint visit and observation shall be prepared by PMC and submit to client. Contractor shall cure all the defects within stipulated period as informed by the PMC / Client and hand over the facilities to the user and Client.

9.4 Final Bill:

- i. The Contractor shall submit final bill within one month after completion/ handing over of the entire work and the same shall be paid within 3 months, from the date of certification by the PMC if it is in order. Disputed items and claims, if any, shall be excluded from the final bill and settled separately later on, as per the decision obtained in the dispute resolution procedure specified herein. Final bill shall be submitted with documents as specified in the Check list of Final bills. The Client shall not pay the final bill of the Contractor without the required documents mentioned in checklist.
- ii. The Contractor shall submit five sets of as built drawings with soft copies along with final bill to the PMC. The PMC shall verify the same, and submit it to the Client with his certification, without which the final bill shall not be considered for payment.
- iii. The recoveries if any, which remains as dues from the Contractor after payment of bills, shall be recovered from the Security deposit or by encasing Bank guaranties and if any balance recoveries still remain the same shall be collected as arrears of land revenue through the Collector of the District.
- iv. Disposal of dismantled material or excess quantity of excavated materials shall be entirely the responsibility of Contractor. No claim or extra item shall be sanctioned or paid to contractor against these works. The final bill shall not be processed unless the dismantled or excess excavated materials is disposed of by the contractor.

- v. For the purpose of handing over authorized representative of contractor / PMC / Client user shall visit the site and ensure the completion in all respect as per agreed terms and condition of contract. Unless and until note of such handing over received to the client, final bill will not be paid. Contractor shall be responsible to cure such defect within stipulated period.

9.5 Payment for Additional Work:

Proposals for additional work as per instructions of the Client shall be submitted by PMC within 30 days along with all supporting documents. Bills for extra items/additional quantities executed, approved by the Client shall be submitted by PMC within 10 days of completion of additional works along with supporting documents, photos, etc. Bills for any other claim if accepted by the Client shall be paid separately. The payment of bills for the original work shall not be withheld for want of decision on the execution of additional quantities/extra items or any other claim. No extra work shall be done without prior approval by the MHDC/PMC.

9.6 Payment Schedule

Name of Work:

Construction of Affordable Housing Project of 1104 EWS and 1444 LIG Tenements Under Pradhan Mantri Awas Yojana (U) on Kh. No 51 & 52, Khairi, Tal. Kamptee, Dist. Nagpur.

Note: Payment will be made on the cost as derived by following form and Respective Percentage.

PAYMENT SCHEDULE					
Sr.	Particulars	Up to Plinth Level	Stilt / Ground Floor	From 1st Floor to Terrace Floor	Total % of Approved Tender Cost
		%	%	%	%
1	Demolition, Dismantling, Land Levelling Fixing of TBM, Baseline, Centre line checking, finalization working layout of site and approval from PMC / Client.	0.5	--		0.50
2	Excavation for foundation individual excavation for footing including cutting, shifting of excavated material, if necessary, Antitermite treatments, filling in plinth, soling, providing plinth protection, PCC, RCC up to plinth including Piling work/Footing, column, plinth beam etc. complete and approval from PMC / Client.	14.5			14.50
Total up to Plinth (Substructure)					15.00
3	Super-structure RCC work for columns, beams, slabs, lintels, pardi, Chajja, canopy, fins water tanks, landings, and waist slabs of staircase including steel etc.			32.57/ No. of floors (From 1st Floor to Terrace floor)	40.00

	complete and approval from PMC / Client.				
4	Brick/AAC/ masonry for walls, RCC partition walls/parapet walls including RCC/ CRC jail etc. complete and approval from PMC / Client.	--	0.63	5.37/ No. of floors from 1st Floor to Terrace floor)	6.00
5	Door and Window shutters with frame, structural steel work, MS grills, M.S. Gates etc. complete and approval from PMC / Client.	--	0.02	6.68 / No. of floors from 1st Floor to Terrace floor)	6.70
6	Water-Proofing to terrace, WC, bath, Chajja, staircase roof cabin, etc. complete and approval from PMC / Client.	--	0.00	1.50 / No. of floors from 1st Floor and above floors. And 0.50 for Terrace)	2.00
7	Flooring in all rooms, entrance and passage, WC, bath, staircase, dado, skirting, window sills, etc and approval from PMC / Client. cupboards, granite platform, shelf, water trough, etc. complete and approval from PMC / Client.	--	0.5	9.00 / No. of floors from 1st Floor to Terrace floor)	9.50
8	Internal Painting, etc. complete and approval from PMC / Client.	--	0.05	0.7 / No. of floors from 1st Floor to Terrace floor)	0.75
9	External Painting etc. complete and approval from PMC / Client.	--	0.05	1.40 / No. of floors from 1st Floor to Terrace floor)	1.45
10	Provide and Fixing Fire Fighting System, Fire Alarm and approval from PMC / Client	--	0.06	.94/ No. of floors from 1st Floor to Terrace floor)	1.00

11	Internal Electrification with service connection, fittings, fixtures, etc. complete and approval from PMC / Client.	--	0.86	5.14 / No. of floors from 1 st Floor to Terrace floor)	6.00
12	Internal Plumbing works including water supply, drainage, water supply and sanitary fittings like WC pans, wash hand basins etc, complete and approval from PMC / Client.	--	0.4	4.2/ No. of floors (From 1 st Floor to Terrace floor)	4.6
Total of All Superstructure					78.00
Infrastructure works					
13	Amenity Building	--	1.28/No. of floors from 1 st Floor to Terrace floor)		1.28
14	Fitness Centre	--	0.30/No. of floors from 1 st Floor to Terrace floor)		0.30
15	Creche-RG5	--	0.22/No. of floors from 1 st Floor to Terrace floor)		0.22
16	Creche-RG2	--	0.22/No. of floors from 1 st Floor to Terrace floor)		0.22
17	Infrastructure Plumbing Works	--			0.59
18	Infrastructure Fire Fighting Works	--			0.37
19	Infrastructure Electrical Works	--			0.69
20	Internal Road and Parking	--			0.29
21	Compound wall/Security Cabin	--			0.34
22	U.G. Tanks	--			0.36
23	Storm Water Drain	--			0.19
24	Parks and Gardens	--			0.15
Total for Infrastructure Works					5.00

30	Payment shall be released on completion of project in all respect, including submission of Royalty Clearance Certificate, a necessary certificate from electrical inspector and MSEB work completion certificate, occupancy certificate, submission of as built drawings, successful handing over. Taking the properties on PRB by PWD civil and electrical. Site clearance conveyance of excess excavated debris material at any lead and all balance works that not covered above etc. complete and approval from PMC / Client.	--	2.00		2.00
				Total	2.00
				Total for Project	100.00

NOTES:

- I. If Contractor has adopted Aluminium Shuttering RCC monolithic structure / technology or adopts and alternate technology in which form finish RCC surface is achieved which will not require further finishing by internal or external plaster in that case percentage (i.e., Internal/External plaster and all cladding, elevational works etc. complete and approval from PMC/ Client) will be suitably added to **Sr. no 3 of 9.6.**
- II. If Contractor has adopted Alternative construction materials / technology apart from Aluminium Shuttering RCC monolithic structure / technology than in that case payment schedule (9.6) may be revised by MHDC considering work sequence and technology requirements etc.
- III. Contractor shall deposit along with final bill; the bank guarantees from Nationalized / Schedule Bank.
 - a) Equivalent to 1% of cost for waterproofing including the cost of urinals, W.C. units, flooring tiles etc. or Minimum Rs.5.00 Lakhs.
 - b) Equivalent to 1 % of cost for anti-termite work or Minimum Rs.2.00 Lakhs.
 - c) Both Bank guarantees from Nationalized / Schedule Bank shall be for 10 years to

the Client.

(Note: Refer clause no.3.3.12)

- IV. If the defects are not rectified the bank guarantee will be encased and Client will carry out the rectification of defects. Contractor shall submit guarantee bond on Rs.1000/- stamp paper for each work i.e., waterproofing and ant termite.
- V. For those items with warranty given by the manufacturer, the warranty received by the Contractor being purchaser of those items shall be transferred back-to-back in the name of the Client so as to be able to take benefit of warranty.
 - a) Five years external paint guarantee bond shall be submitted by the Contractor from manufacturing company
 - b) Five years guarantee bond for sanitary fittings shall be submitted by the Contractor from manufacturing company.
 - c) Five years fully comprehensive free AMC for following items.
 - i. Elevators / Lifts.
 - ii. Firefighting and fire alarm systems.
 - iii. Street lighting
 - iv. Computer Networking systems/Intercom and Telephone systems/ Electrical System
 - v. Water pumping system.
 - vi. Solar Power Generation System.
 - vii. C.C.T.V. System
- VI. The payment schedule percentage as stipulated in the above table is based on tender cost. Further the payment schedule percentage based on accepted tender cost shall be prepared by PMC and approved by Client.
- VII. The bills shall be submitted by the Contractor to the PMC and with intimation to the Client. The percentage of payment schedule as per work done by the Contractor along with all prescribed documents, duly certified by the PMC.
- VIII. No advance shall be paid against material brought on site of work.
- IX. No Payment shall be made for substandard quality/ workmanship of the work and for non-submission of test result and other documents, etc.
- X. The Contractor shall establish Site office and field laboratory as per approved drawing by PMC within 2 months from date of work order. Further delay shall attract compensation as decided by the PMC/Client and the payment from Sr. No. 3 (9.6 payment schedule) onwards shall not be made.
- XI. For any extra item out of this tender, the quantities will be paid at current SSR.
- XII. The whole works shall be carried out by good engineering practices manuals.

CHAPTER NO. 10

10 Form of Offer

10.1 Memorandum of Understanding:

(a)	Name of Work	Construction of Affordable Housing Project of 1104 EWS and 1444 LIG Tenements Under Pradhan Mantri Awas Yojana (U) on Kh. No 51 & 52, Khairi, Tal. Kamptee, Dist. Nagpur.
(b)	The amount of Earnest Money to be Deposited.	Rs 91,46,897/- (Ninety-One Lacs Forty-Six Thousand Eight Hundred and Ninety-seven only) should be paid through online gateway mode.
(c)	Amount of Security Deposit to be deposited.	Security Deposit– 1% (One Percent) of the tendered cost offered by Tenderers and Accepted by Client. 50% of total security deposit shall be in the form of Demand Draft/ FDR/ Bank Guarantee/RTGS/NEFT of Nationalized Bank/ Scheduled Bank payable at Mumbai. 50% of total S.D. shall be deducted from the R.A. bills @ the rate of 5 % .
(d)	Period of Contract	Period allowed for the completion of whole work from the date of written work order to commence the work including monsoon period.

10.1.1 I/We agree that the offer of the contractor shall remain open for acceptance for a minimum period of 120 days and an extendable period as per request of the client from the date fixed for opening of the Bid and thereafter, until it is withdrawn by me/us by notice in writing duly addressed to the authority opening the tenders and sent by registered post A.D. otherwise delivered at the office of such authority.

Earnest Money Deposit (EMD) amounting to **Rs 91,46,897/- (Ninety-One Lacs Forty-Six Thousand Eight Hundred and Ninety-seven only)** in the form of Demand Draft/ FDR/ Bank Guarantee/RTGS/NEFT of Nationalized Bank/ Scheduled Bank payable at Mumbai to MHDC.

10.1.2 The amount of EMD shall not bear interest and shall be liable to be forfeited by the Client, should I/we fail to abide by the Tender conditions including the stipulation to maintain the validity of the bids for the period mentioned hereinabove, sign and

complete the contract documents as specified in the time limit laid down in Clause No.12.5 Termination of Contract.

10.1.3 If this tender is accepted, I/We hereby agree to abide by and fulfil all the terms and provisions of the Conditions of Contract annexed hereto so far as applicable and in default thereof to forfeit and pay to the Client the sums of money mentioned in the said Conditions.

10.1.4 Tender form fee including GST amounting to Rs. 5,900/- should be paid through online gateway mode and Earnest Money Deposit (EMD) amounting to **Rs 91,46,897/- (Ninety-One Lacs Forty-Six Thousand Eight Hundred and Ninety-seven only)** should be paid through online gateway mode. The said EMD is to be absolutely forfeited by the Client should I/We do not deposit the full amount of Security Deposit specified in the above memorandum, in accordance with Clause No. 10.1.1(b) and (c) of the said conditions, otherwise the said sum of Rs. -----/- shall be refunded.

Note: Account details of MHDC wherein EMD & SD etc. has to be deposited may be obtained by the concord bidder by contacting MHDC office during working hours.

Signature of Bidder (Before submission of Tender)

Address:

Date the day of 20

Signature

Designation

Address

(Signature of Witness to Bidder's signature)

Managing Director & C.E.O

Maharashtra Housing Development Corporation Limited, Mumbai

(Signature of the officer by whom accepted.)

10.2 Proforma of Agreement

(On Stamp of Appropriate Value as Per the Maharashtra Stamp Act)

Article of Agreement (hereinafter referred to as the “**Contract**”) made at this Day Month Year

Between

Maharashtra Housing Development Corporation Limited a Government of Maharashtra undertaking, incorporated under the Companies Act. 2013 and having its registered office at Sakhar Bhavan, Office No. 32, 3rd floor, Ramnath Goenka Marg, Nariman Point, Mumbai, Maharashtra 400021. (Hereinafter called “**the Client**”) of the ONE PART and whose registered office is situated at **Mumbai**.

And

M/s. (herein after called the “**Contractor**”) of the OTHER PART. **As the context may require, the Client and the Contractor, are hereinafter collectively called the "Parties" and individually as a "Party"**)

WHEREAS:

The PMC/Client being desirous of having provided and executed certain works mentioned, enumerated or referred to in the Tender Notice, Instructions to Tenderer's General Conditions of Contract, Special Conditions of Contract, specifications, technical report, schedule of quantities, for works drawings and other documents constituting the “TENDER” and acceptance thereof Copies thereto are annexed, all of which are designed to form part of this Contract and are included in the term “CONTRACT” wherever herein used.

AND WHEREAS:

The Client accepted the Tender of the Contractor for the provision and the execution of the said work at the Lump sum cost of Rs. _____ + at the percentage _____ (Above/Below) of Rs. _____ (Estimated cost put to Tender).

AND WHEREAS:

The Contractor has deposited with the Client the sum of Rs. /- (Rupees only) being the initial Security Deposit payable by him at the rate of 0.5% of the tendered value and undertakes to pay the deducted amount from the bills payable to him provided that the deduction does not exceed 5 % (Five Percent) of the billed amount to be retained until the expiry of the defect liability period of **5 (Five)** years for the due observance and performance of this Contract.

NOW THIS AGREEMENT WITNESSES AND IT IS HEREBY AGREED BY AND BETWEEN THE PARTIES AND DECLARED AS FOLLOWS:

- 1) The documents which form the part of the Contract will be the Tender Notice. Detail Tender Notice include Corrigendum to Tender Notice, instruction to tenderers, General Conditions of Contract, Specifications, Drawings, items of work, Special Conditions of Contract and Common Set of Deviation and other documents including acceptance of Tender and Contractor's letter for acceptance thereof and work order, etc.
- 2) In consideration of the payments to be made to the Contractor for the work to be executed by him, the Contractor shall and will duly provide, execute and complete the said work on or before the dates mentioned in the time schedule of completion of work attached to the Tender documents and shall maintain the same at his own cost for the defects liability period thereafter and perform all such acts and things in the Contract mentioned or described or which are to be implied there from or may be reasonably necessary for the completion of the said works at the times and the manner subject to the terms and conditions or stipulations mentioned in the contract.
- 3) In consideration of the due provision, execution and completion of the said work, the Client does hereby agree with the Contractor that the Client will pay to him at the quoted cost and such other sums as may become payable to the Contractor under the provisions of the contract, such payment to be made at such time and in such manner as provided in this Contract.
- 4) In consideration of the due provision, execution and completion of the said work, the Contractor does hereby agree to pay to the Client the amount as may be due to the Client and such other sum may become payable to the Client towards loss, if any as set forth in the said Conditions of Contract, such payments to be made at such time and in such manner as provided in the Contract.

IN WITNESS WHEREOF THE PARTIES HERETO HAVE SET AND SUBSCRIBED THEIR HANDS AND SEALS TO THESE PRESENTS IN DUPLICATE ON THE DAY AND YEAR FIRST MENTIONED ABOVE.

SIGNED AND DELIVERED FOR AND ON

BEHALF OF Maharashtra Housing Development Corporation Limited, Mumbai – 400021.	SIGNED AND DELIVERED OF CONTRACTOR
---	------------------------------------

In the presence of witness

1. Signature _____

Designation_

Address _____

2. Signature _____

Designation_

Address _____

11 Price Variation Clause (Price Adjustment)

Star rates shall be provided for in the SBD for cement and steel. The difference between purchase price of contractor and the star rates shall be recoverable from the contractors or payable to the contractors.

1) Formula for cement component:

$$V1 = QC \times (C1 - CO)$$

Where,

V1 = Amount of Price variation in Rupees to be allowed for cement component.

QC = Quantity of cement in M.T. used in the work during month under consideration.

CO = Basic rate of cement in rupees per M.T. as mentioned in Below Table.

C1 = Current market price in rupees per M.T. of Cement excluding GST during the month under consideration.

2) Formula for TMT FE — 500 Component:

$$V2 = QS \times (S1 - SO)$$

Where,

V2 = Amount of Price Variation in Rupees to be allowed for TMT FE 500 Steel component.

QS = Quantity of steel in M.T. used in the work during month under consideration.

SO = Basic rate of steel in rupees per metric ton as mentioned in Below Table.

S1 = Current market price in rupees per M.T. of steel excluding GST, as during the month under consideration out estimated cost excluding GST.

Basic Star Rate (Excluding GST)

Sr. No.	Component	Amount	Unit
1	Cement	Rs. 6,000/-	Per Metric Tonne
2	Steel FE 500	Rs. 61,000/-	Per Metric Tonne
3	Structural Steel	Rs. 62,575/-	Per Metric Tonne

Total amount due to price variation

$$V = V1 + V2$$

12 Conditions of Contract

12.1 Definition of Work/Project:

The expression "Work" or "Works" or 'Project' were used in these conditions, shall unless there be something in the subject or context repugnant to such constructions, be construct to mean the work or works contracted to be executed under or in virtue of the contract, whether temporary or permanent and whether original, altered, substitute do additional.

12.2 Acceptance of Conditions & Conditional Tenders Liable for Rejection:

All the conditions mentioned in the contract shall be binding on the Contractor(s). Any conditional offer will be liable for rejection. The Contractor(s) who does not accept the conditions shall not be allowed to tender for further works and his name shall be removed from the list of Contractor(s).

12.3 Contractor's Invalid Conditions

If subsequently it is found that the contractor has quoted his offer with some conditions which is not in conformity with the bid document then such conditions shall be treated as invalid and the contract conditions only shall be binding.

12.4 Security Deposit

The successful tenderer whose tenders may be accepted (here in after called "the Contractor(s)" which expressions shall unless excluded by or repugnant to the context include his / her, executors, administrators and assigns) shall

- a) Within Fifteen (15) days of the receipt by him of the notification / letter of the acceptance of his tender by post or email deposit with the Client in the form of Demand Draft/ FDR/ Bank Guarantee/RTGS/NEFT of Nationalized Bank/ Scheduled Bank a sum sufficient to make up the 50% of the total Security Deposits specified in the tender clause 1.8(B)
- b) Remaining 50% of the Security Deposit shall be deducted at the rate of 5% (Five percent) of all money till full Security Deposit is recovered. Such deduction shall be held by the Client by way of Security Deposit. Provided always that, in the event of the Contractor(s) deducting a sufficient sum from every such payment as last aforesaid until the full amount of the Security Deposit is made up.

All compensation or other sums of money payable by the Contractor(s) to the Client under the terms of this contract may be deducted from part of his Security Deposit or from any sums which may be due or may become due to the Contractor(s) on any account whatsoever, and in the event of his Security Deposit being reduced by reason of any such deduction as aforesaid, the Contractor(s) shall within ten (10) days thereafter, make good in cash as aforesaid any sum or sums which may have been deducted from his Security Deposit or any part thereof.

The Security Deposit referred to shall not be converted into interest-bearing securities. If the amount of the Security Deposit to be paid in a lump sum within the period specified at (a) above is not paid, the tender /contract already accepted shall be considered as cancelled and Earnest Money Deposit shall be forfeited. The Security Deposit lodged by a Contractor (s) shall be refunded after the expiry of defect liability period of Five (5) years.

The Security Deposit lodged by the Contractor(s) shall be returned to him on the expiry of the period of guarantee after satisfying conditions mentioned above after deducting there from the amount of expenses, if any, due to the Client under this Agreement.

12.5 Termination of Contract

ACTION WHEN TOTAL SECURITY DEPOSIT IS FORFEITED

In any case in which under any clause of this contract, the Contractor shall have rendered himself liable to pay compensation amounting to the whole of this security deposit (whether paid in one sum or deducted in instalment) or in the case of abandonment of the work owing to serious illness or death of the Contractor, or breach of the Contract or insolvency, or the Contractor fails to maintain the required rate of progress of work, or inordinate delay, or dissolution of the Contractor's firm, or any other cause, the Client shall have the power to adopt any of the courses stated herein-below, as the Client may deem best suited to the interest of the Client.

- a) To terminate the contract (for which 30 days termination notice in writing to the Contractor shall be conclusive evidence of the termination) and in that case the Security Deposit of the Contractor shall stand forfeited and be absolutely at the disposal of the Client;

Note: - On such termination, a joint measurement shall be carried out by the PMC/Client or its representative and the Contractor or its representative. If Contractor fails to participate in such joint measurement despite written communication by the Client, the measurement taken by Client or its representative shall be binding on the Contractor. Payment calculations for the work done by the Contractor shall be done on quantum merit basis.

- b) To carry out the work or any part of the work by the Client debiting the Contractor with the cost of completion of the balance work, expenditure incurred on tools and plant, and charges on additional supervisory staff including the fees payable to Project Management Consultant and Architect and other agencies employed for getting the un-executed part of the work completed and crediting him with the value of the work done by the Client in all respects in the same manner and balance work not to be higher than contract rates, as if it had been carried out by the Contractor under the terms of this contract. The certificate of the Client as to the costs and other allied expenses so incurred and as to the value of the work so done by the Client shall be final, conclusive and binding on the Contractor.
- c) To order that the work of the Contractor be measured up and to take such part thereof as shall be un-executed out of his hands, and to give it to another Contractor(s) to complete, in which case all expenses incurred on advertisement for fixing a new contracting agency, additional supervisory staff including the cost of work charge establishment and the cost of the work executed by the new agency will be debited to the Contractor and at the value of the work done or Executed through the new Contractor shall be credited to the Contractor in all respects and in the same manner and at the same rates as if it had been carried out by the Contractor under the term of his contract. The certificate of the PMC/Client as to the total cost of the work and other expenses incurred as aforesaid for or in getting the unexecuted work done by the new Contractor and as to the value of the work so done shall be final, conclusive binding on the Contractor.

12.6 Action for Unsatisfactory Progress of any Particular Portion of the Work

In case the contract is terminated in accordance with Clause 12.5 (a) above, the Contractor shall not be entitled to recover or be paid any sum for any work there to actually performed by him under this contract unless and until the Client shall have certified in writing the performance of the such work and the amount payable to him

in respect thereof and he shall only be entitled to be paid the amount so certified. In the event of either of the courses referred to in clause 12.5 (b) or (c) being adopted and the cost of the work executed by the Client or through a new Contractor and other allied expenses exceeding the value of such work credited to the Contractors, the amount of excess shall be deducted from any money due to the Contractor, by the Client under the contract or otherwise howsoever from his Security Deposit or the sale proceeds of his assets movable or immovable as may be applicable thereof. Provided, however the Contractor shall have no claim against the Client even if the certified value of the work done by the Client or through a new Contractor exceeds the certified cost of such work and allied expenses, provided always that whichever of the three courses mentioned in Clauses 12.5 (a), (b), (c) here in above is adopted by the Client, the Contractor shall have no claim to compensation for any loss sustained by him by reason of having purchased or procured any materials/machines, or entered into any engagements or made any advance on account of or with a view to the execution of the work in the performance of the contract.

12.7 Compensation if action not taken under Clause 12.5 AND 12.6.

If the progress of any particular portion of the work is unsatisfactory, the PMC / Client shall not withstand that the general progress of the work is in accordance with the conditions mentioned in clause 12.12, been titled to take action under clause 12.5(b) after giving the Contractor 10 days' notice in writing. The Contractor will have no claim for compensation, for any loss sustained by him owing to such action

12.8 Power Regarding Contractor's Plant

In any case in which any of the powers conferred upon the Client by clause 12.5 and 12.6 here of shall have become exercisable and the same shall not have been exercised, the non-exercise thereof shall not constitute a waiving of any of the conditions here of and such powers shall notwithstanding be exercisable in the event of any future case of default by the contractor for which under any clause here of he is declared liable to pay compensation amounting to the whole of his security deposit and the liability of the contractor for past and future compensation shall remain unaffected. In the event of the Client taking action under sub-clause (a) or (c) of clause 12.5, he may, if he so desires, take possession of all or any tools, plant, materials and stores, in or upon the works or the site there of or belonging to the contractor, or procured by him and intended to be used for the execution of the work or any part thereof, paying or allowing for the same in account at the contract rates,

or in the case of contract rates not being applicable at current market rates, to be certified by the Client whose certificate there of shall be final. In the alternative, the Client may, after giving notice in writing to the contractor or his clerk of the work, foreman or other authorized agent require him to remove such tools and plants, materials, or stores from the premises within a time to be specified in such notice, and in the event of the contractor failing to comply with any such requisition, the Client may remove them at the contractor's expenses or sell them by auction or private sale on account of the contractor and at his risk in all respects, and the certificate of the Client as to the expenses of any such removal and the amount of the proceeds and expense of any such sale shall be final and conclusive against the Contractor.

12.9 Compensation Under Workmen's Compensation Act

The Contractor shall be responsible for and shall pay any compensation to his workmen payable under the Workmen's Compensation act, 1923 (VIII of 1923), (hereinafter called the said Act) for injuries caused to the workmen, if such compensation is payable / paid by the Government as principal under sub-section (1) of Section 12 of the said act on behalf of the Contractor, it shall be recoverable by the Client from the Contractor under sub-section (2) of the said section. Such compensation shall be recovered in the manner laid down in Clause 1 above. Work compensation insurance policy shall be in force till the expiry of defect liability period of work i.e., 5 years w.e.f. date of completion of project.

12.10 Compensation Under Medical Aid, Safety & Apprentices Act

- A. The Contractor shall be responsible for and shall pay the expenses of providing medical aid to any workman who may suffer a bodily injury as a result of an accident whether such accident takes place on a holiday or on a working day. If such expenses are incurred by Client the same shall be recoverable from the Contractor forthwith and be deducted without prejudice to any other remedy of the Client from any amount due or that may become due to the Contractor. Certificate of the medical practitioner and authenticated Architect cum PMC about amount of expenses actually incurred on providing such medical aid shall be final, binding and conclusive against the Contractor(s).
- B. The contractor shall provide all necessary personal safety equipment and first aid apparatus available for the use of the persons employed on the site and shall maintain the same in condition suitable for immediate use at any time shall comply with the following regulations in connection therewith.

- a) The workers shall be required to use the equipment's so provided by the Contractor and the Contractor shall take adequate steps to ensure proper use of the equipment by those concerned.
 - b) When work is carried on in proximity to any place where there is a risk of drowning all necessary equipment shall be provided and kept ready for use and all necessary steps shall be taken for the prompt rescue of any person in danger.
 - c) Adequate provision shall be made for prompt first aid treatment of all injuries likely to be sustained during the course of the work.
- C. THE APPRENTICES ACT** The Contractor shall duly comply with all the provision of "The Apprentices Act, 1961" (III of 1961) the rules made there under and the orders that may be issued from time to time under the said Act and the said Rules and on his failure or neglect to do so he shall be subjected to all the liabilities and penalties provided by the said Act and said Rules. (Government Circular No. CAT-6076/3336/ (400)/ Bldg.2 dt. 16-8-1985) CST-1086/CR-243 Ka-Building-2/ Mantralaya, Bombay 400 032, dated 11 Sept. 1987)

12.11 Insurance Cover for the Work - Delated

~~Without limiting his obligations and responsibilities under any other clause above, the Contractor(s) shall prior to commencement of work(s), insure in the joint names of the Client and the Contractor(s) against all loss or damage from whatever cause arising, other than the excepted risks for which he is responsible under the terms of the contract and in such manner that the Client and Contractor are covered for the period of contract including extension of time limit or after completion of work & further Five (5) year defect liability period as stipulated in contract hereof and for any loss or damage occasioned by the Contractor(s) in the course of any operations carried out by him for the purpose of complying with his obligations under Clauses as below:-~~

- ~~a) The works for the time being executed to the value as shown on the last issued payment certificate plus ten (10) percent thereon (to allow for any additional costs and professional fees resulting from the loss or damage together with the materials on site for the Client in the works at their replacement cost.~~
- ~~b) The Constructional Plant and other things brought on to the site by the Contractor(s) to the replacement value of such constructional Plant and other things.~~

- ~~e) It shall be the responsibility of the Contractor(s) to notify the insurance company of any change in the nature and extent of the works and to ensure the adequacy of the insurance cover at all times during the duration of the contract.~~
- ~~d) If the Contractor(s) receives instructions from the Client to insure against War Risk such insurance, shall be at the cost of the Client and shall be into the joint names of the Contractor and the Client.~~
- ~~e) In these conditions the expression "Period of Maintenance" shall mean the period of maintenance named in the tender calculated from the date of completion of the Works certified by the Client in accordance with Clause 12.8 a certificate issue by the Client under the said Clause from the respective dates so certified and in relation to the Period of Maintenance the expression "the works" shall be construed accordingly.~~
- ~~f) The Works shall at, or as soon as practicable, after the expiry of period of maintenance, be delivered to the Client in the condition required by the contract, normal wear and tear excepted, which shall be to the satisfaction of the Client. At the date of completion of the Works (as certified under Clause 12.14 here of), the Contractor shall finish the outstanding Works, if any, as soon as practicable and make good any and all defects, imperfections, shrinkages, faults as may be required by the Client in writing during the Period of maintenance or within thirty (30) days after its expiration as a result of an inspection made by or on behalf of the Client prior to its expiration.~~
- ~~g) All such work shall be carried out by the Contractor (s) at his expense if the necessity thereof shall, in the opinion of the Client, be due to the use of materials or workmanship not in accordance with the contract or neglect or failure on the part of the Contractor(s) to comply with any obligation expressed or implied on the Contractor(s)' part under the contract. If in the opinion of the Client, such necessity shall be due to any other cause the value of such work shall be ascertained and paid for as if it were additional work.~~
- ~~h) If the Contractor(s) fails to do any such required work as aforesaid, the Client shall be entitled to employ and pay any other person(s) to carry out the same and if such work which in the opinion of the Client, the Contractor(s) was liable to do at his own expense under the contract, then all expenses consequent thereon or incidental thereto shall be recoverable from the Contractor(s) by the Client or may be deducted by the Client from any amounts payable or due or which may become due to the Contractor(s).~~
- ~~i) The Contractor(s) shall, if required by the Client in writing, search under the directions of the PMC/Client for the cause of any defects imperfections or faults appearing during the progress of the Works or in the Period of maintenance.~~

~~Unless such defects, imperfections or faults shall be the ones for which the Contractor(s) is liable under the contract, the cost of the Works carried out by the Contractor(s) in searching as aforesaid shall be borne by the Contractor(s) and he shall in such case repair, rectify and make good such defect, imperfection or fault at his own expenses in accordance with the provision of (f) to (h) herein-above.~~

12.12 A. Compensation for Delay of Work:

The date stipulated for carrying out the work as mentioned in the tender shall be strictly observed by the Contractor(s) and shall be reckoned from the date on which the order to commence work is given to the Contractor(s). The work shall throughout the stipulated period of the contract be proceeded with all due diligence (time being deemed to be the essence of the contract on the part of the Contractor(s), and if the Contractor(s) makes/make default therein, it shall pay as compensation the sum which may be determined by the Client and further to ensure good progress of the Work during the execution.

The quantity of work to be done within the specified time would be as follows:

1. 1/4 (25%) of the work (in financial terms) within 3/8 time (37%).
2. 4/10 (40%) of the work (in financial terms) within 1/2 time (50%).
3. 3/4 (75%) of the work (in financial terms) within 3/4 time (75%).
4. 100% work (in financial terms) within stipulated time period.

All the Work including buildings and infrastructural amenities shall be complete in all respects within **the specified period of the contract**, including monsoon.

In the event of the Contractor(s) failing to comply with the aforesaid schedule, it shall be liable to pay as compensation an amount equal to one (1) percent of the said Total tendered amount, or such amount as the Client may decide, as but not less than Rs.1000 /- (Rs. One thousand) per day or more amount up to which the quantity of work shall remain incomplete, provided the amount to be recovered under this clause and clause no. 12.13, shall not exceed ten (10) percent of the Total tendered amount of the whole work. This amount is in addition to the amount of liquidated damages mentioned in clause 12.13.

12.12 B Incentive Clause for Early Completion of Work:

~~In case, the contractor completes the work ahead of stipulated date of completion or justified extended date of completion as determined under~~

~~relevant clauses, a bonus @ 1% (one percent) of the tendered value per month computed on per day basis, shall be payable to the contractor, subject to a maximum limit of 2% (two per cent) of the tendered value. Provided that justified time for extra work shall be calculated on pro-rata basis as cost of extra work X stipulated period /tendered value. The amount of bonus, if payable, shall be paid along with final bill after completion of work.~~

12.12 C Mobilization Advance:

Mobilization advances not exceeding 5% of the tendered value may be given, if requested by the contractor in writing within one month of the work order. Such advance shall be in two or more installments to be determined by the Engineer-in-Charge at his sole discretion. The first installment of such advance shall be released by the Engineer-in-Charge in this behalf. The second and subsequent installments shall be released by the Engineer-in-Charge only after the contractor furnishes a proof of the satisfactory utilization of the earlier installment to the entire satisfaction of the Engineer-in-Charge.

Before any installment of advance is released, the contractor shall execute a Bank Guarantee Bonds not more than 8 in number from Nationalized / Scheduled Bank/Cooperative Bank for the amount equal to 110% of the amount of advance and valid for the period till recovery of advance. This Bank Guarantee shall be kept renewed from time to time to cover the balance amount and likely period of complete recovery.

12.12 D Interest & Recovery:

This mobilization advance mentioned in 12.12 C above bear simple interest at the rate of 10 percent per annum and shall be calculated from the date of payment to the date of recovery, both days inclusive, on the outstanding amount of advance. Recovery of such sums advanced shall be made by the deduction from the contractor's bills commencing after first 10 % of the gross value of the work is executed and paid, on pro-rata percentage basis to the gross value of the work billed beyond 10 % in such a way that the entire advance is recovered by the time 80% of the gross value of the contract is executed and paid, together with interest due on the entire outstanding amount up to the date of recovery of the installment.

12.13 Action to Recover Liquidated Damages.

The work comprised in the tender shall be commenced immediately upon the issue of work order or its acceptance and as soon as possession of the site is handed over. The whole work including all **additions, variations and alteration** as aforesaid subject to relevant clause of the contract agreement (but excluding such, if any, as may have been postponed by an order from the MHDC shall be completed in every

respect by the contractor (s) within the stipulated time, and if for any reason whatsoever (other than wilful obstruction or default on the part of the MHDC or its staff) and except as hereinafter provided the whole of such work could not be finished to the satisfaction of the MHDC, within the stipulated or approved extension periods, the contractor shall pay to the MHDC by way of ascertained liquidated damages for such default the sum at the below mentioned rates or more per day till such default continues, and the amount of such liquidated damages if any may be deducted from any unpaid dues to the contractor(s) or if the liquidated damages not materialized from dues to be paid to the contractor, it may be recovered from the Contractor(s) through legal process by the MHDC, **provided the amount to be recovered under this clause and clause no. 12.12 A, shall not exceed ten (10) percent of the Total tendered amount of the whole work.**

The rate of liquidated damages to be recovered from the contractor is given below.

Sr.No.	Scope of work	Rate of liquidated damages
1	Up to 1 Crore	Rs.2000/- per day
2	Above 1 Crore to 5 Crores.	Rs.3000/- per day
3	Above 5 Crore to 10 Crores.	Rs.5000/- per day
4	Above 10 Crore to 25 Crores.	Rs.10000/- per day
5	Above 25 Crore.	Rs.20000/- per day

Further, if the contractor achieves the subsequent milestone the compensation recovered for earlier delay shall be refunded.

12.14 Action & Compensation for Inferior Quality of Materials / Workmanship:

If at any time before the security deposit or any part thereof is refunded to the contractor, it shall appear to the PMC/Client or his subordinate in charge of the work, that any work has been executed with unsound, imperfect or unskilled, defective workmanship or with materials of inferior/substandard/fake quality, or that any materials or articles provide by him for the execution of the Work are not suited to the Work, or are of a quality inferior to that contracted for, or are otherwise not in accordance with the contract, it shall be lawful for the PMC/Client to intimate this fact in writing to the Contractor(s), and then notwithstanding the fact that the work, materials or articles complained of may have been inadvertently passed, certified and paid for, the Contractor shall be bound forthwith to rectify or remove and reconstruct the work so specified in whole or in part, as the case may require, or if so required,

shall remove the materials or articles so specified and provide other proper and suitable materials or articles at his own charge and cost, and in the event of his failing to do so within a period to be specified by the PMC/Client in the written intimation aforesaid, the Contractor(s) shall be liable to pay compensation at the aforesaid, the Contractor(s) shall be liable to pay compensation at the rate of one percent or such lower percentage as approved by the Client on the amount of the lump sum for every day not exceeding ten days during which the failure so continues, however the compensation so levied will not exceed 10% of the amount of the Contract. In the event of any such failure as aforesaid, the PMC/Client may rectify or remove and re- execute the work or remove and replace the materials or articles complained of as the case may be, at the risk and expense in all respects of the Contractor(s). If the PMC/Client consider that any such inferior work or materials as described above may be accepted or made use of, it shall be within his discretion to accept the same at such reduced rates as he may fix thereof.

12.15 Penalties/ Fine

1. **Rs.25,000/-** (Rupees Twenty-Five Thousand) per day or more amount shall be recovered, if construction programme in PERT/CPM or BAR CHART is not submitted within a month's period from date of work order & the said amount shall not be refunded to contractor.
2. **Rs. 10,000/-** (Rupees Ten Thousand) per day failure to comply with the provision in the Safety Code shall make the Contractor liable to pay to the Client as compensation for each default and the same shall be payable up to the date the said corrective measures are affected and accepted by the PMC/Client. The decision of the Client shall be final and binding in this regard.
3. **Rs.25,000/-** (Rupees Twenty-Five Thousand) per day failure to complete sample flat within three months as per written direction and approval of building material such as flooring tile, door window material, sanitary and plumbing fitting, electrical fixture, kitchen platform etc. (Not applicable for Cyber Crime Office).

12.16 Works to be executed in accordance with Specifications Drawings, Orders, Etc.

The Contractor(s) shall Execute and Complete the entire work/project as per approved drawings and details. The Contractor shall rectify, as ordered by the PMC/Client any errors, which may be found therein during the execution of the said work, and shall provide all plants, labours and materials for the purpose. The Contractor shall also provide all plants, labour and materials which may be necessary and requisite for the Works. All materials and workmanship are to be the best of their

respective kinds. The Contractor(s) shall complete the Work, which in all respects should be as per the specifications given by the Client. The Work should necessarily be completed within the date of completion as specified herein and, in the manner, as desired by the Client. The complete set (s) of all the drawings and copy of contract documents shall be kept at the Site office.

12.17 Alternation in drawings or specifications, etc. not to invalidate contracts:

The Client shall have the power to make any alterations in, or additions to, the original specifications, drawings, designs and instructions that may appear to him to be necessary or advisable during the progress of the work. While making such alterations, the Client may require certain items of the Work to be omitted, from being done or any additional item of Work to be done. The Contractor shall be bound to carry out the work/omit the work in accordance with any such instructions which may be given to him/them in writing signed by the Client and such alterations shall not invalidate the Contract. And any additional work which the contractor may be directed to do in the manner above specified as part of the work shall be carried out by the contractor on the same conditions in all respects on which he agreed to do the main work, The additional or curtailed work shall be carried out or omitted at the current District schedule of rate of P.W.D. prevailing of that year at the time of execution or deletion. If such rate is not available in the contract or DSR then such work shall be carried out or omitted at the rates mutually agreed upon between the Client and the Contractor(s). No work shall be executed before prior approval of the rate.

Rates for works not entered in estimate or schedule of rate of the district.

If the additional or altered work, for which no rate is entered in the schedule of rates of the division, is ordered to be carried out before the rates are agreed upon, then the Contractor shall within seven (7) days of the date of receipt by him of the order to carry out the work, inform the Client of the rate which he/ they propose to charge for such class of work, and if the Client does not agree to this rate he shall send a notice in writing and then the Client shall be at liberty to cancel his order to carry out such a class of work, and arrange to carry it out in such manner as he may consider advisable; provided always that if the Contractor(s) shall commence work or incur any expenditure in regard hereto before the rates have been determined as lastly herein before mentioned, then in such case, he/they shall only be entitled to be paid in respect of the work carried out or expenditure incurred by him / them prior to the date of such determination of the rate as aforesaid according to such rate or rates as may be fixed by the Client. In the event of disputes, the decision of the Client shall be final.

Where, however, the work is to be executed according to the designs, drawings and specifications recommended by the contractor and accepted by the Client the alterations above referred to shall be within the scope of such designs, drawings and specifications appended to the tender.

The time limit for the completion of the work shall be extended in the proportion that the increase in its cost occasioned by alterations or additions bears to the cost of the original contract work, and the certificate of the Client as to such proportion shall be conclusive.

12.18 No Claim or Compensation for Alteration in or Restriction of Work:

1. If at any time after the execution of the contract documents the PMC/Client shall for any reason what-so-ever (other than default on the part of the contractor for which the Client is entitled to rescind the contract) desire that the whole or any part of the work specified in the tender should be suspended for any period or that the whole or part of the work should not be carried out at all, he shall give to the contractor a notice in writing of such desire and upon the receipt of such notice the contractor shall forthwith suspend or stop the work wholly or in part as required, after having due regard to the appropriate stage at which the work should be stopped or suspended so as not to cause any damage or injury to the work already done or endanger the safety thereof, provided that the decision of the Engineer as to the stage at which the work or any part of it could be or could have been safely stopped or suspended shall be final and conclusive against the contractor. The contractor shall have no claim to any payment or compensation whatsoever by reason of or in pursuance of any notice aforesaid, on account of any suspension stoppage or curtailment except to the extent specified herein after.
2. Where the total suspension of the work ordered as aforesaid continued for a continuous period exceeding 90 days, the contractor shall be at liberty to withdraw from the contractual obligations under the contract so far as it pertains to the unexecuted part of the work by giving a 10 day's prior notice in writing to the PMC/Client, within 30 Days of the expiry of the said period of the 90 days, of such intention and requiring the PMC/Client to record the final measurement of the work already done and to pay final bill. Upon giving such notice the contractor shall be deemed to have been discharged from his obligations to complete the remaining unexecuted work under his contract. On receipt of such notice the PMC/Client shall proceed to complete the measurements and make such

payment as may be finally due to the contractor within a period of 90 days from the receipt of such notice in respect of the work already done by the contractor. Such payment shall not in any matter prejudice the right of the contractor to any further Compensation under the remaining provision of this clause.

3. Where the PMC/Client required the contractor to suspend the work for a period in an excess of 30 days at any time or 60 days in the aggregate, the contractor shall be entitled to apply to the PMC/Client within 30 days of resumption of work after such suspension for payment of compensations to the extent of pecuniary loss suffered by him in respect of working machinery remained idle on the site or on the account of his having and to pay the salary or wages of labour engaged by him during the said period of suspension provided always that the contractor shall not be entitled to any claim in respect of any such working machinery salary or wages for the first 30 days whether consecutive or in the aggregate or such suspension or in respect or any suspension whatsoever occasioned unsatisfactory work or any other default on his part. The decision of the Client in this regard shall be final and conclusive against the contractor.

12.19 Claim for Compensation for Delay in Starting the Work

No compensation shall be allowed for any delay caused in the starting of the work on account of acquisition of land or in the case of clearance works, on account of any delay in according to sanction of estimates.

As far as possible, clear site shall be made available to the Contractor. However, in case, there is some obstruction, the Contractor shall start the work on the remaining unobstructed area, and thereafter remove the said obstructions and commence construction of the said area.

12.20 Action when Contractor(S) becomes Defaulter:

If the Contractor(s) shall become bankrupt or if bankruptcy or liquidation proceedings are commenced against it or likely to be commenced against it, or if it enters into any arrangement with or makes any assignment/compromise for the benefit of its creditor/s or is likely to do so, or if a liquidator or receiver is appointed for its assets or it has suspended or delayed the performance of its part of the contract. The Client shall have the right to terminate to the Contract. Further, it shall be lawful for the Client to enter upon the Site, and take possession of the Works and to employ any other person or persons to carry out and complete the same, and to authorize such

person(s) to use the plant, materials and any other property of the Contractor(s) upon the Works.

Further, the Client shall not render itself liable for any civil or criminal action by doing the aforesaid.

The costs and charges which may be incurred in carrying on and completing the said Works shall be payable to the Client by the Contractor(s), and may be set off by the Client against any moneys due or to become due to the Contractor(s). The use of patented materials, methods and design will, however, be restricted only for the purpose of completion of the incomplete works.

12.21 Action where no Specifications

In the case of any class of work for which there is no such specification as is mentioned in Chapter-6 such work shall be carried out in accordance with specifications in Red Book or NBC or IS specifications and in the event, there being no specifications then in such case work shall be carried out in all respect in accordance with all instructions and requirements of the PMC/Client.

12.22 Time Limit for Unforeseen Claims:

Under no circumstances whatever shall the Contractor be entitled to any compensation from Client on any account unless the Contractor shall have submitted claim in writing to the PMC/ Client within one month of the date of such claim occurring.

12.23 Claim for Quantities (Variation) & Variation not to Vitiating Contract

For any addition or curtailment of item/scope/quantity of the work shall be carried out or omitted at the current District schedule of rate of P.W.D. prevailing of that year at the time of execution or deletion. If such rate is not available in the contract or DSR then such work shall be carried out or omitted at the rates mutually agreed upon between the Client and the Contractor(s). No such work shall be executed before prior approval of the rate.

12.23 A) Sufficiency of Schedule of Quantities:

The Contractor shall be deemed to have satisfied himself before tendering as to the correctness and sufficiency of this tender for the works which shall cover all his obligations under the contract and all matters and things necessary for the proper completion of the works. Any such discrepancy if observed is to be brought to the

notice of the Engineer-In-charge before commencement of the work. However, if any discrepancy is noted during the execution of the works the same shall be decided by the **Engineer- In-charge and their decision shall be final and binding on the contractor.**

12.24 Extension of Time on Account of Alterations Etc:

Provided nevertheless that if the Contractor(s) shall be of the opinion that it is are entitled to any extension of time on account of the works being altered, varied or added to or on account of any delay by reason of any inclement weather or any other causes beyond the control of the Contractor(s) or in consequence of any orders to that effect from the Client (which orders the Client is hereby empowered to make), then in any such case or cases, it shall be competent for the Client by an order in writing to extend the stipulated period for final completion by such period or periods as it may deem reasonable and the Contractor(s) shall thereupon complete the Work within such extended period or periods as aforesaid.

Further that the Contractor(s) shall not be entitled to any extension of time unless it shall, within thirty days (30 days) after the happening of the event in respect of which it considers itself to be entitled to any such extension of time, give to the Client written notice of such claim for the extension of time and of the ground or grounds therefore and of the period thereof, unless in any case the Client shall in his discretion, dispense with the requirement of such notice and certify the extension of time provided further that in cases in which any extension of time has been granted, the aforesaid provisions relating to payment of damages for default in completion of the work within the time so extended shall apply.

1. The definition of added, varied & alter shall mean as under: -
 - a. Added: Any extra work not included in the tender. Added works mean other than not included in the tender costing not more than 10% of the work.
 - b. Varied: Major change and/or shifting of location(s) after one month of the anticipated date of commencement of that activity.
 - c. Alter: Any major change in plan or specifications after one month of the anticipated date of commencement of that activity.
2. The BAR/ PERT Chart shall be modified at every mile stone as stated under clause no.12.12 by keeping original contract period to complete the project within contract period by deploying more labour and machinery.

3. All independent work activity shall be completed within contract period such as road, compound wall, storm water drain, street light & tree plantation etc.
4. Work activities which are coming in critical path of PERT chart will be considered for extension if applicable.
5. Liquidated damages/compensation will be levied at any stage as specified under clause no. 12.13 milestone or even after completion of project on the basis of Bar Chart/ PERT Chart & % of payment against work completed without show cause notice to Contractor.
6. No extension on time limit will be considered for following difficulties expressed by Contractor.
 - a. Delay of work during festival period such as Deepavali, Durga Pooja, Ramjan etc.
 - b. Delay of work due to approval of material.
 - c. Delay of work due to delay in payment to contractor.
 - d) Delay of work due to heavy rain.

12.25 Priority of Documents

In case of any discrepancy between the drawings and the specifications, the Client's decision shall be final decision which of the two is to be followed.

Between the General Conditions of Contract and Special Conditions of Contract, if any the later shall prevail.

12.26 Method of Payment:

Payment of the Contractor(s) shall be made by Real Time Gross Settlement ("R.T.G.S.") with any Nationalized / Scheduled Bank with which the Client is having an account.

12.27 Advances:

All amounts whatsoever which the contractor is liable to pay to the Client in connection with the execution of the work shall be deemed to be arrears of land revenue and the Client may without prejudice to any other rights and remedies of the Client recover the same from the contractor as arrears of land revenue.

12.28 Bills to be on Printed Form

The Contractor shall submit all bills on the printed forms prescribed by the Client, to the PMC. The charges to be made in the bills shall always be as per the schedule of payment specified in the tender or in the case of any extra work ordered in

pursuance of these conditions and not mentioned or provided for in the tender, at the cost approved by the Client for such work.

12.29 Bills to be Submitted Monthly

A bill shall be submitted by the Contractor in each month or before the date fixed by the PMC/Client for all work executed in the previous month, and the PMC shall take or cause to be taken the requisite measurements for the purpose of having the same verified and the claim, so far as it is admissible, shall be adjusted, if possible, within 15-20 days from the presentation of the bill only after satisfactory submission of relevant documents/certificates. If the contractor does not submit the bill within the time fixed as aforesaid, the PMC shall measure up the said work in the presence of the contractor or his duly authorized agent whose counter signature to the measurements list shall be sufficient warrant, and PMC/Client may prepare a bill from such list which shall be binding on the contractor in all respect.

12.30 Payment at Reduced Rates

Sub-standard Structural / Construction Work / Item shall be summarily rejected. In cases, where manufactured products are required to be used in the work/ project, and they are not as per specifications (due to non- availability) then the Client may accept the same at reduced payment for use of such products.

12.31 Payment on Interim Certificate to be regarded as Advances

The Contractor shall on submitting a monthly bill therefore, be entitled to receive payment proportionate to the part of the work then approved by PMC and passed by the PMC / CLIENT, whose certificate if such approval and passing of the sum so payable shall be final and conclusive against the Contractor. The contractor (s) shall be paid at the time of Running Account (R.A.) bills a sum equal to the total value of work done since the last payment according to the schedule of payment attached minus standard deduction at that stage. All such interim payments shall be regarded as payments by way of advance against the final payments only and not as payments for work actually done and completed, and shall not preclude the PMC/CLIENT from requiring any bad, unsound, imperfection unskilful work to be removed or taken away and reconstructed or re-erected nor shall any such payment be considered as an admission of the due performance of the Contractor any part thereof in any respect or the occurring of any claim, nor shall it conclude, determine or affect in any other way the powers of the PMC/CLIENT as to the final settlement and adjustment of the accounts or otherwise, or in any other way vary or affect the

Contract. The final bill shall be submitted by the Contractor within one month of the date fixed for the completion of the work. Otherwise, the PMC/Client's certificate of the measurements and of the total amount payable for the work shall be final and binding on all.

Such retained amount will be released in the next alternate R.A. Bill, provided the PMC certifies that such particular items is executed as per specifications stipulated in the tender and all the test results are received and are satisfactory and the quality of work is as per standard specifications and all directions issued by the MHDC have been complied in total.

Payment will be released only when the items is complete. If the item of Work is not complete for all buildings, then proportionate payment based on the Built-up area (percentage of the Item x Built up area of building/ Total Built up area) only will be released. Interest on delay payment from client is not entitled to contractor.

12.32 Completion Certificate in the form of Final Bill:

The certificate of the Client hereinafter referred to showing the final balance due or payable to the Contractor(s) shall be conclusive evidence of the works having been duly completed and that the Contractor(s) shall be entitled to receive payment of the final balance in accordance with such certificate, but without prejudice to the liability of the Contractor(s) under the provision of **clause 10**. Format of Completion Certificates are annexed herewith as **Form No.17.3**

12.33 Final Certificate:

On Completion of the work, the Contractor shall be furnished with a certificate by the PMC/Client of such completion, but no such certificate shall be given nor shall the work be considered to be completed until the Contractor shall have remove from the premises on which the work shall have been executed, all scaffolding, surplus materials and rubbish and clean off the dirt from all wood work, doors, windows, walls, floors, etc., or upon which the work has been executed, or of which he/they had possession for the purpose of executing the work, or until the work has been measured by the PMC or jointly by PMC and the Contractor, until they have received the approval of the Client, the said measurements being binding and conclusive on the Contractor. If the Contractor fails to comply with the requirements of this clause as to the removal of scaffolding, surplus materials and rubbish and cleaning of dirt on or before the date fixed for the completion of the work, the Client may, at the expense

and risk of the Contractor remove or arrange for a third party to remove such scaffolding, surplus materials and rubbish, and dispose of the same as he thinks fit and clean off such dirt as aforesaid. Further, the Client shall not render itself liable for any civil or criminal action by doing the aforesaid and the Contractor shall forthwith pay the amount of all expenses so incurred, but shall have no claim in respect of any such scaffolding or surplus materials as aforesaid, except for any actually realized sum by the sales thereof.

If the Contractor fails to pay such charges, it shall be deducted by the Client from the dues to be paid to the Contractor, or from security deposit.

12.34 Maintenance and Repairs during Guarantee Period:

Unless if otherwise stated differently, the Contractor(s) shall guarantee the satisfactory performance of all services and amenities for Five **(05) years** after the date of completion of work and carry out necessary repairs/improvements that are pointed out within this period for sub- standard/defective but acceptable work.

For the entire period of defect liability, starting from the date of handing over of work to the user department, Contractor shall maintain staff necessary for rectification of defects, if any. Such staff shall include one mason, one plumber and one electrician, who shall hold such qualifications as directed by the Client. Contractor shall deposit the bank guarantee equivalent to 1% of tender cost for waterproofing and anti-termite work for 10 years including the cost of urinals, w.c. units, flooring tiles, dado etc. to the Client. (Note: Refer clause no. 9.7 (III) & (IV). If the defects are not rectified the bank guarantee will be encashed and **Client will** carry out the rectification of defects.

12.35 Decision of PMC/Client shall be Final:

Except where otherwise specified in the Contract, the decision of the Client shall be final, conclusive and binding on the Contractor upon all questions relating to the meaning of the specifications, designs, drawings and technical instructions here-in-before mentioned and as to the quality of workmanship or materials used on the work, or as question(s), claim(s), right(s), matter(s) out of or relating to the contract, instructions, orders, or the contract conditions, or otherwise concerning the execution or failure to execute the same, whether arising, during the progress of the works, or after the completion or abandonment thereof.

12.36 Direction and Control of PMC/Client

- 1) Except where otherwise specified in the contract and subject to the powers delegated to the Engineer by the Client, the decision of the PMC for the time

being shall be final, relating to the meaning of the specifications, designs, drawings and instructions herein before mentioned and as to the quality of workmanship or materials used on the work, or as to any other question, claim right, matter or things whatsoever, if any way arising out of correlating to the contract, designs, drawings, specifications, estimates, instructions, orders, or other conditions, or otherwise concerning the works, or the execution, or failure to execute the same, whether arising during the progress s of the work, or after the completion or abandonment thereof –

- 2) If the Contractor is not satisfied with the decision/ order passed by the Engineer as a foresaid, the Contractor may within thirty days of receipt by him of any order, appeal against it to the Executive Engineer, concerned with the contract work or project.
- 3) If the Contractor is not satisfied with the order passed by the Executive Engineer as a foresaid, the Contractor may, within thirty day so receipt by him of any such order appeal against it to the Director (Technical) MHDC who, if convinced that Prima-facie the Contractors claim rejected by Executive Engineer is not frivolous and that there is some substance in the claim of the Contractor as would merit a detailed examination and decision by the Director (Technical) may be taken.
- 4) If the contractor is not satisfied by the order passed by the Director (Technical) as a foresaid, the contractor may within thirty days of receipt by him of any such order, appeal against it to the M.D. & C.E.O (MHDC). He may refer the dispute to the Technical Advisory Committee (TAC) constituted by M.D. & C.E.O. The committee shall give its advice to the M.D. & C.E.O. (MHDC). However, the decision of M.D. & C.E.O. shall be final and binding on both the parties.

12.37 Control over Works & Removal of Employee:

The Client shall have at all times access to the works which are to be entirely under its control. The Client may require the Contractor/Contractor's employee to be remove from the works if such person is, in his opinion, incompetent or misconducts himself, and thereon the Contractor(s) shall forthwith comply with every such requirement of the Client.

The Contractor shall ensure that while on duty, its employees conduct themselves in an appropriate manner and shall not be under the influence of liquor or other intoxicants and in the event any employee is found to be under the influence of the same, the Client shall have the right to demand the Contractor to replace the said employee immediately and a penalty may be levied on the Contractor.

12.38 Work to be open to Inspection.

CONTRACTOR OR RESPONSIBLE AGENT TO BE PRESENT:

All works under or in course of execution or executed in pursuance of the contract shall at all times be open to the inspection and supervision of PMC/ Client and his subordinates, and the Contractor shall at all times during the usual working hours, and at all other times at which reasonable notice of the intention of the PMC/ Client and his subordinates to visit the work shall have been given to the contractor, either himself be present to receive orders and instructions, or have a responsible agent duly accredited in writing present for that purpose. Orders given to the Contractor's duly authorized agent shall be considered to have the same force and effect as if they had been given to the Contractor himself.

12.39 Notice to be given before work is covered up

The contractor shall give not less than five days' notice in writing to the PMC/Client or his subordinate in charge of the work before covering up or otherwise placing beyond the reach of measurement any work in order that the same may be measured and correct dimensions thereof taken before the same is so covered or placed beyond the reach of measurement and shall not cover up or place beyond the reach of measurement any work without the consent in writing of the PMC/Client or his subordinate in charge of the work, and if any work shall be covered up or placed beyond the reach of measurement, without such notice having been given consent obtained the same shall be uncovered at the contractor's expenses, and in default thereof, no payment or allowance shall be made for such work or for the materials with which the same was executed.

12.40 Quality Assurance Plan

The contractor shall prepare a quality assurance plan (QAP) and get the same approved from the PMC/Client within 30 days from the date of work order. The QAP shall include a) an organization chart of his technical personnel to be deployed on the work along with their qualifications, job descriptions defining the functions of reporting, supervising, inspecting and approving, b) list of tools, equipment's and the machinery and instrumentation which he proposes to use for the construction and for testing in the field and / or in the laboratory and monitoring, c) written method statements detailing his exact proposals of execution of the work in accordance with the specifications, etc. The contractor shall deploy the personnel and equipment on the field as per the approved chart and list respectively and shall modify supplement

the organization chart and the list of machinery, equipment etc. as per the direction of the PMC/Client as and when required as per site conditions. The quality of the work shall be properly documented through certificate, records, checklists and log books of results etc. Such records shall be compiled from the beginning of the work and be continuously updated and supplemented and this shall be the responsibility of the contractor. The forms shall be got approved from the PMC/Client. Where the work is to be done on lump sum basis on contractor's design the contractor shall also prepare and submit a maintenance manual giving procedure for maintenance, with the periodicity of maintenance of works including inspections, tools and equipment's to be used, means of accessibility for all parts of the structure. He shall also include in the manual, the specifications for maintenance work that would be appropriate for his design and technique of construction. This manual shall be submitted within the contract period.

12.41 Testing of Material or any type of Concrete Element:

All the tests for testing the quality, durability, strength and soundness of materials of construction, precast pre-stress, post stress, component, if any, and the building structure as a whole as specified under various chapters herein shall be carried out by the Contractor(s) to the satisfaction of the PMC/Client at the Contractor(s)' cost.

12.42 Minimum Age and the Payments of Fair Wages:

- i. No Contractor(s) shall employ any person who is under the age of eighteen (18) year.
- ii. No Contractor(s) shall employ donkeys or other animals with breaching of string or thin rope. The breaching must be at least three inches wide and should be of tape (Newer).
- iii. No animal suffering from sores, lameness or emaciation or which is immature shall be employed on the work.
- iv. The Client or his representative is authorized to remove from the work any person or animal found working which does not satisfy the conditions mentioned herein above and the Client shall not be liable for any delay caused in the completion of the work by such removal.
- v. The Contractor(s) shall pay fair and reasonable wages as per the existing acts to the workmen employed by him.
- vi. Contractor shall provide drinking water facilities to the workers. Similar amenities shall be provided to the workers engaged on large work in urban areas.

- vii. Contractor to take precaution against accidents which take place on account of labour using loose garments while working near machinery.

12.43 Work on Sundays or Weekly Off:

No work shall be done on weekly off without the sanction in writing of the Client /PMC. The weekly off shall be declared by the Contractor on the commencement of the work.

12.44 Employment of Scarcity Labour.

If Government declares a state of scarcity or famine to exist in any village situated within 10 miles of the work, the contractor shall employ upon such parts of the work, as are suitable for unskilled labour, any person certified to him by the PMC/Client, or be any person to whom the PMC/Client may have delegated this duty in writing to be in need of relief and shall be bound to pay to such person wages not below the minimum which Government may have fixed in this behalf. Any disputes which may arise in connection with the Implementation of this clause shall be decided by the Client whose decision shall be final and binding on the contractor.

12.45 Employment of Local Labour:

The contractor shall employ the unskilled labour to be employed by him on the said work only from locally available Labours and shall give preference enrolled under Maharashtra Government and Self Employment Departments Scheme. Provided, however, that if the required unskilled labour is not available locally, the contractor shall in the first instance employ such number of persons as is available and thereafter may with previous permission, in writing of the PMC of the said work, obtained the rest of requirements of unskilled labour from outside the above scheme.

12.46 Employment of Female Labour

The employment of female labours on works in neighbourhood of soldier's barracks should be avoided as far as possible. The contractor shall employ the labours with the nearest Employment Exchange.

12.47 Employment of Famine Labour:

The contractor shall employ any famine, convict or other labour of a particular kind or class if ordered in writing to do so by the PMC/Client.

12.48 Carpenter/Black-Smith Etc.

The contractor shall engage apprentice such as brick layer, carpenter, wiremen, plumber, as well as blacksmith by recommended by the State Apprenticeship Advisor Director of Technical Education, Dhobi Talaw, Mumbai: 400001. In the construction work (as per Government of Maharashtra, Education Government No. TSA/5170/T5689, dated 7.7.1972).

12.49 Minimum Wages

Wages to be paid to the skilled and unskilled labours engaged by the contractor. The contractor shall pay the labourer's skilled and unskilled according to the wages prescribed by the Minimum Wages Act of 1948 applicable to the area in which the work of the Contractor is in progress.

- 1) The Contractor shall comply with the provisions of the Apprentices Act, 1961, and the rules and orders issued there under from time to time. If he fails to do so, his failure will be a breach of the contract and the Client, May in his discretion may cancel the contract. The contractor shall also be liable, for any pecuniary liability arising out on account of any violation by him of the provision of the Act.
- 2) The contractor shall pay the laborers skilled and unskilled according to the wages prescribed by the Minimum W ages act of the 1948 applicable to the area in which the work is in progress.

12.50 Government Circular No. Cat-1284/ (120)/Bldg.2 DT.14.8.85.

The Contractor shall duly comply with all the provisions of the Contract Labour (Regulation and Abolition) Act, 1970 (37of1970) and the Maharashtra Contract Labour (Regulation and Abolition) Rules, 1971 as amended from time to time and all other relevant statute sand statutory provisions concerning payment of wages particularly to work men employed by the Contractor and working on the site of the work. In particular the Contractors shall pay wages to each worker employed by him on the site of the work at the rates prescribed under the Maharashtra Contract Labour (Regulation and Abolition) Rules, 1971. If the Contractor fails or neglects to pay wages at the said rates or makes short payment and the Client makes such payment of wages in full or part there of less paid by the Contractor as the case may be the amount so paid by the Client to such workers shall be deemed to be an arrear soft and revenue and the Client shall been titled to recover the same as such from the contract or or deduct the same from the amount payable by the Client to the Contractor here under or from any other amount/s payable to him by the Client.

12.51 Cess on Building & Construction Labour Welfare.

(Regulation of Employment & Conditions of Services Act 1996 and The Cess Act of 1996 and as Amended Time to Time)

All the establishments who carry any building or other construction work and which are covered under this Act. are required to pay cess at the rate as stipulated in the said Act. or as may be modified by the Government. The Employer of the establishment is required to provide safety measures at the building or construction work and other welfare measures, such as Canteens, First-Aid facilities, Ambulance, Housing accommodations for workers near the work place etc. The Employer to whom the Act applies has to obtain a registration certificate from the Registering Officer appointed by Government.

12.52 Conditions for Malaria Eradication Anti Malaria and other Health Measures

- A. The antimalarial and other health measures shall be as directed by the joint Director (Malaria and Filarial) of Health Service, Pune.
- B. Contractor shall see that mosquito prone conditions are not created so as to keep vector population to minimum level.
- C. Contractor shall carry out anti-malaria measures in the area as per guidelines prescribed under National Malaria Eradication Program and as directed by the Joint Director (M&F) of Health Services, Pune.
- D. In case of default in carrying out prescribed anti-malaria measures resulting in increase in malaria incidence, contractor shall be liable to pay to Statutory Authority the amount spent by it on antimalarial measures to control the situation in addition to fine.
- E. RELATIONS WITH PUBLIC AUTHORITIES.

The Contractor shall make sufficient arrangements for draining away the sewage water as well as water coming from the bathing and washing places and shall dispose of this water in such a way as not to cause any nuisance. He shall also keep the premises clean by employing sufficient number of sweepers. The Contractor shall comply with all rules, regulations, bye- laws and directions given from time to time by any local or public authority in connection with this work and shall pay fees or charges which are leviable on him without any extra cost of Client.

(Government of Maharashtra P.W.D. Resolution No. CAT-1086/CR- 243/D-Bldg.2 dated 11.9.1987.)

12.53 Responsibilities of Contractor(S) for Damage by Fire, etc.:

From the date of commencement of the Works up to the date of on which the Works are handed over to the Client, the work shall be the responsibility of the Contractor. The Contractor(s) shall be held responsible for any damage done to the same by fire or any other cause, and it shall be liable to make good all such damage and to carry out any repairs which may be rendered necessary to the same by fire or other causes, due to any such cause and restore the Works to a condition equivalent to its condition before such damage.

If the Contractor fails to do so, the Client may arrange for the repair or replacement of the damage and shall deduct the costs for the repairs or replacements from the pending bills of the Contractor, security deposit or any other amount due to the Contractor under the Agreement, without affecting the other rights and remedies of the Client under this Agreement.

Further, the Contractor covenants that, it will not store in or around the Site any article of combustible or inflammable or dangerous nature. If any such article is needed for the execution/completion of the Work, the Contractor shall take all necessary precautions and measures to prevent any fire or damage from such articles.

The Contractor shall indemnify the **Client** from any claims for injuries to any third person (including the employees of the Contractor), for structural damage whether to any part/s of the Work and/or adjoining or neighbouring structures, damage to any property whether of the Client, Contractor and/or third party, due to from any neglect or default, want of proper care or misconduct on the part of the Contractor(s) or of anyone in its employment during the execution of the Work.

12.54 Measure for Prevention of Fire.

The contractor shall not set fire to any standing jungle, trees, brush-wood or grass without a written permit from the PMC. When such permit is given, and also in all cases when destroying cut or drag trees brushwood, grass etc. by fire, the contractor shall take the necessary measure to prevent such fire spreading to or otherwise damaging surrounding property.

12.55 Liabilities of Contractor(S) for any Damages done in/out side of the Work Area:

Compensation for all damages done intentionally or unintentionally by contractor's labour whether in or beyond the limits of the Client's property including any damage

caused by spreading of fire shall be estimated by the PMC or such other officer as he may appoint and the estimate of the PMC subject to the decision of the Client on appeal shall be final and the contractor shall be bound to pay the amount of the assessed compensation on demand, failing which, the same will be recovered from the contractor as damages or deducted by the PMC from any sums that may be due or become due from the Client to contractor under this contract or otherwise. The contractor shall bear the expenses of defending any action or other legal proceeding that may be brought by any persons for injury sustained by him owing to neglect of precautions to prevent the spread of fire and he shall pay any damages and cost that may be awarded by the court in consequences.

12.56 Contractor Liable for Damage done and for Imperfections.

- a) If during the period of 12 (Twelve) months from the date of completion as certified by the PMC/ Client pursuant to clause 12.7 of the contract, and if in the opinion of the PMC/ Client of the said work is defective in any manner whatsoever, the Contractor shall forthwith on receipt of notice in that behalf from the Client, duly commence execution and completely carry out at his cost in every respect all the work that may be necessary for rectifying and setting right the defects specified the rein including dismantling and reconstruction of unsafe portions strictly in accordance with and in the manner prescribed and under the supervision of the PMC/Client . In the event of the Contractor failing or neglecting to commence execution of the said rectification work within the period prescribed therefore in the said notice and /or to complete the same as a fore said as required by the said notice, the PMC/Client may get the same executed and carried out departmentally or by any other agency at the risk, on account and at the cost of the Contractor. The Contractor shall forthwith on demand pay to the Client the amount of such cost, charges and expenses sustained or incurred by the Client of which the certificate of the PMC/ Client shall be final and binding on the Contractor. Such costs, charges and expenses shall be deemed to be arrears of land revenue and in the event of the Contractor failing or neglecting to pay the same on demand as a fore said without prejudice to any other right sand remedies of the Client; the same may be recovered from the contractor as arrears of land revenue. The Client shall also be entitled to deduct the same from any amount which may then be payable or which may there after become payable by the Client to the contractor either in respect of the said work or any other work what so ever or from the amount of security deposit retained by Client.

For the entire period of defect liability, starting from the date of handing over of work to the user department, Contractor shall make available staff necessary for rectification of defects, if any. Such staff shall include mason, plumber, carpenter electrician, etc., who shall hold such qualifications as directed by the Client.

- b) For item/work provided with manufacturer/applicators warranty, refer Chapter 9, clause 9.7

12.57 A) Contractor to Supply Plant, Ladders Scaffoldings, etc.

Liabe for Damages arising from Non-Provisions of Lights, Fencing Etc.

The Contractor shall supply at his own cost all materials, plant, tools appliances, implements, ladders, cordage, tackles, scaffolding and temporary works requisite or proper for the proper execution of the work, whether in the original, altered or substituted from, and whether included in the specification or other documents forming part of the contract referred to in these conditions or not and which may be necessary for the purpose of satisfy in go complying with the requirements of the PMC/ Client as to any matter as to which under these conditions he is entitled to be satisfied, or which he is entitled to require together with the carriage there fore to and from the work. The Contractor shall provide suitable scaffolds and working platforms, gangways and stairways and shall comply with the following regulations in connection therewith-

- a) Suitable scaffolds shall be provided for work men for all works that cannot be safely done from ladder or by other means.
- b) A scaffold shall not be constructed, taken down or substantially altered except.
 - i. Under the supervision of a competent and responsible person; and
 - ii. As far as possible by competent workers possessing adequate experience in this kind of work.
- c) All scaffolds and appliances connected therewith and all ladders shall-be of sound material.
- d)
 - i. Be of adequate strength having regard to the load sands trains to which they will be subjected, and
 - ii. Be maintained in proper condition Scaffolds shall be so constructed that no part thereof can be displaced in consequence of normal use.
- e) Scaffold shall not be over-loaded and so far, as practicable the load shall be evenly distributed.

- f) Before installing lifting gear on scaffolds special precautions shall be taken to ensure the strength and stability of the scaffolds.
- g) Scaffolds shall be periodically inspected by competent person.
- h) Before allowing scaffold to be used by his workmen the Contractor shall whether the scaffold has been erected by his workmen or not, take steps to ensure that it complies fully with the regulations herein specified.
- i) Working platform, gangway, stairways shall-
 - i. Be so constructed that no part thereof can sag unduly on unequally.
 - ii. Be so constructed and maintained, having regard to the prevailing conditions as to reduce as far as practicable risks of person stripping or slipping and
 - iii. Be kept free from any unnecessary obstruction
- j) In the case of working platform, gangways, working place and stairways at a height exceeding 3.00 meters,
 - i. Every working platform and every gangway shall be closely boarded unless other adequate measures are taken to ensure safety,
 - ii. Every working platform and gangway shall have adequate width; and
 - iii. Every working platform and gangway, working place and stairway shall be suitably fenced.
- k) Every opening in the floor of a building or in a working platform shall expect for the time and to the extent require to allow the excess of persons or the transport or shifting of material be provided by suitable means to prevent the fall of persons or material
- l) When persons are employed on a roof where there is a danger of falling from a height exceeding 3.00 meters (to be specified) suitable precautions shall be taken to prevent the fall by provided by suitable means to prevent the fall of persons or material.
- m) Suitable precautions shall be taken to prevent persons being struck by articles which might fall from scaffolds or other working places.
- n) Safe means of access shall be provided to all working platform and other working places.
- o) The Contractor/(s) will have to make payments to labours as per Minimum Wages Act.

12.57 B) Contractor to Supply Hoisting Machine

The Contractor shall comply with the following regulations as regards the Hoisting Appliances to be used by him: -

- a) Hoisting machines and tackles, including their attachments, anchorages and supports shall.
 - i. Be of good mechanical construction, sound material and adequate strength and free from patent defect;
 - ii. Be kept in good repair and in good working order.
- b) Every rope used in hoisting or lowering materials or as a means of suspension shall be of suitable quality and adequate strength and free from patent defect.
- c) Hoisting machines and tackles shall be examined and adequately tested after erection on the site and before use and be re- examined in position at intervals to be prescribed by the PMC/Client.
- d) Every chain, ring, hook, shackle, swivel and pulley block used in hoisting or lowering materials or as a means of suspension shall be periodically examined.
- e) Every crane driver or hoisting appliance operator shall be properly qualified.
- f) No person who is below the age of 18 years shall be in control of any hoisting machine, including any scaffold, or give signals to the operator.
- g) In case of every hoisting machine and of every chain, ring, hook, shackle, swivel and pulley block used in hoisting or lowering or as a means of suspension, the safe working load shall be ascertained by equate means
- h) Every hoisting machine and all gear referred to in preceding regulation shall be plainly marked with the safe working load.
- i) In the case of a hoisting machine having a variable safe working load each safe working load and the condition under which it is applicable shall be clearly indicated.
- j) No part of any hoisting machine or any gear referred to in regulation (h) above shall be loaded beyond the safe working load except for the purpose of testing.
- k) Motors, gearing transmissions, electric wiring and other dangerous parts of hoisting appliances shall be provided with efficient safeguards.
- l) Hoisting appliances shall be provided with such means as will reduce to a minimum the risk of the accidental descent of the load.
- m) Adequate precaution shall be taken to reduce to a minimum the risk of any part of a suspended load becoming accidentally displaced.

12.57 C) Contractor to Supply Fencing and Light

The Contractor shall also supply without charge the requisite number of persons with the means and materials necessary for the purpose of setting out work and counting, weighing and assisting in the measurement or examination at any time and from time to time of the work or the materials, failing which the same may be

provided by the PMC/CLIENT at the expense of the contractor and the expenses may be deducted from any money due to the Contractor under the contractor from his security deposit or the proceeds of sale thereof, or of sufficient portion thereof. The Contractor shall provide all necessary fencing and lights required to protect the public from accident, and shall also be bound to bear the expenses of defence of every suit, action or other legal proceeding, that may be brought by any person for injury sustained owing to neglect of the above precautions, and to pay any damage sand cost which may be awarded in any such suit, action or proceedings to any such person, or which may with the consent of the Contractor be paid for compromising any claim by any such person.

12.58 Sum Payable by Way of Compensation.

Considered as Reasonable Compensation without reference to Actual Loss

All sums payable by the Contractor(s) by way of compensation under any of these conditions shall be considered as a reasonable compensation to be applied to the use of the Client without reference to the actual loss or damage sustained, and whether any damage has or has not been sustained.

12.59 Entering upon or Commencing any Portion of Work

The contractor shall not enter upon or commence any portion of work except with the written authority and instructions of PMC/Client. Failing such authority, the contractor shall have no claim to ask for measurements of or payments for work.

12.60 Changes in Constitution of Firm to be Notified

Any change in the constitution of the contracting firm shall be forthwith notified by the Contractor(s) to the Client for his information & record. If such a change in the constitution of the contracting firm of the Contractor is not notified to the Client within fifteen (15) days, then the same shall be held as breach of contract and a legal proceeding shall follow.

12.61 Work not to be Sub-Late:

The contract shall not be assigned or sublet without the written approval of the PMC/Client.

12.62 Guarantee for Durability of the Structure for Design Life:

- a) The contractor shall ensure that the structure of building constructed for the tenements are durable and shall last for a period of design life (Please refer N.B.C)
- b) Design and drawings for the building shall conform to the requirement stipulated in chapter no.4 'Designs Criteria' and Conditions stipulated therein.

12.63 Taxes Rates and Cess:

The Goods and Service Tax is implanted w.e.f. 01/07/2017 in India replacing other indirect taxes like VAT, Service Tax etc. as per Government of Maharashtra, Finance Department, G.R. No GST-1017/C, R.81/karadhan-1, dt.19/08/2017. The bidders are directed to quote their bids by Exclusion of Goods and Service Tax-2017 on the estimated cost. The tendered cost shall be inclusive of existing and prospective of all taxes, rates and cesses if any and shall be exclusive of Goods **and Service Tax Act-2017** and implication thereof. No reimbursement of any taxes/ rates/ cess/ GST will be given by the client.

12.64 Quarry Fees, Royalties, etc.:

All taxes, quarry fees, royalties, octroi duties and ground rent for stacking materials, if any, shall be paid by the Contractor.

12.65 Durability of the Structure:

- 1) The structure shall be designed for loads and stresses as mentioned in **IS 456-2000** as per provision of DSR. The RCC design of building shall be based on consideration of earthquake forces as per prevailing relevant IS code and seismic coefficient shall be assumed for calculation of earthquake forces as per prevailing relevant IS codes. The importance factor of building shall be considered as 1.0.
- 2) For any type of foundation required for building in case the soil and sub-soil water at site has high sulphate and chloride content, the contractor shall have to take utmost precautions to obtain a dense concrete of low permeability.
- 3) The contractor shall observe method of controlling while constructing various mechanisms of wear and deterioration of concrete so as to make the structure give satisfactory service and safety throughout the life without undue maintenance.
- 4) In observing the design controls, various code of practice and design guides, which give recommendations on durability, shall be followed. The minimum

cement content, water cement ratio, minimum grade of concrete, minimum cover to reinforcement for exposure conditions of environment in city shall be strictly followed.

- 5) The mix design and concrete of various grades, shall be as per latest relevant IS specifications keeping in view both strength and durability. The exposure condition shall be got approved from MERI / Government Institutes before start of work.
- 6) The concrete shall be of low permeability to be durable. For this purpose,
 - I. The concrete shall be strong and dense, with sufficient low water cement ratio,
 - II. The concrete shall be compacted thoroughly,
 - III. Sufficient hydration of the cement shall be ensured through proper curing methods,
 - IV. The aggregate cement ratio for given aggregate shall be sufficient to provide adequate workability with low water cement ratio so that concrete can be compacted with means available with the contractor. Use of plasticizer for workability without increasing the water cement ratio.
- 7) The curing affects primarily the concrete in the cover to the reinforcement and by definition this is the concrete that protects the reinforcement for corrosion by the ingress of aggressive agent. The contractor shall therefore give very special attention to the curing of concrete. The period of curing mentioned in IS code shall be strictly adhered to. The water use for curing shall be potable.
- 8) The contractor shall give special attention to maintain the specified cover during execution as this is very important to transfer the forces in reinforcement, to provide fire resistance to the steel and to protect an alkaline environment at the surface of the steel.
- 9) The contractor shall ensure in the design and construction that there shall be no excessive deflection, creep, and cracks in the structure beyond permissible limits.

12.66 Settlement of Disputes:

Except where otherwise provided in the contract, all questions and disputes relating to the meaning of the specifications, design, drawings and instructions herein before mentioned and as to the quality of workmanship or materials used on the work or as to any other question, claim, right, matter or thing whatsoever in any way arising out of or relating to the contract, designs, drawings, specifications, estimates, instructions, orders or these conditions or otherwise concerning the works or the execution or failure to execute the same whether arising during the progress of the work or after the cancellation, termination, completion or abandonment thereof shall be dealt with as mentioned hereinafter.

If the contractor considers any work demanded of him to be outside the requirements of the contract, or disputes any drawings, record or decision given in writing by the Engineer-in-Charge on any matter in connection with or arising out of the contract or carrying out of the work, to be unacceptable, he shall promptly within 15 days request the Director (Technical) in writing of written instruction or decision. Thereupon the Director (Technical) shall give his written instructions or decision within a period of one month from the receipts of the contractor's letter.

If the Director (Technical) fails to give his instructions or decision in writing within the aforesaid period or if the contractor is dissatisfied with the instructions or decision of Director (Technical) decision appeal to the Managing Director & C.E.O, MHDC who shall afford an opportunity to the contractor to be heard, if the later so desires, and to offer evidence in support of his appeal. The Managing Director & C.E.O, MHDC shall give his decision within 03 months of receipt of the Contractor(s) Appeal and decision of Managing Director & C.E.O, MHDC is final and binding. There is no provision for arbitration.

12.67 Jurisdiction

The agreement shall be governed by the Indian Law for the time being in force and the Courts in Mumbai alone will have jurisdiction to deal with matter arising there from.

13 Additional General Specification

13.1 Preamble

The specification of items of building and infrastructure work are included in chapter no. 6. '**Schedule-C**'. The specification of works of various items which are not covered in Red-Book and which are necessary for completion of work are covered in this chapter. Following Specifications listed are representative only and not exhaustive.

13.2 Pre-Construction Activities

13.2.1 Inspection of sites:

The contractor shall visit and examine the site and satisfy as to the nature and correct dimensions of work and facilities for obtaining material and shall obtain generally his own information on all matters affecting the execution of work. No extra charge made in consequence of any misunderstanding or incorrect information on any of these points on the ground of insufficient description will be allowed. All expenses incurred by the contractors in connection with obtaining information for submitting this tender including his visit to site and efforts in compiling the tender shall be borne by the tenderer and no claims for reimbursement thereof shall be entertained.

13.2.2 Compliance to local laws:

The contractor shall conform to the provisions of any Act of the Legislature relating to the work and to the Regulation of Bye Laws of any authority. He shall also obtain the permission of the Municipality or any other Authorities if required under the existing rules.

13.2.3 Site cleaning:

All the rates quoted are inclusive of removal of rubbish / debris collected during the progress of work, rejected material and clearance of site before and after the work is completed. Not more than one truck load shall be stored that to for not more than 3 days. The contractor shall arrange to remove the same immediately. If the contractor is failing to remove the same and the expenses shall be recovered from the contractor and he will not dispute for the expenses so incurred.

13.2.4 Contractor's responsibility:

The work will from time to time be examined by PMC/Client, but such examination shall not in any way exonerate the contractor from the obligation to remedying any defects which may be observed at any stage of the work or after the same is completed.

13.3 General Specifications for Interior Furnishing Work

13.3.1 General:

- a) The Contractor is advised to study all the drawings in detail, including the intent of the design. All necessary clarifications should be sought by the Contractor prior to submission of the tender.
- b) All work shall be carried out in the best workmanship like manner and wherever, specifications for any items are not given, the relevant Indian Standard Institution specifications shall be applicable and followed
- c) The specifications are for work to be done, item to be supplied and materials to be used in the works as shown and defined on the drawings and described herein, to the satisfaction of the PMC/Client.
- d) The workmanship is to be the best possible and of a high standard. The contractor shall take all steps immediately to make up deficiency if any noticed by the PMC/Client. The materials to be provided by the contractor shall be in accordance with the samples already got approved from the PMC/Client by the contractor and in conformity with specifications and approved list called upon to do so by the PMC/Client.
- e) Samples of all materials are to be submitted to the PMC/Client for their approval before the contractor orders of deliver the material to the site. Samples together with their packing are to be provided free of charge by the contractor and should any materials be rejected they will be removed from the site at the contractor's expense. All samples will be required to submit specimen finishes of colors, fabrics etc. for the approval of the PMC/Client before proceeding with the work.
- f) The contractor shall be responsible for providing and maintaining temporary overages required for the protection of finished work. Contractor shall also to clean out all wood shavings, cut ends and other waste from all parts of the works before coverings or infillings are constructed.
- g) The contractor shall maintain uniform quality and consistency in workmanship throughout the execution of the work.

13.3.2 Materials:

1) PLYWOOD

All ply or block board shall be of Termite proof marine ply, National Ply / Bhutan Ply/ Hunsur Ply/ DONYI Ply /Duro Ply as approved by the PMC/Client. All exposed blockboard edges shall have Cedar Wood lipping of 10 mm thick unless otherwise specified. When decorative ply/block board is used for panelling, the same shall be of matching grains.

2) SCREWS:

All screws shall be of steel screws of nettle fold make confirming to IS - 1365.

3) HINGES:

Brass hinges shall be used and get the approval from PMC/Client for quality and make.

4) TEAKWOOD:

The teakwood should be of the best quality available in India such as Burma, Hunsur or Dandelli. It should be well seasoned and free from gas, knots, wraps, cracks and other defects. All wood work shall be planed and neatly truly finished to the exact dimensions. All joints shall be neat and strong, truly and accurately fitted and glued before being fitted together.

5) DOOR CLOSURES:

Door closures shall be of Godrej/ Everite, Hardwin and before using PMC/Client's approval shall be taken.

6) LOCKS:

Locks shall be as approved by the PMC/Client.

7) DRAWER SLIDING CHANNELS:

The makes used here shall be of Eral Bihari Company (EBDCO), Efficient Gadless (EG) and FIGO. Before using approval of PMC/Client shall be taken.

8) WHEEL CASTORS:

The makes used shall be of Renollo/Classic, Nicholson and before using approval of PMC/Client shall be taken.

9) ADHESIVES:

The adhesives used shall be Fevicol only.

10) GLASS:

The glass shall be modi float clear glass.

11) LAMINATE SHEETS:

All laminates will be of maple or green ply or Formica having a thickness of 1.5mm. The colours will be selected and approved by the PMC/Client.

12) VENEER:

The veneer shall be White Cedar of 4 mm thickness of approved make. The designs will be selected and approved by the PMC/Client.

13) BUILT IN-JOINERY:

Where joinery work is specified to be built-in, it shall be the responsibility of the Contractor to ensure that the joinery works are set in plumb and true in line and shall not be damaged or displaced by subsequent operations.

14) FURNITURE:

The contractor shall first prepare a sample piece and flat and the same shall be got approved by the PMC/Client. The Contractor will be expected to do all the modifications to the sample for which no claim will be entertained. Whenever miter joints are used, dovetailing shall be introduced.

15) TRIM:

Mouldings shall be of first-class cedar, true to detail, clear and sharply defined. It shall be securely fastened with finished nails and screws well set in position and nail holes puttied with putty coloured to match trim.

16) POLISHING:

a. MELAMINE POLISH:

All unevenness shall be rubbed down to smoothness with sand paper and the surface shall be well dusted. The pores in the wood shall be filled up with filler made of a paste whitening in water on methylated spirit (with a suitable pigment like burnt sienna or amber if required).

b. APPLICATION OF MELAMINE POLISH:

A pad of woollen cloth covered by a fine cloth shall be used to apply the polish. The pad shall be moistened with polish and rubbed hard on the surface on a series of overlapping circles applying the polish sparingly but uniformly over the entire area to give an even surface. A trace of linseed oil on the face of the pad may be added which shall facilitate this operation. The surface shall be allowed to dry and one more coat shall be applied and shall be left for drying. After drying the French polish, two coats of melamine polish shall be sprayed. The finished surfaces shall show a uniform texture and a high gloss.

13.3.3 Protection of Work:

The Contractor shall be responsible for the temporary doors and closing in of openings necessary for the protection of the work during progress. Contractor shall also provide and maintain any other temporary covering required for the protection of finished/unfinished wood work that may be damaged during the progress of the work if left unprotected.

13.4 Non-Destructive Tests:

Minimum of five (5) % of structural elements shall be tested by non-destructive test, such as ultrasonic test, pulse velocity test I.S. 1331 Part I and rebound hammer test I.S.1331 Part II probe penetration, and maturity test. The minimum strength at nine and half (9.5) % confidence shall be $1.1 \times$ characteristic strength. This test shall not be accepted for testing the concrete, where cubes have shown below required strength.

13.4.1 Acceptance Criteria for Concrete:

In order to ensure proper control on the quality of concrete, acceptance criteria as laid down in I. S. Specification No.I.S.456-2000 has to be followed. Random samples from fresh concrete should be taken as specified in IS-1199, and cubes should be made, cured and tested as specified in IS-516. The test strength of the sample shall be the average of the strength of three specimens. The individual variation should not be more than fifteen (15) %.

13.4.2 Frequency of Sampling:

The minimum frequency of each sampling concrete shall be in accordance with the following.

Quantity of concrete in the No. of Samples

Work Cum	
----------	--

1 – 5	1
6 – 15	2
16 – 30	3
31 – 50	4
51 and above	4 sets plus one additional set of each Additional 50 m ³ or part thereof

13.5 Street Light:

Making M- 20 grade reinforced cement concrete foundation by considering the safe soil bearing capacity at site as 10 T/Sq. M at 1.5 m depth including supply of steel, concrete, excavation and fixing provided nut bolts with the help of template, duly plastered as per design complete (for 9m & 10m high octagonal /conical GI pole).

Erecting & terminating FR XLPE insulated, galvanised steel formed wire armoured (strip) cable 1100 V, 4 core 16 sq. mm. aluminium conductor complete erected with glands & lugs, on wall/ trusses/pole or laid in provided trench/ pipe as per specification.

Erecting steel tubular swaged pole 165.1 x 139.7 x 114.3 mm dia with 4.5 x 4.5 x 3.65 mm thickness (5+2+2 m) length respectively and total 9 m long with pole cap, base plate in provided foundation (weight 147 kg) as per specification.

Erecting Street light bracket for erection of Single fitting made from 40 mm. dia 'B' class G.I. pipe 1.0 m in length along with pole cap of 300 mm length 100 mm dia duly welded as per specification.

Erecting Street light bracket for erection of Double fitting made from 40 mm. dia 'B' class G.I. pipe 1.0 m in length along with pole cap of 300 mm length 100 mm dia duly welded as per specification.

Erecting integrated LED street light fitting 70-75W IP65 & IK08 class having single piece pressure die-cast aluminium housing, having system lumens output of Min. 7700 Lumens, min. efficacy of 110 lumen/W, CRI>70, CCT up to 6500K, THD<10%, p.f. >0.95, operating range of 140-270V, inbuilt surge protection of 10 kV, Life class of 50,000 Hrs. at L70B50, including driver complete with 3 Years warranty as per specification.

Erecting three phase, 415V, street light control panel up to the max load of 12kW, TPN MCB 63A, powder coated CRCA 14 SWG sheet, outdoor type, having IP54, IK10 protection, suitable rating contactor, 24 hrs. astronomical time switch with minimum 5 years battery back-up, auto/manual selector switch, ON/OFF push buttons, indicator lamps, control wiring, metering device, etc. for automatic operation, with overcurrent, short circuit, earth fault protection on provided iron frame / CC foundation.

Supplying & erecting FRP box of size 150 x 125 x 100 mm, 2.7 mm thick complete on pole/wall as per specification.

Erecting & marking FPMCB 40A to 63A, with rated –Short - circuit breaking capacity (Icn) 10kA in provided distribution board as per specification.

13.6 Additional Technical Specifications for STP

A. Sewage Treatment Plant (STP) – MBBR Technology

Item Description	Specification
Bar Screen Chamber	
Quantity	01 No.
Capacity	3m3
Oil & Grease Trap	
Quantity	01 No.
Capacity	2.5 m3
MOC	RCC
Equalization tank	
Quantity	01 No.
Capacity	170 m3
MOC	RCC
STP equipment & tank foundations	
Quantity	As per requirement
MOC	RCC
MBBR tank	
Quantity	02 No.
Capacity	125 m3 x 2 no= 250 m3
MOC	RCC
Tube Settler tank	
Quantity	02 No.
Capacity	65 m3x 2 No.= 130 m3
MOC	RCC
Filter feed tank	
Quantity	01 No.
Capacity	50 m3
MOC	RCC

Treated water tank	
Quantity	01 Nos.
Size	200 m3
MOC	RCC
Mechanical, Electrical, Instrumentation Items: AWT Scope	
Item Description	Specification
Bar screen	Installed at screen chamber
Quantity	01 No. (Coarse Screen)
Type	Manual
MOC	SS -304
Dimension	1200 mm x 1200 mm
Specification	3 mm thk. SS bar placed at 10 mm c-c distance; screen will be placed inclined with MS angle support.
Raw Sewage Transfer Pump	Installed in Equalization tank
Quantity	03 Nos. (2W+1SB)
Capacity	35 m ³ /hr each
Head	8- 10 M
Type	Submersible sewage pump/ Horizontal centrifugal self-priming
Air blower	For aeration to MBBR and equalization tank
Quantity	3 Nos. (2W+1SB)
Type	Twin lobe air blower with motor
Capacity	750 m3/hr @ 0.5 kg/cm2
Coarse bubble diffusers	For air dispersion inside equalization tank
Quantity	01 Lot
Type	Coarse Bubble Diffuser
Size	80 mm Dia round
MOC	Membrane- EPDM, Body- PVC
Fine bubble diffusers	For air dispersion inside MBBR tank
Quantity	01 Lot

Type	Fine Bubble Diffuser
Size	63 mm Diameter, 1000 mm long/ disc type 300mm dia
MOC	Membrane- EPDM, Body- PVC
MBBR media	Filled in MBBR tank
Quantity	1 Lot
Media	Black, MOC-PP- SA.250-500 m2 / m3 ,15mm x22mm dimension of each piece
Tube settler media	Filled in Tube settler tank
Quantity	1 Lot
Supports for placing TS media	MS – epoxy painted angles
MOC	PVC
Sludge recirculation and transfer pump	For Transferring activated biomass (sludge) to Recycle and sludge dewatering unit
Quantity	01 Nos. (1w)
Capacity	35 m3/ hr.
Head	8-10 M
Type	Horizontal Centrifugal self- priming
Filter Feed Pump	For feeding to filters for filtration
Quantity	03 Nos.
Pump Capacity	65 m3/hr. (32.5m3/her each x 2 working+1 standby)
Head	30 Mtr.
Sludge recirculation and transfer pump	For Transferring activated biomass (sludge) to Recycle and sludge dewatering unit
Quantity	01 Nos. (1w)
Capacity	35 m3/ hr.
Head	8-10 M
Type	Horizontal Centrifugal self- priming

Filter Feed Pump	For feeding to filters for filtration
Quantity	03 Nos.
Pump Capacity	65 m3/hr. (32.5m3/hr. each x 2 working+1 standby)
Head	30 Mtr.
Multigrade sand filter	
Flow rate	65 m3/hr
Dimensions	2100 x 2000
Vessel MOC	MSEP
Inlet outlet accessories	Valves, PG, Sampling Point, Interconnecting & Backwash Piping, Air vent
Media	Fine /coarse, sand, gravels, pebbles
Activated Carbon Filter	
Flow rate	65 m3/hr
Dimensions	2100 x 2000
Vessel MOC	MSEP
Inlet outlet accessories	Valves, PG, Sampling Point, Interconnecting & Backwash Piping, Air vent
Media Qty	Activated carbon, gravels, pebbles
Sludge Dewatering Centrifuge	
Quantity	01 No.
Type	Basket Type
Dimension	18" dia
MOC	MS Body, SS basket
Chlorine Dosing System	
Quantity	01 Set (Tank - 01 No., Pump - 01 No)
Pump Capacity/ Make	0 - 20 LPH
Tank Capacity	200 Liters
MOC	PVC / HDPE
Type of Dosing	Manual

Instruments	
Flow Meter	Water meter dial type
Quantity	01 No. (At Outlet)
Pressure Gauge	2.5" standard glycerin filled pressure gauge
Quantity	01 Lot
Level Switch	Float type level switch
Quantity	01 Lot
Rota Meter	Acrylic body rotameter
Quantity	01 No.
Pipe, fitting & Valves	
Quantity	01 Lot (as per requirement)
MOC of Water Line Piping	UPVC (sch.40)
MOC of Air Line Piping	MS/GI-C class
Valves on uPVC line	UPVC (sch.40)
Valves on MS/ GI line	CI butterfly/ ball valve
Pipe supports	U clamp
MBBR media retention screen	Retaining MBBR media carryover
Quantity	01 No.
Type	GI/ MS perforated screen with opening <10 mm
TS overflow launder weir plate	Retaining floating scum carryover
Quantity	01 No.
Type	MS/FRP scum arresting weir plate

B. STP Specification

Parameters	Inlet Values	Expected Outlet Values Below MPCB Limits
Flow	1450 m ³ /day	-
pH	6-9	6-8
COD	600-800 mg/l	<60mg/l
BOD	250-400 mg/l	< 10 mg/l
TSS	200-250 mg/l	< 10 mg/l
O&G	50-30 mg/l	< Nil
TN	50-30 mg/l	< 10 mg/l

TP	15-5 mg/l	< 1 mg/l
FC		100 MPN/100ml

13.7 Additional Technical Specifications for WTP

13.7.1 Raw water Parameter

Sr. No	Parameters	Unit	Value
1.	pH	-	6.5-8.5
2.	Turbidity	NTU	10
3.	TSS	mg/l	<20
4.	E Coli	/100ml	Present
5.	Total coliform bacteria	/100ml	Present

13.7.2 Outlet Water Quality

Sr. No	Parameters	Unit	Value
1.	pH	-	6.5-8.5
2.	Turbidity	NTU	<1 NTU
3.	TSS	mg/l	BDL
4.	E Coli	/100ml	Absent
5.	Total coliform bacteria	/100ml	Absent

13.7.3 Plant Operating Data

Sr. No	Description	Unit	Value
1	Plant Capacity	m ³ /hr.	27
2	Operating Hrs.	Hours	20

Mechanical, Electrical, Instrumentation Items: AWT Scope		
S. N.	Item Description	Specification
1.	Raw water transfer Pump	
	Quantity	02 No. (1w+1SB)
	Capacity	27 m ³ /hr
	Head	35 M
	Type	Horizontal centrifugal pump
2.	Chlorine Dosing System	
	Quantity	01 Set (Tank - 01 No., Pump - 02 No)

	Pump Capacity/ Type	0 - 20 LPH/ Metering pump
	Tank Capacity	300 Liters
	MOC- Pump	PP
5	Pressure sand filter	
	Quantity	01 No.
	Diameter	1400
	HOS	1800
	Material of Construction	MSEP
	Media	Multigrade Sand
	Media Quantity	As per requirement
6	Activated Carbon Filter	
	Quantity	01 No.
	Diameter	1400
	HOS	1800
	Material of Construction	MSEP
	Media	Multigrade Sand
	Media Quantity	As per requirement
7	UV treatment system	
	Quantity	01 No.
	Flow	27 m3/hr
8.	Electrical control panel	1 nos.
	Type	Relay based control panel
9	Instruments	
A	Flow Meter	Digital water meter
	Quantity	01 No. (At Outlet)
B	Pressure Gauge	4" standard glycerin filled pressure gauge
	Quantity	01 Lot
C	Level Switch	Float type level switch
	Quantity	02 Lot
D	Rota Meter	Acrylic body rotameter
	Quantity	01 No.
10.	Pipe, fitting & Valves	
	Quantity	01 Lot (as per requirement)
	MOC of Water Line Piping (FOR TREATMENT PLANT)	UPVC (sch.40)
	Valves on uPVC line	UPVC
	Pipe supports	U clamp/ Z Angle

13.8 Additional Technical Specifications for OWC

Organic Waste Converter Plant – OWC - 2200Kg/Day

Sr. No.	Description	Qty.
1.	Organic waste composting machine “ECM-200” having 500 lit. Container capacity, with All contact parts in SS 304. Main Motor 3 Phase, 415 V AC, 10 HP, 1440 RPM, Foot mounted Chopper Motor (2 nos.) 3 Phase, 415 V AC, 2 HP, 2880 RPM, Flange mounted Gear box reduction ratio 15:1 *Arrangement for material movement Trolley with 3” Castor wheels Overall Dimensions 2000 x 1300 x 1700 (L x W x H in mm) Approximate Net Weight 1100 Kg	1 Unit
2.	Compost Curing System comprising of the following, • Fabricated rack in M.S. material Size 12’ L * 4’ W * 8’ H, duly Painted / powder coated in Oxford blue / Azure blue shed. • Plastic crates with Perforations, size, 540 mm × 360 mm × 300 mm. 900 Nos.	9 Nos.

14 Labour Regulations

14.1 Preamble:

During continuance of the Contract, the Contractor shall abide at all times by all existing labour enactments and rules made there under, regulations, notifications and bye laws of the State or Central Government or local authority and any other labour law (including rules), regulations, bye laws that may be passed or notifications that may be issued under any labour law in future either by the State or the Central Government or the local authority.

Salient features of some of the major labour laws that are applicable to construction industry are given below.

The Contractor shall keep the employer indemnified in case any action is taken against the employer by the competent authority or account of contravention of any of the provision of any Act or rules made there under, regulations or notifications including amendments. If the Client is caused to pay or reimburse, such amounts as may be necessary to cause or observe, or for non - observance of the provisions stipulated in the notifications / bye laws / Acts / Rules / Regulations including amendments, if any, on the part of the Contractor, the Client shall have the right to deduct any money due to the Contractor including his amount of performance security. The Client shall also have right to recover from the Contractor any sum required or estimated to be required for making good the loss or damage suffered by the Client. The Employees of the Contractor and the Sub-Contractor in no case shall be treated as the Employees of the Client at any point of time.

- 14.1.1 Contractor shall take out necessary Insurance Policy/Policies (viz. Contractor All Risk Insurance Policy, Erection All Risks Insurance Policy etc. as decided by the Directorate of Insurance) so as to provide adequate insurance cover for execution of the awarded contract work for total contract value and complete contract period compulsorily from the "Directorate of Insurance, Maharashtra State, Mumbai" only. Its postal address for correspondence is "264, MHADA, First Floor, Opp. Kalanagar, Bandra (East), Mumbai – 400051". (Telephone No. 26590403/26590690 and Fax No. 26592461). Similarly, all workmen appointed to complete the contract work are required to insure under workmen's compensation Insurance policy. Insurance Policy/Policies taken out from any other company will not be accepted. If any contractor has effected insurance with any other Insurance Company, the same will not be accepted and the amount of premium calculated by the Directorate of Insurance will be recovered directly from the amount payable to the contractor for the

executed contract work. The Director of Insurance reserves the right to distribute the risks of insurance among the other insurers.

14.2 Employees State Insurance Contribution:

- 14.2.1 The Contractor and sub-contractor shall obtain ESI Code number from the concerned office of Employees State Insurance Corporation and a copy thereof be supplied to the client.
- 14.2.2 The Contractor and sub-contractor shall be responsible to deposit employer and employee contribution to ESIC. A copy thereof be supplied to Client.
- 14.2.3 In case Client receives any complaint for non-payment of ESI Contribution or any demand has been raised and recovered by ESIC from client, in such circumstances the client may withheld due payment of Contractor and sub-contractor or may recover it from security deposit amount.

14.3 Employees Provident Fund and Misc. Provisions Act 1952:

- 14.3.1 The contractor and sub-contractor shall deposit the PF Contribution of employer and employee before the concerned PF office and a copy of PF Return shall be supplied to the Client.
- 14.3.2 In case Client receives any complaint about PF Contribution not deposited by the Contractor or sub-contractor in such circumstances the Client may withheld due payment of Contractor and sub-contractor.
- 14.3.3 In case any demand raised by PF Authority the client shall have liberty to withheld the payment of Contractor and sub-contractor or may recover it through due process of law.

14.4 Weekly Day of Rest:

Every worker shall be given a weekly day of rest, which shall be fixed and notified at least 10 days in advance. A worker shall not be required or allowed to work on the weekly rest day unless he has or shall have substituted rest day on one of the five days immediately before or after the rest, provided that no substitution shall be made which shall result in the worker working for more than 10 days consecutively without a rest day for a whole day. Where in accordance with the foregoing provision, a worker works on the rest day and has been given a substitute rest day, he shall be paid wages for the work done on the weekly rest day at the overtime rate of wages.

14.5 Display of Notice Regarding Wages, Weekly Day, Day of Rest Etc.:

The Contractor shall before he commences his work on contract, display and continue to display and correctly maintain in a clean and legible condition in conspicuous places on the works, notices in English and in a local Indian language, spoken and understood by majority of workers, information regarding the rate of fair wages, the hours of works for which such wages payable, the weekly rest day workers are entitled to and name and address of the Inspecting Officer. The contractor shall send a copy of such notice to the Inspection Officer.

14.6 Fixation of Wage Periods:

The contractor shall fix wage periods in respect of wages payable. No wage period shall normally exceed one week.

14.7 Register of Workmen:

A register of workmen shall be maintained in Exhibit - I and kept at the work site or as near to it as possible and the relevant particulars of every workman shall be entered therein within three days of his employment.

Note:

A copy of attendance registers or extract or attendance register shall be provided by Contractor and sub-contractor to the Client

14.8 Employment Card:

The contractor shall issue an employment card in Exhibit - II to each worker on the day of worker's entry into his employment. If a worker has already had any such card with him issued by the previous employer, the contractor shall merely endorse that employment card with relevant entries. On termination of employment, the employment card shall again be endorsed by the contractor and returned to the worker.

14.9 Register of Wages Etc.:

- i. A register of wages cum muster roll in Exhibit - III shall be maintained and kept at the work site or as near to it as possible.
- ii. A wage slip in Exhibit - IV shall be issued to every worker employed by the Contractor at least a day prior to the disbursement of wages.

Note:

The Contractor and sub-contractor shall produce original register of wages at the time of disbursement of wages and client may inspect the payment register at any time.

14.10 Preservation of Registers:

The register of workmen and the register of wages cum muster roll required to be maintained under these regulations shall be preserved for 3 (three) years after the date on which the last entry is made therein.

14.11 Appeal against Decisions of Inspecting Officer:

Any person aggrieved by a decision of the Inspecting Officer, may appeal against such decision to the Regional Labour Commissioner concerned, within 30 days from the date of the decision forwarding simultaneously a copy of his appeal to the Client. The decision of the Regional Labour Commissioner shall be final and binding upon the Contractor and the Workmen.

14.12 Representation of Parties:

- i. A workman shall be entitled to be represented in any investigation or enquiry under the regulations by an officer of a registered trade union of which he is a member or by any officer of a Federation of Trade Union to which the said Trade Union is affiliated or where the workman is not a member of any registered trade union, by an officer of a registered trade union connected with or by any other workman employed in the industries in which the workman is employed.
- ii. A contractor shall be entitled to be represented in any investigation or enquiry under these regulations by an officer of an association of contractors of which he is a member of or by an officer of a Federation of associations of contractors to which the said association is affiliated or where the contractor is not a member of an association of Government by an officer of association of employers, connected with or by any other employer engaged in the industries in which the contractor is engaged.
- iii. No party shall be entitled to be represented by a legal practitioner in any investigation or enquiry under these regulations.

14.13 Inspection of Registers and other Documents:

The Contractor shall allow Inspection of the registers and other documents prescribed under these regulations by Inspecting Officer and the Client or his authorized representative at any time and by the worker / labour or his agent on receipt of the due notice at a convenient time.

14.14 Amendments:

Department may from time to time add to or amend these regulations and issue such directions as it may consider necessary for the proper implementation of these

regulations or for the purpose of removing any difficulties which may arise is the administration thereof.

The Contractor shall employ at least 80% of the total number of unskilled labour to be employed by him on the said work from, out of the persons ordinarily / locally residing.

14.15 Model Rules for Labour Welfare:

Definitions:

a) Work Place

"Work Place" means a place at which on an average 20 or more workers are employed.

b) Large Work Place

"Large Work Place" means a place at which on an average 500 or more workers are employed.

14.16 Accommodation for Labour:

The Contractor(s) shall during the progress of works provided, create and maintain necessary temporary living accommodation and ancillary facilities for labour at his own expenses and to standards and scales as approved by the Client.

14.17 Scale of Accommodation in Latrines and Urinals:

There shall be provided within the premise of every work place, latrines and urinals in an accessible place and the accommodation, separately for each of these shall not be for men and women and these shall not be less than following scales:

No. of Seats

(a)	Where number of persons does not exceed 50	--	2
(b)	Where number of persons exceeds 50 but does Not exceed 100	--	3
(c)	For additional person 3 per 100 or part thereof	--	

In particular cases, the Client shall have the power to vary the scale, where necessary.

14.18 Latrines and Urinals:

- i. Except in work places provided with water flushed latrines connection with a water borne sewage system, all latrines shall be provided with receptacles on dry earth system which shall be cleaned at least four times daily and at least twice

during working hours and kept in a strictly sanitary condition. Receptacles shall be tarred inside and outside at least once a year.

- ii. If women are employed, separate latrines and urinals, shown from those for men and women marked in the vernacular in conspicuous letters "For Women Only" and "Men Only" shall be provided on the scale laid down in rule 6. A poster showing the figure of a man and of a woman shall also be exhibited at the entrance to latrines for each sex.
- iii. There shall be adequate supply of water close to latrine and urinals.

14.19 Construction of Latrines:

Inside wall shall be constructed of masonry or other non-absorbent material and shall be cement washed inside and outside at least once a year. The dates of cement washing shall be noted in a register maintained for the purpose and kept available for inspection. Latrines shall have at least thatched roof. The latrines to be kept clean and dry and odour free (in hygienic condition) all the time.

14.20 Provision of Shelter During Rest:

At every work place, there shall be provided free of cost four suitable sheds, two for males and two others for rest, separately for use of men and women labour. Height of each shelter shall not be less than 3 meters from floor level to lowest part of roof. Shed shall be kept clean and the space provided shall be on the basis of at least 0.5 Sqm. per head.

14.21 Creches:

- i. At a place at where 20 or more women workers are ordinarily employed, there shall be provided at least one hut for use of children under the age of 6 years of such women. Huts shall not be constructed of a standard lower than that of thatched roof, mud floor and walls with wooden planks spread over mud floor and covered with matting.
- ii. Huts shall be provided with suitable and sufficient openings for light and ventilation. There shall be adequate provision of sweepers to keep the places clean.
- iii. There shall be two Dais in attendance. Sanitary utensils shall be provided to the satisfaction of local medical, health and municipal or cantonment authorities. Use of huts shall be restricted to children, their attendants and mothers of children.
- iv. Where the number of women workers is more than 25 but less than 50, the contractor shall provide at least one hut, one Dai to look after children of women

workers, size of crèche(s) shall vary according to the number of woman workers employed. (See para i above)

- v. Crèche(s) shall be properly maintained and necessary equipment like toys etc. provided.

14.22 Canteen

- a) Cooked food canteen on moderate scale shall be provided for the benefit of workers wherever it is considered necessary.
- b) Planning, setting and erection of the canteen structures shall be approved by the Client and the whole of such temporary accommodation shall at all times during the continuance of the works be kept tidy and in a clean and sanitary condition to the satisfaction of the Client at the Contractor's expenses. The Contractor shall confirm generally to satisfy requirements of local medical, health and municipal or cantonment authorities and at all times adopt such precautions, as may be necessary to prevent soil pollution of the site.
- c) On completion of the works in full, such temporary structures shall be cleared away, all rubbish burnt, excreta or other disposal pits or trenches filled in and effectively sealed off and the whole of site left clean and tidy to the entire satisfaction of the Client at the contractor's expenses.

14.23 Enforcement:

The Inspecting Officer mentioned in prevailing para or any other officer nominated in this behalf by the Client shall report to the Client all cases of failure comply with the provisions of these rules either wholly or in part, specifying the penalties to be levied for such breach of these rules, either wholly or in part, specifying the penalties to be levied for such breach of these provisions.

The sum to be levied, as penalty shall, however, be fixed in accordance with the provisions of Clause 12 in this contract.

14.24 Interpretation:

On any question as to the application, interpretations or effect of these rules and decision of the Chief Labour Commissioner or Deputy Chief Labour Commissioner shall be final and binding.

EXHIBIT- I - REGISTER OF WORKMEN

i)	Name and Address of Contractor	:	
ii)	Number and date of the Contract	:	
iii)	Name and Address of the Department awarding the Contract	:	
iv)	Nature of Contract and Location of Work	:	
v)	Duration of the Contract	:	

Sr. No.	Name and Surname of the Worker	Age and Sex	Father's / Husband's Name	Nature of Employment / Designation
1	2	3	4	5

Permanent Home Address of Employee (Village, District)	Present Address	Date of Commencement of Employment	Date of Termination or Leaving of Employees	Signature or Thumb Impression of the Employees	Remarks
6	7	8	9	10	11

EXHIBIT - II - EMPLOYMENT CARD

i)	Name and Sex of the Worker	:	
ii)	Father's / Husband's Name	:	
iii)	Address	:	
iv)	Age and Date of Birth	:	
v)	Identification Marks	:	

Particulars of next of kin - wife / husband and children, if any, or of dependent next of kin in case the worker has no wife / husband or child.

Name: .

Full Address of Dependents: _____

(Specify Village, District and State)

Sr. No.	Name and Address of Employer, Specify whether Contractor or a sub-contractor	Particulars of Location of work-site and description of work done.	Total period which employed during From To	Actual Number of Days Worked	
1	2	3	4	5	6

Leave Taken (No. of days should be specified)	Nature of work done by the worker	Wage period	Wage Rate (with particulars of unit in case of piece work)	Total wage earned by the worker during the period shown under col. 5	Remark	Signature of the Employer
7	8	9	10	11	12	13

Note: For a worker employed at one time on piece work basis and at another on daily wages, relevant entries in respect of each type of employment should be made separately.

EXHIBIT - III: - REGISTER OF WAGES - CUM - MUSTER ROLL

i)	Name and Address of Contractor	:	
ii)	No. and date of the Contract	:	
iii)	Name and Address of the Department awarding the Contract	:	
iv)	Nature of Contract and Location of Work	:	
v)	Duration of the Contract	:	
vi)	Wage Period		

Sr. No.	Name and Surname of the Worker	Father's / Husband's Name	Sex	Designation Nature of Work	Daily Attendance (No. of Units Worked 1, 2, 3, 5,6,7 up to 31)	Total Unit	Attendance
1	2	3	4	5	6	7	8

Fair Wages		Wage Paid			Overtime Worked		
Payable basic other allowance	D.A.	Basic	D.A. and other allowances	Date	No. of Hours	Overtime Earned	Total Wages Paid
9	10	11	12	13	14	15	16

Deductions from wages					Net Wages	Date of payment	Signature or Thumb Impression of the worker	Re-marks
Fines	Deduction for Damage or Loss	House Rent	Recovery of Advance	Other Deductions				
17	18	19	20	21	22	23	24	25

Reasons to be recorded in Column 25.

EXHIBIT - IV: WAGE SLIP

	Name of the Contractor	:	
	Place	:	
1	Name of Worker with Father's / Husband's Name	:	
2	Nature of Employment	:	
3	Wage Period	:	
4	Rate of Wages Payable	:	
5	Total Attendance / Units of Work Done	:	
6	Dates on which overtimes Worked	:	
7	Overtime Wages	:	
8	Gross Wages Payable	:	
9	Total Deductions (Including Nature of Deductions)	:	
10	Net Wages Payable	:	

Contractors Signature/Thumb Impression

Employee's Signature/Thumb Impression

CONTRACTOR**MANAGING DIRECTOR & CEO**

15 Construction Safety Practices

15.1 Safety Code:

15.1.1 The Contractor shall follow the practices given in the SP.-70 “Hand Book on Construction safety practices and relevant codes” referred to herein after as “The Safety Code” Published by the BIS. The Contractor shall comply and take all precautions at Project site to avoid accidents and for safety of workers and public at the site of work. The Contractor shall be liable in the event of any accident on account of its negligence/omission (including its employees / workmen / subcontractors) for not following the Safety Code. The Contractor shall indemnify the PMC/Client against all/any claims arising due to non- compliance of the Safety Code requirements and shall be bound to bear the cost of litigation, including attorney fees, and the costs which may be awarded in any such suit or proceedings to any affected person by a competent court or tribunal. The Contractor shall be liable to settle all claims arising out of any accidents/damages at site of work.

15.1.2 The safety practices in the Safety Code shall be brought to the notice of all workmen and all those concerned by putting up a legible notice in the language that would be understood by the workmen and the concerned parties, on a notice board at a prominent place at the Site. Persons responsible for ensuring compliance with the safety code shall be named therein by the Contractor.

- i. To ensure effective enforcement of the rules /regulations and practices relating to safety, arrangements made by the Contractor shall be open to inspection by the PMC/Client or its authorized representatives and the inspecting officers. In the event of noticing any default/noncompliance by them with regard to the aforesaid, the same, shall be brought to the notice of the Contractor or its authorized representative who shall ensure and take corrective measures, the same shall accordingly be modified by as per the Safety Code
- ii. Failure to comply with the provision in the Safety Code shall make the Contractor liable to pay to the PMC/Client as compensation an amount of Rs. 50/- (Rupees Fifty Only) per day for each default and the same shall be payable up to the date the said corrective measures are affected and accepted by the PMC/Client. The decision of the Client shall be final and binding in this regard.

16 Standard Format

16.1 (Format of Declaration)

I, (Name of Contractor/Authorized person), Aged about years, residing at (Postal Address) do hereby Declare that, I am the proprietor/Partner/Authorized Representative of
.. (Name of Company/Firm/Joint Venture) Registered at

I do hereby declare that, the documents submitted in Eligibility-cum- Technical Bid of the tender document for the work of _are true, correct and complete. In case the contents of Eligibility-cum-Technical Bid and other documents pertaining to the tender submitted by me/our Company/Firm/Joint Venture are found to be incorrect or false, my/our tender shall be liable for rejection and I shall be liable for action under the relevant provisions of Indian Penal Code and other relevant laws.

Signature of Authorized person

Applicant/Tenderer Name

Address.....

Place :-

.....

Date:-

E-mail.....

Mobile No.....

Website.....

16.2 Declaration of the Contractors:

I / we_ hereby solemnly affirm and declare that, I / we have made myself / ourselves thoroughly conversant with the local laws, local conditions, such as soil/ sub soil strata, topography, regarding availability of all materials such as, stones, murum, sand, bricks, metal, cement, steel water and all other requisite matters and availability of skilled and unskilled labour on which, I /we have based my /our rates for this work. The drawings, specifications, conditions and materials required for this work have been carefully studied and understood by me/ us before submitting this Tender. I/we undertake to use only such materials which are recognized to be of the best quality by the appropriate industrial standards and which shall be duly approved by the Maharashtra Housing Development Corporation Ltd. i.e., Client, or its duly authorized representative in writing before the commencement of any "Work" as mentioned in the Tender and to abide by the decision of the Client in all respects and at all times.

Solemnly affirmed at _____)

On this ____ day of ____ 20 ____)

(Signature of the Contractor) Identified by

Signature of the Advocate

16.3 Indemnity Bond:

((On Stamp Paper of Appropriate Value as Per the Maharashtra Stamp Act)

This Indemnity Bond is made on this day of....., by M/s_a company/ partnership firm/ sole proprietor having their registered office at, (hereinafter referred to as the “*Contractor*”) in favour of Maharashtra Housing Development Corporation Limited, Mumbai, having its registered office at 3rd Floor Office No. 32, Sakhar Bhavan, Ramnath Goinka Road, Nariman Point, Mumbai 400021. (Hereinafter referred to as the “*Client*”).

WHEREAS:

The Contractor and the Client have entered into an agreement dated for

NOW THIS DEED WITNESSETH THAT:

- a) The Contractor does hereby covenant with the Client that the Contractor will indemnify and keep indemnified the Client (and each of the officers, directors, employees, advisors and Affiliates) for any action, suit, litigation or claim of any nature whatsoever made by any person of any nature whatsoever in respect of the work done by the Contractor by virtue of any act of commission or omission of the Contractor.
- b) The Contractor agrees to compensate the Client against any loss, costs, charges and expenses including attorney fees that the Client may incur or suffer on account of such action, suit, litigation claim or any dispute.
- c) The Contractor shall also indemnify the Client (and each of the officers, directors, employees, advisors and Affiliates) for making good any claim/penalty/loss or damages, including costs (including counsel fees and legal cost) thereof, in respect of any breach or violation of any provisions of any law, including labour laws governing the employees of the Contractor. In case of any failure to make good the above/any losses/expenses to the Client, the same shall be deducted from the amounts to be paid to Contractor, as per the bills raised, or may be deducted from the security deposit or from any payments to be made to the Contractor under the aforesaid Agreement.
- d) The Contractor shall be solely indemnify the Client (and each of the officers, directors, employees, advisors and Affiliates), from costs or liabilities, arising therefrom (including counsel fees and legal cost) for any claims, against the client for any accidents, injuries, death and/or damages caused to any individual/s and/or property of any third party, due to negligence of the Contractor or its employees, during performance of their duties under the Agreement.

Dated this ____ day of _____

CONTRACTOR

MANAGING DIRECTOR & CEO

IN WITNESS WHEREOF the Contractor hereto have hereunto set and subscribed their respective hands the day and the year hereinabove written.

SIGNED, SEALED AND DELIVERED by the within named Contractor:

Mr.

(1) _____

in the presence of

(1) _____

(2) _____

16.4 Form of Bank Guarantee Bond (EMD)

(On Stamp of Appropriate Value as Per the Maharashtra Stamp Act)
(For EMD)

In consideration of the Maharashtra Housing Development Corporation Ltd., Mumbai
(hereinafter referred to as "*Client*") having agreed to exempt -----

----- (hereinafter referred to as "*Contractor*")

from depositing with the Client, in cash the sum of Rs. ----- (Rupees: -----
----- only) being the amount of EMD payable by
the Contractor to the Client under the terms and conditions of the Agreement (hereinafter
referred

to as "*Agreement*") dated the ----- day of ----- 20__ and
made

between the Client and the Contractor for -----

----- as security for the due observance and performance by the Contractor of the terms
and conditions of the said Agreement, on the Contractor furnishing to the Client, a Guarantee
in the prescribed form of a Nationalized Bank / Scheduled Bank in India, being in fact these
presents in the like sum of Rs .- ----- (Rupees -----
----- only). We ----- Bank / Limited
registered in India under ----- Act and having one of our local head
offices at do hereby:

1. Guarantees the Client:

- a) Due performance and observance by the Contractor of the term's covenants and conditions on the part of the Contractor contained in the said Agreement.
- b) Due and punctual payment by the Contractor to the Client of all sums of money including but not limited to any and all losses, damages, costs, charges, penalties and expenses arising out of the default of the Contractor in respect to the said Agreement.

2. Undertake to pay the Client on demand and without demur the entire amount of the claim but with a maximum of the guarantee amount, within (72) seventy-two hours of such claim being made by the Client and notwithstanding any dispute or disputes raised by the Contractor(s) in any suit or proceeding filed in any court or Tribunal or pending arbitration relating thereto the said sum of Rs. ----- (Rupees only)

Our guarantee hereunder being absolute and unequivocal we agree that:

- a) The guarantee herein contained shall remain in full and effect during the subsistence of the said Agreement and that the same shall continue to be enforceable till all the dues of the Client under or by virtue of the said Agreement(s) have been duly paid

and its claims satisfied or discharged and till the Client certifies that the terms and conditions of the said Agreement have been fully properly carried out by the Contractor and all financial claims of the Client are met in full.

- b) We shall not be discharged or released from the liability under this Guarantee by reasons of:
- i. Any change in the constitution of the Bank or of the Contractor, or
 - ii. Any arrangement entered into between the Client and the Contractor with or without our consent;
 - iii. Any forbearance or indulgence shown to the Contractor;
 - iv. Any variation in the terms covenants or conditions contained in the said Agreement;
 - v. Any time given to the Contractor, or
 - vi. Any other conditions or circumstance under which, in law, a surety would be discharged.
- c) Our liability herein under shall be joint and several with that of the Contractor as if we were principal debtors in respect of the said of Rs.----- (Rupees ----- only) and
- d) We shall not revoke this guarantee during its currency except with the previous consent in writing of the Client.

IN WITNESS WHEREOF the common seal ----- has been here into affixed this ----- day of -----20---

The common seal of ----- was pursuant to the resolution of the Board of Directors of the Company dated the ----- day of ---- 20_ herein affixed in the presence of -----who, in token thereof, have here to set their respective hands in the presence of:

1) -----

2) -----

16.5 Form of Bank Guarantee Bond:(SD)

(On Stamp Paper of Appropriate Value as per The Maharashtra Stamp Act)
(For Security Deposit)

In consideration of the Maharashtra Housing Development Corporation Ltd., Mumbai
(hereinafter referred to as "*Client*") having agreed to exempt-----

----- (hereinafter referred to as
"*Contractor*") from depositing with the Client, in cash the sum of Rs. -----
(Rupees: ----- only) being the amount of
security deposit payable by the Contractor to the Client under the terms and conditions of
the Agreement (hereinafter referred to as "*Agreement*") dated the ----- day of ----
----- 20_ and made between the Client and the Contractor for -----

-----as security for the due observance and performance by the
Contractor of the terms and conditions of the said Agreement, on the Contractor furnishing to
the Client, a Guarantee in the prescribed form of a Nationalized Bank / Scheduled Bank in
India, being in fact these presents in the like sum of Rs.----- (Rupees -----
----- only). We -----
Bank / Limited registered in India under ----- Act and having one of
our local head offices at do hereby:

1. Guarantees the Client:
 - a) Due performance and observance by the Contractor of the terms, covenants and conditions on the part of the Contractor contained in the said Agreement.
 - b) Due and punctual payment by the Contractor to the Client of all sums of money including but not limited to any and all losses, damages, costs, charges, penalties and expenses arising out of the default of the Contractor in respect to the said Agreement.
2. Undertake to pay the Client on demand and without demur the entire amount of the claim but with a maximum of the guarantee amount, within (72) seventy-two hours of such claim being made by the Client and notwithstanding any dispute or disputes raised by the Contractor(s) in any suit or proceeding filed in any court or Tribunal or pending arbitration relating thereto the said sum of Rs. ----- (Rupees only) Our guarantee hereunder being absolute and unequivocal we agree that:
 - a) The guarantee herein contained shall remain in full and effect during the subsistence of the said Agreement and that the same shall continue to be enforceable till all the dues of the Client under or by virtue of the said Agreement(s) have been duly paid and its claims satisfied or discharged and till the Client certifies that the terms and conditions of the said Agreement have been fully properly carried out by the Contractor and all financial claims of the Client are met in full.
 - b) We shall not be discharged or released from the liability under this Guarantee by reasons of:
 - i) Any change in the constitution of the Bank or of the Contractor, or
 - ii) Any arrangement entered into between the Client and the Contractor with or without our consent;
 - iii) Any forbearance or indulgence shown to the Contractor;

- iv) Any variation in the terms covenants or conditions contained in the said Agreement;
- v) Any time given to the Contractor, or
- vi) Any other conditions or circumstance under which, in law, a surety would be discharged.

- c) Our liability herein under shall be joint and several with that of the Contractor as if we were principal debtors in respect of the said of Rs.-----
(Rupees ----- only) and
- d) We shall not revoke this guarantee during its currency except with the previous consent in writing of the Client.

IN WITNESS WHEREOF the common seal ----- has been here into affixed this ---- day of -----20--- The common seal of ----- was pursuant to the resolution of the Board of Directors of the Company dated the ---- day of 20____ herein affixed in the presence of _____ who, in token thereof, have here to set their respective hands in the presence of:

1) -----

2) -----

16.6 Guarantee Bond for Anti-termite / Water Proofing Treatment

(On Stamp Paper of Appropriate Value as Per the Maharashtra Stamp Act)
(Model Proforma)

Dated: -

To,
Maharashtra Housing Development Corporation Ltd.,
Mumbai

Sub.: Construction of -----.

Sir,

I/We ----- have executed and completed the work of Anti-termite / Water Proofing Treatment (hereinafter referred to as "*treatment*") for Building No. ----- on date.

I/We hereby guarantee that, the treatment shall be valid for a period of 10 years (hereinafter referred to as the "Guarantee Period") immediately commencing from the date of completion of building as aforesaid. In the event of any defaults arising during the Guarantee period I/ We will immediately rectify all such defects at my/our own cost to the entire satisfaction of Maharashtra Housing Development Corporation Ltd., Mumbai (hereinafter referred to as "Client").

However, in case the efforts to rectify the same are not initiated by us within two (2) days from the knowledge of the occurrence of such defects, the Client reserves the right to rectify such defects and recover the entire cost of the same from us.

I/ We undertake to pay such cost, without demur to the Client within five (5) days from the demand for such amount, in the manner as shall be directed by the Client.

The aforesaid guarantee can be enforced within Ten (10) years of the expiry date of the guarantee, from the date of execution.

Signature of Contractor

16.7 A Check list for Final Bill Processing:

Name of work:			
Sr. No.	Particulars	Information In the form of Yes/No	Remarks
A)	Technical		
1	Whether as built drawing is submitted by the contractor?		
2	Whether the project is completed within stipulated time period?		
3	Whether structural stability certificate is submitted by Contractor / PMC		
4	Whether water supply completion certificate is submitted by Contractor/PMC?		
5	Whether drainage completion certificate is submitted by Contractor/PMC?		
6	Whether electrification completion certificate is submitted by Contractor/PMC?		
B)	Finance		
7	Whether water proofing guarantee is submitted by the contractor?		
8	Whether anti- termite guarantee is submitted by the contractor?		
9	Whether all other required guarantees in respect of fixtures such as electrical (e.g. pump, lights, transformer, fans) are submitted by the contractor?		
10	Whether requisite security deposit is recovered from contractor? i.e., as mentioned in clause no. 1.8 B		
11	Whether penalty (if applicable) is recovered from contractor?		
12	Whether excess/ saving statement (if any) is finalized?		
13	Whether no due certificate is submitted by contractor?		
C)	Clearance		
14	Whether occupancy certificate is obtained?		
15	Whether requisite procedure for transferring the building on PRB register of PWD is completed?		

Note: This Checklist shall be verified and certified by PMC

APPENDIX - I

List of documents to be attached with R. A. Bills / Final Bill

Sr. No.	Document	Attached Yes / No
1.	Letter of Contractor to PMC with copy to CLIENT (No. and Date) (Form B - 1)	
2.	Letter of PMC to CLIENT with copy to contractor (No. and Date) (Form B - 2)	
3.	Submission of Running Account Bill No. --- (No. and Date) (Form B - 3)	
4.	Certificate of PMC with bill and in MB (Form B - 4)	
5.	Certificate of Contractor with bill (Form B - 5)	
6.	Certificate by PMC regarding deployment of key staff and other staff (Form B - 6)	
7.	Certificate by EE / DE / PE regarding deployment of key staff and other staff by PMC (Form B - 7)	
8.	FORM NO 67 (P.W. 466) (Form B - 8)	
9.	Abstract of Bill (Form B - 9)	
10.	Account of Work (Form B - 10)	
11.	Payment Schedule as per Tender (Form B - 11)	
12.	Payment Schedule Abstract (Form B - 12)	
13.	Details about withheld amount (Form B - 13)	
14.	Reasons of Delay of work (Form B - 14)	
15.	Abstract of test reports with copies of test reports. (Form B - 15)	
16.	Test Frequency Chart (Form B - 16)	
17.	Sample Approved During Period of Bill (Form B - 17)	
18.	Abstract of Steel Receipt and Challans (Form B - 18)	
19.	Steel Consumption Statement (Form B - 19)	
20.	Abstract of Cement Receipt (Form B - 20)	
21.	Cement Consumption Statement (Form B - 21)	
22.	Bar Chart (Physical) (Form B - 22)	
23.	Bar Chart (Financial)	
24.	Another document attached if any	
25.	Progress Report (in pictorial diagram of each building)	
26.	Photographs	
27.	Acquittance (Form B - 23)	

Note: Every page shall be signed by PMC and Contractor.

FORM B - 1

Date:

To,
PMC

Subject:

Ref: Agreement No. Date of Commencement:

Respected Sir,

Please find enclosed here with the R.A. Bill for the above said work for your kind consideration, approval please.

Encl:

- 1] Bill.
- 2] Prescribed Bill forms.
- 3] Lab Test Reports.
- 4] Bar Chart.
- 5] Steel and Cement Challan
- 6] CAR Policy
- 7] Bank Guarantee

For

Contractor

Copy submitted to: MD & C.E.O, MHDC, Mumbai.

FORM B - 2

Date:

To
Managing Director & C.E.O
Maharashtra Housing Development Corporation Limited,
3rd Floor Office No. 32,
Sakhar Bhavan, Ramnath Goalka Road,
Nariman Point,
Mumbai 400 021

Sub: - Submission of R.A. Bill No.

Name of Work: - _____

Ref:

- 1) Agreement No.
- 2) Contractor's Letter No. Date _____

Respected Sir,

Please find enclosed herewith R.A. Bill No. for the above said work of the Agency. -----
-----, for your consideration, approval and payment please.

Thanking You,

Yours truly,

for:

PMC

[]

Encl:

- 1] Contractor's Bill.
- 2] Prescribed Bill forms.
- 3] Lab Test Reports.
- 4] Bar Chart.
- 5] M.B. No.

FORM B - 3**SUBMISSION OF RUNNING ACCOUNT BILL NO.**

Name of Work:

Name of Contractor:

Name of P.M.C.:

Agreement No.:

R.A. Bill No.:

Date:

Sr. No.	Descriptions	Particular
1	Running Account Bill No. and date of submission	
	a. Period of bill	
	b. Date of recording	
	c. M.B. No.	
	A. For	
	d. Date of bill submitted by the contractor to PMC	
	e. Date of bill submitted by the PMC to MHDC	
2	Name of Architect Cum PMC	
3	Name of Contractor	
4	Date of Work Order	
5	Agreement No.	
6	Time Limit for completion of work	
7	Scheduled date of Completion	
8	Extension of Time Limit, (if any) and their letter no.	
9	Compensation levied, (if any)	
10	Estimate Cost of work	
11	Scope of work including infrastructure of work	
12	Tender Cost of work	
13	Date of completion after extension	
14	Up to date amount of Bill- R.A.	
	A) For	

Sr. No.	Descriptions	Particular
15	Amount of This R.A. Bill No.	
	A) For	
A	In percentage of tender amount	
	A) For	
B	In amount	
	A) For	
17	Security Deposit	
I	a. In form of Bank Guarantee No. BG. No. Rs. /- valid up to / / b. CAR Policy No. Valid up to / / Note: - PMC should verify whether the Bank Guarantee/CAR policy is expiring within months from the date of recording. If so, he should not send the bill unless the Bank Guarantee/ CAR Policy is extended by the Contractor. If PMC submitted the bill of contractor after expiry of Bank Guarantee or CAR policy the equivalent amount will be recovered from bill of PMC. PMC should collect Xerox copy of B.G. and CAR Policy from Contractor. Attach the Xerox copy.	
	a. Total amount of security deposit to be recovered through R.A. Bills.	
	b. Total amount of security deposit recovered up to previous bill No.	
	c. Amount of the security deposit to be recovered in this R.A. bill _ 5% Of	
	d. Outstanding of security deposit.	
18	Recovery of compensation if any and their letter No.	
	a. Amount of compensation recovered up to previous bill. No.	
	b. Amount of compensation to be recovered in this bill.	
	c. Total amount recovered against compensation (a+b)	
19	Progress of Works	
	a. % of Work to be done as per target i.e. as per contract agreement	
	b. Actual % of work done till the date of this bill	

Sr. No.	Descriptions	Particular
	c. % of time consumed as per total time period.	
	d. Whether extension is demanded.	
	e. Whether PMC has recommended in detail	
	f. Whether PMC has approved the extension.	
20	Whether bill has been duly signed by the contractor or his authorized person copy of authority letter should be attached.	
	a. Name of persons signing the bill on behalf of contractor and PMC and Architect and their specimen signature.	
	b. Whether signature is duly stamped by authority person.	
21	P.M.C. cum Architects certificate as per MHDC format. (Whether attached)	
22	Whether arithmetic accuracy of Measurements and found correct.	

FORM B - 4

CERTIFICATE OF P.M.C.

Name of work: - _____

R. A. BILL No. / Extra Item No. _____

It is certified that the work measured and recommended for payment of this Running Account Bill of Contractor are fully consistent with the type, quality and specification prescribed in the agreement entered in to with the contractor/ Approved Drawings/ specifications.

Note: For any deviation in specification and unsatisfactory quality of work proposed for payment, P.M.C., shall be responsible.

For

(PMC)

FORM B - 5

CERTIFICATE OF CONTRACTOR WITH BILL

Name of Work: _____

I have satisfied myself by PMC that the value of work done up to date excluding the measured-up additions and alterations is not less than Rs. Only conforming with the contractor's agreement and that with the exceptions of authorized additions and alterations, the work has been done according to the prescribed specifications.

The detail measurements of authorized additions and alterations up to date are recorded in
M.B No. _____ Page no.: _____ Date: _____

For

For

(Contractor)

(PMC)

FORM B - 6

**CERTIFICATE BY PMC REGARDING DEPLOYMENT OF KEY STAFF AND OTHER
STAFF**

Name of Work: _____

It is certified that the Key Staff and other Staff as per manning schedule is deployed by me
on the above-mentioned work site.

For

(PMC)

FORM B - 7

**CERTIFICATE BY EE/DE/PE REGARDING DEPLOYMENT OF KEY STAFF AND OTHER
STAFF BY PMC.**

Name of Work: _____

It is certified that the Key Staff and other Staff as per manning schedule is deployed by PMC
on the above-mentioned work site.

For

(PE/DE/EE)

FORM B - 8

Form 67 (P.W. 466)

R.A. Bill No.

Form 67 (P.W.466)

RUNNING ACCOUNT BILL (INTERMEDIATE)

To be used for "Intermediate payments" to the Contractor.

Sr. No.	Particular	Remarks
1	Name of Architect	
2	Name of P.M.C.	
3	Name of Contractor	
4	Name of Work	
5	Serial No. of this Bill	
6	No. and date of previous bill for this work	
7	Reference of agreement	
8	Accepting Authority and execution of contract agreement	Managing Director & C.E.O., Maharashtra Housing Development Corporation Ltd.
9	Date of written order to commence work	
10	Date of Completion as stipulated in the contract	
11	Extension granted up to	

For

(Contractor)

For

(PMC)

FORM B - 9

ABSTRACT OF BILL

Name of Work:

Name of Contractor:

Name of P.M.C.:

Agreement No.:

R.A. Bill No.:

Sr. No.	Particulars	Up to date Amount	Amount as per Previous Bill	Amount as per this bill since previous
1	2	3	4	5

For

For

(Contractor)

(PMC)

CONTRACTOR

MANAGING DIRECTOR & CEO

FORM B - 10

ACCOUNT OF WORK

Name of Work:

Name of Contractor:

Name of P.M.C.:

Agreement No.:

R.A. Bill No.:

Sr. No.	Description	Amount

For

For

(Contractor)

(PMC)

CONTRACTOR

350

MANAGING DIRECTOR & CEO

FORM B - 11**PAYMENT SCHEDULE AS PER TENDER**

Name of Work :

Name of Contractor :

Name of P.M.C. :

Agreement No. :

R.A. Bill No. :

Sr. No.	Particulars	Type/ Location	Up to plinth level	Gr. Floor	1st Floor	2nd Floor	Terrace	Total
			%	%	%	%	%	%

For

For

(Contractor)

(PMC

CONTRACTOR

MANAGING DIRECTOR & CEO

FORM B - 12**PAYMENT SCHEDULE ABSTRACT**

Name of Work :

Name of Contractor :

Name of P.M.C. :

Agreement No. :

R.A. Bill No. :

Sr. No.	Particular		% as per Schedule	Considered in Previous Bill	Considered Since Previous Bill %	Update
1		Up to plinth				
		Gr. floor				
		Roof				
2		Up to plinth				
		Gr. floor				
		Roof				

For

For

(Contractor)

(PMC)

CONTRACTOR

MANAGING DIRECTOR & CEO

FORM B - 13**DETAIL ABOUT WITHHELD AMOUNTS**

Name of Work :

Name of Contractor :

Name of P.M.C. :

Agreement No. :

R.A. Bill No. :

Sr. No.	Particulars	Amount in Rs.	Reason of Withheld Amount
1	Amounts withheld till previous bill		
2	Amounts proposed to be withheld in this bill		
3	Amounts proposed to be released in this bill		
4	Total amount withheld till this bill		

For

For

(Contractor)

(PMC)

FORM B - 14

REASONS OF DELAY OF WORK IF ANY

Name of Work:

Name of Contractor:

Name of P.M.C.:

Agreement No.:

R.A. Bill No.:

Sr. No.	Month and Year	Description	No of Days

For

For

(Contractor)

(PMC)

CONTRACTOR

MANAGING DIRECTOR & CEO

FORM B - 15**ABSTRACT OF TEST REPORTS**

Name of Work:

Name of Contractor:

Name of P.M.C.:

Agreement No.:

R.A. Bill No.:

Sr. No	Name of material use for work in this bill	Date of receipt	Quantity received	Unit	Testing frequency as per agreement	No. Of tests carried out	Remark about test value
1	2	3	4		5	6	7
1							
2							
3							
4							
5							
6							

For

For

(Contractor)

(PMC)

CONTRACTOR**MANAGING DIRECTOR & CEO**

FORM B - 16

TEST FREQUENCY CHART

Name of Work:

Name of Contractor:

Name of P.M.C.:

Agreement No.:

R.A. Bill No.:

Sr. No.	Item Description	Test	Frequency (MIN)	Unit	Quantity			No of test Required	No of test		Total Test	Remarks
					In this Bill	Up to last	Up to Date		Outside lab	Field lab		

For

For

(Contractor)

(PMC)

CONTRACTOR

MANAGING DIRECTOR & CEO

FORM B - 17**SAMPLES APPROVED DURING PERIOD OF BILL**

Name of Work:

Name of Contractor:

Name of P.M.C.:

Agreement No.:

R.A. Bill No.:

Sr. No.	Name of material	As per agreement		As per approval		Date of approval
		Brand	Details of materials such as thickness, Diameter length, width weight etc.	Brand	Details of materials such as thickness, Diameter length, width weight etc.	
1	2	3	4	5	6	7

For

For

(Contractor)

(PMC)

CONTRACTOR**MANAGING DIRECTOR & CEO**

FORM B - 18**ABSTRACT OF STEEL RECEIPT**

Name of Work:

Name of Contractor:

Name of P.M.C.:

Agreement No.:

R.A. Bill No.:

Sr. No.	Date	INV. NO	Make of steel	Unit	DIAMETER OF BAR					Total	Remarks
					8mm	10mm	12mm	16mm	20mm		
					Steel Received from previous RA Bill						

For

For

(Contractor)

(PMC)

CONTRACTOR

MANAGING DIRECTOR & CEO

FORM B - 19

STEEL CONSUMPTION STATEMENT

Name of Work:

Name of Contractor:

Name of P.M.C.:

Agreement No.:

R.A. Bill No.:

Sr. No.	Description	Diameter of Bar						Total Consumption
		8mm	10mm	12mm	16mm	20mm	25mm	32mm

For

For

(Contractor)

(PMC)

CONTRACTOR

MANAGING DIRECTOR & CEO

FORM B - 20

ABSTRACT OF CEMENT RECEIPT

Name of Work:

Name of Contractor:

Name of P.M.C.:

Agreement No.:

R.A. Bill No.:

Sr. No	Date	Inv. No	Make	Quantity	Unit	Total	Remark
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

For

For

(Contractor)

(PMC)

CONTRACTOR

MANAGING DIRECTOR & CEO

FORM B - 21**CEMENT CONSUMPTION STATEMENT**

Name of Work:

Name of Contractor:

Name of P.M.C.:

Agreement No.:

R.A. Bill No.:

Sr. No	Description of Items	Executed Quantity	Unit	Theoretical ratio of consumption	Consumption in Bags		Remarks
					Theoretical	Actual	
1							
2							
3							
4							

For

For

(Contractor)

(PMC)

FORM B - 22

361

CONTRACTOR

MANAGING DIRECTOR & CEO

BAR CHART

Sr. No.	Particulars	Month/ Weeks	duration	January				February				March.....			
				1	2	3	4	1	2	3	4	1	2	3	4
1															
2															
3															
4															
5															
6															
7															

For

For

(Contractor)

(PMC)

ACQUITANCE

Name of Work: - _____

R.A. BILL NO. _____

Received Rs.

.....

as an intermediate payment in connection with the contract with the M.D. & C.E.O., MHDC Ltd., Mumbai

Date:

Witness:

Signature of contractor and stamp

Paid by me cash/ or cheque No.....

Dated initials of the person

Actually, making the person

CONTRACTOR

MANAGING DIRECTOR & CEO

16.8 Check list for Handing over of the Project:

Name of work:			
Sr. No.	Particulars	Information In the form of Yes/No	Remarks
A)	Technical		
1	Whether water connection is obtained?		
2	Whether electrical connection is obtained?		
3	Whether Sewage Connection is obtained?		
4	Whether electrical fixtures are working?		
5	Whether sanitary fixtures are working?		
6	Whether external colour is as per Client's approval?		
7	Whether landscape is as per Client's approval?		
8	Whether infrastructure works such as Road, GSR, ESR, Storm water drain, Compound wall, etc. are completed in all respect?		
9	Whether inventory of whole project is completed?		
10	Whether all work is executed as per approved drawings?		
11	Whether completion certificate is submitted by Architect / PMC / Contractor / their sub consultant		
B)	Clearance		
12	Whether Occupation certificate is obtained?		
13	Whether Firefighting NOC is obtained?		
14	Whether Lift NOC is obtained?		
15	If any		

Note: This Checklist shall be verified and certified by PMC

Appendix-II TREE PLANTATION DETAILS			
Sr.No.	Names		
A	TREES		
1	Amaltas		
2	Bakul		
3	Coconut tree (without trunk)		
4	Desi Mango		
5	Fishtail Palms		
6	Gulmohar		
7	Jackfruit		
8	Neem		
9	Peltophorum Yellow		
10	Royal palms (without brown trunk)		
11	Saptparni		
12	Tamhan or Queen flower		
13	Kadamba		
14	Mussaenda frandosa		
15	Areca Catechu - Betel nut palm		
B	SHRUBS		
1	Euphorbia Carcasana		
2	Nerium Oleander		
3	Ficus Panda		
4	Canna Variegata		
5	Bougainvillea glabra		
6	Bambusa Vulgaris		
7	Tecoma gaudichaudi		
C	GROUND COVERS		
1	Alternanthera Red (15 to 16 nos per sqm)		
2	Nerium Oleander (15 to 16 nos per sqm)		
3	Ficus Panda (15 to 16 nos per sqm)		
D	CLIMBERS		
1	Allamanda Violacea (7 to 8 nos per sqm)		
E	OTHER ITEMS RELATED TO PLANTATIONS		
1	Cost of digging of each pit	--	--
2	Cost of putting good soil, manure per pit	--	--
3	Cost of Tree Guard per unit	--	--

17 Special Condition of Contract

17.1 General

- 17.1.1 The site infrastructural works for this project shall be carried out by the All the approvals of infrastructure work shall be obtained within 6 months from the date of work order of main work and will be completed along with building works.
- 17.1.2 Though the scope of agreement covers entire work, the work shall be restricted to the Approval of local authority/GOM/MHDC & firm demand from public. If the work of tenements is required to be restricted due to any reason in future, no compensation whatsoever nature shall be payable.
- 17.1.3 Sufficient Precaution form design point of view and execution point of view (like providing safety belt to working labourers and also providing safety nets) are to be taken to enable smooth execution of multi-storied buildings without any untoward incident.
- 17.1.4 The well trained, qualified, experienced structural Engineer and site Engineers should be deployed full time by the contractor and the contractor shall be held responsible for the quality, stability and workmanship of the work.
- 17.1.5 The lifts shall be maintained by the Contractor for **5 years** after the date of receipt of occupation Certificate.
- 17.1.6 And fulfilment of all other obligations of the Contractor in accordance with the provisions of this Contract Agreement and matters incidental thereto or necessary for the any or all of the obligations of the Contractor as per the relevant Acts such as The Real Estate (Regulation and Development) Act, 2016 (RERA), Building Commission and provisions under environmental Clearance.

17.2 Special Condition of Contract for MEP

- Electrical
 - Scope of Work:
The scope for Electrical Building work is from Meter Room Isolator to Flat Internal point wiring, building common area is from meter room to common area lighting, lifts, Water supply Pumps as indicated in Electrical Main SLD.
 - Flat Internal

Flat internal wiring is in casing capping with flush type switches and sockets. Separate conduit of 25mm dia PVC HMS for each flat should be provided for service mains.

- Point Wiring

Point wiring including all fittings and accessories like PVC Trunking (casing capping) ISI mark, 1.2 mm thick, minimum 20 mm width, Switch boards, Fan Hook, modular type switch, sockets. Lopping is as per Indicated in typical flat Layout.

No of Point wiring Points - EWS: 11, (Light – 7, Fan – 2, Bell Point – 1, T.V. point -1) T.V. point is part of T.V. power socket switch board (T.V. SB). – 1 no's

- Power Point 6A, 6/16A & 16A

EWS Flat-

6A Power Point on Switchboard: 2 Nos. (1 Bed Room, 1 Living)

6A Individual Power Point: 3 Nos. (2 Living, 1 Kitchen)

6/16, An Individual Power Point: 1 Nos. (1 Kitchen)

LIG Flat-

6A Power Point on Switchboard: 3 Nos. (1 Bed Room, 1 Living, 1 Master Bedroom)

6A Individual Power Point: 3 Nos. (2 Living, 1 Kitchen)

6/16, An Individual Power Point: 1 Nos. (1 Kitchen)

- DB

SPN DB - 12 Ways SPNDB Surface Mounted.

SPN DB – 8 ways SPNDB Surface Mounted.

Incomer - DP RCCB 25A – 1 Nos.

Outgoings - MCB 10A - 3 Nos.

Outgoings – MCB 16A – 1 Nos.

- LED Light Fixtures

Supply & Installation of Bulkhead in Lift Shaft Lighting (Bulkhead).

- Building Common Area

For DB Scheduled ref. LDB of Common Area --. 12 Ways SPNDB Surface Mounted. Surface Mounted Located at Ground Floor.

Lighting is connected mains by 2x2.5 sq.mm F.R.L.S copper PVC insulated wire till first light point from further sub mains by 2x1.5 sq.mm F.R.L.S copper PVC insulated wire.

No of wires in one conduit is as per technical specifications.

- Meter room

Meter room wiring is as per standards. Contractor to consider all details, like Meter room isolator (MCCB), Bus bar, DP Isolator, required trunking / conduiting, clamping as and where required including all equipment's and associates has to be consider/provide to complete meter room electrical work as required by utility company.

- Infra Electrical from SS to Meter Room Incomer

The Electrical infra work are including LT Feeder Pillars, Cables from Substation up to LT feeder Pillar, Meter room Incomer cable from Feeder Pillar. All above Infra Work will be as per Approval of MSEDCL.

Contractor to consider all Civil Work for Feeder Pillar, Danger Board, Concrete Foundation.

Feeder Pillar, LT Cable, Gland, as per MSEDCL specification,

Electrical infrastructure including Commissioning of Electrical Energy meter has to be carried out as per the approved MSEDCL specifications.

- Electrical Contractor Criteria

- Pre-Requirement Submittals at the Time of Agreement with Successful Bidder

Tenderer should have Electrical sub-contractor company profile, Class A & above registration in PWD / CPWD, written agreement with main contractor along with list of similar works carried out by him in last five years.

- Criteria of Subcontractors

For Electrical services if sub-contractors are proposed by the Bidder must have executed similar type of respective works (Except Solar PV Project) value of the work should not be less than **Rs. 1600000/- (Sixteen Crore Only)** of Electrical installation work including Street light, Distribution Transformer, Feeder Pillar etc. services given in the tender. Each such subcontractor proposed for the work must have written approval of their eligibility from Engineer (Elec.), MHDC.

- Third Party Inspection (Inspection, Testing and Inspection Certificates):

The MHDC Representative or duly authorized representative shall have at all reasonable times free access to the Bidder's premises or works and shall have the power at all reasonable times to inspect and examine the materials and workmanship of the works during its manufacture or erection, if part of the works is being manufactured or assembled at other premises or works, the Bidder shall obtain permission to inspect as if the works were manufactured or assembled on the Bidder's own premises or works. Inspection may be made at any stage of manufacture, dispatch or at site at

the option of the MHDC and the equipment if found unsatisfactory due to bad workmanship or quality, material is liable to be rejected. All equipment being supplied shall conform to type tests and shall be subject to routine tests in accordance with requirements stipulated under respective sections. Bidder shall submit the type tests reports for approval. The Bidder shall intimate the MHDC Representative the detailed program about the tests at least (30) days in advance in case of domestic supplies. If for any item type test were pending payment would be made on successful completion of type/routine test(s) actually carried out as per the MHDC instructions. The Bidder shall give the MHDC thirty (30) days written notice of any material being ready for testing. Such tests shall be to the Bidder's account. The Engineer-in-Charge of MHDC and PMC unless witnessing of the tests is virtually waived will attend such tests within thirty (30) days of the date of which the equipment is notified as being ready for test/inspection, failing which the Bidder may proceed. With the test which shall be deemed to have been made in the presence of MHDC Representative and he shall forthwith forward to the Engineer's Representative duly certified copies of tests in triplicate.

The MHDC within (30) days from the date of inspection as defined shall inform in writing to the Bidder of any objection to any drawings and all or any equipment and workmanship which in his opinion is not in accordance with the Contract. The Bidder shall give due consideration to such objections and make the necessary modifications accordingly.

When the factory tests have been completed at the Bidder's or Sub Bidder's works, the MHDC shall issue a certificate to this effect within fifteen (15) days after completion of tests but if the tests are not witnessed by the MHDC, the certificate shall be issued within fifteen (15) days of receipt of the Bidder's Test certificate by the MHDC. Failure of the issue such a certificate shall not prevent the Bidder from proceeding with the works. The completion of these tests or the issue of the certificate shall not bind the MHDC to accept the equipment should, it, on further tests after erection, is found not to comply with the Specification. The equipment shall be dispatched to site only after approval of test reports and issuance of material inspection clearance certificate by the MHDC. For tests whether at the premises or at the works of the Bidder or of any Sub Bidder, the Bidder except where otherwise specified shall provide free of charge such items as labour, materials, electricity, fuel, water, stores, apparatus and instruments as may be required by MHDC

Representative or this authorized representative to carry out effectively such tests of the equipment in accordance with the Specification. The inspection by MHDC Representative and issue of Inspection Certificate thereon shall in no way limit the liabilities and responsibilities of the Bidder in respect of the agreed quality assurance program forming a part of the Contract. The MHDC will have the right of having at his own expenses any other tests(s) of reasonable nature carried out at Bidder's premises or at site or in any other place in addition of aforesaid type and routine tests to satisfy that the material complies with the specifications. The MHDC Representative reserves the right for getting any field tests not specified in respective sections of the technical specification conducted on the completely assembled equipment at site. The testing equipment for these tests shall be provided by the Bidder. All expenses relevant to the preparation and performance of testing, inspection and preparation of any test reports or certificates shall be borne by the Bidder/seller EXCEPT for the salaries, fees, traveling, lodging and boarding expense of the MHDC and PMC. However, if the visit duration of MHDC is extended for the reasons not attributable to MHDC, the cost of the extended period of visit shall be borne by the Bidder/seller. The Bidder/sellers shall carry out tests related to performance tests as described in the specifications and specified in the order. All such performance tests shall be at supplier costs. Supplier shall also provide all the tests certificates and documents as demanded by the Inspector for his satisfaction that the order has been executed as per PO specifications. All such certificates, documents in original shall be submitted to the MHDC before dispatch of material. The goods shall be dispatched from suppliers shop only after written confirmation from MHDC. The contractor shall consider all cost towards inspection of goods by MHDC & PMC at factory / manufacturers works prior to shipping for 2 to 3 persons. (Travelling (Air / Train AC/ or Vehicle) / stay etc. complete).

○ Fee, Permits & Tests: -

The Bidder shall obtain all sanctions and permits required for the above said works from all the relevant authorities before & after the execution of the works. On completion of the work, the Bidder shall obtain N.O.C from concerned authorities including, Electrical Inspector (PWD). The original of the same shall be delivered to the MHDC through Engineer-in-Charge. The MHDC shall have full power regarding the equipment/ materials get tested by authorized/ recognized independent agency at the Bidder's expense in order to prove their soundness and adequacy. The Bidder will rectify the defects/

suggestions pointed out by independent agency through MHDC at Bidder's expense. The installation shall comply in all respects with the requirements of Indian Electricity Act 2003, CEA 2010, Indian Electricity Rules Regulation 2010 and other related Laws and Regulations (for F.F. etc.), IS 732 as amended up to date, there under and special Requirements, if any, of the State Electricity Boards etc. The Bidder shall be liable to furnish the list of authorized licensed persons/ employed/ deputed to carry out the works/ perform the assigned duties to fulfil the requirement of Rule No.3 of regulation 2010 as amended up to date.

- Lifts

- Passenger Auto Lift

As per Schedule C

- LIFT CONTRACTOR

Lift contractor/Manufacture should have been installed at least 50 lifts in India Govt /Semi-Govt organization. The Lift Contractor shall have valid Electrical Contractor License & Lift Contractor license from Chief Electrical Inspector, Industries, Energy & Labour Department, Mumbai to erect & maintain Two speed/ Variable speed Automatic Door Lifts in Maharashtra State.

- Plumbing

- Flat Internal

- Kitchen sink without drain board: 600x450 mm bowl depth 178 mm
 - C.P. sink cock with raised J shaped swinging casted spout
 - Providing and fixing white vitreous China pedestal type water closet (European type W.C. pan) with seat and lid, 10-liter low level white P.V.C. flushing cistern, including flush pipe, with manually controlled device (handle lever), conforming to IS:7231, with all fittings and fixtures complete, including cutting and making good the walls and floors wherever required: W.C. pan with ISI marked white solid plastic seat and lid.
 - Providing and fixing Anglo-Indian type white glazed earthenware water closet pan with trap with UPVC seat and lid with chromium plated brass hinges and rubber buffers including UPVC and vent pipe up to the outside face of the wall with flushing cistern and necessary fitting including cutting and making good to the walls and floors testing etc. complete (prior approval of sample and brand by Ex Engineer is necessary before use).
 - Bib Cock (in WC area).
 - Wall Hung Wash Basin with C.I. brackets, wall mounted 15 mm long body C.P. brass pillar tap, 32 mm C.P. brass waste of standard pattern, including

painting of fittings and brackets, cutting and making good the walls wherever require: White Vitreous China Flat back wash basin size 450x 300 mm with wall mounted 15 mm long body C.P. brass pillar tap.

- Terrace Looping

ISI marked UPVC pipe and fittings SCH 80, UV stabilized as per IS-4985, including all UPVC plain & threaded fittings, fixing on the wall with Hi-tech clamp with EPDM (FLAT) rubber, Z bracket 5mm thick.

- Pipes

All Plumbing Pipe on wall vertical/Horizontal runs have suitable pipe support as per specification of pre-fabricated GI coted MS Brackets.

- Water Supply

- Internal Cold-Water Supply - CPVC SDR - 11
- Cold Water Supply Down take - CPVC SCH-40
- Cold Water Terrace Ring Main -UPVC pipe and fittings SCH 80, UV stabilized as per IS-4985
- Rising Main (From UG Tank to OHT) - GI B Class (Medium Grade).

- Drainage

- Internal Drainage (In toilet, Bath & Kitchen)- SWR Poly Vinyl Chloride (PVC) class B 6Kg/Cm² pipes.
- Drainage Down take in shaft till Ground Floor Ceiling (In toilet, Bath & Kitchen) - SWR Poly Vinyl Chloride (PVC) class B 6Kg/Cm² pipes.
- Ground Floor Ceiling suspended to First Chamber/Gully Tap near shaft - Cast Iron LA Class Pipe.

- Rain Water

- Rain water Down Take from Terrace to Ground Floor Ceiling - P.V.C. Rain water pipes of 75/160mm outer diameter and having wall thickness of 2.2 to 2.7 mm.

- Pumps & Water Distribution

- The total water is stored in UGWT located at ground floor. There will be two compartments in the UGWT, each will have.
 - Domestic Water tank,
 - Flushing Water tank.
- The potable water from municipal water or various sources of water supply shall be collected in domestic water tank.
- Through a set of a submersible pump water will transfer from domestic UGWT to Firefighting OHT. Domestic OHT Capacity is half day storage.

- Total Pumping Time shall be 2.5 to 3 Hr/Day for to transfer Water from UG to OHT.
 - Water from Fire Fighting OHT will be over flow to Domestic Water OHT tank this is in order to keep water fresh, clean & to prevent stagnation of water in firefighting OHT.
 - Water from Domestic water tank flows through pipes by gravity to the ring main at the Terrace level and in turn the same will feed to Wash basin, Health faucet.
 - The sizing of the entire distribution network is based on the simultaneous use of fixture unit's demand.
 - A shut off valve is provided within the battery limit of each block, which serves for the purpose of maintenance of any utility. Individual toilets will be provided with main control valve for isolation and maintenance of the same.
- Pumps Specification

Particulars	Details
Pump Type	Submersible Bore well type
MOC	
Suction and Stage Casing	FG 200 Cast Iron
Suction Strainer	AISI 304
Impeller	NORYL
Shaft	AISI 410
Shaft Sealing	Oil Seal
Motor	Water Filled Submersible Type
Pump Diameter	Six Inch
Cable Included	3 meters
Accessories required	MS Stand for Pump horizontal mounting
Head	65m
Flow	To be calculated. (LPM)
Make	Kirloskar, KSB, Willo

- Fire Fighting
Firefighting system is as per Fire NOC.

- Pipes
 - Above Ground - GI C Class Pipe.
 - Below Ground - GI C Class Pipe with coating of Bitumen Paint & 4mm thick Wrapping as per IS 10221 for underground fire pipe of Fire Fighting System.
 - Note: Dually painted with one coat of zinc chromate & 2 coat of Enamel Paint (Fire Red).

- Valves
 - Gun metal SS single outlet hydrant valve fitted with necessary accessories.
 - Cast iron double flange sluice valve
 - Cast iron double Wafer Type flange butterfly valve
 - Double flange Plate Type NRV as per approved by Fire CFO.
 - End line Strainer for +ve suction with Drain connection.
- Firefighting accessories
 - Wall mounting swinging Hose reel drum fitted with 19 mm dia. 30m high pressure polypropylene (Polyhose) long fitted with necessary accessories
 - Fire Shaft Door size as per layout for housing Floor Hydrant valve, hose pipe, hose reel and branch pipe made from 16 SWG sheet and angle iron 25 mm. x 25 mm. x 4 mm. having front doors with viewing glass (8"x6") and locking arrangement with necessary fixing material such as rubber bidding etc. duly painted in post box red colour.
 - R.R.L. Hose pipe, 63mm dia, 15m in length, fitted with necessary accessories.
 - SS branch Pipe 63 mm Ø fitted with 20 mm dia detachable hexagonal nozzle.
 - Air Vessel of 300 mm Ø 1.5m in height M.S. tank installed inside pump house.
 - Brass orifice plate having 6 mm. thickness, 140 mm outer dia of 140 mm at every single outlet hydrant valve
 - Round bottom FIRE Bucket of 9 Liters capacity as per IS: 2546 made out of 24-gauge G.I. sheet with extra handle at bottom duly painted white inside and red outside with FIRE Mark.
 - Provision of Fire Escape Chute provisions as per the requirement as per vendor specifications.
 - Floor mounting Stand for keeping 4 Nos. of FIRE buckets 1500mm in length, 900mm in height frame made out of 30x30x4 mm angle iron with cross supports for legs, welded with 4 hooks and duly painted with one coat of red lead and two coats of silver paint
 - ABC powder type fire extinguisher as per IS : 13849 of 6 kg capacity with necessary clamp.

- Sign boards made out of 3mm thick "Opaque" PVC foam board with Laptop cut, PVC non-reflective self-adhesive vinyl painted foam board, complete with mirror fasteners.
- Check Fire NOC for more details.
- Fire sprinkler system in Commercial Building.
- Fire Alarm.
 - As per NBC 2016-chapter 4 Fire & Life Safety Table No. -7
- Public addressable
 - As per NBC 2016-chapter 4 Fire & Life Safety Table No. -7

CHAPTER NO. 18

18 Forms

18.1 Form I A: Details of list of works in hand as on submission of Tender

Sr. No.	Name of the work	Agreement No.	Tendered Amount/ Project Cost	Date of commencement	Stipulated date of Completion	Value of work already done (Rs. In lakhs)	Value of balance work (Rs. in lakhs)	Value of balance work to be done during this contract period.	Probable date of completion	Remarks
1	2	3	4	5	6	7	8	9	10	11

Note: -

- i. Works in hand shall be supported by certificate issued by agency / authority/Executive engineer. In case private work if any by client / Architect.
- ii. Any wrong information/misinformation in this regard would invite disqualification and legal action

Seal and Signature of Bidder

Signature of Chartered Accountant

18.2 Form I B: List and Details of Tendered Works

Sr. No.	Name of the work	Name and address of the client	Tendered Amount Rs. In lakh	Time Limit	Probable date when decision expected	Other relevant date if any
1	2	3	4	5	6	7

Note: -

- i. Any wrong information/misinformation in this regard would invite disqualification and legal action

Seal and Signature of Bidder

18.3 Form II: List of Plant and Machinery:

List of Plant and Machinery immediately available with the Tenderer:

Sr. No.	Name of equipment	No. of Units	Kind and make	Capacity	Age and Condition	Present location	Remarks
1	2	3	4	5	6	7	8

Note: -

- i. Any wrong information/misinformation in this regard would invite disqualification and legal action

Seal and Signature of Bidder

18.4 Form III: Details of work of similar type and magnitude:

Details of work of similar type and magnitude carried out by contractor during last five years
i.e., from 2017 to 2022

Sr. No.	Name of work	Name and Address of the organization for whom the work was done	Place and Country	Agreement No.	Date of commencement	Tendered cost	Total cost of work done	Date of completion	Principle Features & Built-up area (sq. Mtr.)
1	2	3	4	5	6	7	8	9	10

Note: -

- The similar works completed shall be supported by the certificated issued by officer not below the rank of Executive Engineer.
- In case of the private works, the work done certificate under the hand and seal of Owner / Architect / Project Management Consultant shall be submitted along with Commencement certificate / letter of approval or completion / occupancy certificate of Municipal Corporation, Municipal Council or Town Planning Authority, any Government Authority concern.”.
- Any wrong information / misinformation in this regard would invite disqualification and legal action.

Seal and Signature of Bidder

18.5 Form IV: Details if Technical Personals available

Details of Technical Personal available with the contractor since Five years:

Sr. No.	Name of Personnel with age	Designation	Qualifications	Whether working in field or office	Experience of execution of similar work	Length of service of the persons who is working with the tenderer (in Years)	Remarks
1	2	3	4	5	6	7	8

Note: -

- Project in charge from technical person shown in above table shall be available on site of work during execution.
- Any wrong information/misinformation in this regard would invite disqualification and legal action

Seal and Signature of Bidder

18.6 Form No- V: Statement Showing Payment Received

Statement showing payment received (i.e., receipt) in all classes of civil engineering construction works only during last five years.

Sr. No.	Name of Work	Name of Client and Address, PAN No.	Amount put to tender / tendered cost.	Agreement No.	Date of Commencement	Type and Nature of Work	Amount received for the work done during each of last five years.					Amount of work still remaining	Expected date of completion.
							16-17	17-18	18-19	19-20	20-21		
1	2	3	4	5	6	7	8	9	10	11	12	13	14
1													
2													
3													
4													
5													
						Grand Total							

Sign & Seal of Chartered Accountant

Seal & Sign of Bidder.

Note: -

- Form V shall be certified by the Chartered Accountant on each page with seal and unique document identification number (UDIN) by Institute of Chartered Accountant of India (ICAI) under signature of Chartered Accountant as per above, If not found certified UDIN by ICAI then it will not be considered.
- Advance receipt shall not be considered as amount received for the work done.
- Any wrong information / misinformation/ incomplete information in this regard would invite disqualification and legal action.

The bid capacity will be calculated for the maximum turnover from all types of civil engineering works (total receipts) as per form V.

CHAPTER NO. 19

19 Drawings

19.1 Contract drawings: -

The Contract Drawings provided for tendering purpose with the tender documents shall be used as a reference only. Contractor should visualize the nature and type of work contemplated and to ensure that the rates and prices quoted by him take due consideration of the complexities of work involved during actual execution / construction as experienced contractors in the field.

The tendered rates/prices for the work shall be deemed to include the cost of preparation, supply and delivery of all necessary drawings, prints, tracings and negatives which the contractor is required to provide in accordance with the contract.

19.2 Documentation:

If so, ordered by the Engineer-in-charge, the contractor will prepare drawings of the work as constructed and will supply original and three copies to the Engineer who will verify and certify these drawings.

Final as constructed drawings shall then be prepared by the contractor and applied in triplicate along with a micro-film of the same to Engineer for record and reference purpose at the contractor's cost.

**MAHARASHTRA HOUSING
DEVELOPMENT CORPORATION
LIMITED MUMBAI**

(GOVERNMENT OF MAHARASHTRA UNDERTAKING)

**TENDER FORM
(TENDER DRAWING)**

**Construction of Affordable Housing Project of 1824 EWS Tenements Under Pradhan
Mantri Awas Yojana (U) on Kh. No 59/2 & 60, Bhilgaon, Tal. Kamptee, Dist. Nagpur
(Package – 2)**

ARCHITECT, RCC AND MEP LAYOUTS

ARCHITECT SITE PLANS

BUILDING FLOOR PLANS

BUILDING ELEVATIONS

BUILDING SECTIONS

TYPICAL UNIT PLANS

MEP ALL FLOOR BUILDING PLANS

RCC ALL FLOOR BUILDING PLANS

LINK: